MINIMAL OBSTRUCTIONS FOR THE SIZE RAMSEY NUMBER OF STAR FORESTS, I: CERTIFICATES, ARROWING GRAPHS AND THE REDUCTION TO FIVE DEGREES

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Abstract. We prove a reduction of below-budget Ramsey hosts for arbitrary pairs of star forests to contained minimal obstructions of degrees 4, 6, 8, 10, and 12, using the certificate and structural inputs marked for completion in this draft. Exact star-pool residuals preserve the entire targets, including unit components and multiplicities. An edge-minus-matching potential gives the diagonal lower bound up to an explicit matching deficit, and equality permits an exact normalization preserving a prescribed maximum matching. Deletion identities restrict the universal graph input to a class of regular factor-critical graphs. The resulting criterion is an equivalence; the emptiness of the five obstruction classes remains an additional input. The main result of this paper was obtained with the assistance of the Ansatz system, an automated proof-search and verification system.

[author/blocker] No author, affiliation, or email was supplied.

[metadata/blocker] The Mathematics Subject Classification remains to be supplied.

[prime/blocker] [GAP: The complete certificate definition and several substantial technical predecessors are absent from the supplied material. Their required conclusions are recorded explicitly below as pending inputs. Consequently this draft does not yet give an unconditional, self-contained proof of the reduction.]

[instruction/conflict] The requested one-sentence chapter proof notices cannot replace the supplied mathematical proofs without losing content. Those notices are retained, and the supplied supporting arguments are included in Section 5. Missing technical developments remain flagged.

[compile/blocker] Compilation, citation and reference resolution, and the requested page limits have not been verified in this environment.

1. Introduction

Background. The size Ramsey problem for two star forests asks for the least number of host edges that force one of two prescribed monochromatic forests. For positive integers

$$n_1 \geq \cdots \geq n_s \geq 1, \qquad m_1 \geq \cdots \geq m_t \geq 1,$$

the proposed value is the diagonal sum

$$\widehat{r}\left(\bigsqcup_{i=1}^{s} K_{1,n_i}, \bigsqcup_{j=1}^{t} K_{1,m_j}\right) = \sum_{k=2}^{s+t} \max_{\substack{1\leq i\leq s,\ 1\leq j\leq t \\ i+j=k}} (n_i + m_j - 1). \tag{1.1}$$

This formulation is recorded in [DJK+21, Conjecture 1]. The union of stars whose capacities are the diagonal maxima gives the corresponding upper bound, as explained at the beginning of [DJK+21, Section 2].

Known cases include lists whose entries are all odd, and the cases of one star or two equal stars against a star forest with component sizes at least two; see [DJK+21, Theorems 2.3–2.5]. The arguments developed here retain arbitrary repetitions, gaps, parities, and unit components. Their

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graph-theoretic setting draws on matching structure and prescribed-degree subgraphs [LP86, Lov70, FLL+23].

This first companion develops the certificate framework and the reduction needed to study the lower bound in (1.1). The intended main paper derives the formula after the five obstruction classes have been excluded. The second companion uses the obstruction structure and deletion identities established here.

[cite/blocker] [GAP: Bibliographic entries for the intended main paper and second companion were not supplied. The requested series citation labels Main and II cannot yet be inserted as resolved citations.]

1.1. **Contained minimal obstructions.** The first result keeps the obstruction inside the actual host. We use the full-target notation of Setup 2.2, the certificate quantities of Notation 3.1, and the graph properties of Definitions 3.2 and 3.3. In particular, $\mathcal{A}_q$ contains all nonempty actual suffix pairs after legal leading-component deletions totaling the integer q.

Theorem A (Contained minimal obstructions)**.** *Let F_1, F_2 be arbitrary nonempty star forests with positive nonincreasing integer component-size lists*

$$n_1, \dots, n_s, \qquad m_1, \dots, m_t, \qquad s, t \in \mathbb{Z}_{\geq 1}.$$

Put

$$M = s + t - 1, \qquad \ell_k = \max_{\substack{1 \leq i \leq s,\ 1 \leq j \leq t \\ i+j=k}} (n_i + m_j - 1) \quad (2 \leq k \leq s + t).$$

Fix an integer q with $0 \leq q < M$, and put

$$K = M - q, \qquad L_q = \sum_{k=q+2}^{s+t} \ell_k.$$

Let $\mathcal{A}_q$ contain every ordered nonempty actual suffix pair after legal leading-component deletions totaling q. Use the generic universal-allocation certificate costs, their row maxima c^*_θ, their sum Σ, and

$$\mathrm{ex} = c_{0,0} - \nu, \qquad \mathrm{gap} = \Sigma - e.$$

Let tr *be the top missing-allocation run, and put*

$$h_-(D) = \max\left(4, \left\lfloor \frac{D}{2} \right\rfloor + 1\right) \qquad (D \in \mathbb{Z}_{\geq 0}).$$

A proper subgraph may delete vertices, edges, or both.

(i) *Let H be a finite simple graph that arrows every entire prescribed-color member of $\mathcal{A}_q$, and assume that $e(H) < L_q$. There are an actual connected component C of H and a subgraph F of C such that*

$$\mathrm{ex}(C) \geq K - \nu(H) \geq 4, \qquad \mathrm{gap}(C) \geq 1,$$

and F is a totally critical closed d-lump satisfying

$$d \in \{4, 6, 8, 10, 12\}, \qquad d \leq \Delta(H), \qquad \mathrm{gap}(F) = 1, \qquad \mathrm{tr}(F) + \mathrm{ex}(F) \geq h_-(d) + 1.$$

Every proper subgraph of F satisfies T_-. Every proper subgraph J with $\mathrm{ex}(J) \geq 4$ is LP-good, meaning $\Sigma(J) \leq e(J)$.

The graph F need not be induced in H. No inequality $\mathrm{ex}(F) \geq K - \nu(H)$ is asserted.

Write $|V(F)| = 2k + 1$, where $k \in \mathbb{Z}_{\geq 0}$. If $\mathrm{ex}(F) \geq 4$, then for every edge $a \in E(F)$ and every integer $\theta \geq 0$,

$$r_\theta(a) := c^*_\theta(F) - c^*_\theta(F - a) \in \{0, 1\}, \qquad c^*_0 = c_{0,0}.$$

These drops satisfy

$$r_\theta(a) = 0 \quad (\theta \geq d + 1), \qquad r_d(a) = 1.$$

Moreover,

$$\nu(F-a)=k, \qquad \operatorname{ex}(F-a)=\operatorname{ex}(F)-r_0(a),$$

and

$$R(a):=\sum_{\theta=0}^{d} r_\theta(a)\geq 2, \qquad \operatorname{gap}(F-a)=2-R(a)\leq 0.$$

Thus every edge strictly lowers some row below d; the row may depend on the edge.

If $d=4$, every proper subgraph J of F satisfies

$$\operatorname{gap}(J)>0 \implies \operatorname{gap}(J)=1, \qquad \operatorname{tr}(J)+\operatorname{ex}(J)\leq 3.$$

Every such proper subgraph with $\operatorname{ex}(J)\geq 3$ is LP-good.

(ii) *Let H be a finite simple graph. Assume that every totally critical closed d-lump subgraph F of H with*

$$d\in\{4,6,8,10,12\}, \qquad \operatorname{gap}(F)=1,$$

and with every proper subgraph satisfying T_-, obeys

$$\operatorname{tr}(F)+\operatorname{ex}(F)\leq h_-(d).$$

If $e(H)<L_q$, then H has an edge coloring avoiding both entire prescribed targets of some member of $\mathcal{A}_q$.

Every such coloring extends over the unchanged canonical q-star pool of capacities $\ell_2,\dots,\ell_{q+1}$ to avoidance of both full original targets. It also extends across any specified actual outer q-vertex set with residual graph H, preserving every residual edge color.

No natural-degree, head, potential, exact-boundary, or equality-profile restriction on H is imposed. This is a contained-subgraph obstruction and conditional avoidance theorem. It does not establish the universal five-degree bounds or preservation of the ambient degree.

The minimality in Theorem A includes spanning edge deletions. This permits the row-drop conclusions while preserving containment in an actual connected component. The passage from induced to arbitrary subgraphs accounts for the possible loss of inducedness.

1.2. **The universal graph input.** The second result restricts the universal graph question to graphs satisfying all available deletion identities. The class $\mathcal{L}_{\mathrm{rig}}$ is defined in Definition 7.7. Its definition includes the five allowed degrees, the excess restrictions, exact vertex drops, incident-edge batch drops, and a conditional ledger for vertex sets.

Theorem B (Universal criterion on the rigid class)**.** *Use the generic universal-allocation certificate quantities*

$$c_\theta^*, \quad c_0^*=c_{0,0}, \quad \Sigma, \quad \operatorname{ex}=c_{0,0}-\nu, \quad \operatorname{gap}=\Sigma-e, \quad \operatorname{tr}.$$

For every integer $D\geq 0$, put

$$h_-(D)=\max\left(4,\left\lfloor\frac{D}{2}\right\rfloor+1\right), \qquad h_0(D)=\max\left(4,\left\lceil\frac{D}{2}\right\rceil+1\right).$$

Let T_- and T_0 be the properties of Definition 3.3, and let $\mathcal{L}_{\mathrm{rig}}$ be the class of Definition 7.7. The following assertions are equivalent:

(i) *Every finite simple graph satisfies T_-.*
(ii) *Every finite simple graph satisfies T_0.*
(iii) *For every $d\in\{4,6,8,10,12\}$, every totally critical closed d-lump F with $\operatorname{gap}(F)=1$ satisfies*

$$\operatorname{tr}(F)+\operatorname{ex}(F)\leq h_-(d).$$

Here total criticality means that every vertex deletion strictly lowers every positive row $1,\dots,d$.

(iv) *The class $\mathcal{L}_{\mathrm{rig}}$ is empty.*

(v) *Every* $F \in \mathcal{L}_{\mathrm{rig}}$ *satisfies*

$$\mathrm{tr}(F) + \mathrm{ex}(F) \le h_-(d), \qquad d = \Delta(F).$$

Thus the five run-plus-excess bounds, in increasing degree order, are

$$4, \quad 4, \quad 5, \quad 6, \quad 7,$$

and they may be restricted to $\mathcal{L}_{\mathrm{rig}}$ *without changing the universal graph input.*

Under these equivalent assertions, every finite simple graph C *satisfying*

$$\mathrm{ex}(C) \ge h_0(\Delta(C))$$

is LP-good.

There is also a local statement. Let F_1, F_2 *be arbitrary full nonempty positive star forests, with sorted integer component-size lists as in Setup 2.2. For any integer* q *with* $0 \le q < M$*, every finite simple graph* H *satisfying*

$$H \longrightarrow (F_R, F_B) \quad \text{for every } (F_R, F_B) \in \mathcal{A}_q, \qquad e(H) < L_q,$$

contains a subgraph $F \in \mathcal{L}_{\mathrm{rig}}$ *with* $\Delta(F) \le \Delta(H)$*.*

Consequently, if all subgraphs F *of* H *belonging to* $\mathcal{L}_{\mathrm{rig}}$ *satisfy*

$$\mathrm{tr}(F) + \mathrm{ex}(F) \le h_-(\Delta(F)),$$

then $e(H) < L_q$ *guarantees a coloring avoiding both entire targets of some actual member of* $\mathcal{A}_q$*. In particular, absence of all such subgraphs is sufficient. Every such coloring extends over the unchanged canonical* q*-star pool and across any specified actual outer* q*-vertex set to avoidance of both unchanged full originals, preserving all residual edge colors.*

The contained graph need not be induced. None of the equivalent universal conditions, the emptiness of $\mathcal{L}_{\mathrm{rig}}$*, or the unrestricted Ramsey formula is asserted.*

The threshold in the LP-goodness conclusion is inclusive. The equivalence between T_- and T_0 is a statement about their validity on all finite simple graphs; it is not a pointwise equivalence at odd maximum degree.

Sketch of the proofs. The pool theorem assigns the largest target component that fits each pool star. Exchanges between a pool position and the core only trim or discard a core requirement. Theorem 4.2 therefore gives an equivalence for entire copies. Lemma 4.3 provides the corresponding extension across an actual deleted vertex set, including copies that meet that set through leaves.

Theorem 4.4 applies the potential $e - \nu$ to every available tail family. Under equality, Theorem 4.5 preserves the restriction of any initially prescribed maximum matching. Theorem 4.8 then replaces successive full-degree deletions by whole isolated stars without changing either edge count or matching number.

The assembly first extracts an induced failure inside a connected component. Minimizing among its subgraphs gives Theorem A. Theorems 6.2 and 6.3 impose the conditions defining $\mathcal{L}_{\mathrm{rig}}$. A minimal failure belongs to that class, which proves the converse implications in Theorem B.

Structure of the paper. Section 2 fixes notation. Sections 3, 4, 5, 6, and 7 give the five chapter interfaces in order. Section 5 retains the supplied supporting proofs and identifies their pending premises. Section 6 proves the assembly results. Section 7 describes their use in the series.

Acknowledgements.

Remark 1.1 (Automated assistance). The main result of this paper was obtained with the assistance of the Ansatz system, an automated proof-search and verification system. [ack/blocker] No verification statement was supplied; independent re-verification is not claimed. Because automated systems have limitations, it is possible that we have missed related references in the literature, and we welcome comments from experts.

Remark 1.2. This is the first technical companion to [Main]. Its results are the certificate framework and the reduction of the size Ramsey formula to the emptiness of five classes of minimal obstructions. All statements of this paper were produced and verified within the Ansatz automated proof-search and verification system, and each was additionally reviewed by two independent automated audits; this is not a substitute for checking by a human referee. The granular lemmas lying below the level presented here are recorded in the verified fact record [Rec].

2. Preliminaries

This section fixes the notation used by the chapter statements and the assembly. We use the usual terminology for star forests and matchings [DJK+21, LP86].

Notation 2.1 (Graphs and integer parameters)**.** All graphs are finite and simple. The quantities $e(J)$, $\nu(J)$, $d_J(v)$, and $\Delta(J)$ denote edge count, maximum matching size, vertex degree, and maximum degree. We put $\Delta(\varnothing) = 0$. For a vertex set X, the graph $J - X$ is induced on $V(J) \setminus X$. For an edge set S, the graph $J - S$ retains every vertex and deletes exactly the edges of S. In particular, $J - Q$ deletes only edges when Q is a matching.

Disjoint union is denoted by $\sqcup$. Containment means ordinary subgraph containment unless inducedness is stated. The empty forest occurs in every graph. Every nonempty target component has a positive number of edges.

All component sizes, capacities, indices, thresholds, matching budgets, deletion counts, degrees, and run lengths are integers. Unless a more restrictive domain is displayed, counting parameters lie in $\mathbb{Z}_{\geq 0}$.

Setup 2.2 (Full targets and actual suffixes)**.** Let $s, t \in \mathbb{Z}_{\geq 1}$, and let

$$n_1 \geq \cdots \geq n_s \geq 1, \qquad m_1 \geq \cdots \geq m_t \geq 1$$

be integer lists. Put

$$F_1 = \bigsqcup_{i=1}^{s} K_{1,n_i}, \qquad F_2 = \bigsqcup_{j=1}^{t} K_{1,m_j}, \qquad M = s + t - 1.$$

We also write $F_R = F_1$ and $F_B = F_2$ when the prescribed colors are emphasized. The notation $G \to (F_1, F_2)$ means that every red-blue edge coloring of G contains a red F_1 or a blue F_2.

For every integer k with $2 \leq k \leq M + 1$, put

$$\ell_k = \max_{\substack{1 \leq i \leq s,\ 1 \leq j \leq t \\ i+j=k}} (n_i + m_j - 1), \qquad L = \sum_{k=2}^{M+1} \ell_k.$$

For every integer q with $0 \leq q \leq M$, put

$$P_q = \sum_{k=q+2}^{M+1} (\ell_k - 1).$$

Empty sums are zero.

For integers $0 \leq \alpha < s$ and $0 \leq \beta < t$, write

$$F_1^{\alpha} = \bigsqcup_{i=\alpha+1}^{s} K_{1,n_i}, \qquad F_2^{\beta} = \bigsqcup_{j=\beta+1}^{t} K_{1,m_j}.$$

For $0 \leq q < M$, define

$$\mathcal{A}_q = \left\{ (F_1^{\alpha}, F_2^{\beta}) : \alpha + \beta = q,\ 0 \leq \alpha < s,\ 0 \leq \beta < t \right\},$$

$$K = M - q, \qquad L_q = \sum_{k=q+2}^{M+1} \ell_k, \qquad T_q = \bigsqcup_{k=2}^{q+1} K_{1,\ell_k}.$$

Every member of $\mathcal{A}_q$ consists of actual nonempty suffix forests. No unit component or repeated component is removed except by its specified leading cut.

3. Certificates and costs

The goal of this section is to fix the certificate quantities appearing in Theorems A and B. Matching partitions and deletion questions provide related background [TY19, QLS+19, QLS+19b, LM24].

Notation 3.1 (Universal-allocation certificate quantities)**.** For a finite simple graph J and integers $A, B \geq 0$, let $c_{A,B}(J)$ denote the intended generic universal-allocation certificate cost. Its defining requirement is a bounded-degree/bounded-matching edge partition for *every* allocation into two nonnegative integer matching budgets of the specified total. An existential choice of one allocation is not the defining requirement.

Write $c^*_\theta(J)$ for the row maxima, where $\theta \in \mathbb{Z}_{\geq 0}$, and put

$$c^*_0(J) = c_{0,0}(J), \qquad \Sigma(J) = \sum_{\theta \geq 0} c^*_\theta(J),$$

$$\mathrm{ex}(J) = c_{0,0}(J) - \nu(J), \qquad \mathrm{gap}(J) = \Sigma(J) - e(J).$$

The intended sum defining Σ is finite. A graph is *LP-good* when $\Sigma(J) \leq e(J)$.

[math/blocker] [GAP: The exact admissibility predicate defining the bounded-degree/bounded-matching certificate partition, and the precise indexing of its row maxima, were not supplied. This notation preserves the intended quantities and their universal quantifier but does not supply the missing definition.]

Definition 3.2 (Top missing-allocation run)**.** Let J be a finite simple graph with $c = c_{0,0}(J) \geq 1$. The integer $\mathrm{tr}(J)$ is the maximum length of a consecutive interval of integers

$$x \in \{0, 1, \ldots, c-1\}$$

for which there is no edge partition

$$E(J) = P_R \sqcup P_B$$

satisfying

$$\nu(P_R) \leq x, \qquad \nu(P_B) \leq c - 1 - x.$$

Here the matching numbers refer to the spanning subgraphs with the indicated edge sets.

Definition 3.3 (Gap criteria and closed lumps)**.** For every integer $D \geq 0$, put

$$h_-(D) = \max\left(4, \left\lfloor \frac{D}{2} \right\rfloor + 1\right), \qquad h_0(D) = \max\left(4, \left\lceil \frac{D}{2} \right\rceil + 1\right).$$

A finite simple graph J satisfies T_- when

$$\mathrm{gap}(J) \geq 1 \implies \mathrm{gap}(J) = 1 \quad \text{and} \quad \mathrm{tr}(J) + \mathrm{ex}(J) \leq h_-(\Delta(J)).$$

The property T_0 uses h_0 in place of h_-.

For an integer $d \geq 0$, a *closed d-lump* is a connected d-regular graph of odd order whose deletion of any vertex leaves a perfect matching.

A positive-gap graph J is *totally critical* when deleting any maximum-degree vertex strictly lowers every row c^*_θ with $1 \leq \theta \leq \Delta(J)$. In a regular graph this condition concerns every vertex.

Setup 3.4 (Pending elementary certificate properties)**.** The arguments below require the following properties of the intended certificate costs.

(i) All row costs are nonnegative integers, and

$$c^*_\theta(J) = 0 \qquad (\theta > \Delta(J)).$$

If $D \geq 1$, $\Delta(J) \leq D$, and a matching covers every vertex of degree D in J, then

$$c^*_D(J) = 0.$$

(ii) For every subgraph $J' \subseteq J$ and every $\theta \in \mathbb{Z}_{\geq 0}$,

$$c^*_\theta(J') \leq c^*_\theta(J).$$

For every $v \in V(J)$,

$$0 \leq c^*_\theta(J) - c^*_\theta(J - v) \leq 1.$$

(iii) The excess is nonnegative. The quantity

$$\mathrm{ps}(J) := \mathrm{ex}(J) - \mathrm{gap}(J)$$

is nonnegative, so

$$\mathrm{gap}(J) = \mathrm{ex}(J) - \mathrm{ps}(J).$$

(iv) If $c_{0,0}(J) \geq 1$, then $\mathrm{tr}(J)$ is defined and $\mathrm{tr}(J) \geq 1$.

[math/blocker] [GAP: Proofs of these certificate properties, including the covering-matching vanishing criterion, are absent. They are recorded as pending premises, not as additional proved results.]

4. Arrowing graphs: tails, pools, and potential

In this section, we state the transfer and potential results used in the assembly. Their supplied proofs are retained in Section 5.

Definition 4.1 (Greedy residual of a star pool)**.** Let F be a finite star forest with positive component sizes, allowing F to be empty. For an integer $b \geq 0$, let D_b act on its component-size multiset by deleting one largest remaining size at most b, if such a size exists, and doing nothing otherwise.

For an integer $N \geq 0$ and a vector $c = (c_1, \ldots, c_N) \in \mathbb{Z}^N_{\geq 0}$, let $\mathrm{res}_c(F)$ be the actual star forest represented by the multiset after applying

$$D_{c_1}, \ldots, D_{c_N}.$$

Put

$$P(c) = \bigsqcup_{\substack{1 \leq i \leq N \\ c_i > 0}} K_{1,c_i}.$$

Zero-capacity positions are ignored.

Theorem 4.2 (Exact residuals for star pools)**.** *Let F, N, and c be as in Definition 4.1.*

(i) *The operators D_b commute on positive integer multisets. Consequently $\mathrm{res}_c(F)$ is independent of the order of the capacities and of choices between equal-sized target components.*

(ii) *For every finite simple core graph J disjoint from $P(c)$,*

$$J \sqcup P(c) \supseteq F \quad \Longleftrightarrow \quad J \supseteq \mathrm{res}_c(F).$$

The empty forest is present in every graph.

More precisely, $\mathrm{res}_c(F)$ embeds in every complementary forest $F - I$ obtained by assigning a feasible subset I of the target components to distinct pool stars. The residual is itself one of these feasible complements.

(iii) *If F is nonempty and $P(c)$ avoids F, then $\mathrm{res}_c(F)$ is isomorphic to the greatest-common remnant $\mathrm{rem}(F, P(c))$. This means that it embeds in every feasible complementary forest and is greatest under subgraph containment among star forests with that property.*

In this star-pool setting, the remnant is attained as an actual complement, and the transfer is an equivalence for every core.

(iv) *Let F_R, F_B be nonempty star forests. Let*

$$Q = \bigsqcup_{i=1}^{N} K_{1,d_i}, \qquad d_i \in \mathbb{Z}_{\geq 1},$$

and let $G = H \sqcup Q$ for any finite simple core H. For every integer vector r with $0 \leq r_i \leq d_i$, put

$$F_R(r) = \mathrm{res}_r(F_R), \qquad F_B(r) = \mathrm{res}_{d-r}(F_B).$$

Then

$$G \to (F_R, F_B) \quad \Longleftrightarrow \quad H \to (F_R(r), F_B(r))$$

for every r for which both residuals are nonempty.

If no such r exists, the condition on H is vacuous and Q itself arrows the original pair. Equivalently, G has a coloring avoiding both full targets if and only if some such r admits a core coloring avoiding both actual residuals. Coloring any r_i edges of pool star i red and its remaining edges blue gives a full avoiding extension.

Every assertion includes unit components and preserves existence and avoidance. No equality between full-copy counts and residual-copy counts is asserted.

Proof. The complete development is given in Section 9. □

Lemma 4.3 (Tail persistence with arbitrary cross-edge colors)**.** *Use Setup 2.2. Let G be a finite simple graph. Choose integers*

$$0 \leq \alpha < s, \qquad 0 \leq \beta < t, \qquad q = \alpha + \beta \geq 1,$$

and let $X = A \sqcup B \subseteq V(G)$ satisfy $|A| = \alpha$ and $|B| = \beta$.

Every edge coloring of $G - X$ avoiding a red F_1^α and a blue F_2^β extends to a coloring of G avoiding both full targets as follows:

(i) *Color edges from A to $G - X$ and edges within A red.*
(ii) *Color edges from B to $G - X$ and edges within B blue.*
(iii) *Assign each edge between A and B either color independently.*

Every nontrivial connected red subgraph meeting X contains a vertex of A. Every nontrivial connected blue subgraph meeting X contains a vertex of B. Hence

$$G \to (F_1, F_2) \quad \Longrightarrow \quad G - X \to (F_1^\alpha, F_2^\beta)$$

for every choice of the cross-edge colors, including when target copies meet X through leaves.

Proof. The complete development is given in Section 9. □

Theorem 4.4 (The edge-minus-matching potential)**.** *Let $\mathcal{A}$ be a finite nonempty family of target pairs. For every $\gamma \in \mathcal{A}$, let $s_\gamma, t_\gamma \in \mathbb{Z}_{\geq 1}$, and let*

$$n_1^\gamma \geq \cdots \geq n_{s_\gamma}^\gamma \geq 1, \qquad m_1^\gamma \geq \cdots \geq m_{t_\gamma}^\gamma \geq 1$$

be integer lists. Let F_1^γ, F_2^γ be the respective vertex-disjoint unions of stars with these edge counts. Put

$$N = \max_{\gamma \in \mathcal{A}} (s_\gamma + t_\gamma - 1).$$

For every integer k with $2 \leq k \leq N + 1$, define

$$\overline{\ell}_k = \max_{\gamma \in \mathcal{A},\ \substack{1 \leq i \leq s_\gamma,\ 1 \leq j \leq t_\gamma \\ i+j=k}} (n_i^\gamma + m_j^\gamma - 1).$$

Let G be a finite simple graph satisfying

$$G \to (F_1^\gamma, F_2^\gamma) \qquad \textit{separately for every } \gamma \in \mathcal{A}.$$

Then

$$e(G) - \nu(G) \geq \sum_{k=2}^{N+1} (\bar{\ell}_k - 1).$$

For a single pair as in Setup 2.2, every Ramsey host satisfies

$$e(G) \geq L + \nu(G) - (s + t - 1).$$

Consequently,

$$\nu(G) \geq s + t - 1 \quad \Longrightarrow \quad e(G) \geq L.$$

Every host with $e(G) < L$ instead satisfies

$$\nu(G) \leq s + t - 2, \qquad L - e(G) \leq s + t - 1 - \nu(G).$$

All assertions allow arbitrary repetitions, gaps, parities, and unit target components.

Proof. The complete development is given in Section 9. □

Theorem 4.5 (Tight deletion with a prescribed maximum matching)**.** *Use Setup 2.2. Let $G \to (F_R, F_B)$ satisfy*

$$e(G) - \nu(G) = L - M.$$

Fix any maximum matching Q of G. Starting with $X_0 = \varnothing$, for every integer q with $0 \leq q < M$, choose any maximum-degree vertex v_q of

$$J_q = (G - Q) - X_q,$$

and put $X_{q+1} = X_q \cup \{v_q\}$. Such a vertex exists at every step. Write

$$H_q = G - X_q, \qquad Q_q = \{xy \in Q : x, y \notin X_q\}.$$

Regardless of the choices among maximum-degree vertices,

$$Q_q \text{ is a maximum matching of } H_q, \qquad e(J_q) = e(H_q) - \nu(H_q) = P_q \quad (0 \leq q \leq M),$$

and

$$\Delta(J_q) = d_{J_q}(v_q) = \ell_{q+2} - 1 \quad (0 \leq q < M).$$

Let $\epsilon_q = 1$ if Q_q covers v_q, and let $\epsilon_q = 0$ otherwise. Then

$$\nu(H_q) - \nu(H_{q+1}) = \epsilon_q, \qquad d_{H_q}(v_q) = \ell_{q+2} - 1 + \epsilon_q.$$

Choosing a Q_q-covered vertex among the maximum-degree vertices of J_q whenever one exists also makes v_q a maximum-degree vertex of H_q.

Finally, H_M consists of Q_M and isolated vertices, and

$$M - \nu(G) = \big|\{q \in \{0, \ldots, M-1\} : \epsilon_q = 0\}\big| - |Q_M|.$$

Thus the entire tight chain may preserve the restriction of any initially prescribed maximum matching without exchanging that matching.

Proof. The complete development is given in Section 9. □

Setup 4.6 (Pending binary-tail inputs)**.** For two positive nonincreasing integer lists with every entry in $\{1, 2\}$, the following inputs are required.

(i) For every legal total leading cut, the family of nonempty actual suffix pairs has a member whose diagonal profile attains the coordinatewise envelope of the profiles of the entire family.

(ii) If a graph of maximum degree at most two arrows a nonempty pair of such binary star forests, its edge count is at least the sum of that pair's complete diagonal profile.

[math/blocker] [GAP: The binary envelope-attainment argument and the binary single-pair lower-bound proof were not supplied. Neither input is replaced here by an unproved named theorem.]

Theorem 4.7 (Payment of a degree-two residual). *Use Setup 2.2, and let $G \to (F_R, F_B)$. Let $X \subseteq V(G)$ have integer size $q < M$, and put $H = G - X$. If $\Delta(H) \le 2$, then*

$$e(H) \ge \sum_{k=q+2}^{M+1} \ell_k.$$

If, in addition,

$$e(G) - e(H) \ge \sum_{k=2}^{q+1} \ell_k,$$

then $e(G) \ge L$, with the entire original diagonal budget accounted for.

Consequently, assume that

$$e(G) - \nu(G) = L - M, \qquad \nu(G) < M.$$

Consider any zero-surplus maximum-degree deletion chain of length M: at step i, the selected maximum-degree vertex has potential drop $\ell_{i+2} - 1$, and the terminal potential is zero. Such chains are supplied by Theorem 4.5 with its stated tie preference. Let q be the first index at which the selected deletion preserves matching number. This index exists, and

$$\Delta(H_q) = \ell_{q+2} - 1 \ge 3.$$

In particular, a degree-two first matching-preserving step cannot occur.

Proof. The complete development is given in Section 9. □

Theorem 4.8 (Canonical pools and exact host normalization). *Use Setup 2.2. For every integer q with $0 \le q < M$ and every finite simple core H disjoint from T_q,*

$$H \sqcup T_q \to (F_R, F_B) \iff H \to (F'_R, F'_B) \text{ for every } (F'_R, F'_B) \in \mathcal{A}_q.$$

From any avoiding coloring of $H \sqcup T_q$, one can retain every edge color of H, choose a member of $\mathcal{A}_q$ avoided by that core coloring, and make every pool star monochromatic while continuing to avoid both full original targets. Conversely, every avoiding core coloring of any member of $\mathcal{A}_q$ extends by monochromatic pool stars to an avoiding coloring of the original pair.

For any finite simple graph G and any $X \subseteq V(G)$ with $|X| = q < M$, put

$$G^\star_X = (G - X) \sqcup T_q.$$

Then

$$G \to (F_R, F_B) \implies G^\star_X \to (F_R, F_B).$$

An avoiding coloring of $G^\star_X$ can be pulled back explicitly to a full avoiding coloring of G. With

$$c_X = e(G) - e(G - X), \qquad h_X = \nu(G) - \nu(G - X),$$

the exact identities are

$$e(G^\star_X) = e(G) - c_X + \sum_{j=2}^{q+1} \ell_j, \qquad \nu(G^\star_X) = \nu(G) - h_X + q.$$

Finally, assume that

$$G \to (F_R, F_B), \qquad e(G) - \nu(G) = L - M, \qquad \nu(G) < M.$$

Fix any maximum matching Q. There are an integer q with $0 \le q < \nu(G)$ and a set X of size q such that, for $H = G - X$,

$$\nu(H) = \nu(G) - q, \qquad e(H) = e(G) - \sum_{j=2}^{q+1} \ell_j, \qquad \Delta(H) = \ell_{q+2} - 1 \ge 3.$$

The restriction of Q to H is maximum. The replacement $G_X^\star$ has exactly the same e, ν, and $e-\nu$ as G, and it still arrows the original pair.

This replacement is obtained by successively detaching full-degree vertices of degrees $\ell_2, \dots, \ell_{q+1}$ as whole isolated stars. It has the explicit avoiding-color pullback described above. No unrestricted matching-rank rigidity or endpoint escape is asserted.

Proof. The complete development is given in Section 9. □

5. Degree splitting and covering matchings

The goal of this section is to identify the degree-splitting input used in Theorems 4.4 and 4.5. Coloring and degree-decrease results provide the relevant background [Bro41, DJK+21, AHD+25].

Setup 5.1 (Pending covering-matching split)**.** Let J be a finite simple graph, let $A, B \in \mathbb{Z}_{\geq 0}$, and put $D = A + B$. The required implication is the following:

$$\Delta(J) \leq D, \quad \text{some matching covers every vertex of degree } D$$
$$\implies E(J) = E_R \sqcup E_B, \qquad \Delta(J[E_R]) \leq A, \qquad \Delta(J[E_B]) \leq B.$$

In this display, $J[E_R]$ and $J[E_B]$ mean spanning graphs with those edge sets, not vertex-induced subgraphs.

[math/blocker] [GAP: The covering-matching degree-split proof is missing. The cited coloring literature supplies context, but the supplied citation descriptions do not identify this precise implication as a published theorem. The regular-completion, orientation, and covering-matching developments required for the full chapter must be inserted.]

6. Closed lumps and LP-goodness at high degree

In this section, we state the proper-subgraph result for degree four and record the structural inputs needed for the five-degree assembly. Equimatchable graphs and factor-critical structures are studied in [AAE+16, EK13]; related star-factor and even-regular questions appear in [KWY07, ZZD+24, ZZD+26].

[cite/blocker] The supplied record for KWY07star gives the second author's forename as Yang. The checked arXiv record gives Yunjian Wu. The bibliography uses the checked name; the supplied reference record requires correction.

Setup 6.1 (Pending sharp-profile extraction at degree four)**.** The following are the required consequences of the sharp-profile argument.

(i) For graphs of maximum degree at most four, the sharp profile agrees with the generic profile: the restricted sharp thresholds begin at six, and costs above the maximum degree vanish.
(ii) Every graph J of maximum degree at most four satisfies $\mathrm{gap}(J) \leq 1$.
(iii) If such a graph J has positive gap and

$$\mathrm{tr}(J) + \mathrm{ex}(J) \geq 4,$$

then it contains a nonempty induced closed 4-lump. At this threshold, the cycle alternatives C_5 and C_7 in the underlying extraction are excluded.

[math/blocker] [GAP: The sharp-profile definition, its gap bound, and the extraction argument excluding the two cycle alternatives were not supplied.]

Theorem 6.2 (Proper subgraphs of a connected quartic graph)**.** *Let Q be any connected 4-regular finite simple graph. Let J be any proper subgraph of Q, obtained by deleting vertices, edges, or both. Inducedness is not assumed. Then*

$$\mathrm{gap}(J) \geq 1 \quad \implies \quad \mathrm{gap}(J) = 1, \qquad \mathrm{tr}(J) + \mathrm{ex}(J) \leq 3.$$

Consequently, every proper subgraph J with $\mathrm{ex}(J) \geq 3$ is LP-good. Every proper subgraph satisfies T_-.

In particular, if Q is a closed 4-lump satisfying

$$\mathrm{gap}(Q) = 1, \qquad \mathrm{tr}(Q) + \mathrm{ex}(Q) \geq 5,$$

then Q violates T_-, while every proper subgraph of Q satisfies T_-.

Proof. The complete development is given in Section 11. □

Domination arguments are among the methods associated with the degree restrictions. The supplied literature identifies ordinary domination, weighted greedy methods, quartic replacement arguments, and random selection as background [ASG21, BK14, LS04, Zam21]. These references do not replace the certificate-specific conclusions required next.

Setup 6.3 (Pending excess exclusions)**.** The assembly uses the following inputs.

(i) Every closed 4-lump of excess at most two satisfies

$$\mathrm{gap} > 0 \quad \Longrightarrow \quad \mathrm{gap} = 1, \quad \mathrm{tr} + \mathrm{ex} \leq 4.$$

Every positive-gap closed 4-lump of excess three satisfies $\mathrm{tr} \leq 1$.

(ii) Every induced-minimal T_--failure of degree six has excess at least five.

(iii) Every induced-minimal T_--failure of degree eight has excess at least five.

(iv) Every induced-minimal T_--failure of degree ten or twelve has excess at least six.

(v) A closed 12-lump whose every proper induced subgraph satisfies T_- cannot have excess seven.

[math/blocker] [GAP: The proofs of these degree-specific excess exclusions were not supplied. They are necessary premises for Theorem 6.2.]

[math/blocker] [GAP: The ten excess-window, threshold, and split-achievability results requested for use by the second companion were not supplied as complete statements and proofs. Only the consequences explicitly present in the supplied arguments are recorded here. The missing ten results cannot be reconstructed from their names.]

7. The gap criterion and induced-minimal failures

This section specifies the remaining graph inputs used in the assembly and defines the restricted obstruction class in Theorem B. Each pending input is distinguished from a proved conclusion.

Setup 7.1 (Pending universal threshold transfer)**.** The universal assertions that every finite simple graph satisfies T_- and that every finite simple graph satisfies T_0 are equivalent. Either universal assertion implies that every finite simple graph C satisfying

$$\mathrm{ex}(C) \geq h_0(\Delta(C))$$

is LP-good. This is an equivalence of universal assertions, not a pointwise equivalence at odd maximum degree.

[math/blocker] [GAP: The proof of the universal floor-to-ceiling transfer and its inclusive LP-goodness consequence was not supplied.]

Setup 7.2 (Pending induced-minimal failure extraction)**.** Every finite simple graph J violating T_- contains an induced subgraph F such that

$$d := \Delta(F) \in \{4, 6, 8, 10, 12\},$$

F is a totally critical closed d-lump,

$$\mathrm{gap}(F) = 1, \qquad \mathrm{tr}(F) + \mathrm{ex}(F) \geq h_-(d) + 1,$$

and every proper induced subgraph of F satisfies T_-.

[math/blocker] [GAP: The induced-minimal failure theorem and the degree closures restricting its degree to the displayed five values were not supplied.]

Setup 7.3 (Pending proper-subgraph minimality input)**.** Let F be any finite simple graph violating T_- such that every proper subgraph satisfies T_-. The following conclusions are required for every such F:

(i) The graph F is a totally critical closed even d-lump, with
$$\mathrm{gap}(F) = 1, \qquad \mathrm{tr}(F) + \mathrm{ex}(F) \geq h_-(d) + 1.$$
(ii) Every proper subgraph J with $\mathrm{ex}(J) \geq 4$ is LP-good.
(iii) If $|V(F)| = 2k+1$ and $\mathrm{ex}(F) \geq 4$, then for every edge a and integer $\theta \geq 0$,
$$r_\theta(a) := c^*_\theta(F) - c^*_\theta(F-a) \in \{0,1\}.$$
The drops vanish for $\theta \geq d+1$, and $r_d(a) = 1$. Moreover,
$$\nu(F-a) = k, \qquad \mathrm{ex}(F-a) = \mathrm{ex}(F) - r_0(a),$$
$$R(a) := \sum_{\theta=0}^{d} r_\theta(a) \geq 2, \qquad \mathrm{gap}(F-a) = 2 - R(a) \leq 0.$$
Thus an additional strict drop occurs below row d, possibly in different rows for different edges.

[math/blocker] [GAP: The proper-subgraph minimality theorem, its excess-four LP bound, and its edge-deletion conclusions were not supplied with proofs. The quantifier is every minimal failure, not only one globally chosen graph.]

Setup 7.4 (Pending extraction from an actual-family host)**.** Use Setup 2.2, and fix an integer $0 \leq q < M$. If a finite simple graph H arrows every entire prescribed-color member of $\mathcal{A}_q$ and satisfies $e(H) < L_q$, then an actual connected component C of H satisfies
$$\mathrm{ex}(C) \geq K - \nu(H) \geq 4, \qquad \mathrm{gap}(C) \geq 1,$$
and C contains an induced subgraph J violating T_-.

[math/blocker] [GAP: The certificate-to-host extraction proof giving both the component excess bound and an induced failure was not supplied. No assumption on the ambient maximum degree may be added to this input.]

Setup 7.5 (Pending hereditary excess bound)**.** If every induced subgraph of a finite simple graph F, including F itself, satisfies T_-, and $\mathrm{gap}(F) = 1$, then
$$\mathrm{ex}(F) \leq 3.$$

[math/blocker] [GAP: The hereditary excess bound was not supplied with a proof. It is needed to deduce emptiness of the restricted class from the universal criterion.]

Theorem 7.6 (Exact five-degree gap-one input)**.** *Use the universal-allocation certificate costs and the quantities of Notation 3.1 and Definitions 3.2 and 3.3. In particular, each cost requires its bounded-degree/bounded-matching partition for every nonnegative integer allocation of the total matching budget.*

The following assertions are equivalent:

(i) *Every finite simple graph satisfies T_-.*
(ii) *Every finite simple graph satisfies T_0.*
(iii) *The assertion B_5 holds: for every $d \in \{4,6,8,10,12\}$, every totally critical closed d-lump F with $\mathrm{gap}(F) = 1$ satisfies*
$$\mathrm{tr}(F) + \mathrm{ex}(F) \leq \max(4, d/2+1).$$

The five bounds in increasing degree order are $4, 4, 5, 6, 7$.

Under any of these equivalent assertions, every finite simple graph C *with*

$$\mathrm{ex}(C) \geq h_0(\Delta(C))$$

is LP-good. This theorem establishes the equivalence and conditional implication only. It asserts neither B_5 *nor a universal top-run or LP-goodness theorem.*

Proof. The complete development is given in Section 12. □

Definition 7.7 (The rigid obstruction class)**.** The class $\mathcal{L}_{\mathrm{rig}}$ consists of the finite simple graphs F satisfying all of the following conditions.

The graph F is a closed d-lump with

$$d \in \{4, 6, 8, 10, 12\}, \qquad \mathrm{gap}(F) = 1,$$

and every proper subgraph of F satisfies T_-. Write

$$|V(F)| = 2k + 1, \qquad k = \nu(F), \qquad E = \mathrm{ex}(F), \qquad p_d = \begin{cases} 3, & d = 4, \\ 4, & d \neq 4. \end{cases}$$

The excess restrictions are

$$E \geq 4 \quad (d = 4), \qquad E \geq 5 \quad (d = 6, 8), \qquad E \geq 6 \quad (d = 10, 12),$$

together with $E \neq 7$ when $d = 12$.

The following three deletion conditions are also required.

(a) For every $v \in V(F)$ and every integer $0 \leq \theta \leq d$,

$$c^*_\theta(F - v) = c^*_\theta(F) - 1.$$

(b) For every $v \in V(F)$ and every nonempty set S of edges incident with v, put

$$r_\theta(S) = c^*_\theta(F) - c^*_\theta(F - S),$$

where $F - S$ retains all vertices. Then

$$r_\theta(S) \in \{0, 1\} \quad (0 \leq \theta \leq d), \qquad r_d(S) = 1, \qquad \sum_{\theta=0}^{d-1} r_\theta(S) \geq |S|.$$

(c) For every nonempty $X \subseteq V(F)$, put

$$j = |X|, \qquad J = F - X, \qquad b = j + \nu(J) - k.$$

Whenever $b \leq E - p_d$, require

$$c^*_d(J) = 0, \qquad 1 \leq c^*_\theta(F) - c^*_\theta(J) \leq j \quad (0 \leq \theta < d).$$

Writing

$$t_\theta = j - \big[c^*_\theta(F) - c^*_\theta(J)\big],$$

require

$$\sum_{\theta=0}^{d-1} t_\theta \leq e(F[X]).$$

8. Certificates, costs and the LP-goodness criterion

[author/blocker] No author, affiliation, email, or subject classification was supplied.

[prime/blocker] Complete unconditional proofs cannot be certified: the supporting assertions identified in Setup 8.9 were not supplied with proofs or applicable published references. Arguments using those assertions are conditional.

[compile/blocker] Compilation has not been run. The checks for LaTeX errors and unresolved references remain outstanding.

[instruction/conflict] The requested length must yield to preservation of all ten statements and their proof steps. The standalone front matter is retained; the chapter body begins below.

Certificate costs separate maximum-degree restrictions from matching budgets. We fix these quantities, establish their elementary relations, and then organize the matching, threshold, and potential arguments in dependency order. No result of an earlier chapter is assumed available; the precise additional assertions needed by the supplied arguments are recorded in Setup 8.9.

Remark 8.1. The main result of this paper was obtained with the assistance of the Ansatz system, an automated proof-search and verification system. [ack/blocker] No independent verification statement was supplied. Because automated systems have limitations, it is possible that we have missed related references in the literature, and we welcome comments from experts.

8.1. **Setup.** This section fixes the certificate notation and states every additional premise required by the arguments below.

Notation 8.2. All graphs are finite and simple. For a graph H, write

$$n(H) = |V(H)|, \qquad e(H) = |E(H)|,$$

and let $\Delta(H)$, $\nu(H)$, $\alpha(H)$, and $\tau(H)$ denote its maximum degree, maximum matching cardinality, independence number, and minimum vertex-cover cardinality, respectively. The maximum degree and matching number of an empty edge set are zero.

For $U \subseteq V(H)$, the graph $H - U$ is obtained by deleting U, and $H[U]$ is the induced subgraph on U. Write $H - v = H - \{v\}$. The open neighborhood of U is

$$N_H(U) = \{w \in V(H) \setminus U : w \text{ is adjacent to a vertex of } U\}.$$

For disjoint vertex sets U, W, let $e(U, W)$ count the edges between them, and put $e(U) = e(H[U])$ when the ambient graph is fixed. Let $o(H)$ be the number of odd-order connected components of H, including isolated vertices.

For an edge set P, the quantities $\Delta(P)$ and $\nu(P)$ refer to the graph formed by its edges. Adding isolated vertices to that graph changes neither quantity. A graph is factor-critical if it has at least one vertex and deleting any vertex leaves a perfect matching. The empty matching is perfect in the empty graph.

All capacities, matching budgets, threshold indices, component orders, and counting parameters are integers. In particular, $A, B, x, y, c, \theta \in \mathbb{Z}_{\geq 0}$ whenever these symbols denote certificate parameters.

Definition 8.3 (Certificates and costs)**.** Let H be a graph and let $A, B, x, y \in \mathbb{Z}_{\geq 0}$. An $(A, B; x, y)$-certificate is a partition

$$E(H) = Q_R \sqcup Q_B \sqcup P_R \sqcup P_B$$

such that

$$\Delta(Q_R) \leq A, \qquad \Delta(Q_B) \leq B, \qquad \nu(P_R) \leq x, \qquad \nu(P_B) \leq y.$$

Any part may be empty. Define $c_{A,B}(H)$ to be the least $c \in \mathbb{Z}_{\geq 0}$ for which every $x, y \in \mathbb{Z}_{\geq 0}$ with $x + y = c$ admit such a certificate. Its existence is proved in Theorem 8.10. Put

$$c^*_\theta(H) = \max_{\substack{A,B \in \mathbb{Z}_{\geq 0} \\ A+B=\theta}} c_{A,B}(H), \qquad \Sigma(H) = \sum_{\theta \geq 0} c^*_\theta(H),$$

and

$$\mathrm{ex}(H) = c_{0,0}(H) - \nu(H), \qquad \mathrm{ps}(H) = e(H) - \nu(H) - \sum_{\theta \geq 1} c^*_\theta(H).$$

The quantity $\mathrm{ps}(H)$ is the potential slack. Define

$$\mathrm{gap}(H) = \Sigma(H) - e(H).$$

The graph H is LP-good if $\Sigma(H) \leq e(H)$.

Definition 8.4 (Matching splits and barriers)**.** A pair $x, y \in \mathbb{Z}_{\geq 0}$ is split-achievable in H if

$$E(H) = P_R \sqcup P_B, \qquad \nu(P_R) \leq x, \quad \nu(P_B) \leq y.$$

This is exactly an $(0, 0; x, y)$-certificate, because its degree-restricted parts must be empty.

A vertex set $X \subseteq V(H)$ is attaining, or an optimal barrier, if

$$\nu(H) = |X| + \sum_C \left\lfloor \frac{|V(C)|}{2} \right\rfloor = \frac{n(H) + |X| - o(H - X)}{2},$$

where C ranges over all components of $H - X$.

Definition 8.5 (Good remainders and top missing runs)**.** For $\theta \in \mathbb{Z}_{\geq 1}$, a graph J is θ-good if $\Delta(J) \leq \theta$ and some matching covers every vertex of degree exactly θ. A graph of maximum degree at most $\theta - 1$ is θ-good, using the empty matching.

If $c_{0,0}(H) \geq 1$, put $\ell = c_{0,0}(H) - 1$. The top missing-allocation run $\mathrm{tr}(H)$ is the largest number of consecutive integers in a block

$$\{a, a+1, \ldots, b\} \subseteq \{0, 1, \ldots, \ell\}$$

such that $(x, \ell - x)$ is not split-achievable for every x in the block. We use this quantity only when $c_{0,0}(H) \geq 1$.

Definition 8.6 (Sharp costs)**.** For $\theta \in \mathbb{Z}_{\geq 0}$, let

$$C_\sharp(\theta, H) = \begin{cases} \max\limits_{\substack{A,B \in 2\mathbb{Z}_{\geq 0} \\ A+B=\theta}} c_{A,B}(H), & \theta \in \{6, 8, \ldots, 26\}, \\ c^*_\theta(H), & \text{otherwise.} \end{cases}$$

Define the sharp slack by

$$\psi_\sharp(H) = e(H) - \nu(H) - \sum_{\theta \geq 1} C_\sharp(\theta, H).$$

Setup 8.7 (Star-forest targets)**.** Let $s, t \in \mathbb{Z}_{\geq 1}$, and let

$$n_1 \geq \cdots \geq n_s \geq 1, \qquad m_1 \geq \cdots \geq m_t \geq 1$$

be integer sequences. Let

$$F_1 = \bigsqcup_{i=1}^{s} K_{1,n_i}, \qquad F_2 = \bigsqcup_{j=1}^{t} K_{1,m_j}.$$

Put

$$M = s + t - 1, \qquad \ell_k = \max_{\substack{1 \leq i \leq s,\ 1 \leq j \leq t \\ i+j=k}} (n_i + m_j - 1) \quad (2 \leq k \leq s+t), \qquad L = \sum_{k=2}^{s+t} \ell_k.$$

The relation $H \to (F_1, F_2)$ means that every red-blue coloring of $E(H)$ contains a red copy of F_1 or a blue copy of F_2; copies are ordinary subgraphs. The size Ramsey number is the minimum number of edges of such a graph H.

Theorem 8.8 (Tutte–Berge identity [QLS+19, Theorem 1.6]). *Every graph H has an attaining set. Equivalently,*

$$\nu(H) = \min_{X \subseteq V(H)} \frac{n(H) + |X| - o(H - X)}{2}.$$

Setup 8.9 (Required supporting assertions). The following statements record precisely the additional premises used below. They are requested results of the earlier development of the series, but no earlier chapter, proof, or applicable published citation was supplied for them. Their inclusion here is an assumption register, not a claim that they have been established.

(1) If $A, B \in \mathbb{Z}_{\geq 0}$, $D = A + B \geq 1$, $\Delta(J) \leq D$, and a matching covers every degree-D vertex of J, then

$$E(J) = Q_R \sqcup Q_B, \qquad \Delta(Q_R) \leq A, \quad \Delta(Q_B) \leq B.$$

[math/blocker] [GAP: the matching-cover degree-decomposition theorem is required.]

(2) For $S \subseteq V(J)$, let $u_S(J)$ be the minimum number of vertices of S left uncovered by a matching. Let $q_S(J - X)$ count the odd components of $J - X$ whose vertices all belong to S. The required prescribed-set formula is

$$u_S(J) = \max_{X \subseteq V(J)} \big(q_S(J - X) - |X|\big).$$

The cubic application also requires the following refinement. If $\Delta(J) = 3$, S is its set of degree-three vertices, $\kappa = u_S(J) \geq 1$, and X has maximum cardinality among the maximizing sets, then, with $b = |X|$, the collection $\mathcal{K}$ of odd components of $J - X$ contained in S satisfies

$$|\mathcal{K}| = b + \kappa.$$

Every $K \in \mathcal{K}$ is factor-critical, all its boundary edges end in X, and

$$\nu(J) = b + \sum_{Q} \nu(Q),$$

where Q ranges over all components of $J - X$. The required threshold bound is

$$c_3^*(J) \leq \kappa.$$

[math/blocker] [GAP: the prescribed-set matching formula and these maximizing-set consequences are required.]

(3) If J is connected, $n(J) \geq 2$, $D \in \mathbb{Z}_{\geq 2}$, and $\Delta(J) \leq D$, then for every integer

$$1 \leq r \leq \left\lceil \frac{n(J) - 1}{D} \right\rceil$$

there is an independent set I with

$$|I| = r, \qquad |N_J(I)| \leq (D - 1)r + 1.$$

[math/blocker] [GAP: the connected independent-set neighborhood bound is required.]

(4) In Setup 8.7, if $H \to (F_1, F_2)$, put

$$\sigma = e(H) - \nu(H) - (L - M), \qquad \delta = M - \nu(H).$$

The required inequalities are

$$e(H) = L + \sigma - \delta, \qquad \sigma \geq \mathrm{ps}(H) \geq 0, \qquad \mathrm{ex}(H) \geq \delta, \qquad L - e(H) \leq \mathrm{ex}(H) - \mathrm{ps}(H).$$

[math/blocker] [GAP: the certificate-to-arrowing inequalities are required.]

(5) In Setup 8.7, if

$$e(H) - \nu(H) = L - M, \qquad \nu(H) < M,$$

then H has a red-blue coloring avoiding both a red F_1 and a blue F_2. [math/blocker] [GAP: the zero-surplus coloring theorem is required.]

(6) For every pair of sequences in Setup 8.7, there is a graph with exactly L edges arrowing (F_1, F_2). [math/blocker] [GAP: the star-forest upper-bound construction is required.]

(7) The required complete-graph calculations are

$$\mathrm{ex}(K_5) = 1, \qquad \mathrm{gap}(K_5) \le 0.$$

For every odd integer $k \ge 7$, the required bound is

$$\mathrm{gap}(K_k) \le \mathbf{1}_{\{a \text{ odd}\}} - \frac{a(a-1)}{2}$$

for an integer $a \ge 2$. [math/blocker] [GAP: these complete-graph cost and gap bounds are required; the parameterization of a was not supplied.]

(8) If Q is factor-critical, $n(Q) \ge 3$, and $\mathrm{ps}(Q) = 0$, then Q is connected and regular of positive even degree D. For $s = \alpha(Q)$, either

$$n(Q) = 2s + 1, \qquad \text{or} \qquad D = 4, \quad n(Q) = 2s + 3.$$

Moreover, $s = 1$ forces $Q \cong K_3$, and $s = 2$ forces $Q \cong C_5$. [math/blocker] [GAP: the preliminary zero-slack structural classification is required.]

(9) Under the assumptions of item 8, let S be a maximum independent set, put $G = Q - S$, and set $k = (n(Q) - 1)/2$. The required conclusions are

$$\mathrm{ps}(G) = 0, \qquad \nu(G) = k + 1 - |S|,$$

and every maximum-cardinality attaining set X of G satisfies $|X| \le 1$. [math/blocker] [GAP: the maximum-independent-set deletion equalities and barrier bound are required.]

(10) Every graph of maximum degree at most three with at most nine degree-three vertices is 3-good. [math/blocker] [GAP: the small cubic-exposure obstruction bound is required.]

(11) The required cycle calculations are

$$\nu(C_3) = c_{0,0}(C_3) = 1,$$

$$\mathrm{ex}(C_5) = \mathrm{ex}(C_7) = 1, \qquad \mathrm{gap}(C_5) > 0, \quad \mathrm{gap}(C_7) > 0,$$

and

$$\mathrm{ps}(C_m) \ge 1 \qquad \text{for every odd integer } m \ge 9.$$

[math/blocker] [GAP: these odd-cycle cost and gap calculations are required.]

(12) If K is an odd-order factor-critical graph with exactly one vertex of degree two and every other vertex of degree three, then

$$\mathrm{ps}(K) \ge 2.$$

[math/blocker] [GAP: the almost-cubic potential-slack bound is required.]

(13) If J has actual maximum degree $D \ge 4$ and $\psi_\sharp(J) = 0$, then a matching covers every degree-D vertex of J. The same conclusion is required when $\mathrm{ps}(J) = 0$ and every induced subgraph of J has generic gap at most one. [math/blocker] [GAP: the sharp and hereditary generic zero-slack coveredness theorems are required.]

[instruction/conflict] No result number in Companion I or an earlier chapter is available for Setup 8.9. Assigning such a citation would invent a dependency.

[cite/blocker] The ledger supplies no original reference for the vertex-partition lemma attributed to Lovász or for König's theorem. Their supplied proofs are retained below; the missing prescribed-set premise is separately flagged.

8.2. **Certificate operations.** In this section, we prove the certificate identities and the potential inequality used by every later threshold argument.

Theorem 8.10 (Basic certificate properties)**.** *For every graph H and all $A, B \in \mathbb{Z}_{\geq 0}$, the following hold.*

(a) *$c_{A,B}(H)$ is well defined. Every pair of nonnegative integers with sum c has an (A,B)-certificate if and only if $c \geq c_{A,B}(H)$.*

(b)
$$c_{A,B}(H) = c_{B,A}(H),$$
and
$$c_{A,B}(H) \leq c_{A',B'}(H) \quad \text{if } 0 \leq A' \leq A,\ 0 \leq B' \leq B.$$
In particular, $c^*_\theta(H) \leq c_{0,0}(H)$.

(c) *If $H = H_1 \sqcup H_2$ is a vertex-disjoint union, then*
$$c_{A,B}(H) \leq c_{A,B}(H_1) + c_{A,B}(H_2),$$
and
$$c^*_\theta(H) \leq c^*_\theta(H_1) + c^*_\theta(H_2).$$

(d) *If $A + B \geq 1$, $\Delta(H) \leq A + B$, and a matching covers every vertex of degree $A + B$, then $c_{A,B}(H) = 0$. The matching condition is automatic when $\Delta(H) \leq A + B - 1$. Consequently,*
$$c^*_\theta(H) = 0 \qquad (\theta \geq \Delta(H) + 1),$$
and $\Sigma(H)$ is finite.

(e) *For every vertex v,*
$$c_{A,B}(H) \leq c_{A,B}(H - v) + 1, \qquad c^*_\theta(H) \leq c^*_\theta(H - v) + 1.$$

(f) *If H' is a subgraph of H, then*
$$c_{A,B}(H') \leq c_{A,B}(H).$$

(g)
$$\nu(H) \leq c_{0,0}(H) \leq \tau(H), \qquad 0 \leq \mathrm{ex}(H) \leq \tau(H) - \nu(H).$$

(h)
$$\sum_{\theta \geq 1} c^*_\theta(H) \leq e(H) - \nu(H), \qquad \textit{equivalently } \mathrm{ps}(H) \geq 0.$$

(i)
$$\Sigma(H) - e(H) = \mathrm{ex}(H) - \mathrm{ps}(H).$$
Thus H is LP-good if and only if $\mathrm{ps}(H) \geq \mathrm{ex}(H)$. Also,
$$\Sigma(H) \leq e(H) + \mathrm{ex}(H),$$
and every graph with $\tau(H) = \nu(H)$ is LP-good.

All these quantities depend only on the edge set; isolated vertices change none of them.

Proof. Parts (d), (h), and the consequences using them rely on Setup 8.9(1). The remaining certificate manipulations are direct.

For (a), take $c = 2\nu(H)$. Every pair $x + y = c$ has $x \geq \nu(H)$ or $y \geq \nu(H)$. Assigning all edges to the corresponding P-part gives a certificate, so the admissible set of c's is nonempty. It therefore has a least element.

Assume every pair of sum c is certified, and let $x + y = c + 1$. If $x \geq 1$, a certificate for $(x - 1, y)$ also certifies (x, y). If $x = 0$, a certificate for $(0, c)$ also certifies $(0, c + 1)$. Induction and minimality prove (a).

For (b), exchanging (Q_R, P_R) and (Q_B, P_B) exchanges the capacities and matching budgets. The pairs of a fixed sum are closed under this exchange, giving symmetry. Increasing either capacity preserves every certificate, which proves monotonicity and the bound by $c_{0,0}$.

For (c), combine certificates of H_1 and H_2 part by part. Maximum degree takes the larger value on a vertex-disjoint union, while matching number is additive. Put $c_i = c_{A,B}(H_i)$, and let $x + y = c_1 + c_2$. Set

$$x_1 = \min(x, c_1), \quad y_1 = c_1 - x_1, \quad x_2 = x - x_1, \quad y_2 = y - y_1.$$

Then $x_2 \geq 0$. If $x_1 = x$, then $y_2 = c_2$; if $x_1 = c_1$, then $y_2 = y$. Thus all four budgets are nonnegative, and

$$x_i + y_i = c_i \qquad (i = 1, 2).$$

The certificates exist by (a), and their union certifies (x, y). Maximizing over $A + B = \theta$ proves the threshold inequality.

For (d), put $D = A + B \geq 1$. The required degree decomposition in Setup 8.9(1) partitions the edges into Q_R, Q_B with the prescribed degree bounds. Taking both P-parts empty certifies the only pair of total zero. If $\Delta(H) \leq D - 1$, there are no degree-D vertices, so the empty matching meets the hypothesis. This proves the vanishing assertion and finiteness.

For (e), put $c = c_{A,B}(H - v)$, and let $x + y = c + 1$. If $x \geq 1$, take a certificate of $H - v$ for $(x - 1, y)$ and add every edge at v to P_R. A matching uses at most one of the added edges, so its red size is at most x. If $x = 0$, use a certificate for $(0, c)$ and add the edges at v to P_B. This proves the cost inequality. Maximization gives its threshold version.

For (f), intersect each certificate part with $E(H')$. Neither maximum degree nor matching number increases under edge restriction.

For the lower bound in (g), a total $c < \nu(H)$ cannot certify $(0, c)$ at capacities $(0, 0)$. The parts Q_R, Q_B, P_R would be empty, forcing $P_B = E(H)$. For the upper bound, take a minimum vertex cover W, and let $x + y = |W|$. Partition

$$W = W_R \sqcup W_B, \qquad |W_R| = x, \quad |W_B| = y.$$

Let P_R contain all edges with an endpoint in W_R, and let P_B contain the remaining edges. Every edge of P_B meets W_B. Disjoint edges use distinct cover vertices, so

$$\nu(P_R) \leq x, \qquad \nu(P_B) \leq y.$$

This proves (g).

For (h), induct on $n(H)$. If H is edgeless, all positive thresholds vanish by (d), and $e(H) - \nu(H) = 0$. Otherwise put $\Delta = \Delta(H) \geq 1$. Thresholds above Δ vanish.

First suppose a maximum matching misses a vertex v of degree Δ. That matching lies in $H - v$, so $\nu(H - v) = \nu(H)$. Parts (e) and the induction hypothesis give

$$\begin{aligned} \sum_{\theta \geq 1} c^*_\theta(H) &\leq \sum_{\theta=1}^{\Delta} \big(c^*_\theta(H - v) + 1\big) \\ &\leq e(H - v) - \nu(H - v) + \Delta \\ &= e(H) - \nu(H). \end{aligned}$$

The second inequality also uses nonnegativity of every threshold cost.

Now suppose every maximum matching covers every degree-Δ vertex. A single maximum matching then covers all these vertices, so (d) gives $c^*_\Delta(H) = 0$. Choose a degree-Δ vertex v. Deleting it

loses at most one matching edge, and hence $\nu(H-v) \geq \nu(H) - 1$. Therefore

$$\begin{aligned}\sum_{\theta\geq 1} c^*_\theta(H) &\leq \sum_{\theta=1}^{\Delta-1} \big(c^*_\theta(H-v)+1\big)\\ &\leq e(H-v) - \nu(H-v) + \Delta - 1\\ &\leq e(H) - \Delta - \nu(H) + 1 + \Delta - 1\\ &= e(H) - \nu(H).\end{aligned}$$

These cases are exhaustive.

For (i), expand the definitions:

$$\begin{aligned}\Sigma(H) - e(H) &= c_{0,0}(H) + \sum_{\theta\geq 1} c^*_\theta(H) - e(H)\\ &= \big(c_{0,0}(H) - \nu(H)\big) - \left(e(H) - \nu(H) - \sum_{\theta\geq 1} c^*_\theta(H)\right)\\ &= \mathrm{ex}(H) - \mathrm{ps}(H).\end{aligned}$$

The LP-goodness criterion and the bound on $\Sigma(H)$ follow from (h). If $\tau(H) = \nu(H)$, part (g) gives $\mathrm{ex}(H) = 0$, while (h) gives $\mathrm{ps}(H) \geq 0$.

Finally, all certificate conditions involve only edges and their incidences. Isolated vertices affect none of the four certificate bounds, $e(H)$, or $\nu(H)$. They therefore affect none of the stated costs or slacks. □

8.3. **Odd components and matching splits.** The goal of this section is to obtain the matching decomposition and splitting bounds used in the excess estimates.

Lemma 8.11 (Maximum attaining sets)**.** *Every graph G has an attaining set of maximum cardinality. For every such set S, each component C of $G - S$ has odd order and is factor-critical. In particular,*

$$\nu(G) = |S| + \sum_C \frac{|V(C)| - 1}{2}.$$

Proof. Put $N = n(G)$ and $q = \nu(G)$. Theorem 8.8 supplies an attaining set. There are finitely many subsets of $V(G)$, so one has maximum cardinality.

For every $X \subseteq V(G)$, a matching has at most $|X|$ edges incident with X. Its remaining edges lie in components of $G - X$, with at most $\lfloor |V(C)|/2 \rfloor$ edges in each. Hence

$$q \leq |X| + \sum_C \left\lfloor \frac{|V(C)|}{2} \right\rfloor = \frac{N + |X| - o(G-X)}{2}.$$

The identity subtracts one from every odd component order before dividing by two. Thus

$$\phi(X) := o(G-X) - |X| \leq N - 2q,$$

with equality exactly for attaining sets. In particular, $\phi(S) = N - 2q$.

Suppose a component C of $G - S$ has even order. Choose $v \in V(C)$. The graph $C - v$ has odd total order, so it has at least one odd component: the parity of its order is the parity of its number of odd components. The other components are unaffected, and therefore

$$\phi(S \cup \{v\}) \geq o(G-S) + 1 - (|S| + 1) = \phi(S).$$

The upper bound on ϕ forces equality. This contradicts maximality of $|S|$. Every component of $G - S$ consequently has odd order.

Fix such a component C and $v \in V(C)$. Suppose $J = C - v$ has no perfect matching. Its order N_J is even, and

$$q_J := \nu(J) \leq N_J/2 - 1.$$

If $N_J = 0$, the empty matching already contradicts the supposition. By Theorem 8.8, there is $T \subseteq V(J)$ such that

$$q_J = |T| + \sum_B \left\lfloor \frac{|V(B)|}{2} \right\rfloor = \frac{N_J + |T| - o(J - T)}{2}.$$

Thus

$$o(J - T) - |T| = N_J - 2q_J \geq 2.$$

Set $S' = S \cup \{v\} \cup T$. The added vertices lie in C, so

$$|S'| = |S| + 1 + |T| > |S|.$$

The old odd component C is replaced by at least $|T| + 2$ odd components, while all others persist. Hence

$$\phi(S') \geq o(G - S) - 1 + (|T| + 2) - (|S| + 1 + |T|) = \phi(S).$$

Again equality is forced, contradicting maximality. Thus $C - v$ has a perfect matching for every v. Replacing the floors in the attaining identity by $(|V(C)| - 1)/2$ proves the final formula. □

Remark 8.12. The decomposition in [QLS+19b, Theorem 1.6] has factor-critical odd components and perfectly matchable even components. Lemma 8.11 proves the all-odd conclusion for every maximum-cardinality attaining set from the matching identity.

Theorem 8.13 (Matching splits and the first possible excess two)**.** *Let H be a graph.*

(I) *The cost $c_{0,0}(H)$ is well defined,*

$$\nu(H) \leq c_{0,0}(H), \qquad \operatorname{ex}(H) \geq 0,$$

and every pair of total $c \in \mathbb{Z}_{\geq 0}$ is split-achievable if and only if $c \geq c_{0,0}(H)$.

(II) *If $n(H) \leq x + y + 1 + \max(x, y)$, where $x, y \in \mathbb{Z}_{\geq 0}$, then (x, y) is split-achievable. Consequently,*

$$c_{0,0}(H) \leq c \quad \text{whenever} \quad c + \left\lceil \frac{c}{2} \right\rceil + 1 \geq n(H).$$

(III) *If H has an optimal barrier X such that every component of $H - X$ has at most ten vertices or exactly twelve vertices, then*

$$c_{0,0}(H) \leq \nu(H) + 1, \qquad \operatorname{ex}(H) \leq 1.$$

(IV) *The inequality $\operatorname{ex}(H) \leq 1$ holds if H has at most ten vertices, if every component has at most ten vertices, or if $\nu(H) \leq 4$. Consequently, $\operatorname{ex}(H) \geq 2$ implies $\nu(H) \geq 5$ and the existence of a component with at least eleven vertices.*

(V) *If $n(H) = 11$ and $\operatorname{ex}(H) \geq 2$, then H is connected,*

$$\nu(H) = 5, \qquad c_{0,0}(H) = 7, \qquad \operatorname{ex}(H) = 2,$$

and $(3, 3)$ is not split-achievable. Equivalently, every red-blue coloring has a red four-edge matching or a blue four-edge matching. Conversely, $n(H) = 11$, $\nu(H) = 5$, and failure of $(3, 3)$ imply $\operatorname{ex}(H) = 2$.

(VI)

$$\nu(K_{11}) = 5, \qquad c_{0,0}(K_{11}) = 7, \qquad \operatorname{ex}(K_{11}) = 2.$$

Proof. Write $n = n(H)$.

For (I), every pair of total $2\nu(H)$ has a coordinate at least $\nu(H)$, so assigning all edges to that color proves existence. If $c < \nu(H)$, the pair $(0, c)$ fails because its red part must be empty. If every pair of total c works, then a pair (x, y) of total $c+1$ inherits a split from $(x-1, y)$ when $x \geq 1$, and from $(0, c)$ when $x = 0$. Induction and minimality prove the assertion.

For (II), exchange the colors if necessary so that $x \geq y$. Choose

$$|S| = \min(n, 2x+1), \qquad T = V(H) \setminus S.$$

Then

$$|T| = \max(0, n - 2x - 1) \leq y.$$

Put the edges inside S in P_R, and all remaining edges in P_B. A red matching uses at most $2x+1$ vertices, so it has at most x edges. Every blue edge meets T, so a blue matching has at most $|T| \leq y$ edges. For a pair of total c,

$$\max(x, y) \geq \lceil c/2 \rceil .$$

The displayed condition on c therefore makes every pair of that total achievable, and (I) applies.

For (III), let $Q_1, \ldots, Q_p$ be the components of $H - X$, and put

$$q_i = |V(Q_i)|, \qquad w_i = \lfloor q_i/2 \rfloor.$$

Optimality gives

$$|X| + \sum_i w_i = |X| + \frac{n - |X| - o(H-X)}{2} = \nu(H). \tag{8.1}$$

For $q \in \{1, \ldots, 10, 12\}$, set

$$f(q) = \left\lfloor \frac{q}{2} \right\rfloor + 2 + \left\lceil \frac{\lfloor q/2 \rfloor + 1}{2} \right\rceil .$$

Its values for $q = 1, \ldots, 10$ are

$$3, 4, 4, 6, 6, 7, 7, 9, 9, 10,$$

and $f(12) = 12$. Thus $q \leq f(q)$ in every required case. If $r + b = w_i + 1$, then

$$r + b + 1 + \max(r, b) \geq w_i + 2 + \left\lceil \frac{w_i + 1}{2} \right\rceil = f(q_i) \geq q_i.$$

Part (II) therefore splits Q_i at every pair

$$(r, w_i + 1 - r), \qquad 0 \leq r \leq w_i + 1.$$

Now let $x + y = \nu(H) + 1$. If $x \geq \nu(H)$, put every edge in red. Otherwise list pieces $\pi_1, \ldots, \pi_N$: first the vertices of X, each with weight one, and then the components Q_i, with weights w_i. Their total weight is $\nu(H) > x$. Let W_k be the sum of the first k weights, with $W_0 = 0$, and choose the least j with $W_j > x$. Put

$$r = x - W_{j-1}, \qquad 0 \leq r \leq w(\pi_j) - 1.$$

Call earlier pieces red and later pieces blue. If π_j is a vertex of X, call it blue. If $\pi_j = Q_i$, choose a split

$$E(Q_i) = P_R^i \sqcup P_B^i, \qquad \nu(P_R^i) \leq r, \quad \nu(P_B^i) \leq w_i + 1 - r.$$

Let P_R contain every edge incident with a red vertex of X, every edge of a red component, and P_R^i if a component straddles the cut. Put all remaining edges in P_B. Every edge either meets X or lies in one component of $H - X$.

A red matching has at most one edge for each red vertex of X, at most w_k edges inside each red component Q_k, and at most r edges in the straddling component. Its size is therefore at most

$$W_{j-1} + r = x.$$

A blue matching is bounded by the blue vertices of X, the weights of blue components, and the blue matching budget of the straddling component. If $\pi_j = Q_i$, this gives

$$\nu(P_B) \le \nu(H) - W_j + w_i + 1 - r = \nu(H) - W_{j-1} + 1 - r = y.$$

If π_j is a vertex of X, then $W_j = W_{j-1} + 1$, so $W_{j-1} = x$. The blue pieces have total weight $\nu(H) - x = y - 1$, which also suffices. Part (I) proves (III).

For (IV), Theorem 8.8 gives an optimal barrier X. If every component of H has at most ten vertices, then so does every component of $H - X$, and (III) applies. This includes $n(H) \le 10$. If $\nu(H) \le 4$, (8.1) gives $w_i \le 4$, hence

$$q_i \le 2w_i + 1 \le 9.$$

Apply (III) again. The remaining assertions are contrapositives.

For (V), assume $n = 11$ and $\mathrm{ex}(H) \ge 2$. Part (IV) gives $\nu(H) \ge 5$, while $\nu(H) \le \lfloor 11/2 \rfloor = 5$. A disconnected graph on eleven vertices has all components of order at most ten, so H is connected. Since

$$7 + \lceil 7/2 \rceil + 1 = 12 \ge 11,$$

part (II) gives $c_{0,0}(H) \le 7$. Thus $c_{0,0}(H) = 7$ and $\mathrm{ex}(H) = 2$. Some pair of total six fails by (I). Every such pair with maximum coordinate at least four succeeds by (II), so the failing pair is $(3, 3)$. Failure means exactly that every partition has a color class of matching number at least four. Conversely, failure of $(3, 3)$ gives $c_{0,0}(H) > 6$, while (II) gives $c_{0,0}(H) \le 7$. If $\nu(H) = 5$, the excess is two.

For (VI), five disjoint edges exist in K_{11}, and no matching can have more. By (V), it remains to exclude a split with both matching numbers at most three.

We first verify the multipartite matching calculation used here. A complete multipartite graph on N vertices, with largest part of size a, has a matching of size

$$\min\big(\lfloor N/2 \rfloor, N - a\big).$$

If $a \ge N - a$, match all vertices outside a largest part to distinct vertices inside it. Otherwise $a < N/2$, so $a \le h := \lfloor N/2 \rfloor$. Order the vertices with each part consecutive, and match position i to position $i + h$ for $1 \le i \le h$. These pairs are disjoint. A part containing both endpoints would have at least $h + 1 > a$ vertices, so every pair is an edge.

Suppose $E(K_{11}) = P_R \sqcup P_B$, with both matching numbers at most three. Apply Theorem 8.8 to the red spanning graph G_R. Some X satisfies

$$\frac{11 + |X| - o(G_R - X)}{2} = \nu(G_R) \le 3.$$

If p is the number of components of $G_R - X$, then

$$p \ge o(G_R - X) \ge |X| + 5.$$

These components partition $N = 11 - |X|$ vertices, so $|X| \le 3$. Their largest order is at most

$$N - (p - 1) \le N - |X| - 4.$$

Every edge between two different red components is blue. Thus the blue graph contains the complete multipartite graph with those components as parts. The calculation above gives

$$\nu(P_B) \ge \min\left(\left\lfloor \frac{11 - |X|}{2} \right\rfloor, |X| + 4\right) \ge 4,$$

a contradiction. □

Remark 8.14. The barrier proof assigns weight one to each barrier vertex and weight $(|V(C)| - 1)/2$ to each odd component furnished by Lemma 8.11. A red budget cuts this list in at most one component, and the vertex-count split spends at most one extra unit there. Thus excess two first occurs on eleven vertices, where it is the matching-Ramsey condition $H \to (4K_2, 4K_2)$.

[cite/blocker] The supplied interpretation attributes the complete-graph matching-Ramsey formula to Cockayne–Lorimer, but no corresponding ledger entry was supplied. The required K_{11} case is proved above.

8.4. **Partitions and low thresholds.** In this section, vertex partitions convert small induced degrees into upper bounds for complementary thresholds.

Theorem 8.15 (Lovász partitions and threshold pairs)**.** *Let G be a graph on n vertices.*

(I) *Let $m \in \mathbb{Z}_{\geq 1}$ and $a_1, \dots, a_m \in \mathbb{Z}_{\geq 0}$. If*

$$\sum_{i=1}^{m} (a_i + 1) \geq \Delta(G) + 1,$$

then $V(G)$ has a partition into possibly empty sets $V_1, \dots, V_m$ such that $\Delta(G[V_i]) \leq a_i$. In particular, capacities $a, b \in \mathbb{Z}_{\geq 0}$ with $a + b \geq \Delta(G) - 1$ admit such a two-part partition.

(II) *If $m \in \mathbb{Z}_{\geq 1}$, $\theta_1, \dots, \theta_m \in \mathbb{Z}_{\geq 1}$, and*

$$\sum_i \theta_i \geq \Delta(G) + 1,$$

then

$$\sum_i c^*_{\theta_i}(G) \leq (m-1)n.$$

In particular,

$$c^*_\theta(G) + c^*_{\theta'}(G) \leq n \quad \text{if } \theta, \theta' \geq 1, \quad \theta + \theta' \geq \Delta(G) + 1.$$

(III) *Let $D \in \mathbb{Z}_{\geq 1}$ satisfy $\Delta(G) \leq D$, and assume $G - v$ has a perfect matching. Then*

$$c^*_\theta(G) + c^*_{\theta'}(G) \leq n + 1 \quad \text{if } \theta, \theta' \geq 1, \quad \theta + \theta' \geq D.$$

If $\deg_G(v) \leq D - 1$, the upper bound improves to n.

(IV) *If G is factor-critical and $D \in \mathbb{Z}_{\geq 1}$ satisfies $D \geq \Delta(G)$, then n is odd,*

$$\nu(G) = \frac{n-1}{2}, \qquad c^*_\theta(G) \leq \frac{n+1}{2} \quad (\theta \geq 1,\ 2\theta \geq D).$$

If some vertex has degree at most $D - 1$, then

$$c^*_\theta(G) + c^*_{\theta'}(G) \leq n \quad (\theta, \theta' \geq 1,\ \theta + \theta' \geq D),$$

and

$$c^*_\theta(G) \leq \frac{n-1}{2} = \nu(G) \quad (\theta \geq 1,\ 2\theta \geq D).$$

Proof. We use Theorem 8.10(d),(e). They imply that, whenever $G - U$ is θ-good,

$$c^*_\theta(G) \leq |U|. \tag{8.2}$$

The remainder has zero cost by part (d), and reinserting the vertices of U costs at most $|U|$ by part (e). This use retains the dependency in Setup 8.9(1).

For (I), the case $m = 1$ follows by taking $V_1 = V(G)$. Let $m \geq 2$. Among all partitions $\mathcal{P} = (V_1, \dots, V_m)$, choose one minimizing

$$\Phi(\mathcal{P}) = \sum_i \frac{e(G[V_i])}{a_i + 1}.$$

Suppose $u \in V_i$ has $d_i \geq a_i + 1$ neighbors in V_i. Write $d_j = |N_G(u) \cap V_j|$. Then

$$\sum_{j \neq i} d_j \leq \Delta(G) - a_i - 1 \leq \sum_{j \neq i} (a_j + 1) - 1 < \sum_{j \neq i} (a_j + 1).$$

Some $j \neq i$ therefore has $d_j \leq a_j$. Moving u into V_j changes the potential by

$$-\frac{d_i}{a_i+1} + \frac{d_j}{a_j+1} \leq -1 + \frac{a_j}{a_j+1} < 0,$$

contradicting minimality. Thus every induced degree satisfies its bound. The two-part statement follows by setting $a_1 = a, a_2 = b$.

For (II), apply (I) with $a_i = \theta_i - 1$. It gives

$$\Delta(G[V_i]) \leq \theta_i - 1.$$

The induced graph $G[V_i]$ is θ_i-good, so (8.2) gives

$$c^*_{\theta_i}(G) \leq n - |V_i|.$$

Summing yields $mn - n = (m-1)n$. The pair statement is the case $m = 2$.

For (III), fix a perfect matching M of $G - v$, and put

$$a = \theta - 1, \qquad b = \theta' - 1.$$

Then $a + b \geq D - 2$. Let G' have vertex set $V(G) \setminus \{v\}$ and edge set $E(G - v) \setminus M$. Every vertex loses one matching edge, so

$$\Delta(G') \leq D - 1.$$

Part (I) gives a partition $V(G) \setminus \{v\} = V_1 \sqcup V_2$ with induced G'-degrees at most a, b.

For $H_1 = G[V_1]$ and $u \in V_1$,

$$\deg_{H_1}(u) = |N_{G'}(u) \cap V_1| + \mathbf{1}_{\{\text{the } M\text{-partner of } u \text{ lies in } V_1\}} \leq a + 1 = \theta.$$

Equality requires the matching partner to lie in V_1. Consequently $M \cap E(H_1)$ covers every degree-θ vertex, so H_1 is θ-good. The same argument makes $G[V_2]$ θ'-good. Applying (8.2),

$$c^*_\theta(G) \leq |V_2| + 1, \qquad c^*_{\theta'}(G) \leq |V_1| + 1.$$

Their sum is at most $n + 1$.

If $\deg_G(v) \leq D - 1$, instead let G'' have all vertices of G and edge set $E(G) \setminus M$. Vertices other than v lose one edge, while v already has degree at most $D - 1$. Thus $\Delta(G'') \leq D - 1$, and (I) partitions all of $V(G)$ into V_1, V_2 with the same bounds a, b. For $u \neq v$, equality at the restored degree bound again requires its matching partner inside the part. If $v \in V_1$, its degree in $G[V_1]$ is at most $a < \theta$, because it has no matching edge; the other part is treated in the same way. Both induced graphs are good, and

$$c^*_\theta(G) \leq |V_2|, \qquad c^*_{\theta'}(G) \leq |V_1|.$$

This proves the bound n.

For (IV), a perfect matching of $G - v$ shows that $n - 1$ is even and supplies $(n-1)/2$ edges. The vertex bound on a matching gives equality for $\nu(G)$. Apply (III) with $\theta' = \theta$ to obtain

$$2c^*_\theta(G) \leq n + 1.$$

If a vertex has degree at most $D - 1$, the improved part of (III) gives the pair bound n. Taking equal thresholds and using integrality and odd n gives

$$c^*_\theta(G) \leq \lfloor n/2 \rfloor = (n-1)/2.$$

□

Theorem 8.16 (König's theorem and subcubic thresholds)**.** *For a graph G, let $\lambda(G)$ be the minimum size of a vertex set U such that $G - U$ is a disjoint union of paths and even cycles, with isolated vertices counted as paths. Equivalently, $\Delta(G - U) \leq 2$ and $G - U$ has no odd cycle.*

(I) *Every bipartite graph B satisfies $\tau(B) = \nu(B)$.*
(II) *If $\Delta(G) \leq 3$, then $c^*_1(G) \leq \nu(G)$.*
(III) *Every graph G satisfies $c^*_2(G) \leq \lambda(G)$.*

Proof. For (I), use the prescribed-set formula recorded in Setup 8.9(2); this is an unresolved premise of this proof. Let B have bipartition (P, Q), and let u_P be the minimum number of vertices of P uncovered by a matching. Choose a maximizing set X^* with

$$u_P = q_P(B - X^*) - |X^*|.$$

A component of $B - X^*$ contained in P is a single vertex, because P is independent. Let P' be the set of these isolated P-vertices. Then

$$|P'| = q_P(B - X^*), \qquad N_B(P') \subseteq X^*.$$

The set

$$W = (P \setminus P') \cup N_B(P')$$

is a vertex cover: an edge pq, with $p \in P$, is covered by p if $p \notin P'$, and otherwise by $q \in N_B(P')$. Hence

$$\tau(B) \leq |P| - |P'| + |N_B(P')| \leq |P| - (|P'| - |X^*|) = |P| - u_P.$$

A matching leaving exactly u_P vertices of P uncovered has $|P| - u_P$ edges, because each matching edge contains exactly one vertex of P. Thus

$$\nu(B) \geq |P| - u_P \geq \tau(B).$$

Every matching uses distinct cover vertices, so $\tau(B) \geq \nu(B)$, proving equality.

For (II), choose a partition $V(G) = V_1 \sqcup V_2$ maximizing the number of crossing edges. If a vertex had at least two neighbors in its own part, it would have at most one in the other part. Moving it would increase the crossing count. Consequently,

$$Q := E(G[V_1]) \cup E(G[V_2])$$

has maximum degree at most one. The remaining edge set $R = E(G) \setminus Q$ is bipartite. Part (I) supplies a vertex cover W of R with

$$|W| = \nu(R) \leq \nu(G).$$

For $x + y = |W|$, partition $W = W_R \sqcup W_B$ with sizes x, y. Let P_R be the edges of R meeting W_R, and let $P_B = R \setminus P_R$. Every blue edge meets W_B, so the matching bounds are x, y. Thus

$$(Q, \varnothing, P_R, P_B)$$

is a $(1, 0; x, y)$-certificate. Theorem 8.10(a),(b) gives

$$c_1^*(G) = c_{1,0}(G) = c_{0,1}(G) \leq |W| \leq \nu(G).$$

For (III), choose U attaining $\lambda(G)$, and put $F = E(G - U)$. Alternating edge colors along paths and even cycles partitions F into two matchings; also $\Delta(F) \leq 2$. Given $x + y = |U|$, partition $U = U_R \sqcup U_B$ with sizes x, y. Let P_R contain all edges meeting U_R, and let P_B contain the remaining edges meeting U_B. Every edge outside F meets U, and the two matching bounds follow by counting distinct vertices in U_R, U_B.

For capacities $(2, 0)$, take $Q_R = F, Q_B = \varnothing$. For $(0, 2)$, reverse these choices. For $(1, 1)$, use the two alternating edge classes of F. All three capacity pairs certify every allocation of total $|U|$, so

$$c_2^*(G) \leq |U| = \lambda(G).$$

□

8.5. **Connected degree bounds for excess.** The goal of this section is to use isolated vertices after deletion to control both the cost and the location of missing allocations.

Theorem 8.17 (Degree bounds and missing-allocation intervals). (I) *Let H be connected on $n \geq 2$ vertices, and let $D \in \mathbb{Z}_{\geq 2}$ satisfy $\Delta(H) \leq D$. Put*

$$N = \lfloor n/2 \rfloor, \qquad \epsilon = n - 2N.$$

For an integer $e \geq 0$, define

$$c = N + e, \qquad x^* = \lfloor c/2 \rfloor, \qquad t^* = x^* - 2e - (1 - \epsilon).$$

If either $t^* \leq 0$, or

$$t^* \geq 1, \qquad (D-1)t^* + 1 \leq x^*, \qquad t^* \leq \left\lceil \frac{n-1}{D} \right\rceil,$$

then $c_{0,0}(H) \leq N + e$.

(II) *If H is connected of order $2k+1$, where $k \in \mathbb{Z}_{\geq 1}$, and $\Delta(H) \leq D$ for $D \in \mathbb{Z}_{\geq 2}$, then*

$$c_{0,0}(H) \leq k + \left\lceil \frac{(D-2)k + 2}{3D - 2} \right\rceil.$$

If $\nu(H) = k$, the last ceiling also bounds $\mathrm{ex}(H)$.

(III) *Let H be connected of order $2k+1$, with $\nu(H) = k$, and let $D \in \mathbb{Z}_{\geq 3}$ satisfy $\Delta(H) \leq D$. Assume $E := \mathrm{ex}(H) \geq 1$, and put*

$$u = \max\left(0, \left\lfloor \frac{2E - 3}{D - 2} \right\rfloor\right).$$

Every integer $x \in [0, k + E - 1]$ for which $(x, k + E - 1 - x)$ fails lies in

$$[\,2E - 1 + u,\ k - E - u\,].$$

Consequently,

$$1 \leq \mathrm{tr}(H) \leq k - 3E + 2 - 2u, \qquad k \geq 3E - 1 + 2u.$$

More generally, for every integer B, if $\mathrm{tr}(H) + E \geq B + 1$, then

$$k \geq B + 2E - 1 + 2u.$$

No LP-goodness, gap, regularity, minimum-degree, or factor-criticality hypothesis is imposed.

Proof. This proof uses the independent-set witness in Setup 8.9(3).

We first spell out the isolation mechanism. If $|W| = x$ and $H - W$ has at least $n - x - 2y - 1$ isolated vertices, put all edges meeting W in red. A red matching has at most x edges. The blue graph then has at most $2y + 1$ nonisolated vertices, so its matching number is at most y. If the required number of isolated vertices is nonpositive, the same conclusion follows directly from $n - x \leq 2y + 1$.

For (I), fix $x + y = c$, and exchange colors so that $x \leq y$, hence $x \leq x^*$. If $y \geq N$, put every edge in blue, since $\nu(H) \leq N$. Assume $y \leq N - 1$. In particular, $x \leq y < n$. Put

$$a = 2e + (1 - \epsilon), \qquad t = x - a.$$

Since $y = N + e - x$ and $n = 2N + \epsilon$,

$$t = n - x - 2y - 1, \qquad t \leq t^*.$$

If $t \leq 0$, any x-vertex set W works by the isolation mechanism.

If $t \geq 1$, then $t^* \geq 1$. The assumed inequality for t^* is equivalent to

$$(D-2)x^* + 1 \leq (D-1)a.$$

Because $D \geq 2$ and $x \leq x^*$, this implies

$$(D-2)x+1 \leq (D-1)a, \qquad \text{hence} \qquad (D-1)t+1 \leq x.$$

Also $t \leq \lceil (n-1)/D \rceil$. The required witness supplies an independent set I of size t with

$$|N_H(I)| \leq (D-1)t+1 \leq x.$$

Enlarge its neighborhood to an x-vertex set W avoiding I. This is possible because

$$n-t = x+2y+1 \geq x.$$

Deleting W isolates I, so the isolation mechanism certifies (x,y). Every pair of total c has now been treated, and Theorem 8.13(I) gives the conclusion.

For (II), set

$$e_0 = \left\lceil \frac{(D-2)k+2}{3D-2} \right\rceil, \qquad x^* = \left\lfloor \frac{k+e_0}{2} \right\rfloor, \qquad t^* = x^* - 2e_0.$$

The denominator is positive, and

$$(3D-2)e_0 \geq (D-2)k+2.$$

If $t^* \leq 0$, apply (I). Otherwise,

$$(D-2)x^*+1 \leq \frac{(D-2)(k+e_0)+2}{2} \leq 2(D-1)e_0,$$

which is equivalent to $(D-1)t^*+1 \leq x^*$. Moreover,

$$t^* \leq \frac{k-3e_0}{2} \leq \frac{2k-3}{3D-2} < \frac{2k}{D}.$$

The middle inequality substitutes the lower bound on e_0. The strict inequality follows from

$$2k(3D-2) - D(2k-3) = 4k(D-1)+3D > 0.$$

Thus $t^* \leq \lceil 2k/D \rceil$, and (I) applies. Subtracting $\nu(H) = k$ gives the excess bound when that matching hypothesis holds.

For (III), put

$$\ell = k+E-1 = c_{0,0}(H)-1, \qquad L_0 = 2E-2+u.$$

We show that every pair of total ℓ with minimum coordinate at most L_0 succeeds. Exchange colors so that $x \leq y$ and $x \leq L_0$. If $x \leq 2E-2$, then

$$y = \ell - x \geq k-E+1, \qquad 2k+1 \leq \ell+1+y.$$

Theorem 8.13(II) applies.

Otherwise put $t = x-2E+2$, so $1 \leq t \leq u$. In particular, $u \geq 1$ and $E \geq 2$. Its definition gives

$$(D-2)u \leq 2E-3,$$

and therefore

$$(D-1)t+1 \leq t+2E-2 = x.$$

Theorem 8.13(II), applied at total $k+\lceil k/3 \rceil$, gives

$$E \leq \lceil k/3 \rceil \leq (k+2)/3,$$

because

$$k+\lceil k/3 \rceil + \left\lceil \frac{k+\lceil k/3 \rceil}{2} \right\rceil + 1 \geq 2k+1.$$

Consequently,

$$u \leq \frac{2E-3}{D-2} \leq \frac{2k-5}{3(D-2)} < \frac{2k}{D}.$$

For the last inequality,

$$6k(D-2) - D(2k-5) = 4k(D-3) + 5D > 0.$$

The witness in Setup 8.9(3) supplies an independent t-set with neighborhood size at most x. Enlarge that neighborhood to W of size x, avoiding the independent set. Here again

$$t = n - x - 2y - 1, \qquad n - t = x + 2y + 1 \geq x,$$

so the enlargement is possible and the isolation split succeeds.

Every missing pair therefore has both coordinates at least

$$L_0 + 1 = 2E - 1 + u.$$

Its red budget lies in

$$[\,2E - 1 + u,\ \ell - (2E - 1 + u)\,] = [\,2E - 1 + u,\ k - E - u\,].$$

This integer interval has $k - 3E + 2 - 2u$ elements. Some allocation is missing at total $c_{0,0}(H) - 1$ by Theorem 8.13(I), so $\mathrm{tr}(H) \geq 1$. Every missing run lies in the displayed interval, proving

$$1 \leq \mathrm{tr}(H) \leq k - 3E + 2 - 2u.$$

Thus $k \geq 3E - 1 + 2u$. Finally,

$$\mathrm{tr}(H) + E \leq k - 2E + 2 - 2u.$$

Combining this with $\mathrm{tr}(H) + E \geq B + 1$ proves the final bound. □

8.6. **Components and the LP-goodness criterion.** In this section, the maximum rule for excess and the additive lower bound for slack reduce the mixed-host inequality to one component.

Proposition 8.18 (Excess and slack under disjoint union)**.** *If $H = H_1 \sqcup H_2$ is a vertex-disjoint union, then*

$$\mathrm{ex}(H) \leq \max\{\mathrm{ex}(H_1), \mathrm{ex}(H_2)\}, \qquad \mathrm{ps}(H) \geq \mathrm{ps}(H_1) + \mathrm{ps}(H_2).$$

If H has at least one edge, then

$$\mathrm{ex}(H) \leq \max_C \mathrm{ex}(C), \qquad \mathrm{ps}(H) \geq \sum_C \mathrm{ps}(C),$$

where C ranges over its components containing an edge.

Proof. Put $N_i = \nu(H_i)$. Matching number and edge number add over a vertex-disjoint union. By Theorem 8.10(c),

$$c^*_\theta(H) \leq c^*_\theta(H_1) + c^*_\theta(H_2).$$

Summing over positive thresholds and subtracting from $e(H) - \nu(H)$ proves the slack inequality.

For excess, put

$$\epsilon = \max\{\mathrm{ex}(H_1), \mathrm{ex}(H_2)\}, \qquad c = N_1 + N_2 + \epsilon.$$

Let $x + y = c$. If $x \geq N_1$, color all edges of H_1 red. The remaining pair $(x - N_1, y)$ has total

$$N_2 + \epsilon \geq c_{0,0}(H_2),$$

so Theorem 8.10(a) certifies it in H_2. If $x < N_1$, then

$$y = N_1 + N_2 + \epsilon - x > N_2 + \epsilon \geq N_2.$$

Color all edges of H_2 blue. The remaining pair $(x, y - N_2)$ has total $N_1 + \epsilon \geq c_{0,0}(H_1)$, so it is certified in H_1. Combining the two certificates proves that every pair of total c works in H. Hence

$$c_{0,0}(H) \leq c, \qquad \mathrm{ex}(H) \leq \epsilon.$$

Isolated vertices affect none of these quantities by Theorem 8.10. Delete them and induct on the number of remaining components. □

Theorem 8.19 (Mixed hosts at level zero)**.** *Use Setup 8.7.*

(I) *If $H \to (F_1, F_2)$ and every component C of H containing an edge satisfies*

$$\mathrm{ex}(C) \le 1 \qquad \text{or} \qquad \Sigma(C) \le e(C),$$

then $e(H) \ge L$.

(II) *The same conclusion holds if every component with at least eleven vertices has excess at most one or is LP-good.*

(III) *If every connected graph with excess at least two is LP-good, then the size Ramsey number of (F_1, F_2) equals L for every pair of sequences in Setup 8.7.*

Proof. This argument requires Setup 8.9(4)–(6).

For (I), suppose $e(H) \le L - 1$, and put

$$\sigma = e(H) - \nu(H) - (L - M), \qquad \delta = M - \nu(H).$$

The required arrowing inequalities give

$$e(H) = L + \sigma - \delta, \qquad \sigma \ge \mathrm{ps}(H) \ge 0.$$

Thus $\delta \ge \sigma + 1 \ge 1$, and $\nu(H) < M$. If $\delta = 1$, then $\sigma = 0$, so

$$e(H) - \nu(H) = L - M, \qquad \nu(H) < M.$$

The zero-surplus coloring assertion supplies a coloring avoiding both targets, contradicting arrowing. Therefore $\delta \ge 2$. The arrowing inequalities also give

$$\mathrm{ex}(H) \ge \delta \ge 2.$$

In particular, H has an edge: an edgeless graph has $c_{0,0} = 0$ and excess zero.

Choose an edge-containing component C^* maximizing excess. Proposition 8.18 gives

$$\mathrm{ex}(C^*) \ge \mathrm{ex}(H) \ge 2.$$

The hypothesis makes C^* LP-good, and Theorem 8.10(i) gives

$$\mathrm{ps}(C^*) \ge \mathrm{ex}(C^*).$$

Combining the arrowing inequality, Proposition 8.18, and nonnegative component slack,

$$\begin{aligned} L - e(H) &\le \mathrm{ex}(H) - \mathrm{ps}(H) \\ &\le \mathrm{ex}(C^*) - \sum_C \mathrm{ps}(C) \\ &\le \mathrm{ex}(C^*) - \mathrm{ps}(C^*) \\ &\le 0. \end{aligned}$$

This contradicts $e(H) \le L - 1$.

For (II), every component of order at most ten has excess at most one by Theorem 8.13(IV), so (I) applies.

For (III), the stated connected-graph hypothesis makes every edge-containing component either of excess at most one or LP-good. Part (I) gives the lower bound L. The construction required in Setup 8.9(6) gives an arrowing graph with exactly L edges, proving equality. □

8.7. **Positive potential slack.** The goal of this section is to identify deletions that force positive slack and to apply them to one-vertex extensions of odd cliques.

Theorem 8.20 (Covered deletions and odd cliques)**.** (I) *Let J have maximum degree $D \in \mathbb{Z}_{\geq 1}$, and assume a matching covers every degree-D vertex. For every vertex v of degree D, put $t = \nu(J) - \nu(J - v)$. Then*

$$\operatorname{ps}(J) \geq \operatorname{ps}(J - v) + 1 - t.$$

Consequently, every nonregular factor-critical graph of order at least three has potential slack at least one.

(II) *Every odd complete graph K_k, where $k \geq 5$, has $\operatorname{ps}(K_k) \geq 1$.*

(III) *Let $x \in V(G)$, and assume the components $T_1, \dots, T_p$ of $G - x$ are odd complete graphs, where $p \in \mathbb{Z}_{\geq 1}$. If*

$$\alpha(G) \leq p, \qquad \operatorname{ps}(G) = 0,$$

then every T_i has order one or three.

Proof. For (I), deleting a vertex changes matching number by $t \in \{0, 1\}$. Coveredness and Theorem 8.10(d) give $c_D^*(J) = 0$, and all higher thresholds vanish. Subgraph monotonicity gives the same vanishing for $J - v$. At every positive threshold below D, part (e) gives a cost increase of at most one. Since deleting v removes D edges,

$$\begin{aligned}\operatorname{ps}(J) - \operatorname{ps}(J - v) &= D - t - \sum_{\theta \geq 1} \bigl(c_\theta^*(J) - c_\theta^*(J - v)\bigr) \\ &\geq 1 - t.\end{aligned}$$

Let Q be nonregular and factor-critical of order $n \geq 3$. Then n is odd and

$$\nu(Q) = \nu(Q - v) = (n - 1)/2$$

for every vertex v. Its maximum degree is positive, because an edgeless graph of order at least three is not factor-critical. Choose a vertex w whose degree is below the maximum. A perfect matching of $Q - w$ covers all maximum-degree vertices of Q. Applying the deletion inequality to a maximum-degree vertex v, with $t = 0$, gives

$$\operatorname{ps}(Q) \geq \operatorname{ps}(Q - v) + 1 \geq 1.$$

For (II), use the clique calculations in Setup 8.9(7). They give

$$\operatorname{ex}(K_5) = 1, \qquad \operatorname{gap}(K_5) \leq 0.$$

Theorem 8.10(i) yields

$$\operatorname{ps}(K_5) = \operatorname{ex}(K_5) - \operatorname{gap}(K_5) \geq 1.$$

For odd $k \geq 7$, the required bound is

$$\operatorname{gap}(K_k) \leq \mathbf{1}_{\{a \text{ odd}\}} - \frac{a(a-1)}{2}$$

for an integer $a \geq 2$. The right side is -1 for $a = 2$, and at most $1 - 3 = -2$ for $a \geq 3$. Thus $\operatorname{gap}(K_k) \leq -1$. Since $\operatorname{ex}(K_k) \geq 0$, the same identity gives $\operatorname{ps}(K_k) \geq 1$.

For (III), put

$$k_i = |V(T_i)|, \qquad k = \max_i k_i, \qquad h = \deg_G(x).$$

If each clique contained a nonneighbor of x, those p vertices together with x would be independent. Hence x is adjacent to every vertex of at least one T_i, and $h \geq 1$.

A matching uses at most one edge at x. If it uses none, its size is at most $\sum_i (k_i - 1)/2$. If it uses an edge from x into T_j, the internal matching in T_j has at most $(k_j - 1)/2$ edges, and the same

bound holds in every other clique. Conversely, choose any edge at x, perfectly match the remaining even set in its clique, and take near-perfect matchings in all other cliques. Thus

$$\nu(G) = 1 + \sum_i \frac{k_i - 1}{2}. \tag{8.3}$$

Suppose $k \geq 5$. If an order-k clique has no neighbor of x, it is an entire component of G. Part (II), Proposition 8.18, and nonnegative slack would give $\mathrm{ps}(G) \geq 1$. Hence every largest clique touches x.

First suppose $h \geq k$. Vertices of T_i have degree $k_i - 1$ or k_i, according as they avoid or meet x. Therefore $\Delta(G) = h$, and a vertex other than x has degree h only when it lies in an order-h clique and is adjacent to x.

If an order-h clique is entirely adjacent to x, it uses all h neighbors of x. Match x into that clique, perfectly match its remainder, and take near-perfect matchings elsewhere. This covers all degree-h vertices. If no order-h clique is entirely adjacent to x, each has a nonneighbor of x. Match x into any neighboring clique and perfectly match its remainder. In each other order-h clique, choose a near-perfect matching omitting a nonneighbor of x; choose arbitrary near-perfect matchings in the remaining cliques. Again all degree-h vertices are covered.

By (8.3),

$$\nu(G) - \nu(G - x) = 1.$$

Part (I) applied to x gives $\mathrm{ps}(G) \geq \mathrm{ps}(G - x)$. Proposition 8.18 and part (II) give

$$\mathrm{ps}(G - x) \geq \mathrm{ps}(K_k) \geq 1,$$

a contradiction.

Now suppose $h < k$. Since every largest clique touches x, $\Delta(G) = k$. Its degree-k vertices are precisely the neighbors of x in order-k cliques. Each such clique has a nonneighbor of x. There is a clique T_j entirely adjacent to x, and its order is at most $h < k$. Match x into T_j, perfectly match its remainder, and in each order-k clique choose a near-perfect matching omitting a nonneighbor of x. Complete the other clique matchings arbitrarily. This covers all degree-k vertices.

Choose a neighbor v of x in a largest clique. It has degree k. A maximum matching can omit v: match x into the smaller clique T_j, omit v in its clique, and complete the other matchings as in (8.3). Thus $\nu(G - v) = \nu(G)$. Part (I) gives

$$\mathrm{ps}(G) \geq \mathrm{ps}(G - v) + 1 \geq 1,$$

again a contradiction. Therefore $k \leq 3$, and all positive odd k_i belong to $\{1, 3\}$. □

8.8. **Factor-critical graphs of zero slack.** In this section, the supplied structural alternatives are excluded by small replacements in a maximum independent set.

Theorem 8.21 (Factor-critical zero-slack classification)**.** *For every factor-critical graph Q of order at least three,*

$$\mathrm{ps}(Q) = 0 \quad \Longleftrightarrow \quad Q \cong C_3,\ C_5,\ \text{or } C_7.$$

Consequently every other factor-critical graph of order at least three has $\mathrm{ps}(Q) \geq 1$. *In particular, every connected regular factor-critical graph of degree at least four has strictly positive generic potential slack. No bound on order, independence number, excess, or regular degree is imposed.*

Proof. This proof is conditional on Setup 8.9(8)–(11), as well as the premises of Theorem 8.20. All costs and slacks in this proof are generic.

Assume Q is factor-critical, $n = n(Q) \geq 3$, and $\mathrm{ps}(Q) = 0$. The structural assertion in Setup 8.9(8) makes Q connected and regular of positive even degree D. Put $s = \alpha(Q)$. Its alternatives are

$$n = 2s + 1, \qquad \text{or} \qquad D = 4, \quad n = 2s + 3,$$

and $s = 1$ forces K_3, while $s = 2$ forces C_5.

Choose a maximum independent set S, put $G = Q - S$, and let

$$k = (n-1)/2 = \nu(Q), \qquad N_S(v) = N_Q(v) \cap S \quad (v \notin S).$$

The required deletion assertion gives

$$\mathrm{ps}(G) = 0, \qquad \nu(G) = k + 1 - s, \qquad |X| \le 1 \tag{8.4}$$

for every maximum-cardinality attaining set X of G. Every vertex outside S has a neighbor in S, since otherwise S could be enlarged. Thus $\Delta(G) \le D - 1$. Regularity and independence give

$$e(Q) = e(G) + Ds, \qquad e(G) = \frac{D(n-2s)}{2}. \tag{8.5}$$

Step 1. Zero slack forces equality in every independent-set deletion bound.

For $1 \le i \le D-1$, Theorem 8.10(e) gives

$$c_i^*(Q) \le s + c_i^*(G).$$

For any vertex v, a perfect matching of $Q - v$ covers every degree-D vertex there. Parts (d),(e) give $c_D^*(Q) \le 1$. Thresholds above D vanish on Q, and thresholds at least D vanish on G. Therefore

$$\begin{aligned}\sum_{i\ge 1} c_i^*(Q) &\le (D-1)s + \sum_{i\ge 1} c_i^*(G) + 1 \\ &= (D-1)s + e(G) - \nu(G) + 1 \\ &= e(G) + Ds - k \\ &= e(Q) - \nu(Q).\end{aligned}$$

Here we used (8.4), (8.5), and $\mathrm{ps}(G) = 0$. The left side equals the last expression because $\mathrm{ps}(Q) = 0$. Every summed bound has a nonnegative difference, so

$$c_i^*(Q) = s + c_i^*(G) \quad (1 \le i \le D-1), \qquad c_D^*(Q) = 1. \tag{8.6}$$

We will contradict these equalities when $D \ge 4$. We also use (8.2): a θ-good remainder $Q - U$ gives $c_\theta^*(Q) \le |U|$.

Step 2. The alternative $n = 2s + 1$ is impossible when $D \ge 4$.

Equations (8.4) and (8.5) give

$$\nu(G) = 1, \qquad e(G) = D/2 \ge 2.$$

A graph with matching number one and at least two edges consists, apart from isolated vertices, of a star or a triangle. To verify this, choose intersecting distinct edges ab, ac. If every edge meets a, they form a star. Otherwise an edge avoiding a must meet both chosen edges, so it is bc. Any further edge must meet ab, ac, bc, and therefore is one of those triangle edges.

Thus G is a star with $D/2$ edges, or a triangle with $D = 6$, together with isolated vertices. In either case $\Delta(G) \le D - 2$, so $c_{D-1}^*(G) = 0$. Let v be the star center or any triangle vertex. It has at least two neighbors in S, because

$$|N_S(v)| = D - \deg_G(v) \ge 2.$$

Choose distinct $a, b \in N_S(v)$, and delete

$$U = (S \setminus \{a, b\}) \cup \{v\}, \qquad |U| = s - 1.$$

Put $J = Q - U$. Each of a, b has degree exactly $D - 1$, since the only deleted vertex outside S is its neighbor v. They are nonadjacent. In the star case, all other retained vertices have degree at most two because $G - v$ is edgeless. In the triangle case they have degree at most three because $G - v$ has one edge, while $D - 1 = 5$.

Thus the only degree-$(D-1)$ vertices of J are a, b. Each has at least three neighbors outside $\{a, b\}$, so distinct neighbors can be chosen to form a two-edge matching covering them. The graph J is $(D-1)$-good, and

$$c^*_{D-1}(Q) \le s - 1.$$

This contradicts (8.6), which gives $c^*_{D-1}(Q) = s$.

Step 3. We reduce the remaining alternative to two configurations with six residual edges.

Assume $D = 4$ and $n = 2s+3$. The small-independence conclusions of Setup 8.9(8) imply $s \ge 3$. Equations (8.4) and (8.5) become

$$\nu(G) = 2, \qquad e(G) = 6, \qquad \mathrm{ps}(G) = 0. \tag{8.7}$$

Choose a maximum-cardinality attaining set X of G, and put $b = |X| \le 1$. Lemma 8.11 says every component of $G - X$ is odd and factor-critical. Its matching identity gives the number p of components:

$$p = |V(G)| + b - 2\nu(G) = s - 1 + b. \tag{8.8}$$

Suppose first that $b = 0$. Then G has $s-1$ odd factor-critical components. Proposition 8.18 and nonnegative slack make every component have slack zero. Their independence numbers sum to $\alpha(G) \le s$, so every component has independence number one, except possibly one of independence number two. The small-independence structural conclusions imply that the former components are singletons or triangles, and the latter, if present, is C_5. Singletons are allowed separately because the structural assertion concerns order at least three. A C_5 component would make $e(G)$ equal to five plus a multiple of three, contrary to $e(G) = 6$. Thus G consists of two triangles T_1, T_2 and isolated vertices.

Every triangle vertex has exactly two neighbors in S. There are $v_1 \in T_1$, $v_2 \in T_2$ with

$$N_S(v_1) \ne N_S(v_2).$$

Otherwise all six triangle vertices would share the same two neighbors, giving those neighbors degree at least six in Q, contrary to $D = 4$. The union of these two neighborhoods has at least three elements. Choose a three-element subset R of it, and put

$$P = \{v_1, v_2\}, \qquad U = (S \setminus R) \cup P, \qquad J = Q - U.$$

Then $|U| = s - 1$. Every vertex of R loses a neighbor in P, so its degree in J is at most three. The retained triangle vertices have internal degree one and at most two neighbors in R. Other retained vertices outside R have internal degree zero and at most three neighbors in R. Thus J is subcubic.

Every cubic vertex outside R has at least two neighbors in R. Since the sum of degrees in R is at most nine, there are at most $\lfloor 9/2 \rfloor = 4$ such cubic vertices, and at most seven cubic vertices altogether. Setup 8.9(10) makes J 3-good. Hence $c^*_3(Q) \le s - 1$. But $\Delta(G) = 2$, so $c^*_3(G) = 0$, and (8.6) gives $c^*_3(Q) = s$, a contradiction.

Step 4. The configuration with a one-vertex attaining set also admits a saving deletion.

Let $b = 1$, with $X = \{x\}$. By (8.8), $G - x$ has s components. Each has independence number one, because their independent sets combine in G and $\alpha(G) \le s$. They are therefore odd complete graphs. Theorem 8.20(III), applied using $\mathrm{ps}(G) = 0$, shows that each has order one or three. The attaining matching identity and $\nu(G) = 2$ show that exactly one is a triangle. Call it T, and let Z be the set of the other $s-1$ vertices.

Every edge of G is a triangle edge or meets x. Since $e(G) = 6$, the vertex x has degree three in G, and hence a unique neighbor a in S. Put

$$r = |N_G(x) \cap T| \in \{0, 1, 2, 3\}, \qquad |N_G(x) \cap Z| = 3 - r. \tag{8.9}$$

For $v \in T$ and $z \in Z$,

$$|N_S(v)| = 2 - \mathbf{1}_{\{xv \in E(G)\}}, \qquad |N_S(z)| = 4 - \mathbf{1}_{\{xz \in E(G)\}}.$$

The only possible cubic vertices of G are x and the at most three triangle vertices adjacent to it. Setup 8.9(10) makes G 3-good, so $c_3^*(G) = 0$. Equation (8.6) gives

$$c_3^*(Q) = s.$$

We construct U of size $s-1$ leaving a 3-good graph. Choose two vertices $P \subseteq V(G)$ and a three-element set

$$R \subseteq N_S(P) := \bigcup_{v \in P} N_S(v),$$

and put $U = (S \setminus R) \cup P$, $J = Q - U$. Each vertex of R loses a neighbor, so has degree at most three in J. We use two forms.

In Form A, $P = \{x, v\}$ for $v \in T$. Then $G - P$ is one edge and isolated vertices. Its triangle vertices have at most two neighbors in S. Thus J is subcubic, and every cubic vertex outside R has at least two neighbors in R. Counting the at most nine incidences from R gives at most seven cubic vertices in all. The small cubic-exposure assertion makes J 3-good.

In Form B, $P = \{v, z\}$, with $v \in T$, $z \in Z$, and x adjacent to at least one of v, z. Require also

$$R \neq N_S(w) \quad \text{for every remaining } w \in Z \cap N_G(x).$$

These forbidden sets have size three. The retained x has internal degree at most two in $G - P$, and at most one neighbor in R. A retained triangle vertex w has internal degree

$$1 + \mathbf{1}_{\{xw \in E(G)\}}$$

and at most $2 - \mathbf{1}_{\{xw \in E(G)\}}$ neighbors in R. A retained $w \in Z$ adjacent to x keeps at most two of its three S-neighbors, because $R \neq N_S(w)$, together with its possible internal neighbor x. A vertex of Z not adjacent to x has internal degree zero and at most three neighbors in R. Hence $\Delta(J) \leq 3$.

Only x and the two retained triangle vertices can have internal degree two in $G - P$. Thus every cubic vertex outside R has at least two neighbors in R, with at most three exceptions having at least one. If their number is h, then

$$2h - 3 \leq \sum_{u \in R} \deg_J(u) \leq 9.$$

Therefore $h \leq 6$, and J has at most nine cubic vertices. Again the small cubic-exposure assertion makes J 3-good.

We now obtain one of these forms in every case.

If $r = 0$ and some $v \in T$ has $a \notin N_S(v)$, take

$$P = \{x, v\}, \qquad R = \{a\} \cup N_S(v).$$

This is a three-element set, so Form A applies.

If $r = 0$ and $a \in N_S(v)$ for every $v \in T$, then a is adjacent to x and all three triangle vertices. These are all four of its neighbors in Q, so no vertex of Z is adjacent to a. By (8.9), choose $z \in N_G(x) \cap Z$. Its three S-neighbors avoid a. Choose distinct $b, c \in N_S(z)$, choose any $v \in T$, and set

$$P = \{v, z\}, \qquad R = \{a, b, c\}.$$

Then $R \subseteq N_S(P)$, and x is adjacent to z. Every forbidden $N_S(w)$ avoids a, so Form B applies.

Suppose $r \geq 1$ and some $z \in Z$ is not adjacent to x. Choose $v \in T$ adjacent to x. The set $N_S(z)$ has four elements, so $N_S(v) \cup N_S(z)$ has at least four distinct three-element subsets. There are at most $3 - r \leq 2$ forbidden sets from vertices of Z adjacent to x. Choose a three-element subset R avoiding all of them, and take $P = \{v, z\}$. Form B applies.

Finally, suppose $r \geq 1$ and every $z \in Z$ is adjacent to x. Then

$$s - 1 = |Z| = 3 - r.$$

Since $s \geq 3$, this forces $s = 3, r = 1$. The two vertices of Z each have S-neighborhood S. The two triangle vertices not adjacent to x cannot both be adjacent to a, since then a would have five neighbors: x, both vertices of Z, and those two triangle vertices. Choose such a triangle vertex v not adjacent to a. Then

$$N_S(v) = S \setminus \{a\}.$$

Taking $P = \{x, v\}$ and $R = S = N_S(P)$ gives Form A.

These cases exhaust (8.9). Each gives

$$|U| = s - 3 + 2 = s - 1$$

and a 3-good remainder. Thus $c_3^*(Q) \leq s - 1$, contradicting $c_3^*(Q) = s$. The second structural alternative is excluded.

Step 5. The remaining graphs are precisely the three stated cycles.

The degree D is positive and even, and all $D \geq 4$ have been excluded. Hence $D = 2$. A finite connected 2-regular graph is a cycle: following edges closes a cycle, whose vertices already use both incident edges, and connectedness excludes any other vertex. Factor-criticality makes its order odd. Setup 8.9(11) gives positive slack for odd cycles of order at least nine. Thus Q is C_3, C_5, or C_7.

Conversely, deleting any vertex from any of these odd cycles leaves an even-order path, whose alternating edges form a perfect matching. For C_3, the required cycle calculation gives $\nu = c_{0,0} = 1$. Monotonicity bounds both c_1^*, c_2^* by one. Both are positive: at total matching budget zero, capacities $(1, 0)$ would require the whole triangle to be a matching, and capacities $(1, 1)$ would require its edges to partition into two matchings. Neither is possible. Thresholds at least three vanish, so

$$\mathrm{ps}(C_3) = 3 - 1 - 1 - 1 = 0.$$

For C_5, C_7, the required cycle calculations give excess one and positive gap. Since

$$\mathrm{gap} = \mathrm{ex} - \mathrm{ps}, \qquad \mathrm{ps} \geq 0,$$

their integral gap is one and their slack is zero. Finally, nonnegative integral slack is at least one whenever it is nonzero. This proves every stated consequence. □

8.9. **Independence-number bounds.** The goal of this section is to preserve order minus matching number through maximum-degree deletions and finish on paths and cycles.

Theorem 8.22 (Zero slack and independence number)**.** *Let H be a graph.*

(I) *If $\Delta(H) = 3$ and κ is the minimum number of degree-three vertices left unmatched by a matching, then*

$$\mathrm{ps}(H) \geq 2\kappa.$$

(II) *If $\psi_\sharp(H) = 0$, then*

$$n(H) - \nu(H) \leq 2\alpha(H), \qquad n(H) \leq 4\alpha(H).$$

More precisely, repeatedly delete an arbitrary maximum-degree vertex while the current maximum degree is at least three, and let R be the terminal induced graph. Every deletion lowers matching number by exactly one, every intermediate graph has zero sharp slack, and

$$n(R) - \nu(R) = n(H) - \nu(H).$$

Equality $n(H) - \nu(H) = 2\alpha(H)$ implies, for every such deletion sequence, that R is a disjoint union of triangles, with the empty union allowed, and $\alpha(R) = \alpha(H)$. Equality $n(H) = 4\alpha(H)$ additionally implies that H has a perfect matching.

(III) *Every assertion in (II) holds with generic* ps *in place of sharp slack if every induced subgraph J of H, including H, has $\mathrm{gap}(J) \leq 1$. These generic conclusions hold without that hereditary hypothesis when $\Delta(H) \leq 5$.*

Proof. The cubic argument uses Setup 8.9(2) and (12). The deletion argument at degree at least four uses item 13.

Sharp costs inherit subgraph monotonicity and the one-vertex inequality by maximizing over the prescribed capacity pairs. Also

$$0 \le C_\sharp(\theta, J) \le c^*_\theta(J),$$

so Theorem 8.10(h) gives

$$\psi_\sharp(J) \ge \mathrm{ps}(J) \ge 0.$$

For (I), if $\kappa = 0$, use nonnegative generic slack. Assume $\kappa \ge 1$. Let S be the degree-three vertices, and choose X of maximum cardinality among the sets attaining

$$q_S(H - X) - |X| = \kappa.$$

Put $b = |X|$, let $\mathcal{K}$ consist of the odd components of $H - X$ contained in S, and let Y comprise all vertices of the other components. The required prescribed-exposure refinement gives

$$|\mathcal{K}| = b + \kappa,$$

factor-criticality of every $K \in \mathcal{K}$, and the fact that all boundary edges of K end in X.

Let $d(K)$ be the boundary size. Since every vertex of K has degree three in H,

$$3|V(K)| = 2e(K) + d(K),$$

so $d(K)$ is odd. It cannot be zero, which would give a cubic graph of odd order and an odd degree sum. In particular, $X \neq \varnothing$.

If $d(K) = 1$, precisely one vertex has degree two inside K, and all others have degree three. The almost-cubic premise gives

$$\mathrm{ps}(K) \ge 2 = 3 - d(K).$$

If $d(K) \ge 3$, nonnegative slack gives

$$\mathrm{ps}(K) \ge 0 \ge 3 - d(K).$$

Thus every $K \in \mathcal{K}$ satisfies $\mathrm{ps}(K) \ge 3 - d(K)$.

The required matching identity is

$$\nu(H) = b + \sum_Q \nu(Q),$$

over all components Q of $H - X$. Put

$$e_{\mathrm{inc}} = e(H) - \sum_Q e(Q).$$

Repeated vertex deletion and union subadditivity give, for $i = 1, 2$,

$$c^*_i(H) \le b + \sum_Q c^*_i(Q).$$

The required top-threshold bound is $c^*_3(H) \le \kappa$, and thresholds above three vanish. Therefore

$$\begin{aligned}\mathrm{ps}(H) &\ge e_{\mathrm{inc}} - 3b - \kappa + \sum_Q \big(e(Q) - \nu(Q) - c^*_1(Q) - c^*_2(Q)\big) \\ &\ge e_{\mathrm{inc}} - 3b - \kappa + \sum_Q \mathrm{ps}(Q).\end{aligned}$$

The last inequality discards the nonnegative costs $c^*_3(Q)$.

The edge decomposition gives

$$e_{\mathrm{inc}} = e(X) + e(X, Y) + \sum_{K \in \mathcal{K}} d(K).$$

Using $|\mathcal{K}| = b + \kappa$, we obtain the exact identity

$$3b + \kappa - e_{\mathrm{inc}} = \sum_{K \in \mathcal{K}} (3 - d(K)) - 2\kappa - e(X) - e(X, Y).$$

Substitution yields

$$\begin{aligned} \mathrm{ps}(H) \geq 2\kappa + e(X) + e(X, Y) + \sum_{Q \notin \mathcal{K}} \mathrm{ps}(Q) \\ + \sum_{K \in \mathcal{K}} \big(\mathrm{ps}(K) - (3 - d(K))\big) \geq 2\kappa. \end{aligned}$$

This also covers disconnected H.

For (II), let J have zero sharp slack and actual maximum degree $D \geq 3$. If $D \geq 4$, the coveredness premise in Setup 8.9(13) supplies a matching covering all degree-D vertices. If $D = 3$, sharp and generic profiles coincide, so $\mathrm{ps}(J) = 0$; part (I) forces $\kappa = 0$ and gives the same conclusion.

Theorem 8.10(d) now gives $c_D^*(J) = 0$, and all costs at thresholds at least D vanish. By subgraph monotonicity they also vanish in $J - v$. Choose v of degree D, and put

$$h = \nu(J) - \nu(J - v) \in \{0, 1\}.$$

The membership follows because deleting a vertex loses at most its one incident matching edge, and every matching of $J - v$ remains a matching of J. For $1 \leq i < D$, integrality, subgraph monotonicity, and the vertex-deletion inequality give

$$C_\sharp(i, J) - C_\sharp(i, J - v) \in \{0, 1\}.$$

Since deleting v removes exactly D edges,

$$\begin{aligned} \psi_\sharp(J) - \psi_\sharp(J - v) &= D - h - \sum_{i=1}^{D-1} \big(C_\sharp(i, J) - C_\sharp(i, J - v)\big) \\ &\geq 1 - h. \end{aligned}$$

The left side is nonpositive because $\psi_\sharp(J) = 0$ and the child slack is nonnegative. Therefore $h = 1$, the child slack is zero, and every one of the $D - 1$ cost differences equals one.

This argument applies at every deletion and terminates after finitely many steps at an induced graph R with maximum degree at most two. Each step lowers both order and matching number by one, so

$$n(R) - \nu(R) = n(H) - \nu(H).$$

Every component of R is a path or a cycle, including single-vertex paths. For a path on $m \geq 1$ vertices, alternating vertices and edges attain

$$\alpha = \lceil m/2 \rceil, \qquad \nu = \lfloor m/2 \rfloor.$$

Pairing consecutive vertices gives the corresponding upper bounds. Thus $n - \nu = \alpha < 2\alpha$. For an even cycle on $2k$ vertices, alternating choices give $\alpha = \nu = k$, with upper bounds from its consecutive pairs, and again $n - \nu = k < 2k$. For an odd cycle on $2k + 1$ vertices, $k \geq 1$, a matching has at most k edges. An independent set has at most k vertices, since summing the inequalities that each consecutive edge has at most one selected endpoint gives twice its size at most $2k + 1$. Alternating choices attain both bounds. Hence

$$n - \nu = k + 1 \leq 2k = 2\alpha,$$

with equality exactly when $k = 1$, that is, for a triangle.

Matching and independence numbers add over components, and R is induced in H. Therefore

$$n(H) - \nu(H) = n(R) - \nu(R) \leq 2\alpha(R) \leq 2\alpha(H).$$

Equality forces every nonempty component of R to be a triangle and $\alpha(R) = \alpha(H)$. The deletion choices were arbitrary, so this conclusion holds for every sequence. Finally,

$$\frac{n(H)}{2} \le n(H) - \nu(H) \le 2\alpha(H),$$

because $\nu(H) \le n(H)/2$. Thus $n(H) \le 4\alpha(H)$. If equality holds, then $\nu(H) = n(H)/2$, which means that H has a perfect matching.

For (III), the hereditary generic gap hypothesis persists under vertex deletion. At every zero-generic-slack intermediate graph of actual maximum degree at least four, the generic coveredness premise in Setup 8.9(13) supplies the matching. At degree three, part (I) supplies it. The deletion identity becomes

$$\mathrm{ps}(J) - \mathrm{ps}(J - v) = D - h - \sum_{i=1}^{D-1} \big(c_i^*(J) - c_i^*(J - v)\big) \ge 1 - h.$$

Generic nonnegative slack gives $h = 1$, zero child slack, and equality in each cost difference. The path-and-cycle calculation then proves all assertions of (II).

If $\Delta(H) \le 5$, the same degree bound holds for every induced subgraph. Sharp and generic profiles coincide: thresholds below six have identical definitions, and thresholds at least six vanish by Theorem 8.10(d). The unconditional sharp conclusions of (II), subject to their recorded supporting premises, therefore give the generic conclusions without a hereditary gap assumption. □

9. Arrowing graphs: tails, pools and the potential lemma

This chapter establishes the deletion and replacement statements used for star-forest hosts in the reduction. The tails lemma, Theorem 9.26, gives an explicit extension of avoiding colourings; the potential lemma, Theorem 9.28, applies simultaneously to a family of target pairs; and Theorem 9.30 identifies the canonical pools and the exact normalization of a tight host. We then prove the matching, localization, and small-surplus consequences, retaining every positive component of the original targets.

[author/blocker] No author, affiliation, or email was supplied. [addressed-partial: Existing author, affiliation, and email placeholders retained | deferred: authorship details must be supplied by the author.]

[instruction/conflict] The complete-document requirement is implemented by the standalone front matter and bibliography. The chapter format governs the body, so no abstract is included. Preservation of all supplied statements and proofs takes precedence over the requested page range. [acknowledged: The chapter format, standalone front matter, and supplied mathematical content are retained.]

[prime/blocker] Compilation, undefined-citation and undefined-reference checks, and pagination have not been verified. [acknowledged: These checks remain pending at the external compilation gate.]

Remark 9.1 (Assistance and verification). The main results of this paper were obtained with the assistance of the Ansatz system, an automated proof-search and verification system. Because automated systems have limitations, it is possible that we have missed related references in the literature, and we welcome comments from experts. [ack/blocker] No independent proof-verification statement was supplied. [acknowledged: No independent proof-verification claim is added.]

9.1. Setup and prerequisites.

We fix the notation and state explicitly the earlier results used in the proofs.

Setup 9.2 (Earlier certificate framework)**.** We assume, established earlier in this paper, the certificate framework: the costs $c_{A,B}$, c_θ^*, and Σ; the quantities ν, ex, ps, and gap; LP-goodness; and split-achievability.

[math/blocker] [GAP: The certificate definitions, their parameter domains, and their basic cost properties were not supplied. They must be imported verbatim from Chapter 1. No additional properties of these quantities are assumed implicitly below.] [out-of-scope: thm-body-math; env=setup; label=setup:certificates]

Setup 9.3 (Graphs and target lists)**.** All graphs are finite and simple. Copies are ordinary subgraphs and need not be induced. The star $K_{1,d}$, for an integer $d \geq 1$, has d edges. Let $s, t \geq 1$ be integers, and let

$$n_1 \geq \cdots \geq n_s \geq 1, \qquad m_1 \geq \cdots \geq m_t \geq 1$$

be integer lists. Put

$$F_1 = \bigsqcup_{i=1}^{s} K_{1,n_i}, \qquad F_2 = \bigsqcup_{j=1}^{t} K_{1,m_j}, \qquad M = s + t - 1.$$

The notation $G \to (F_1, F_2)$ means that every red-blue colouring of $E(G)$ contains a red F_1 or a blue F_2. The empty forest is present in every graph.

Write $e(H) = |E(H)|$, let $\nu(H)$ be the maximum matching size, and let $\Delta(H)$ be the maximum degree, with $\Delta(H) = 0$ when H has no vertices. For $v \in V(H)$, write $d_H(v)$ for its degree. Vertex deletion removes the vertex and all incident edges. If W is a matching, $H - W$ removes only its edges. Put

$$\Phi(H) = e(H) - \nu(H).$$

All component sizes, indices, capacities, deletion counts, matching counts, and surpluses below are integers.

Notation 9.4 (Diagonals, suffixes, and pools)**.** For every integer k with $2 \leq k \leq M + 1$, put

$$\ell_k = \max_{\substack{1 \leq i \leq s,\ 1 \leq j \leq t \\ i+j=k}} (n_i + m_j - 1), \qquad L = \sum_{k=2}^{M+1} \ell_k.$$

For integers $0 \leq \alpha < s$ and $0 \leq \beta < t$, the actual suffixes are

$$F_1^{\alpha} = \bigsqcup_{i=\alpha+1}^{s} K_{1,n_i}, \qquad F_2^{\beta} = \bigsqcup_{j=\beta+1}^{t} K_{1,m_j}.$$

For every integer $0 \leq q < M$, define

$$\mathcal{A}_q = \big\{(F_1^{\alpha}, F_2^{\beta}) : \alpha + \beta = q,\ 0 \leq \alpha < s,\ 0 \leq \beta < t\big\}, \qquad T_q = \bigsqcup_{k=2}^{q+1} K_{1,\ell_k}.$$

Thus T_0 is empty. For integers $0 \leq q \leq M$, put

$$L_q = \sum_{k=q+2}^{M+1} \ell_k, \qquad P_q = \sum_{k=q+2}^{M+1} (\ell_k - 1),$$

where empty sums are zero. In particular, $L_0 = L$ and $P_0 = L - M$.

Notation 9.5 (Families of pairs)**.** Let $\mathcal{A}$ be a finite nonempty family of ordered pairs of nonempty star forests. A member $\gamma \in \mathcal{A}$ has positive integer lengths s_γ, t_γ and nonincreasing positive integer lists

$$n_1^{\gamma}, \ldots, n_{s_\gamma}^{\gamma}, \qquad m_1^{\gamma}, \ldots, m_{t_\gamma}^{\gamma}.$$

Write $F_1^{\gamma}, F_2^{\gamma}$ for the corresponding forests, and put

$$N = \max_{\gamma \in \mathcal{A}} (s_\gamma + t_\gamma - 1), \qquad \bar{\ell}_k = \max_{\substack{\gamma \in \mathcal{A},\ 1 \leq i \leq s_\gamma,\ 1 \leq j \leq t_\gamma \\ i+j=k}} (n_i^{\gamma} + m_j^{\gamma} - 1) \quad (2 \leq k \leq N + 1).$$

When a degree bound is under discussion, put

$$D = \bar{\ell}_2 - 1, \qquad P = \sum_{k=2}^{N+1} (\bar{\ell}_k - 1).$$

The integer R denotes the length of the initial plateau of the envelope at $D+1$. A terminal maximizing triple is a triple (γ, i, j) satisfying

$$i + j = R + 1, \qquad n_i^\gamma + m_j^\gamma - 1 = D + 1.$$

For an integer $D \geq 0$, let $\delta_D(H)$ be the minimum number of degree-D vertices left uncovered by a matching of H.

For nonnegative integers A, B, define the minimum degree overflow by

$$\Omega_{A,B}(H) = \min_{\chi: E(H) \to \{\text{red},\text{blue}\}} \sum_{v \in V(H)} \big((d_{\text{red}}(v) - A)_+ + (d_{\text{blue}}(v) - B)_+ \big), \qquad x_+ = \max\{x, 0\}.$$

An A/B degree colouring is a colouring with red maximum degree at most A and blue maximum degree at most B.

The published background for even-capacity decompositions is [DJK+21, Lemma 2.1 and Theorem 2.5]. Degree-interval formulations appear in [FLL+23, Theorem 2.37] and their deficiency theory in [Lov70, Theorem 7.3]. These references supply attribution for the earlier development; they do not replace its quantitative matching and overflow statements. The bipartite matching-partition statement used in that development is recorded in [TY19, Theorem 1.1].

[cite/blocker] The earlier-series theorem numbers were not supplied. The locator [I, Theorem X] remains unresolved for each assumed Earlier result below; the local numbering identifies the assumption recorded here, not a verified theorem number in the earlier development. [deferred: to reference_verify]

[math/blocker] [GAP: The Earlier results below record exactly the forms invoked by the supplied proofs. Their full source statements and proofs were not supplied. Their establishment before this chapter, including the absence of circular dependencies, must be checked. Arguments using them are conditional on these prerequisites.] [addressed-partial: Three supplied statements are recorded as numbered assumptions, and the small-surplus theorem is explicitly conditional | deferred: establishment of the earlier prerequisites remains outside this seam repair.]

Earlier result 9.6 (Potential deletion). Assume the following result from the earlier development, with locator [I, Theorem X]. For every finite simple graph H,

$$\Phi(H) \geq 0, \qquad \nu(H) - \nu(H - v) \in \{0, 1\} \quad (v \in V(H)).$$

If $A, B \geq 0$ are integers and H has no A/B degree colouring, there is a maximum-degree vertex v such that

$$\Phi(H) - \Phi(H - v) \geq A + B.$$

The deletion construction uses the loss of at most one matching edge at a vertex. At the degree obstruction where that loss would leave the bound short by one, it chooses a maximum-degree vertex missed by a maximum matching.

Earlier result 9.7 (Attaining matching decompositions). Every finite simple graph H has a vertex set S such that

$$\nu(H) = |S| + \sum_C \left\lfloor \frac{|V(C)|}{2} \right\rfloor,$$

where C ranges over the components of $H - S$. Such an S is called attaining. The published decomposition statement is [QLS+19, QLS+19b, Theorem 1.6]. We additionally assume the earlier-series assertion, with locator [I, Theorem X], that every attaining set of maximum cardinality has only odd, factor-critical complementary components. A graph is factor-critical when deleting any one of its vertices leaves a perfect matching.

Earlier result 9.8 (Robust containment and attaining sets). Assume the following earlier result, with locator [I, Theorem X]. Let J be a nonempty positive-edge star forest, and let $k \geq 0$ be an integer. If $H - X$ contains J for every $X \subseteq V(H)$ with $|X| \leq k$, and

$$\nu(H) < k + \nu(J),$$

then every attaining set S of H satisfies

$$|S| \leq \nu(H) - \nu(J) \leq k - 1.$$

Earlier result 9.9 (A matching against a graph). Assume the earlier greedy-deletion bound, with locator [I, Theorem X]. For every integer $r \geq 1$, every finite simple graph J with at least one edge, and every finite simple graph H,

$$H \to (rK_2, J) \quad \Longrightarrow \quad e(H) \geq e(J) + (r-1)\Delta(J).$$

Earlier result 9.10 (The diagonal-star host). Assume the earlier upper-bound construction, with locator [I, Theorem X]. In Setup 9.3,

$$\bigsqcup_{k=2}^{M+1} K_{1,\ell_k} \;\to\; (F_1, F_2).$$

Earlier result 9.11 (Exact residual operation for a monochromatic pool star). Assume the earlier greedy residual equivalence, with locator [I, Theorem X]. Let a coloured core be disjoint from a monochromatic star. If the star has at least as many edges as the largest remaining component in its colour, processing this star removes exactly that largest component from the target list in that colour and leaves the other list unchanged. Existence of a full target in the combined coloured graph is equivalent to existence of the resulting residual target in the remaining coloured graph. This operation can be iterated over disjoint pool stars; empty residual targets are interpreted by the empty-forest convention.

Earlier result 9.12 (Tight greedy complements). Assume the earlier greedy-complement theorem, with locator [I, Theorem X]. Let $G \to (F_1, F_2)$, assume

$$\Phi(G) = L - M, \qquad \nu(G) < M,$$

and fix a maximum matching Q. Along the permitted greedy deletions, write

$$H_h = G - X_h, \qquad Q_h = Q \cap E(H_h), \qquad J_h = H_h - Q_h, \qquad |X_h| = h.$$

A Q_h-covered vertex of maximum degree in J_h is a permitted choice. At every reached stage with $h < M$, the restriction Q_h is maximum,

$$\nu(H_h) = \nu(G) - h, \qquad e(J_h) = P_h, \qquad \Delta(J_h) = \ell_{h+2} - 1.$$

Each permitted deletion loses one matching edge and preserves these assertions at the next reached stage.

Earlier result 9.13 (Residual hosts of maximum degree two). Assume the earlier low-degree bound, with locator [I, Theorem X]. If $G \to (F_1, F_2)$, $X \subseteq V(G)$, $|X| = q < M$, and $\Delta(G - X) \leq 2$, then

$$e(G - X) \geq L_q.$$

Earlier result 9.14 (Private pendant leaves). Assume the earlier augmentation statement, with locator [I, Theorem X]. Let $S \subseteq V(G)$, and suppose every component C of $G - S$ satisfies

$$\nu(C) = \left\lfloor \frac{|V(C)|}{2} \right\rfloor.$$

Adjoin one distinct new pendant leaf to every vertex of S, obtaining G^+. Then

$$\nu(G^+) = |S| + \sum_C \nu(C),$$

and S is attaining in G^{+}. The components of $G^{+}-S$ are those of $G-S$ and the new isolated vertices.

Earlier result 9.15 (Full-target density localization). Assume the earlier density theorem, with locator [I, Theorem X]. Let $G \to (F_1, F_2)$, put $E_i = e(F_i)$, and let S be any attaining set. Assume

$$d = M - \nu(G) \geq 1, \qquad \eta_1 = \sum_{i=1}^{s} \left(\left\lceil \frac{n_i}{2} \right\rceil - 1 \right).$$

If

$$E_2 > 2t - 3d - 2|S| - 2\eta_1,$$

some component C of $G-S$ satisfies

$$\nu(C) \geq E_2 + d - 1.$$

If also $E_2 \geq s - 2d + 1$, the same component satisfies

$$\nu(C) \geq \max(E_1, E_2) + d - 1.$$

The colour-exchanged assertions also hold.

Earlier result 9.16 (Degrees forced by robust containment). Assume the earlier robustness consequence, with locator [I, Theorem X]. Let C be a finite connected simple graph, let J have at least one edge and no isolated vertices, and let $k, r \geq 0$ and $D \geq 3$ be integers. Assume

$$|V(C)| = |V(J)| + k + r, \qquad \Delta(C) \leq D, \qquad |V(C)| > rD + 1,$$

and that $C - Z$ contains J for every k-element subset $Z \subseteq V(C)$. Then

$$d_C(x) \geq k + 1 - r(D-1) \quad (x \in V(C)), \qquad k + 1 \leq D + (D-1)r.$$

Earlier result 9.17 (Indexed required-matching bound). Assume the earlier indexed bound, with locator [I, Theorem X]. In the family setting of Notation 9.5, let $D = \bar{\ell}_2 - 1 \geq 3$, let H arrow every family member, and assume $\Delta(H) \leq D$. Whenever

$$n_i^{\gamma} + m_j^{\gamma} = D + 2,$$

the minimum number of uncovered degree-D vertices satisfies

$$\delta_D(H) \geq i + j - 1.$$

Earlier result 9.18 (Indexed degree overflow). Assume the earlier full-target overflow bound, with locator [I, Theorem X]. Under the hypotheses of Earlier result 9.17, put

$$a = n_i^{\gamma}, \qquad b = m_j^{\gamma}, \qquad A = a - 1, \qquad B = b - 1.$$

For a triple with $a + b = D + 2$,

$$\Omega_{A,B}(H) \geq i + j - 1.$$

Earlier result 9.19 (Parity bounds for overflow). Assume the following earlier quantitative bounds, with locator [I, Theorem X]. Let $A, B \geq 1$ be integers, put $D = A + B$, and let $\Delta(H) \leq D$. Write $\omega = \Omega_{A,B}(H)$. If one capacity U is odd and the other capacity V is positive and even, then

$$\delta_D(H) \geq \omega + (U-1) \left\lceil \frac{\omega}{V} \right\rceil.$$

If both A and B are odd, and κ counts the whole connected components of H that are D-regular and have odd order, then

$$\omega = \kappa \quad \text{or} \quad \delta_D(H) \geq \omega + \min(A, B) - 1.$$

For even nonnegative capacities A, B, a graph of maximum degree at most $A + B$ has an A/B degree colouring; the positive-capacity case used below follows from [DJK+21, Lemma 2.1]. The earlier development derives this decomposition through regular completion, circuit orientation, and bipartite edge colouring.

Earlier result 9.20 (The zero-surplus family theorem). Assume the earlier zero-surplus theorem, with locator [I, Theorem X]. In Notation 9.5, let $D \geq 3$, let H arrow every member, and assume

$$\Delta(H) \leq D, \qquad \Phi(H) = P.$$

Every terminal maximizing triple has sizes either both positive and even, or D is odd and its unordered size pair is $\{2, D\}$.

Assume additionally that at least one terminal maximizing triple has both sizes at least three. Then D is even and at least six. Exactly R whole connected components are D-regular of odd order, and each has a near-perfect matching. If Q is the union of all other components, then, for every terminal maximizing triple with sizes $a, b \geq 3$, Q has an $(a-1)/(b-1)$ degree colouring. These conclusions require no upper bound on $e(H)$.

Earlier result 9.21 (Minimum degree overflow and indexed target bounds). We assume that the following statement has been established earlier in this series.

For a finite simple graph G and nonnegative integers A, B, let

$$\Omega_{A,B}(G) = \min_{\chi\colon E(G)\to\{\text{red,blue}\}} \sum_{v\in V(G)} \big((d_{\text{red}}(v) - A)_+ + (d_{\text{blue}}(v) - B)_+\big), \qquad x_+ = \max\{x, 0\},$$

where $d_{\text{red}}(v)$ and $d_{\text{blue}}(v)$ are the colour degrees. For disjoint vertex sets S, T, let $e_G(S, T)$ count the edges between them. For a component C, write $e_G(C, T) = e_G(V(C), T)$.

Assume first that $A, B \geq 1$, and put

$$D = A + B, \qquad d_v = d_G(v), \qquad C_D = \sum_{v\in V(G)} (d_v - D)_+,$$

$$a_v = \max\{\min(A, d_v), d_v - B\}, \qquad b_v = \max\{\min(B, d_v), d_v - A\}.$$

For disjoint $S, T \subseteq V(G)$, let $q_{D,A}(S, T)$ count the connected components C of $G - (S \cup T)$ for which every vertex has degree D in G and

$$A|V(C)| + e_G(C, T)$$

is odd. Then

$$\Omega_{A,B}(G) = C_D + \max_{\substack{S,T\subseteq V(G)\\ S\cap T=\varnothing}} \left(q_{D,A}(S, T) + e_G(S, T) - \sum_{v\in S} a_v - \sum_{v\in T} b_v \right). \tag{9.1}$$

If $\Delta(G) \leq D$, then

$$C_D = 0, \qquad a_v = \min(A, d_G(v)), \qquad b_v = \min(B, d_G(v)).$$

Now let

$$F_1 = \bigsqcup_{h=1}^{s} K_{1,n_h}, \qquad F_2 = \bigsqcup_{h=1}^{t} K_{1,m_h},$$

where $s, t \geq 1$ are integers and the size lists are positive, integral, and nonincreasing. Assume $G \to (F_1, F_2)$ for ordinary subgraph copies. For every $1 \leq i \leq s$ and $1 \leq j \leq t$,

$$\Omega_{n_i-1,m_j-1}(G) \geq i + j - 1.$$

Consequently, whenever $n_i, m_j \geq 2$, the maximum in (9.1), with

$$A = n_i - 1, \qquad B = m_j - 1, \qquad D = n_i + m_j - 2,$$

is at least $i + j - 1 - C_D$. If also $\Delta(G) \leq n_i + m_j - 2$, there exist disjoint $S, T \subseteq V(G)$ satisfying

$$q_{n_i+m_j-2,n_i-1}(S, T) + e_G(S, T) - \sum_{v\in S} \min(n_i - 1, d_G(v)) - \sum_{v\in T} \min(m_j - 1, d_G(v)) \geq i + j - 1.$$

All target components, including unit stars, are retained in the arrowing assertion.

Earlier result 9.22 (Quantitative parity bounds for uncovered vertices). We assume that the following statement has been established earlier in this series.

Let $D \geq 3$ be an integer, and let G be a finite simple graph with $\Delta(G) \leq D$. Let $\delta = \delta_D(G)$ be the minimum number of vertices of degree D in G left uncovered by a matching. For positive integer capacities A, B with $A + B = D$, put

$$\omega = \Omega_{A,B}(G).$$

If D is odd, let O and E be the odd and even members of $\{A, B\}$, respectively. Then

$$\delta \geq \omega + (O - 1) \left\lceil \frac{\omega}{E} \right\rceil.$$

In particular, if $\omega > 0$, then

$$\delta \geq \omega + O - 1.$$

If D is even and A, B are odd, put

$$m = \min(A, B), \qquad w = \frac{D}{2} - m,$$

and let κ count the whole connected components of G that have odd order and are D-regular. Then $\omega \geq \kappa$. If $w = 0$, then $\omega = \kappa$. If $w > 0$, then

$$\delta \geq \omega + (m - 1) \left\lceil \frac{\omega - \kappa}{w} \right\rceil.$$

In particular, in the even-degree case with odd capacities,

$$\omega = \kappa \quad \text{or} \quad \delta \geq \omega + m - 1.$$

These bounds concern integer barrier slopes; no factor-criticality assumption is made.

Earlier result 9.23 (Exact equality for a family of target pairs). We assume that the following statement has been established earlier in this series.

Let $\mathcal{A}$ be a finite nonempty family of ordered pairs of nonempty star forests. For each $\gamma \in \mathcal{A}$, let $s_\gamma, t_\gamma \geq 1$ be integers, with positive integer lists

$$n_1^\gamma \geq \cdots \geq n_{s_\gamma}^\gamma \geq 1, \qquad m_1^\gamma \geq \cdots \geq m_{t_\gamma}^\gamma \geq 1,$$

and corresponding forests

$$F_1^\gamma = \bigsqcup_{h=1}^{s_\gamma} K_{1,n_h^\gamma}, \qquad F_2^\gamma = \bigsqcup_{h=1}^{t_\gamma} K_{1,m_h^\gamma}.$$

Put

$$N = \max_{\gamma \in \mathcal{A}} (s_\gamma + t_\gamma - 1),$$

$$\ell_k = \max_{\substack{\gamma \in \mathcal{A},\ 1 \leq i \leq s_\gamma,\ 1 \leq j \leq t_\gamma \\ i+j=k}} (n_i^\gamma + m_j^\gamma - 1) \quad (2 \leq k \leq N + 1),$$

$$D = \ell_2 - 1 \geq 3, \qquad P = \sum_{k=2}^{N+1} (\ell_k - 1).$$

Let R be the length of the initial plateau of the envelope at $D + 1$. Assume that a finite simple graph H arrows every member of $\mathcal{A}$, and that

$$\Delta(H) \leq D, \qquad e(H) - \nu(H) = P.$$

For every $\gamma \in \mathcal{A}$ and valid indices i, j satisfying

$$i + j = R + 1, \qquad n_i^\gamma + m_j^\gamma - 1 = D + 1,$$

put $a = n_i^\gamma$ and $b = m_j^\gamma$. Then either a, b are both positive even integers, or D is odd and

$$\{a, b\} = \{2, D\}.$$

If at least one such terminal maximizing triple has $a, b \geq 3$, then D is even and at least six. The graph H has exactly R whole connected components that have odd order and are D-regular. Each has a near-perfect matching. Let Q be the union of all other components. There is a matching of Q covering all its vertices whose degree in H is D. For every terminal maximizing triple with $a, b \geq 3$, the graph Q has an edge colouring with red maximum degree at most $a-1$ and blue maximum degree at most $b-1$. No upper bound on $e(H)$ or positive matching-deficit hypothesis is imposed.

9.2. **Tails and their complete families.** We prove the deletion statements that keep every legal actual suffix available.

Lemma 9.24 (Vertex deletion and matching trimming). *Let $G \to (F_1, F_2)$ in Setup 9.3.*

(i) *For integers $0 \leq a < s$ and $0 \leq b < t$, and every $X \subseteq V(G)$ with $|X| = a + b$,*

$$G - X \to \left(\bigsqcup_{i=a+1}^{s} K_{1,n_i}, \bigsqcup_{j=b+1}^{t} K_{1,m_j} \right).$$

(ii) *For every matching W in G,*

$$G - W \to (F_1^-, F_2), \qquad F_1^- = \bigsqcup_{\substack{1 \leq i \leq s \\ n_i \geq 2}} K_{1,n_i-1}.$$

(iii) *If $n_s \geq 2$, replacing every n_i by $n_i - 1$ changes the diagonal sum from L to $L - M$. Consequently, if the diagonal-sum lower bound is known for this trimmed pair and G has a matching of size at least M, then $e(G) \geq L$.*

Proof. We first prove the one-vertex implication used in part (i). Let

$$A = \bigsqcup_{h=1}^{u} K_{1,p_h}, \qquad p_1 \geq \cdots \geq p_u \geq 1, \qquad u \geq 2,$$

where all parameters are integers, and let

$$A_{\text{tail}} = \bigsqcup_{h=2}^{u} K_{1,p_h}.$$

Let B be a graph; the implication will be used only when B is a nonempty star forest without isolated vertices. Suppose $Q \to (A, B)$, and let $v \in V(Q)$. If $Q - v$ did not arrow (A_{tail}, B), choose a colouring avoiding both targets and colour all edges incident with v red.

A blue B uses no edge incident with v. Since B has no isolated vertices in the stated application, such a copy would already lie in $Q - v$. A red A avoiding v contains a red A_{tail} in $Q - v$. If a red A meets v in its component of size p_h, delete that component. The remaining $u - 1$ sizes, in decreasing order, are

$$p_1, \ldots, p_{h-1}, p_{h+1}, \ldots, p_u.$$

Their k-th entry is at least p_{k+1} for every integer $1 \leq k \leq u - 1$. Trimming leaves gives A_{tail} in $Q - v$. Both alternatives contradict the chosen colouring. Thus

$$Q - v \to (A_{\text{tail}}, B).$$

Interchanging red and blue gives the implication that deletes the largest blue component. The restriction on isolated vertices concerns only the blue-copy step and holds in every application here.

For part (i), enumerate X in any order. Use the red one-vertex implication at exactly a deletion steps and the blue implication at the other b steps. Before every red step at least two red components

remain, since $a < s$. Before every blue step at least two blue components remain, since $b < t$. After all $a + b$ steps, precisely the first a red and first b blue components have been omitted. This also includes $a + b = 0$.

For part (ii), take any colouring of $G - W$, and colour every edge of W red. A blue F_2 in the extension uses no edge of W, so it is already present in $G-W$. Otherwise there is a red F_1. At most one edge of its i-th star belongs to W, because the star edges share a centre and W is a matching. For each i with $n_i \geq 2$, retain exactly $n_i - 1$ of the red edges left at that centre after removing W. The selected stars remain vertex-disjoint. Omit the original components with $n_i = 1$. This gives F_1^-, including the case where it is empty.

For part (iii), $n_s \geq 2$ ensures that all s trimmed sizes are positive and remain nonincreasing. For every integer $2 \leq k \leq M + 1$, the new diagonal maximum is

$$\max_{i+j=k} \big((n_i - 1) + m_j - 1\big) = \ell_k - 1.$$

The index sets are unchanged. There are M diagonals, so the new sum is $L-M$. If the lower bound holds for the trimmed pair, part (ii) gives

$$e(G - W) \geq \sum_{k=2}^{M+1} (\ell_k - 1).$$

Since deleting W removes exactly $|W|$ edges,

$$e(G) = e(G - W) + |W| \geq L - M + |W| \geq L$$

when $|W| \geq M$. No external theorem is used. □

Lemma 9.25 (Fixed-partition extension)**.** *Let the target lists be as in Setup 9.3. Choose integers*

$$0 \leq \alpha < s, \qquad 0 \leq \beta < t, \qquad q = \alpha + \beta \geq 1.$$

Let G be any finite simple graph, and let $X = A \sqcup B \subseteq V(G)$, where

$$|A| = \alpha, \qquad |B| = \beta.$$

Every colouring of $G - X$ avoiding a red F_1^α and a blue F_2^β extends to a colouring avoiding the full pair by the following rule:

(i) *retain the colours on $G - X$;*
(ii) *colour edges from A to $G - X$, and edges within A, red;*
(iii) *colour edges from B to $G - X$, and edges within B, blue;*
(iv) *colour every edge between A and B red.*

Every connected monochromatic subgraph containing an edge and meeting X contains a vertex of A if red and a vertex of B if blue. Consequently,

$$G \to (F_1, F_2) \quad \Longrightarrow \quad G - X \to (F_1^\alpha, F_2^\beta).$$

The assertion includes unit stars and copies meeting X through leaves.

Proof. This is the simultaneous extension underlying Lemma 9.24(i). The categories of new edges are disjoint and exhaust all edges incident with X, so every new edge receives exactly one colour.

Every red edge incident with B has its other endpoint in A, since edges within B and from B to $G-X$ are blue. A connected red subgraph containing an edge and meeting X therefore either meets A already, or meets B and reaches A along a red edge. No blue edge is incident with A. Hence a connected blue subgraph containing an edge and meeting X contains a vertex of B. Isolated vertices in a colour graph are irrelevant because every target component has positive size.

Suppose the extension contains a red F_1. Each component meeting X contains a vertex of A. Vertex-disjointness implies that at most α components meet X, including components whose

intersection with X consists of leaves. Let $r \le \alpha$ be their number, and list the remaining target indices as

$$i_1 < \cdots < i_{s-r}.$$

At most r indices were omitted, so

$$i_k \le k + r \le k + \alpha \qquad (1 \le k \le s - \alpha).$$

Thus $n_{i_k} \ge n_{k+\alpha}$. Retain the first $s - \alpha$ untouched stars and trim the k-th to $n_{k+\alpha}$ edges. This gives a red F_1^α in $G - X$, a contradiction.

For a blue F_2, at most β components meet X, because each uses a distinct vertex of B. Let $r' \le \beta$ be their number, and let

$$j_1 < \cdots < j_{t-r'}$$

be the untouched indices. For every $1 \le k \le t - \beta$,

$$j_k \le k + r' \le k + \beta, \qquad m_{j_k} \ge m_{k+\beta}.$$

Trimming the first $t - \beta$ untouched stars gives a blue F_2^β in $G - X$, again a contradiction.

The extension therefore avoids both full targets. If G arrows the full pair, an avoiding colouring of $G - X$ would contradict this extension statement. □

Theorem 9.26 (Tails with arbitrary cross-edge colours)**.** *Under the hypotheses of Lemma 9.25, every avoiding colouring of $G - X$ extends to a colouring avoiding the full pair when the edges between A and B are assigned either colour independently. The other new edges are coloured as in that lemma.*

Every connected red subgraph containing an edge and meeting X contains a vertex of A. Every connected blue subgraph containing an edge and meeting X contains a vertex of B. Hence

$$G \to (F_1, F_2) \quad \Longrightarrow \quad G - X \to (F_1^\alpha, F_2^\beta)$$

for every choice of the cross-edge colours, including when target copies meet X through leaves.

Proof. We change only the cross-edge clause of Lemma 9.25. The edge categories still partition all edges incident with X, and every core colour is retained.

Every red edge incident with B has its other endpoint in A, because the edges within B and from B to the core are blue. If a connected red subgraph containing an edge meets X, it either already contains a vertex of A, or contains a vertex of B incident within that subgraph with a red edge. The latter edge reaches A. Thus the subgraph meets A.

Every blue edge incident with A has its other endpoint in B, because edges within A and from A to the core are red. The same connectedness argument shows that a connected blue subgraph containing an edge and meeting X meets B. Neither argument restricts the cross-edge colours.

Suppose a red F_1 occurs. Each of its components meeting X uses a distinct vertex of A, so their number r is at most α. Write the untouched indices as

$$i_1 < \cdots < i_{s-r}.$$

For every integer $1 \le k \le s - \alpha$,

$$i_k \le k + r \le k + \alpha, \qquad n_{i_k} \ge n_{\alpha+k}.$$

Retaining and trimming the first $s - \alpha$ untouched stars gives a red F_1^α in the core. This contradicts the original colouring. No vertex of X was required to be a centre.

If a blue F_2 occurs, at most β components meet X. Let their number be $r_B \le \beta$, and list the untouched indices as

$$j_1 < \cdots < j_{t-r_B}.$$

For $1 \le k \le t - \beta$,

$$j_k \le k + r_B \le k + \beta, \qquad m_{j_k} \ge m_{\beta+k}.$$

Trimming the first $t-\beta$ untouched stars gives a forbidden blue F_2^{β} in the core. Both full targets are absent.

An avoiding colouring of $G-X$, together with this extension, would contradict $G \to (F_1, F_2)$. Positive component sizes ensure that every target component used in the connectedness argument contains an edge. □

Proposition 9.27 (Normalization of the complete suffix family)**.** *In Setup 9.3, fix an integer* $0 \le q \le s+t-2$*, and put*

$$\alpha_0 = \max(0, q-t+1), \qquad \beta_0 = \max(0, q-s+1),$$

$$s_0 = s-\alpha_0, \qquad t_0 = t-\beta_0, \qquad q_0 = q-\alpha_0-\beta_0.$$

Let $F_{1,0} = F_1^{\alpha_0}$ *and* $F_{2,0} = F_2^{\beta_0}$*. Then*

$$s_0, t_0 \ge 1, \qquad 0 \le q_0 \le \min(s_0-1, t_0-1).$$

The map

$$(\alpha, \beta) \longmapsto (\alpha-\alpha_0, \beta-\beta_0)$$

is a bijection from the feasible allocations defining $\mathcal{A}_q$ *onto*

$$\{(\gamma, q_0-\gamma) : \gamma \in \mathbb{Z},\ 0 \le \gamma \le q_0\}.$$

The ordered target forests are identical under this correspondence. Thus $\mathcal{A}_q$ *is the complete natural* q_0*-suffix family of* $(F_{1,0}, F_{2,0})$*, with common component budget*

$$K = s+t-1-q = s_0+t_0-1-q_0.$$

Its entire envelope is

$$(\ell_{q+2}, \dots, \ell_{s+t}),$$

without an assertion that one member attains all entries.

If H *arrows every member of* $\mathcal{A}_q$ *and* $\Delta(H) \le D$*, for an integer* $D \ge 0$*, then* H *contains each full base forest* $F_{1,0}$ *and* $F_{2,0}$ *as an uncoloured subgraph. Every component size in the two base lists is at most* D*. This applies to* $H = G-X$ *whenever* $G \to (F_1, F_2)$*,* $|X| = q$*, and* $\Delta(G-X) \le D$*.*

Proof. A feasible original allocation exists: take

$$\alpha = \max(0, q-t+1), \qquad \beta = q-\alpha.$$

The inequality $q \le s+t-2$ gives $\alpha \le s-1$, and the definition gives $0 \le \beta \le t-1$. Both coordinates are nonnegative. Hence $\alpha_0 \le s-1$; the symmetric argument gives $\beta_0 \le t-1$. Therefore $s_0, t_0 \ge 1$.

Every feasible allocation satisfies

$$\alpha \ge q-(t-1), \qquad \alpha \ge 0,$$

so $\alpha \ge \alpha_0$. Likewise $\beta \ge \beta_0$. Existence of a feasible allocation gives $\alpha_0+\beta_0 \le q$, hence $q_0 \ge 0$. The definitions also give

$$q-\beta_0 \le s-1, \qquad q-\alpha_0 \le t-1.$$

Subtracting α_0 and β_0, respectively, yields

$$q_0 \le s_0-1, \qquad q_0 \le t_0-1.$$

The translated coordinates of a feasible allocation are nonnegative and sum to q_0. Conversely, for any integer $0 \le \gamma \le q_0$, put

$$\alpha = \alpha_0+\gamma, \qquad \beta = \beta_0+q_0-\gamma.$$

Their sum is q, and

$$\alpha \le \alpha_0+q_0 \le s-1, \qquad \beta \le \beta_0+q_0 \le t-1.$$

These operations are inverse.

Deleting γ additional red components from $F_{1,0}$ leaves the original suffix starting at

$$\alpha_0 + \gamma + 1 = \alpha + 1.$$

The corresponding blue identity holds with β. Thus the actual ordered forests agree, including repeated and unit components. Their common budget is

$$(s-\alpha) + (t-\beta) - 1 = s + t - 1 - q = s_0 + t_0 - 1 - q_0.$$

For an integer $2 \le h \le K+1$, a local candidate on diagonal h of a suffix pair has original indices i, j satisfying $i + j = q + h$. Its value is at most ℓ_{q+h}. Conversely, take any valid original i, j with that sum. Since

$$(i-1) + (j-1) = q + h - 2 \ge q,$$

choose

$$\alpha = \max(0, q - (j-1)), \qquad \beta = q - \alpha.$$

Then $0 \le \alpha \le i-1$, $0 \le \beta \le j-1$, and $\alpha + \beta = q$. In particular, $\alpha < s$ and $\beta < t$. The local indices $i - \alpha, j - \beta$ are valid and sum to h. Every original candidate therefore appears in some family member. Taking maxima proves the envelope identity on every diagonal, including the final unit diagonals. The family identity transfers this complete envelope to the normalized description without requiring a single attaining pair.

The normalized allocation $\gamma = 0$ retains all of $F_{1,0}$. In the all-red colouring of H, its nonempty blue target cannot occur, so $F_{1,0}$ embeds. The allocation $\gamma = q_0$, together with the all-blue colouring, forces $F_{2,0}$. A star of size d in an uncoloured copy has a centre incident with d edges of H, hence $d \le \Delta(H) \le D$. These are individual containments. They assert neither a simultaneous disjoint packing of the base forests nor arrowing of their pair when $q_0 > 0$.

Finally, Lemma 9.24(i) gives $G - X$ arrowing every legal member of $\mathcal{A}_q$. The preceding argument applies to $H = G - X$. □

9.3. **The potential lemma.** We use the complete collection of tails to charge one potential contribution at each deletion.

Theorem 9.28 (Potential for a family of target pairs)**.** *Let $\mathcal{A}$, N, and $\bar{\ell}_k$ be as in Notation 9.5. If a finite simple graph G arrows every ordered pair in $\mathcal{A}$, separately, then*

$$e(G) - \nu(G) \ge \sum_{k=2}^{N+1} (\bar{\ell}_k - 1).$$

For a single pair from Setup 9.3,

$$e(G) \ge L + \nu(G) - M.$$

Consequently, $\nu(G) \ge M$ implies $e(G) \ge L$. If $e(G) < L$, then

$$\nu(G) \le M - 1 = s + t - 2, \qquad L - e(G) \le M - \nu(G).$$

All assertions allow repetitions, gaps, either parity, and unit components.

Proof. We use $\Phi(H) = e(H) - \nu(H)$ and Earlier result 9.6. First, every envelope entry is defined. Choose a member whose component budget is N. Its valid sums of indices run through every integer from 2 to $N+1$, since both index sets are nonempty consecutive integer intervals. Thus every finite maximum defining $\bar{\ell}_k$ has a candidate and is attained.

We recursively delete N distinct vertices. Put $X_0 = \varnothing$ and $H_0 = G$. At a stage $0 \le q < N$, let $H_q = G - X_q$, where $|X_q| = q$. Choose a member γ and valid indices i, j such that

$$i + j = q + 2, \qquad n_i^\gamma + m_j^\gamma - 1 = \bar{\ell}_{q+2}.$$

The integers $i-1, j-1$ are legal prefix deletions and sum to q. Lemma 9.24(i), applied to the original member and the entire set X_q, gives

$$H_q \to \left(\bigsqcup_{h=i}^{s_\gamma} K_{1,n_h^\gamma}, \bigsqcup_{h=j}^{t_\gamma} K_{1,m_h^\gamma} \right).$$

Both forests are nonempty.

Put $A = n_i^\gamma - 1$ and $B = m_j^\gamma - 1$. They are nonnegative integers. An A/B degree colouring would contain neither a red n_i^γ-edge star nor a blue m_j^γ-edge star. It would therefore avoid both displayed tails, since these are their respective leading components. Such a colouring is impossible.

Earlier result 9.6 supplies a maximum-degree vertex $v_q \in V(H_q)$ satisfying

$$\Phi(H_q) - \Phi(H_q - v_q) \geq A + B = \bar{\ell}_{q+2} - 1.$$

The degree-colouring obstruction also implies that H_q has an edge, because an edgeless graph admits every pair of nonnegative degree capacities. Set

$$X_{q+1} = X_q \cup \{v_q\}, \qquad H_{q+1} = H_q - v_q.$$

The vertices are distinct, and the recursion is valid through $q = N-1$. At each stage, persistence is applied anew to X_q. No fixed tail pair is required to work at all stages.

Summing the N inequalities gives

$$\Phi(G) - \Phi(H_N) \geq \sum_{k=2}^{N+1} (\bar{\ell}_k - 1).$$

Since $\Phi(H_N) \geq 0$,

$$e(G) - \nu(G) \geq \sum_{k=2}^{N+1} (\bar{\ell}_k - 1).$$

For a singleton family, $N = M = s + t - 1$, and the sum on the right is $L - M$. Rearrangement gives $e(G) \geq L + \nu(G) - M$. If $\nu(G) \geq M$, this yields $e(G) \geq L$. If $e(G) < L$, integrality gives

$$1 \leq L - e(G) \leq M - \nu(G),$$

so $\nu(G) \leq M - 1$. The proof used only positivity and ordering of the component sizes. □

Proposition 9.29 (Residual potentials and tight deletion chains)**.** *Let $G \to (F_1, F_2)$ in Setup 9.3. For every $X \subseteq V(G)$ with integer size $q \leq M$,*

$$\Phi(G - X) \geq P_q.$$

Put

$$\sigma = \Phi(G) - P_0, \qquad \sigma_X = \Phi(G - X) - P_q.$$

Both are nonnegative, and

$$e(G) - e(G - X) = \sum_{k=2}^{q+1} (\ell_k - 1) + \sigma - \sigma_X + \nu(G) - \nu(G - X).$$

If $\sigma = 0$, the maximum-degree deletion construction in Earlier result 9.6 can be iterated for M steps. It gives $H_0 = G$ and $H_{q+1} = H_q - v_q$ such that, for every integer $0 \leq q < M$,

$$\Phi(H_q) = P_q, \qquad d_{H_q}(v_q) = \ell_{q+2} - 1 + \epsilon_q, \qquad \epsilon_q = \nu(H_q) - \nu(H_{q+1}) \in \{0, 1\}.$$

At every such tight residual H_q, each vertex w satisfies

$$d_{H_q}(w) - \big(\nu(H_q) - \nu(H_q - w)\big) \leq \ell_{q+2} - 1.$$

Hence $\Delta(H_q) \leq \ell_{q+2}$, and every vertex of degree ℓ_{q+2} is covered by every maximum matching.

The final residual H_M is a disjoint union of edges and isolated vertices. If

$$z = e(H_M) = \nu(H_M),$$

then

$$M - \nu(G) = |\{q \in \mathbb{Z} : 0 \le q < M,\ \epsilon_q = 0\}| - z.$$

The assertions include unit components and $q = M$.

Proof. Fix $q < M$ and $|X| = q$. Consider all tail pairs indexed by integers

$$a + b = q, \qquad 0 \le a < s, \qquad 0 \le b < t.$$

This family is nonempty because $q \le s + t - 2$ can be distributed between the capacities $s - 1$ and $t - 1$. Each pair has component budget $M - q$, and Lemma 9.24(i) gives its arrowing by $G - X$.

For a local diagonal $2 \le r \le M - q + 1$, each tail entry has original indices i, j with $i + j = q + r$. Conversely, for any such original indices, choose

$$a = \max(0, q - (j - 1)), \qquad b = q - a.$$

Then $a, b \ge 0$, $b \le j - 1$, and $a \le i - 1$, since

$$q \le i + j - 2 = q + r - 2.$$

Thus $a < s$, $b < t$, and the local indices $i - a, j - b$ are valid and sum to r. Every original entry on diagonal $q + r$ occurs in the envelope. Its r-th entry is therefore ℓ_{q+r}, as also recorded in Proposition 9.27. Theorem 9.28 gives

$$\Phi(G - X) \ge \sum_{r=2}^{M-q+1} (\ell_{q+r} - 1) = P_q.$$

When $q = M$, the assertion is $\Phi(G - X) \ge 0$, which follows from Earlier result 9.6. No empty-tail arrowing statement is needed.

The cases just proved give $\sigma, \sigma_X \ge 0$. Also,

$$e(G) - e(G - X) = \Phi(G) - \Phi(G - X) + \nu(G) - \nu(G - X).$$

Substituting $P_0 + \sigma$ and $P_q + \sigma_X$, and using

$$P_0 - P_q = \sum_{k=2}^{q+1} (\ell_k - 1),$$

proves the identity.

Assume $\sigma = 0$. Suppose $H_q = G - X_q$ has been reached with $|X_q| = q < M$ and $\Phi(H_q) = P_q$. Choose i, j attaining ℓ_{q+2}, with $i + j = q + 2$. Lemma 9.24(i) gives arrowing of the tails beginning at i, j. A degree colouring with capacities $n_i - 1, m_j - 1$ would avoid their leading stars and hence both tails. It is impossible, and H_q has an edge.

Earlier result 9.6 gives a maximum-degree vertex v_q with

$$\Phi(H_q) - \Phi(H_q - v_q) \ge \ell_{q+2} - 1.$$

The residual bound already proved gives

$$\Phi(H_q - v_q) \ge P_{q+1} = P_q - (\ell_{q+2} - 1).$$

Together these force equality. Hence $H_{q+1} = H_q - v_q$ is tight. The induction continues through $q = M - 1$, where the lower bound on the next residual is nonnegativity.

For any vertex $w \in V(H_q)$, put

$$\epsilon(w) = \nu(H_q) - \nu(H_q - w) \in \{0, 1\}.$$

Apply the residual bound to $X_q \cup \{w\}$. It gives

$$\Phi(H_q) - \Phi(H_q - w) \le P_q - P_{q+1} = \ell_{q+2} - 1.$$

The left side equals $d_{H_q}(w) - \epsilon(w)$, proving the stated inequality and $d_{H_q}(w) \le \ell_{q+2}$. If equality holds in the latter bound, then $\epsilon(w) = 1$. This is equivalent to every maximum matching covering w: an uncovered maximum matching survives deletion of w, and a matching of $H_q - w$ of size $\nu(H_q)$ is an uncovered maximum matching of H_q. For the selected vertex, the exact potential drop gives

$$d_{H_q}(v_q) = \ell_{q+2} - 1 + \epsilon_q.$$

Finally, $\Phi(H_M) = P_M = 0$ gives $e(H_M) = \nu(H_M)$. A maximum matching therefore contains every edge of H_M. These edges are pairwise disjoint, so the residual consists of edges and isolated vertices. Telescoping the matching losses yields

$$\nu(G) = z + \sum_{q=0}^{M-1} \epsilon_q.$$

There are M indices, and rearrangement gives the final identity. This identity does not assert that its right side is nonpositive. □

9.4. **Canonical pools and exact normalization.** We identify the suffix condition represented by a prefix pool and then detach the full-degree vertices of a tight host.

Theorem 9.30 (Canonical pools and exact host normalization)**.** *Let $F_R = F_1$ and $F_B = F_2$ be as in Setup 9.3. Fix an integer $0 \le q < M$. For every finite simple core H disjoint from T_q,*

$$H \sqcup T_q \to (F_R, F_B) \quad \Longleftrightarrow \quad H \to (A, B) \text{ for every } (A, B) \in \mathcal{A}_q.$$

More precisely, an avoiding colouring of $H \sqcup T_q$ determines a member of $\mathcal{A}_q$ avoided by its unchanged core colouring. The pool stars can then all be made monochromatic while both full targets remain absent. Conversely, every core colouring avoiding any member of $\mathcal{A}_q$ extends by monochromatic pool stars to a colouring avoiding the original full pair.

For any finite simple graph G and any $X \subseteq V(G)$ with $|X| = q < M$, put

$$G^\star_X = (G - X) \sqcup T_q, \qquad c_X = e(G) - e(G - X), \qquad h_X = \nu(G) - \nu(G - X).$$

Then

$$G \to (F_R, F_B) \quad \Longrightarrow \quad G^\star_X \to (F_R, F_B).$$

An avoiding colouring of $G^\star_X$ has an explicit pullback to an avoiding colouring of G. Moreover,

$$e(G^\star_X) = e(G) - c_X + \sum_{j=2}^{q+1} \ell_j, \qquad \nu(G^\star_X) = \nu(G) - h_X + q.$$

Finally, assume

$$G \to (F_R, F_B), \qquad \Phi(G) = L - M, \qquad \nu(G) < M,$$

and fix any maximum matching Q. There are an integer $0 \le q < \nu(G)$ and a set $X \subseteq V(G)$ of size q such that $H = G - X$ satisfies

$$\nu(H) = \nu(G) - q, \qquad e(H) = e(G) - \sum_{j=2}^{q+1} \ell_j, \qquad \Delta(H) = \ell_{q+2} - 1 \ge 3.$$

The restriction of Q to H is maximum. The graph $G^\star_X$ has the same edge count, matching number, and potential as G, and still arrows the original pair. It is obtained by successively detaching full-degree vertices of degrees

$$\ell_2, \dots, \ell_{q+1}$$

as whole isolated stars, with the stated avoiding-colour pullback. This normalization asserts neither unrestricted matching-rank rigidity nor endpoint escape.

Proof. We first prove the constructive equivalence, then the pullback, and finally the tight normalization.

Step 1. Scanning the prefix pool selects an actual suffix pair. Fix an avoiding colouring of $H \sqcup T_q$. Process the q pool stars in their displayed order. Before star h, where $1 \le h \le q$, suppose α red heads and β blue heads have been assigned to earlier pool stars, with

$$\alpha + \beta = h - 1.$$

Each assigned component uses spokes of its colour in its own pool star. Neither original target has been exhausted: otherwise those assigned components alone would form a full monochromatic target in the pool. Thus the next sizes $n_{\alpha+1}$ and $m_{\beta+1}$ exist. Their index sum is $h+1$, so

$$\ell_{h+1} \ge n_{\alpha+1} + m_{\beta+1} - 1.$$

If the current star had fewer than $n_{\alpha+1}$ red spokes and fewer than $m_{\beta+1}$ blue spokes, its capacity would be at most

$$n_{\alpha+1} + m_{\beta+1} - 2,$$

a contradiction. Choose a colour with enough spokes and assign its largest remaining component to this pool star. Increase the corresponding prefix count by one. Exhaustion remains impossible by avoidance.

After all stars, $\alpha + \beta = q$, $\alpha < s$, and $\beta < t$. If the red core contained F_R^α, its components and the chosen red pool components would form the full red target. They are vertex-disjoint because the core and all pool stars are disjoint. Hence this suffix is absent. The corresponding blue suffix is absent by the same disjointness argument. The unchanged core colouring avoids a member of $\mathcal{A}_q$.

Conversely, fix a core colouring avoiding (F_R^α, F_B^β), with legal $\alpha + \beta = q$. Choose any sequence of q colour labels containing α red and β blue labels. Make the successive pool stars monochromatic with these colours. If the preceding counts are α_0, β_0, their sum is $h-1$, and

$$\ell_{h+1} \ge n_{\alpha_0+1} + m_{\beta_0+1} - 1.$$

This capacity is at least the current largest remaining size in the designated colour. The targets are not exhausted, since the final counts are strictly below s, t. Earlier result 9.11 therefore deletes exactly the largest remaining component in the designated colour and none in the other. After all stars, its residual pair is exactly (F_R^α, F_B^β). The full-copy residual equivalence proves avoidance of both full targets.

Using the labels selected by the first scan gives the monochromatic-pool normalization without changing core colours. The two constructions prove the arrowing equivalence. When $q = 0$, the pool is empty and $\mathcal{A}_0$ contains only the original pair, so the conclusion still holds.

Step 2. The pool replacement has an explicit pullback. Let $|X| = q < M$. If G arrows the original pair, Lemma 9.24(i) says that $G - X$ arrows every member of $\mathcal{A}_q$. Step 1 gives arrowing of $G_X^\star$.

Now begin with an avoiding colouring of $G_X^\star$. Step 1 retains its core colours and selects a legal split α, β whose actual tails are absent there. Partition

$$X = X_R \sqcup X_B, \qquad |X_R| = \alpha, \quad |X_B| = \beta.$$

Retain the core colours. Colour every original edge with an endpoint in X_R red. Colour every remaining original edge with an endpoint in X_B blue. These prescriptions partition the edges outside the core, giving red priority to the cross-edges.

In an alleged full red target, at most α vertex-disjoint components meet X_R. Discard them. A remaining positive red component cannot meet X_B, because every edge from X_B to a vertex outside X_R is blue. Thus every remaining component lies in the core. Deleting at most α components from an ordered star forest leaves a forest containing the suffix after its α largest components: the j-th surviving index is at most $\alpha + j$, and the associated size is at least $n_{\alpha+j}$. Trimming spokes gives the forbidden red core suffix.

No blue component can meet X_R, since all its incident edges are red. Discarding the at most β blue components meeting X_B leaves a forest in the core containing the blue suffix. This is also forbidden. The pullback avoids both full targets, including their unit components.

Each pool star has positive capacity, so

$$e(T_q) = \sum_{j=2}^{q+1} \ell_j, \qquad \nu(T_q) = q.$$

Edge counts and matching numbers add over disjoint host components. Substituting the definitions of c_X, h_X proves the replacement formulas.

Step 3. Full-degree deletions preserve both budgets in a tight host. Use the prescribed maximum matching Q and Earlier result 9.12. At a reached stage h, write

$$H_h = G - X_h, \qquad Q_h = Q \cap E(H_h), \qquad J_h = H_h - Q_h.$$

The earlier theorem gives Q_h maximum and

$$e(J_h) = P_h, \qquad \Delta(J_h) = \ell_{h+2} - 1 \quad (h < M).$$

At a Q_h-covered vertex, the degree in H_h exceeds the degree in J_h by one. At an uncovered vertex, the degrees agree. Hence every degree in H_h is at most ℓ_{h+2}.

If a vertex has degree exactly ℓ_{h+2}, it is Q_h-covered; otherwise its degree in J_h would exceed $\Delta(J_h)$. Its degree in J_h is $\ell_{h+2} - 1$, so it is a permitted greedy choice. Delete such a vertex whenever one exists. Each step loses one matching edge, preserves maximality of the restriction of Q, and removes exactly ℓ_{h+2} residual edges.

This process stops at some $q < \nu(G)$. If it continued for $\nu(G)$ steps, the residual matching number would be zero, so the residual graph would be edgeless. But $\nu(G) < M$, and Lemma 9.24(i) gives arrowing of nonempty actual suffix pairs after that deletion. An edgeless graph cannot arrow two positive-edge forests. This contradiction proves the claimed stopping time.

At the stopping stage,

$$\Delta(H_q) \le \ell_{q+2} - 1.$$

Since $J_q \subseteq H_q$ has maximum degree $\ell_{q+2} - 1$, equality holds. Summing the full-degree edge losses and the unit matching losses gives the stated formulas for $e(H_q)$ and $\nu(H_q)$. The restriction of Q remains maximum.

The assumptions give

$$e(G) = L - M + \nu(G) < L.$$

If $\Delta(H_q) \le 2$, Earlier result 9.13 gives

$$e(H_q) \ge \sum_{j=q+2}^{M+1} \ell_j.$$

Adding the already paid prefix would imply $e(G) \ge L$, a contradiction. Thus

$$\Delta(H_q) = \ell_{q+2} - 1 \ge 3.$$

Finally,

$$c_{X_q} = \sum_{j=2}^{q+1} \ell_j, \qquad h_{X_q} = q.$$

The replacement formulas show that e, ν, and Φ are unchanged. Arrowing and the avoiding-colour pullback were proved for every X. At each intermediate deletion, the replacement consists of the surviving core and the stars already detached. This gives the successive whole-star detachment, whose exact suffix conditions are those of Step 1. No further matching-rank conclusion is used. □

9.5. **Matching capacity and localization.** We prove the edge-count and matching consequences of arrowing, followed by localization of the full original targets.

Theorem 9.31 (A matching against a star forest)**.** *Let $s, t \geq 1$ be integers, let $m_1 \geq \cdots \geq m_t \geq 1$ be integers, and put*

$$F_2 = \bigsqcup_{j=1}^{t} K_{1,m_j}.$$

Let $\widehat{R}(sK_2, F_2)$ be the least edge count of a finite simple graph arrowing the indicated pair. Then

$$\widehat{R}(sK_2, F_2) = (s-1)m_1 + \sum_{j=1}^{t} m_j = sm_1 + \sum_{j=2}^{t} m_j.$$

An attaining host is

$$sK_{1,m_1} \ \sqcup \ \bigsqcup_{j=2}^{t} K_{1,m_j}.$$

The same value equals $\widehat{R}(F_2, sK_2)$.

Proof. Apply Earlier result 9.9 with $J = F_2$. This graph has at least one edge, since $t \geq 1$ and all $m_j \geq 1$. Its edge count is

$$e(F_2) = \sum_{j=1}^{t} m_j.$$

Its maximum degree is m_1: a centre of a largest star has that degree, every other centre has degree at most m_1, and every leaf has degree $1 \leq m_1$. Hence every arrowing host has at least

$$\sum_{j=1}^{t} m_j + (s-1)m_1$$

edges.

For the upper bound, use Earlier result 9.10 with $n_i = 1$ for every $1 \leq i \leq s$. The red target is sK_2. On diagonal k, the admissible blue indices form the nonempty interval

$$\max(1, k-s) \leq j \leq \min(t, k-1).$$

Since $n_i + m_j - 1 = m_j$ and the blue list is nonincreasing,

$$\ell_k = m_{\max(1,k-s)}.$$

For $k = 2, \ldots, s+1$, this gives exactly s copies of m_1. For $k = s+2, \ldots, s+t$, it gives one copy of each m_j, $2 \leq j \leq t$. The latter interval and sum are empty when $t = 1$. Thus the diagonal-star host is the one stated, and it has

$$sm_1 + \sum_{j=2}^{t} m_j = (s-1)m_1 + \sum_{j=1}^{t} m_j$$

edges. Earlier result 9.10 gives its arrowing, proving equality.

Interchanging red and blue is a bijection on the colourings of any fixed host. It interchanges the two ordered target conditions. The class of arrowing hosts, and hence its minimum edge count, is unchanged. □

Proposition 9.32 (A matching bound from half-order capacity). *Let $G \to (F_1, F_2)$ in Setup 9.3, and put*

$$T(F_1) = \sum_{i=1}^{s} \left\lfloor \frac{n_i + 1}{2} \right\rfloor, \qquad T(F_2) = \sum_{j=1}^{t} \left\lfloor \frac{m_j + 1}{2} \right\rfloor.$$

Then

$$\nu(G) \geq \min\left(s + t - 1,\ \max\left(T(F_2) + \left\lfloor \frac{s-1}{2} \right\rfloor,\ T(F_1) + \left\lfloor \frac{t-1}{2} \right\rfloor\right)\right).$$

In particular,

$$\sum_{j=1}^{t} \left\lfloor \frac{m_j - 1}{2} \right\rfloor \geq \left\lceil \frac{s-1}{2} \right\rceil$$

implies $\nu(G) \geq s + t - 1$, as does its colour-exchanged counterpart. Unit components are allowed.

Proof. Write $q = \nu(G)$. If $q \geq s + t - 1$, the assertion holds. Suppose $q < s + t - 1$. The all-red colouring supplies F_1, and one edge from each star gives $q \geq s$. The all-blue colouring gives $q \geq t$. Thus $s, t \geq 2$: if $s = 1$, then $q \geq t = s + t - 1$, and the case $t = 1$ gives the same contradiction.

We prove

$$q \geq T(F_2) + \left\lfloor \frac{s-1}{2} \right\rfloor.$$

By Earlier result 9.7, choose an attaining set S with

$$q = |S| + \sum_{C} \left\lfloor \frac{|V(C)|}{2} \right\rfloor,$$

where C ranges over the components of $G - S$. Put $h = |S|$.

For every $X \subseteq V(G)$ with $|X| \leq s - 1$, Lemma 9.24(i), deleting $|X|$ leading red components, leaves a nonempty red suffix against the full F_2. Its all-blue colouring forces F_2 in $G - X$. Hence Earlier result 9.8 applies with $J = F_2$ and $k = s - 1$. Since $\nu(F_2) = t$ and $q < (s-1) + t$, it gives

$$h \leq q - t \leq s - 2.$$

Define

$$r = \left\lfloor \frac{s - 1 - h}{2} \right\rfloor.$$

Inside each component C, pair any $2\lfloor |V(C)|/2 \rfloor$ vertices into $\lfloor |V(C)|/2 \rfloor$ disjoint unordered pairs. These pairs need not be edges. Pairs from distinct components are disjoint, and their total number is $q - h$. Since

$$q - h \geq s - h > r,$$

choose exactly r such pairs. Let r_C count those selected in C, and let Y be their endpoints. Then

$$|Y| = 2r, \qquad \sum_{C} r_C = r, \qquad Y \cap S = \varnothing.$$

The set $X = S \cup Y$ has size $h + 2r \leq s - 1$, so $G - X$ contains F_2.

Each star of this copy is connected and lies within one original component C of $G - S$, even if deletion of Y disconnects that component. Let J_C be the blue target indices assigned to C. These sets partition $\{1, \ldots, t\}$. Vertex-disjointness gives

$$\sum_{j \in J_C} (m_j + 1) \leq |V(C)| - 2r_C.$$

For any finite list of nonnegative integers z_j,

$$\sum_{j} \left\lfloor \frac{z_j}{2} \right\rfloor \leq \left\lfloor \frac{\sum_j z_j}{2} \right\rfloor.$$

To see this, write each z_j as an even integer plus a remainder in $\{0,1\}$. Applying the inequality to the vertex count gives

$$\sum_{j\in J_C}\left\lfloor\frac{m_j+1}{2}\right\rfloor \le \left\lfloor\frac{|V(C)|-2r_C}{2}\right\rfloor = \left\lfloor\frac{|V(C)|}{2}\right\rfloor - r_C.$$

This also holds when J_C is empty. Summing over C yields

$$T(F_2) \le q-h-r.$$

Therefore

$$\begin{aligned} q &\ge T(F_2)+h+\left\lfloor\frac{s-1-h}{2}\right\rfloor \\ &= T(F_2)+\left\lfloor\frac{s-1+h}{2}\right\rfloor \\ &\ge T(F_2)+\left\lfloor\frac{s-1}{2}\right\rfloor. \end{aligned}$$

The equality uses integrality of h, and the final inequality uses $h \ge 0$.

Exchanging the targets gives

$$q \ge T(F_1)+\left\lfloor\frac{t-1}{2}\right\rfloor$$

under the same premise $q < s+t-1$. Thus q is at least the maximum of the two expressions in this case. Together with the case $q \ge s+t-1$, this proves the minimum formula.

For every positive integer m_j,

$$\left\lfloor\frac{m_j+1}{2}\right\rfloor = 1+\left\lfloor\frac{m_j-1}{2}\right\rfloor.$$

Under the stated sufficient condition,

$$T(F_2)+\left\lfloor\frac{s-1}{2}\right\rfloor \ge t+\left\lceil\frac{s-1}{2}\right\rceil+\left\lfloor\frac{s-1}{2}\right\rfloor = s+t-1.$$

The minimum bound gives the conclusion. Interchanging the targets proves the other condition. Every step allows unit components. □

Proposition 9.33 (Localization with the full original target)**.** *Let F_1, F_2 be nonempty positive-edge star forests with integer component counts $s,t \ge 1$, and let $G \to (F_1,F_2)$. Suppose $S \subseteq V(G)$ satisfies*

$$\nu(G) = |S| + \sum_C \nu(C),$$

where C ranges over the components of $G-S$. For any union H of these components, put

$$\mu = \nu(H), \qquad p = s-\nu(G)+\mu.$$

If $p \ge 1$, then

$$H \to (pK_2, F_2).$$

Every colouring of H avoiding this pair extends to a full avoiding colouring of G by colouring every edge outside $E(H)$ red.

The localization applies, in particular, to every attaining set

$$\nu(G) = |S| + \sum_C \left\lfloor\frac{|V(C)|}{2}\right\rfloor.$$

No maximum-cardinality assumption is needed, and every complementary component then satisfies

$$\nu(C) = \left\lfloor \frac{|V(C)|}{2} \right\rfloor.$$

In this attaining setting, put $d = s+t-1-\nu(G) \geq 1$. *If* $\mu \geq t-d$, *and* o *of the selected components have odd order, then* $H - X$ *contains the entire* F_2 *for every* $X \subseteq V(H)$ *of size at most* $\mu - t + d$, *and*

$$e(F_2) \leq \mu + o - d.$$

Thus an odd complementary component has matching number either below $t-d$ *or at least* $e(F_2)+d-1$. *An even complementary component has matching number either below* $t-d$ *or at least* $e(F_2)+d$.

Moreover, if $X \subseteq V(G)$ *satisfies*

$$\nu(G - X) = \nu(G) - |X|$$

and Y *is attaining in* $G - X$, *then* $X \cup Y$ *is attaining in* G. *The preceding full-target conclusions apply to component unions of* $(G - X) - Y$. *If* Y *has maximum attaining cardinality in* $G - X$, *these components are odd and factor-critical. The lifted set* $X \cup Y$ *need not have maximum attaining cardinality in* G. *All assertions have colour-exchanged versions and retain unit components.*

Proof. Assume first the matching-additive equality for S. Fix a colouring of H avoiding red pK_2 and blue F_2, with $p \geq 1$. Retain its colours and colour every other edge of G red. A red matching has at most $|S|$ edges meeting S. Its other edges lie within components of $G - S$. At most

$$\sum_{C \text{ unselected}} \nu(C)$$

lie outside H, and at most $p-1$ lie in H. Its total size is therefore at most

$$|S| + \sum_{C \text{ unselected}} \nu(C) + p - 1 = \nu(G) - \mu + p - 1 = s - 1.$$

Every F_1 contains an s-edge matching, obtained by choosing one edge from each positive component. Thus there is no red F_1. All blue edges lie in H, where F_2 is absent. This proves the extension statement. Arrowing of G excludes such a colouring, hence $H \to (pK_2, F_2)$.

Now let S be attaining in the half-order sense. Take a maximum matching W of G. At most $|S|$ of its edges meet S, and at most $\lfloor |V(C)|/2 \rfloor$ lie internally in a component C. These upper bounds sum to $\nu(G) = |W|$. Every individual count must attain its upper bound; otherwise their total would be smaller than $|W|$. In particular, W has $\lfloor |V(C)|/2 \rfloor$ internal edges in every C. Consequently,

$$\nu(C) \geq \left\lfloor \frac{|V(C)|}{2} \right\rfloor.$$

The opposite inequality follows by counting matching endpoints. Thus equality holds and supplies the matching-additive equality. Neither factor-criticality nor maximality of S was used.

Assume $d \geq 1$ and $\mu \geq t - d$. Then

$$p = s - \nu(G) + \mu = \mu - t + d + 1 \geq 1.$$

For $X \subseteq V(H)$ with $|X| \leq p-1$, colour all edges meeting X red and all other edges blue. Every red matching has size at most $|X| \leq p-1$. Arrowing forces a blue copy of the whole F_2 in $H - X$, proving robustness.

The all-blue colouring of H forces F_2, hence $\mu \geq t$. The all-red colouring forces pK_2, hence $p \leq \mu$. Therefore a set of exactly $p-1$ vertices of H exists. Each selected component has matching number equal to its half-order floor, so

$$|V(H)| = 2\mu + o.$$

Also,

$$|V(F_2)| = e(F_2) + t.$$

Delete exactly $p-1=\mu-t+d$ vertices. Robust containment gives

$$e(F_2)+t\le 2\mu+o-(\mu-t+d)=\mu+o+t-d.$$

Subtracting t proves $e(F_2)\le \mu+o-d$. For a single odd component, $o=1$, giving

$$\mu\ge e(F_2)+d-1$$

whenever $\mu\ge t-d$. For a single even component, $o=0$, giving

$$\mu\ge e(F_2)+d.$$

Failure of the threshold gives the other alternative. Reversing colour names throughout proves the symmetric statements.

For the lifting assertion, the components of $G-(X\cup Y)$ are exactly those of $(G-X)-Y$, and $X\cap Y=\varnothing$. Therefore

$$\begin{aligned}|X\cup Y|+\sum_C\left\lfloor\frac{|V(C)|}{2}\right\rfloor &= |X|+|Y|+\sum_C\left\lfloor\frac{|V(C)|}{2}\right\rfloor\\ &= |X|+\nu(G-X)\\ &= \nu(G).\end{aligned}$$

Thus $X\cup Y$ is attaining for the original graph. All preceding statements apply with its original full targets and deficit. If Y has maximum attaining cardinality in the residual, Earlier result 9.7 gives odd factor-critical complementary components. A prefix of deletions that loses exactly one matching edge per deleted vertex satisfies the displayed exact-loss hypothesis and is included. No assertion of maximum cardinality for $X\cup Y$ is needed, and the localized F_2 was never shortened. □

Theorem 9.34 (Full-target localization after arbitrary deletions)**.** *Let F_1,F_2 be nonempty star forests with positive integer component sizes $n_1,\dots,n_s$ and $m_1,\dots,m_t$, edge counts E_1,E_2, and $M=s+t-1$. Let $G\to(F_1,F_2)$. Choose any $X\subseteq V(G)$ with integer size $q<M$, put $H=G-X$, and assume*

$$\delta=M-q-\nu(H)\ge 1.$$

Let Y be any attaining set of H, and put

$$\eta_1=\sum_{i=1}^{s}\left(\left\lceil\frac{n_i}{2}\right\rceil-1\right).$$

(i) *For every union T of components of $H-Y$ with $\mu=\nu(T)\ge t-\delta$,*

$$T\to\big((\mu-t+\delta+1)K_2,F_2\big).$$

It retains the entire F_2 after every deletion of at most $\mu-t+\delta$ vertices. If o selected components have odd order, then

$$E_2\le\mu+o-\delta.$$

A single odd component has matching number either below $t-\delta$ or at least $E_2+\delta-1$. For an even component, the latter bound is $E_2+\delta$.

(ii) *If*

$$E_2>2t-3\delta-2(q+|Y|)-2\eta_1,$$

some component C of $H-Y$ satisfies

$$\nu(C)\ge E_2+\delta-1.$$

If also $E_2\ge s-2\delta+1$, the same component satisfies

$$\nu(C)\ge\max(E_1,E_2)+\delta-1.$$

Parts (i) and (ii) have colour-exchanged versions. They concern the original full targets, with no exact matching-loss assumption on X, and no edge-budget, maximum-degree, potential-surplus, or maximality assumption on Y.

(iii) *Let C be an odd component of $H - Y$ with matching number*

$$\mu \geq E_2 + \delta - 1.$$

Assume $\Delta(C) \leq D$ for an integer $D \geq 3$, and put

$$k = \mu - t + \delta, \qquad \rho = \mu - E_2 - \delta + 1.$$

If $2\mu + 1 > \rho D + 1$, then every $x \in V(C)$ satisfies

$$d_C(x) \geq k + 1 - \rho(D - 1),$$

and

$$(D - 2)\mu \geq (D - 1)E_2 - t + D(\delta - 2) + 2.$$

In particular, if F_2 has 12 components and 15 edges, $\Delta(C) \leq 4$, $\mu \leq 21$, and $\delta \in \{1, 2\}$, then

$$\mu \geq 16 \quad \text{when } \delta = 1, \qquad \mu \geq 18 \quad \text{when } \delta = 2.$$

All assertions include the original unit components.

Proof. We first construct an auxiliary graph with the effective deficit δ. Put $y = |Y|$ and $S = X \cup Y$. Every component C of $H - Y$ satisfies

$$\nu(C) = \left\lfloor \frac{|V(C)|}{2} \right\rfloor.$$

To verify this, take a maximum matching of H. At most y edges meet Y, and at most $\lfloor |V(C)|/2 \rfloor$ lie internally in C. The attaining equality makes the sum of these bounds exactly $\nu(H)$. Each internal bound is therefore attained. The reverse bound on $\nu(C)$ follows from counting endpoints. Consequently,

$$|S| + \sum_C \nu(C) = q + y + \nu(H) - y = q + \nu(H) = M - \delta. \tag{9.2}$$

These C are precisely the components of $G - S$.

By Earlier result 9.14, attach a distinct new pendant leaf to every vertex of S, retaining all original vertices and edges. Call the resulting graph G^+. Since $G \subseteq G^+$, this graph arrows the unchanged full pair. Earlier result 9.14 and (9.2) give

$$\nu(G^+) = M - \delta,$$

and S is attaining in G^+. The components of $G^+ - S$ are the original components of $H - Y$ and the new isolated vertices. No degree or edge-count property of G^+ is used.

For part (i), apply Proposition 9.33 to G^+, S, and the chosen union T of original components. Its matching target size is

$$s - \nu(G^+) + \mu = s - (M - \delta) + \mu = \mu - t + \delta + 1,$$

which is positive. The proposition gives arrowing against the entire F_2, robustness under every deletion of at most $\mu - t + \delta$ vertices, and

$$E_2 \leq \mu + o - \delta.$$

Taking a single component gives the odd and even alternatives. No new leaf belongs to T. Interchanging the two targets proves the other orientation.

For part (ii), apply Earlier result 9.15 to G^+, its attaining set S, and the full original target lists. The deficit is δ, the half-order excess is the unchanged η_1, and $|S| = q + |Y|$. Its strict hypothesis is exactly the displayed inequality in part (ii). It gives a complementary component with matching number at least $E_2 + \delta - 1$. This lower bound is at least one, since $E_2 \geq 1$ and

$\delta \geq 1$. Thus the component is not one of the new isolated leaves. It lies in the original $H - Y$. The additional hypothesis $E_2 \geq s - 2\delta + 1$ is the additional hypothesis of the same earlier theorem, giving the common maximum-edge bound for this component. Exchanging the target roles gives the symmetric conclusion. The proof used no equality between $\nu(G) - \nu(H)$ and q; the private leaves supply the effective attaining equality.

For part (iii), write $E = E_2$. The component C is odd and its matching number equals its half-order floor, so

$$|V(C)| = 2\mu + 1.$$

The lower bound on μ makes $\rho = \mu - E - \delta + 1$ a nonnegative integer. Since F_2 has t positive components,

$$|V(F_2)| = E + t.$$

Also $E \geq t$, so $k = \mu - t + \delta \geq 0$. Part (i) gives a copy of the entire F_2 after every deletion of k vertices from C. Such deletion sets exist: its all-blue consequence gives $\mu \geq t$, and direct substitution gives

$$2\mu + 1 = (E + t) + (\mu - t + \delta) + (\mu - E - \delta + 1) = |V(F_2)| + k + \rho. \tag{9.3}$$

The graph C is finite and connected. The graph F_2 has an edge and no isolated vertices. The assumed order inequality is

$$|V(C)| > \rho D + 1.$$

Thus Earlier result 9.16, with $r = \rho$, gives

$$d_C(x) \geq k + 1 - \rho(D - 1)$$

and

$$k + 1 \leq D + (D - 1)\rho. \tag{9.4}$$

Substitute $k = \mu - t + \delta$ and $\rho = \mu - E - \delta + 1$ into (9.4). Expansion and rearrangement give

$$\begin{aligned}(D - 2)\mu &\geq (D - 1)E - t + D\delta - 2D + 2 \\ &= (D - 1)E_2 - t + D(\delta - 2) + 2.\end{aligned}$$

For the stated specialization, let $E_2 = 15$, $t = 12$, $D = 4$, and $\delta \in \{1, 2\}$. Under the initial lower bound of part (iii), its order hypothesis is equivalent to

$$\mu < 2(15 + \delta - 1) = 28 + 2\delta.$$

It holds because $\mu \leq 21$. The resulting inequality is

$$2\mu \geq 27 + 4\delta.$$

When $\delta = 1$, integrality gives $\mu \geq 16$. When $\delta = 2$, it gives $\mu \geq 18$. Every augmentation preserved the original lists, and every containment retained their full unit blocks. □

9.6. **Small surplus and maximum-degree vertices.** We apply Theorem 9.28 after deleting uncovered maximum-degree vertices, then use the earlier parity bounds.

The conclusions of Theorem 9.35 are conditional on the indexed matching bound in Earlier result 9.17, the indexed overflow bound in Earlier result 9.21, the parity inequalities in Earlier result 9.22, and the even-capacity assertion of Earlier result 9.19. Its use of Theorem 9.28 also depends on Earlier result 9.6.

Theorem 9.35 (Small-surplus family structure). *Let $\mathcal{A}$ be a finite nonempty family as in Notation 9.5. Write its envelope as $\ell_j = \bar{\ell}_j$, and put*

$$N = \max_{\gamma \in \mathcal{A}}(s_\gamma + t_\gamma - 1), \qquad D = \ell_2 - 1 \geq 3, \qquad P = \sum_{j=2}^{N+1}(\ell_j - 1).$$

Let R be the initial plateau length at $D+1$. Let H arrow every family member, and assume $\Delta(H) \le D$. Put

$$\sigma = e(H) - \nu(H) - P, \qquad \delta_D = \delta_D(H).$$

Then σ is a nonnegative integer,

$$R \le \delta_D \le R + \sigma,$$

and every maximum matching leaves between R and $R+\sigma$ degree-D vertices uncovered.

A terminal maximizing triple (γ, i, j) has

$$i + j = R + 1, \qquad n_i^\gamma + m_j^\gamma - 1 = D + 1,$$

and sizes $a = n_i^\gamma, b = m_j^\gamma$. If $\sigma \in \{0,1\}$, every such triple has sizes either both positive and even, or D is odd and

$$\{a, b\} = \{2, D\}.$$

Assume jointly that $\sigma \in \{0,1\}$ and at least one terminal maximizing triple has both sizes at least three. Then:

(i) *D is even and $D \ge 6$.*

(ii) *If κ counts the whole connected components of H that are D-regular and have odd order, then*

$$R \le \kappa \le R + \sigma,$$

and every counted component has a near-perfect matching.

(iii) *Let Q be the union of the other components. Every maximum matching of Q leaves at most*

$$R + \sigma - \kappa \le \sigma$$

vertices of ambient degree D uncovered.

(iv) *For each terminal maximizing triple with $a, b \ge 3$, the graph Q has an $(a-1)/(b-1)$ degree colouring, possibly depending on the triple.*

Parts (i)–(iv) are asserted only under the joint nonbinary and low-surplus hypothesis. There is no edge-count upper bound or positive matching-deficit hypothesis.

Proof. We first bound the uncovered degree-D vertices for every maximum matching. We then use the indexed overflow and parity bounds to treat $\sigma = 0$ and $\sigma = 1$ together. The component conclusions retain the additional nonbinary hypothesis.

Step 1. The envelope and potential bound control every maximum matching. A member with component budget N supplies a candidate on every diagonal $2, \dots, N+1$. The envelope is nonincreasing. For a candidate on diagonal $j \ge 3$, decrease an index larger than one. The preceding diagonal then has a candidate of at least the same value, because the target lists are nonincreasing. Maximizing over candidates and family members gives

$$\ell_{j-1} \ge \ell_j.$$

Thus $1 \le R \le N$. The first R potential entries are D, and each later entry is at most $D-1$. Consequently,

$$P \le DN - (N - R). \tag{9.5}$$

Theorem 9.28 gives $\sigma \ge 0$, and all terms are integral.

Choose a terminal maximizing triple. Its size sum is $D+2$, and its index sum is $R+1$. Earlier result 9.17, applied to that full member, gives

$$\delta_D \ge R.$$

Fix an arbitrary maximum matching W of H. Let I be its uncovered degree-D vertices and put $k = |I|$. Then $k \ge R$. The set I is independent: an edge between two uncovered vertices could be added to W. Thus deleting I removes exactly Dk edges. It removes no edge of W, so

$$\nu(H - I) = \nu(H).$$

The reverse matching inequality follows from $H - I \subseteq H$. Hence

$$\Phi(H - I) = P + \sigma - Dk. \tag{9.6}$$

Suppose first that $k < N$. Retain every pair of nonempty actual suffixes obtained from an original member by deleting k leading components in total, whenever the split is legal. This family is nonempty because some member has budget $N > k$. Its maximum component budget is $N - k$. Theorem 9.26, applied to I, gives its arrowing by $H - I$.

At a local diagonal $2 \le h \le N - k + 1$, every suffix candidate is an original candidate of index sum $k + h$. Conversely, let original indices i, j satisfy $i + j = k + h$, and put

$$\alpha = \max(0, k - (j - 1)), \qquad \beta = k - \alpha.$$

Then $\alpha, \beta \ge 0$, their sum is k, and

$$\beta \le j - 1, \qquad \alpha \le i - 1,$$

the latter because $k \le i + j - 2$. The suffixes are legal and nonempty. Their local indices $i - \alpha, j - \beta$ sum to h. Thus every original candidate appears, and the new envelope entry is ℓ_{k+h}.

Theorem 9.28 gives

$$\Phi(H - I) \ge P'_k := \sum_{j=k+2}^{N+1} (\ell_j - 1).$$

Since $k \ge R$, the first k potential entries sum to at most

$$RD + (k - R)(D - 1) = Dk - (k - R).$$

Using (9.6),

$$\begin{aligned} 0 &\le \Phi(H - I) - P'_k \\ &= P - P'_k + \sigma - Dk \\ &\le \sigma - (k - R). \end{aligned}$$

Therefore $k \le R + \sigma$.

If $k \ge N$, use nonnegative residual potential, (9.5), and (9.6):

$$0 \le P + \sigma - Dk \le \sigma - D(k - N) - (N - R).$$

Hence

$$\sigma \ge D(k - N) + N - R \ge k - R,$$

since $D \ge 1$ and $k - N \ge 0$. Again $k \le R + \sigma$. This argument does not assume that nonempty suffixes survive when $k \ge N$.

The interval holds for every maximum matching. Since $\delta_D \le k$ for each such matching,

$$R \le \delta_D \le R + \sigma.$$

Step 2. We set up the overflow bounds for both surplus values. Assume $\sigma \in \{0, 1\}$. Step 1 gives

$$R \le \delta_D \le R + \sigma \le R + 1.$$

Fix any terminal triple with sizes a, b. Each full target of each member embeds uncoloured in H: use the all-red or all-blue colouring of that member. Thus

$$a, b \le \Delta(H) \le D.$$

Since $a + b = D + 2$, both $a, b \ge 2$. Put

$$A = a - 1, \qquad B = b - 1, \qquad \omega = \Omega_{A,B}(H).$$

The capacities are positive and sum to D. The indexed inequality in Earlier result 9.21, applied to the selected full target pair, gives

$$\omega \ge i + j - 1 = R. \tag{9.7}$$

Step 3. The degree-colouring and parity bounds restrict the sizes of every terminal maximizing triple.

If a, b were both odd, then A, B would both be even. Earlier result 9.19 would give an A/B degree colouring. It would avoid a red a-edge star and a blue b-edge star, and hence avoid the selected full member. This contradicts arrowing.

If a, b have opposite parity, write e_* for the even size and o_* for the odd size. The odd capacity is $e_* - 1$, and the positive even capacity is $o_* - 1$. The odd-degree inequality in Earlier result 9.22 gives

$$\delta_D \geq \omega + (e_* - 2) \left\lceil \frac{\omega}{o_* - 1} \right\rceil .$$

The ceiling is at least one because $\omega \geq R \geq 1$. Since $\delta_D \leq R + \sigma \leq R + 1$ and $\omega \geq R$, it follows that

$$e_* - 2 \leq 1.$$

The even integer e_* is at least two, so $e_* = 2$. The other size is D, which is odd. This proves the parity alternatives for every terminal triple for both values of σ.

Step 4. Under the joint nonbinary and low-surplus hypothesis, the odd regular components account for all overflow. If no terminal triple has both sizes at least three, none of parts (i)–(iv) is asserted. For the rest of the proof, assume jointly that $\sigma \in \{0, 1\}$ and that a terminal triple has both sizes at least three. Choose any such triple, and use its capacities $A = a - 1$, $B = b - 1$ and overflow $\omega = \Omega_{A,B}(H)$. Step 3 forces both sizes to be even and at least four. Therefore D is even and at least six, and A, B are odd and at least three.

Define κ, Q as in the statement. The even-degree alternative in Earlier result 9.22 gives

$$\omega = \kappa \quad \text{or} \quad \delta_D \geq \omega + \min(A, B) - 1.$$

The second alternative implies

$$\delta_D \geq \omega + 2 \geq R + 2,$$

contradicting $\delta_D \leq R + \sigma \leq R + 1$. Thus $\omega = \kappa$, and (9.7) gives $\kappa \geq R$. Each counted odd D-regular component leaves at least one degree-D vertex uncovered by every matching. Hence

$$\kappa \leq \delta_D \leq R + \sigma,$$

proving the count in part (ii).

For a counted component C, the integer

$$|V(C)| - 2\nu(C)$$

is positive and odd. If it were at least three in one component, a union of maximum matchings over all components of H would leave at least

$$\kappa + 2 \geq R + 2 > R + \sigma$$

degree-D vertices uncovered. This contradicts the upper bound from Step 1. Therefore each counted component has deficiency one and a near-perfect matching.

Take any maximum matching of Q and combine it with maximum matchings of the κ counted components. The union is maximum in H, so it leaves at most $R + \sigma$ degree-D vertices uncovered. Exactly κ of them lie in the counted components. The matching of Q therefore leaves at most

$$R + \sigma - \kappa \leq \sigma$$

such vertices uncovered, proving part (iii). When $\sigma = 0$, these bounds give $\kappa = R$, and every maximum matching of Q covers all its vertices of ambient degree D.

Finally, minimum degree overflow is additive over whole connected components: the colour choices and the vertex-sum objective separate by component. Each counted component has positive integral overflow for the odd capacities A, B. If its overflow were zero, then

$$d_{\text{red}} \leq A, \qquad d_{\text{blue}} \leq B, \qquad d_{\text{red}} + d_{\text{blue}} = D = A + B$$

would force $d_{\mathrm{red}} = A$ at every vertex. The red degree sum would be

$$A|V(C)|,$$

which is odd, contradicting its equality to twice the number of red edges. Thus each counted component contributes at least one to the minimum overflow.

Their κ contributions exhaust $\omega = \kappa$. Every component of Q consequently has minimum overflow zero. Choosing minimizing colourings on these components gives an A/B degree colouring of Q. The terminal triple was chosen arbitrarily among those with $a, b \geq 3$, so this proves part (iv) for every such triple. The choice of a minimizing colouring may depend on the triple.

Every use of the component conclusions was under the explicit nonbinary hypothesis. No step used an upper bound on $e(H)$ or a comparison between $\nu(H)$ and N. □

10. Degree splitting and covering matchings

This chapter develops the degree splittings and matching estimates used with the certificate framework. We assume the certificate definitions and matching results established earlier in the series, record the precise prerequisites in Section 10.1, and then give the complete proofs of the results needed here.

10.1. **Setup.**

We assume, as established earlier in this series, the certificate framework of Chapter 1 and the matching and certificate statements recorded below. These statements are prerequisites; their proofs are not repeated here.

Notation 10.1. Graphs are finite and simple unless a multigraph is explicitly specified. For a graph H, write

$$n(H) = |V(H)|, \qquad e(H) = |E(H)|,$$

and let $d_H(v)$, $\Delta(H)$, $\nu(H)$, and $\alpha(H)$ denote vertex degree, maximum degree, maximum matching size, and independence number. An edgeless graph has maximum degree zero. For an edge set $F \subseteq E(H)$, its degrees and matching number refer to the spanning graph $(V(H), F)$.

For $X \subseteq V(H)$, the graph $H - X$ is obtained by deleting the vertices in X. For $P \subseteq E(H)$, the graph $H - P$ is obtained by deleting only the edges in P. The open neighborhood $N_H(I)$ of a vertex set I is the union of the open neighborhoods of its vertices, and $N_H[I] = I \cup N_H(I)$.

All parameters counting vertices, edges, components, colors, or positions in a list are integers. Their nonnegativity or positivity is specified where they occur. Subgraph containment is ordinary, not necessarily induced. A copy of a graph includes all its vertices, including any isolated vertices specified in that graph.

Definition 10.2 (Required vertices and covering thresholds)**.** Let $D \in \mathbb{Z}_{\geq 1}$ and let $\Delta(H) \leq D$. Set

$$S_D(H) = \{v \in V(H) \mid d_H(v) = D\}.$$

For any prescribed set $S \subseteq V(H)$, let $u_S(H)$ be the minimum number of vertices of S left uncovered by a matching of H. For $X \subseteq V(H)$, let $q_S(H - X)$ be the number of odd-order connected components of $H - X$ whose vertex sets are contained wholly in S. When $S = S_D(H)$, write

$$\delta = u_{S_D(H)}(H).$$

Also put

$$\Phi(H) = e(H) - \nu(H).$$

An edge set $F \subseteq E(H)$ covers S by incidence if every vertex of S is incident with an edge of F. For $D \in \mathbb{Z}_{\geq 2}$, the covering threshold is

$$\theta_D(H) = \min\bigl\{\nu((V(H), F)) \mid F \subseteq E(H),\ F \text{ covers } S_D(H)\bigr\}.$$

The empty cover is permitted when $S_D(H) = \varnothing$.

A graph is factor-critical if deleting any one vertex leaves a perfect matching. It is well-covered if every inclusion-maximal independent set is maximum.

Notation 10.3 (Star forests and arrowing)**.** For $a \in \mathbb{Z}_{\geq 1}$, the graph $K_{1,a}$ is the star with a edges. A star forest is a vertex-disjoint union of stars; isolated vertices are allowed when explicitly stated. For graphs F_1, F_2, H, write

$$H \longrightarrow (F_1, F_2)$$

if every red-blue coloring of $E(H)$ contains a red copy of F_1 or a blue copy of F_2.

Setup 10.4 (Even-degree interpolation from the earlier development)**.** We use the following form of the earlier interpolation result. Let L be a finite loopless multigraph, with parallel edge instances retained separately. Let $E_0, E_1 \subseteq E(L)$, and let $r \in \mathbb{Z}_{\geq 0}$. Assume that

$$d_{E_1}(v) - d_{E_0}(v) = 2r \qquad (v \in V(L)).$$

For every $k \in \mathbb{Z}$ with $0 \leq k \leq r$, there is a spanning subgraph F of L satisfying

$$d_F(v) = d_{E_0}(v) + 2k \qquad (v \in V(L)).$$

The earlier result is formulated for pseudographs and includes this multigraph case. The case $r = 0$ is permitted.

Setup 10.5 (Bipartite coloring and the odd-degree covering criterion)**.** The earlier bipartite edge-coloring result states that a finite simple bipartite graph of maximum degree $r \in \mathbb{Z}_{\geq 0}$ has a proper edge coloring with r colors.

Setup 10.6 (Covering maximum-degree vertices for odd degree)**.** Let $D \in \mathbb{Z}_{\geq 1}$ be odd, and let Q be a finite simple graph with $\Delta(Q) \leq D$. The earlier covering criterion states that if $Q[S_D(Q)]$ contains no two vertex-disjoint odd cycles, then Q has a matching covering $S_D(Q)$.

Setup 10.7 (Exceptional-edge certificate from Chapter 1)**.** Let $s, t \in \mathbb{Z}_{\geq 1}$, and let

$$n_1 \geq \cdots \geq n_s \geq 1, \qquad m_1 \geq \cdots \geq m_t \geq 1$$

be integer lists. Put

$$F_1 = \bigsqcup_{\ell=1}^{s} K_{1,n_\ell}, \qquad F_2 = \bigsqcup_{\ell=1}^{t} K_{1,m_\ell}.$$

Let $i \in \{1, \ldots, s\}$ and $j \in \{1, \ldots, t\}$. Assume that the edge set of a graph has a partition into spanning subgraphs Q_R, Q_B, P_R, P_B such that

$$\Delta(Q_R) \leq n_i - 1, \qquad \Delta(Q_B) \leq m_j - 1, \qquad \nu(P_R) \leq i - 1, \qquad \nu(P_B) \leq j - 1.$$

The exceptional-edge certificate states that coloring $Q_R \cup P_R$ red and $Q_B \cup P_B$ blue avoids both a red F_1 and a blue F_2.

Setup 10.8 (Required-set matching criterion from Chapter 1)**.** Let G be a finite simple graph and $S \subseteq V(G)$. The earlier required-set criterion states that G has a matching covering S if and only if

$$q_S(G - X) \leq |X| \qquad \text{for every } X \subseteq V(G).$$

Setup 10.9 (Ordinary attaining sets from the earlier development)**.** For a graph J, let $o(J)$ denote its number of odd-order connected components. The earlier matching decomposition gives

$$|V(J)| - 2\nu(J) = \max_{X \subseteq V(J)} \big(o(J - X) - |X|\big).$$

Equivalently,

$$\nu(J) = \min_{X \subseteq V(J)} \left(|X| + \sum_{B} \left\lfloor \frac{|V(B)|}{2} \right\rfloor \right),$$

where B ranges over the connected components of $J-X$. A set attaining either equivalent extremum is called an ordinary attaining set. Every ordinary attaining set of maximum cardinality has the property that all components of $J - X$ have odd order and are factor-critical.

Setup 10.10 (Augmentation preserving required coverage)**.** The earlier augmentation result states that any matching of a finite graph can be augmented to a maximum matching while preserving coverage of every previously covered vertex. Consequently, for every prescribed vertex set, a maximum matching can attain the minimum number of uncovered prescribed vertices.

Definition 10.11 (Fractional covers)**.** Let $D \in \mathbb{Z}_{\geq 2}$, let $\Delta(H) \leq D$, and put $S = S_D(H)$. A fractional matching cover of S is a family of edge weights

$$a_e \in [0,1] \qquad (e \in E(H))$$

such that the incident weight sum is at most one at every vertex and exactly one at every vertex of S. Let $w = w(H, D)$ be its minimum total weight.

Setup 10.12 (Fractional potentials from Chapter 1)**.** For the fractional cover problem in Definition 10.11, the earlier potential result gives an optimal vertex potential z with

$$z_v = 0 \quad (v \notin S), \qquad 0 \leq z_v \leq 1, \qquad z_x + z_y \leq 1 \quad (xy \in E(H)), \qquad \sum_{v \in V(H)} z_v = w.$$

Its nonnegative edge slacks

$$s_{xy} = 1 - z_x - z_y$$

satisfy

$$\sum_{xy \in E(H)} s_{xy} = e(H) - Dw.$$

For any optimal fractional cover and any optimal vertex potential, every edge of positive cover weight satisfies $z_x + z_y = 1$.

Setup 10.13 (Half-integral optimal covers from Chapter 1)**.** The earlier fractional-cover result gives an optimal cover whose positive support consists of weight-one matching edges and pairwise vertex-disjoint odd cycles with every cycle edge of weight one-half. The matching edges and the cycles are mutually vertex-disjoint.

Setup 10.14 (Certificate costs and slack from Chapter 1)**.** We retain the generic certificate quantities $c_{0,0}(J)$, $c^*_\vartheta(J)$, and $\Sigma(J)$ from Chapter 1. Their identities used here are

$$\mathrm{ps}(J) = e(J) - \nu(J) - \sum_{\vartheta \in \mathbb{Z}_{\geq 1}} c^*_\vartheta(J), \tag{10.1}$$

$$\mathrm{gap}(J) = \Sigma(J) - e(J) = c_{0,0}(J) - \nu(J) - \mathrm{ps}(J). \tag{10.2}$$

The earlier framework gives $\mathrm{ps}(J) \in \mathbb{Z}_{\geq 0}$.

We also retain the sharp slack $\psi_{\mathrm{sharp}}(J)$. Its profile uses the generic costs except for even–even restrictions at the even integer thresholds $6, 8, \ldots, 26$.

[math/blocker] [GAP: The full definitions of $c_{0,0}$, c^*_ϑ, Σ, and the sharp cost profile were not supplied. Statements involving these quantities are conditional on the Chapter 1 framework and the earlier properties recorded here; these identities do not replace the missing definitions.]

Setup 10.15 (Maximum-degree deletion from the earlier development)**.** Assume that a finite simple graph J satisfies either

$$\psi_{\mathrm{sharp}}(J) = 0,$$

or

$$\mathrm{ps}(J) = 0 \quad \text{and every induced subgraph of } J \text{ has generic gap at most } 1.$$

The earlier deletion result states that deleting any maximum-degree vertex while the current maximum degree is at least 3 lowers the matching number by one and preserves zero slack for the applicable profile. This can be repeated until the remaining graph has maximum degree at most 2. In the generic case, the same conclusion holds without the hereditary gap hypothesis whenever $\Delta(J) \leq 5$.

Setup 10.16 (Independent-set certificate bound from the earlier development)**.** Let Q be a finite simple regular factor-critical graph of odd order $n \in \mathbb{Z}_{\geq 3}$ and positive degree. Let S be a maximum independent set, put $s = |S|$, and set $G = Q - S$. The earlier independent top-degree bound gives

$$\operatorname{gap}(Q) \leq c_{0,0}(Q) + 1 - s - \nu(G) - \operatorname{ps}(G).$$

Setup 10.17 (Robust containment and independent neighborhoods)**.** Let $k, r \in \mathbb{Z}_{\geq 0}$, and let J be a finite simple graph with at least one edge and no isolated vertices. Assume that

$$|V(H)| = |V(J)| + k + r$$

and that $H - X$ contains J for every k-element subset $X \subseteq V(H)$. The earlier independent-neighborhood obstruction states that every independent set $I \subseteq V(H)$ of size $r + 1$ satisfies

$$|N_H(I)| \geq k + 1.$$

Remark 10.18 (Draft status). [author/blocker] No author supplied; the author, affiliation, and email fields are placeholders.

[metadata/blocker] The Mathematics Subject Classification was not supplied.

[cite/blocker] The earlier-series statement numbers were not supplied. Their required forms are recorded above without invented external reference numbers.

[compile/blocker] This source has not been compiled. Zero LaTeX errors and resolved citations and cross-references remain unverified.

[instruction/conflict] The complete supplied mathematics and prerequisite statements are retained. The requested page target remains unverified.

Remark 10.19 (Automated assistance). The main result of this paper was obtained with the assistance of the Ansatz system, an automated proof-search and verification system. [ack/blocker] No verification statement was supplied. Because automated systems have limitations, it is possible that we have missed related references in the literature, and we welcome comments from experts.

10.2. **Degree splitting.** In this section, we prove the degree-splitting results, including the covering-matching criterion in Theorem 10.21. The even-capacity construction is motivated by [DJK+21, Lemma 2.1 and Theorem 2.5]; its regular completion, circuit orientation, and decomposition are proved below.

Proposition 10.20 (Even capacities and an odd-degree condition)**.** *Let $A, B \in \mathbb{Z}_{\geq 0}$, and put $D = A + B$. Let Q be a finite simple graph with $\Delta(Q) \leq D$, and set*

$$S_D(Q) = \{v \in V(Q) \mid d_Q(v) = D\}.$$

Assume either of the following:

(i) A and B are both even;
(ii) D is odd, and $Q[S_D(Q)]$ contains no two vertex-disjoint odd cycles.

Then $E(Q)$ has a red-blue coloring in which every red degree is at most A and every blue degree is at most B.

Proof. We first prove the even-capacity case by embedding Q in a simple regular graph and decomposing an associated bipartite graph. We then use the covering criterion in Setup 10.6 to remove the parity discrepancy when D is odd.

Step 1. We construct a finite simple D-regular graph containing Q.

Assume that A and B are even. If $D = 0$, then Q has no edges, and its empty coloring has both required bounds. Assume $D > 0$, and put $r = D/2 \in \mathbb{Z}_{\geq 1}$.

Start with $T_0 = Q$. Given a finite simple graph T_h with maximum degree at most D, form T_{h+1} from two disjoint copies of T_h. For each vertex z of T_h whose degree is less than D, add the edge joining its two copies. These edges join distinct copies, so they are neither loops nor repetitions of existing edges. A copy of a vertex of degree $d < D$ has new degree $d + 1$; a copy of a vertex of degree D still has degree D.

Induction on $h \in \mathbb{Z}_{\geq 0}$ therefore gives

$$d_{T_h}(\widetilde{z}) = \min\{D, d_Q(z) + h\}$$

for every descendant $\widetilde{z}$ of an original vertex z. Thus T_D is finite, simple, and D-regular. Choosing the first copy at each stage embeds Q as a subgraph of T_D. Write $T = T_D$. If Q has no vertices, then T also has no vertices and the conclusion follows. We henceforth work with the nonempty components of T.

Step 2. We construct a closed trail using every edge of each component of T.

In a finite connected graph whose degrees are all even, start at a vertex incident with an edge and traverse unused incident edges until no continuation is available, never repeating an edge. The trail terminates because the graph is finite. It cannot terminate at a vertex different from its initial vertex. At such a terminal vertex, the used incident edges occur in arrival-departure pairs except for the final arrival. Their number is odd, so the even total degree leaves an unused incident edge, contradicting termination. The trail is therefore closed.

If unused edges remain in the component, some vertex of the closed trail is incident with an unused edge. To see this, either an unused edge already has an endpoint on the trail, or a path from an endpoint of an unused edge to the trail has a first edge entering the trail. That entering edge was not traversed, since its endpoint outside the trail was not visited.

The unused-edge graph still has even degree at every vertex, since a closed trail uses an even number of incident edges at each vertex. Starting at the chosen trail vertex gives another closed trail in the unused-edge graph. Splice it into the first trail at their common vertex. Each splice increases the number of used edges. Finiteness therefore gives a closed trail using every edge of the component exactly once. Apply this construction separately to every component of T.

Step 3. The circuit orientations give the required even-capacity split.

Orient each edge of T in the direction in which its circuit traverses it. Every vertex has r incoming and r outgoing edges, since arrivals and departures are paired and its degree is $2r$.

Form a bipartite graph U with an outgoing copy z_{out} and an incoming copy z_{in} of every vertex $z \in V(T)$. Replace each oriented edge $x \to y$ by

$$x_{\mathrm{out}} y_{\mathrm{in}}.$$

The graph U is finite and simple because T is simple and each edge is oriented once. Every vertex of U has degree r. By Setup 10.5, U has a proper edge coloring with r colors. Its r incident edges at any vertex have distinct colors, so every color occurs exactly once there.

Choose $A/2$ colors, which is possible since A is even and $0 \leq A \leq D = 2r$. Color an edge of T red if its corresponding edge of U has a chosen color, and blue otherwise. At every vertex z, exactly $A/2$ outgoing and $A/2$ incoming edges are red. No edge is both incoming and outgoing at the same vertex, because its endpoints are distinct. Hence the red degree is A, and the blue degree is $D - A = B$. Restricting to the embedded copy of Q can only decrease these degrees. This proves (i).

Step 4. A covering matching reduces (ii) to the even-capacity case.

Assume (ii). Then D is a positive odd integer. Setup 10.6 gives a matching M of Q covering every vertex of degree D. In $Q - M$, each such vertex loses one incident edge, while every other

vertex already had degree at most $D-1$. Thus

$$\Delta(Q-M) \le D-1.$$

Exactly one of A, B is odd, and that integer is at least one. If A is odd, apply (i) to $Q-M$ with the nonnegative even capacities $A-1, B$, whose sum is $D-1$. Restore every edge of M in red. A matching adds at most one red incident edge at each vertex, so the resulting bounds are A, B.

If B is odd, apply (i) with capacities $A, B-1$ and restore M in blue. This gives the same required bounds and exhausts (ii). □

Theorem 10.21 (Degree splitting with a covering matching)**.** *Let $D \in \mathbb{Z}_{\ge 1}$, and let G be a finite simple graph with $\Delta(G) \le D$. Put*

$$S_D = \{v \in V(G) \mid d_G(v) = D\},$$

and let δ be the minimum number of vertices of S_D left uncovered by a matching of G.

(i) The graph G has a matching covering S_D if and only if, for every $A, B \in \mathbb{Z}_{\ge 0}$ with $A+B = D$, its edges have a red-blue coloring with red maximum degree at most A and blue maximum degree at most B.

(ii) There is an edge set P with $|P| \le \delta$ such that $G-P$ admits all the splittings in (i). More precisely, for every such A, B and every $u, v \in \mathbb{Z}_{\ge 0}$ with $\delta \le u+v$, the edge set of G has a partition into spanning subgraphs Q_R, Q_B, P_R, P_B satisfying

$$\Delta(Q_R) \le A, \qquad \Delta(Q_B) \le B, \qquad |E(P_R)| \le u, \qquad |E(P_B)| \le v.$$

(iii) Let $s, t \in \mathbb{Z}_{\ge 1}$, and let

$$n_1 \ge \cdots \ge n_s \ge 1, \qquad m_1 \ge \cdots \ge m_t \ge 1$$

be integer lists, with corresponding vertex-disjoint star forests F_1, F_2. Assume $G \longrightarrow (F_1, F_2)$. If

$$i \in \{1, \dots, s\}, \qquad j \in \{1, \dots, t\}, \qquad D = n_i + m_j - 2 \ge 1,$$

then

$$\delta \ge i + j - 1.$$

In particular, put

$$a = n_1, \qquad b = m_1, \qquad r_1 = |\{i \mid n_i = a\}|, \qquad r_2 = |\{j \mid m_j = b\}|.$$

Every such arrowing graph with $\Delta(G) \le a+b-2$ and $a+b-2 \ge 1$ has minimum uncovered degree-$(a+b-2)$ vertex count at least $r_1 + r_2 - 1$.

Proof. A covering matching permits a regular completion with a perfect matching. We use Setup 10.4 in that completion, then remove one edge at each uncovered required vertex for the general case.

Step 1. A matching covering S_D gives every degree split.

Assume that M is a matching of G covering S_D. Take two disjoint copies of G. For every original vertex x, let x' denote its corresponding vertex in the other copy, and add exactly

$$D - d_G(x)$$

parallel edge instances between x and x'. The resulting finite loopless multigraph H is D-regular.

Use M in both copies. For each original vertex x not covered by M, choose one added edge xx'. Such an edge exists because an uncovered vertex is outside S_D, so $D - d_G(x) \ge 1$. These chosen edges have disjoint endpoint pairs and are vertex-disjoint from both copies of M. Their union with those copies is a perfect matching N of H.

We show that H has a spanning subgraph of degree A for every integer A with $0 \le A \le D$. All interpolation applications retain individual parallel edge instances.

Suppose first that D is even. If A is even, apply Setup 10.4 with

$$E_0 = \varnothing, \qquad E_1 = E(H), \qquad r = D/2, \qquad k = A/2.$$

This gives a subgraph of degree A. If A is odd, then $1 \le A \le D-1$ and $D \ge 2$. Apply the same setup with

$$E_0 = E(N), \qquad E_1 = E(H) \setminus E(N), \qquad r = (D-2)/2, \qquad k = (A-1)/2.$$

The two edge sets have degrees 1 and $D-1$, so their degree difference is $2r = D-2$. Since $0 \le k \le r$, interpolation gives degree $1 + 2k = A$.

Suppose now that D is odd. The graph $H-N$ is $(D-1)$-regular. If A is even, then $0 \le A \le D-1$, and interpolation on $H - N$, with empty lower edge set, full upper edge set,

$$r = (D-1)/2, \qquad k = A/2,$$

gives degree A. If A is odd, then $D-A$ is even and belongs to $[0, D-1]$. The preceding application gives a subgraph of $H - N$ of degree $D - A$; its complement in H has degree A. This also covers $D = 1$, since the required even factor may be empty and $r = 0$ is allowed.

Color the resulting A-factor red and its complement in H blue. The color degrees are A and $D - A = B$. Restrict to the original edge instances in one copy of G. Removing added edges and the other copy cannot increase either color degree. This proves the forward implication in (i).

For the reverse implication, use the split with $A = 1$ and $B = D - 1$. The red edges form a matching. A vertex of original degree D has blue degree at most $D - 1$, so it has a red incident edge. Thus the red matching covers S_D.

Step 2. At most δ edge deletions leave a graph with a covering matching.

Choose a matching M attaining δ, which exists because G is finite. Let $U \subseteq S_D$ be its uncovered required vertices, so $|U| = \delta$. Every $x \in U$ has degree $D \ge 1$. Choose one incident edge for each such x, and let P be the set of chosen edges, counting repeated choices only once. Then

$$|P| \le \delta.$$

No edge of P belongs to M, because it has an endpoint uncovered by M. Hence M remains a matching in $G - P$.

Every vertex of U loses an incident edge and has degree at most $D - 1$ in $G - P$. Every vertex still of degree D was a degree-D vertex of G outside U, and is covered by M. Apply (i) to $G - P$, with the same parameter D.

For fixed A, B, denote the two resulting spanning subgraphs by Q_R, Q_B. If $\delta \le u + v$, then $|P| \le u + v$. Choose $P_R \subseteq P$ of size $\min\{u, |P|\}$, and put $P_B = P \setminus P_R$. Then

$$|P_R| \le u, \qquad |P_B| = \max\{0, |P| - u\} \le v.$$

Regarded as spanning subgraphs, these edge sets give the partition in (ii). This construction includes $U = \varnothing$ and $P = \varnothing$.

Step 3. The exceptional-edge certificate gives the arrowing obstruction.

Suppose, contrary to (iii), that $\delta \le i + j - 2$. Apply (ii) with

$$A = n_i - 1, \qquad B = m_j - 1, \qquad u = i - 1, \qquad v = j - 1.$$

These are nonnegative integers and $A + B = D$. Each exceptional graph has matching number at most its number of edges, so

$$\nu(P_R) \le i - 1, \qquad \nu(P_B) \le j - 1.$$

Setup 10.7 says that coloring $Q_R \cup P_R$ red and $Q_B \cup P_B$ blue avoids both full ordered target forests. This contradicts $G \longrightarrow (F_1, F_2)$. Therefore $\delta \ge i + j - 1$.

For the final assertion, take $i = r_1$ and $j = r_2$. The lists are nonincreasing, so $n_{r_1} = a$ and $m_{r_2} = b$. Applying the result with $D = a + b - 2$ gives $\delta \ge r_1 + r_2 - 1$. □

10.3. **Uncovered required vertices.** The goal of this section is to express the deficiency in Theorem 10.21 through required odd components and then count the matching edges forced by those components.

Theorem 10.22 (Required-vertex deficiency formula)**.** *Let G be a finite simple graph and $S \subseteq V(G)$. Let $u_S(G)$ be the minimum number of vertices of S left uncovered by a matching. For $X \subseteq V(G)$, let $q_S(G - X)$ count the odd-order connected components of $G - X$ whose vertex sets are contained wholly in S. Then*

$$u_S(G) = \max_{X \subseteq V(G)} \big(q_S(G - X) - |X|\big).$$

Equivalently, if

$$\delta_S(G) = \max_{X \subseteq V(G)} \big(q_S(G - X) - |X|\big),$$

then $u_S(G) = \delta_S(G)$. Here $u_S(G)$ counts uncovered required vertices, rather than matching edges.

Proof. All extrema exist because G is finite. Put

$$\delta = \max_{X \subseteq V(G)} \big(q_S(G - X) - |X|\big).$$

Taking $X = \varnothing$ shows that $\delta \geq 0$.

Fix a matching M and a vertex set X. Among the $q_S(G - X)$ required odd components, at most $|X|$ have an M-edge joining them to X, since those matching edges have distinct endpoints in X. Every remaining such component has no matching edge leaving it: all graph edges leaving a component of $G - X$ go to X. Its odd-order vertex set cannot be covered by internal matching edges, which cover vertices in pairs. Hence it contains an uncovered vertex of S. Distinct components give distinct uncovered vertices. Thus M leaves at least $q_S(G - X) - |X|$ required vertices uncovered. Maximizing over X and minimizing over M gives

$$u_S(G) \geq \delta.$$

For the converse, add a set Z of δ new vertices. Join each new vertex to every original vertex of S and to no vertex outside S, adding no other edges. Let H be the resulting finite simple graph, with the original set S still prescribed.

We check Setup 10.8. Let $Y \subseteq V(H)$, and write

$$X = Y \cap V(G), \qquad W = Y \cap Z.$$

If some $z \in Z \setminus W$ remains, every remaining vertex of S is adjacent to z. Every component containing a vertex of S therefore also contains a vertex outside S. No component is contained wholly in S, so the required odd-component count is zero and is at most $|Y|$.

If all new vertices have been deleted, then $W = Z$ and $H - Y = G - X$. The definition of δ gives

$$q_S(H - Y) = q_S(G - X) \leq |X| + \delta = |Y|.$$

When $\delta = 0$, only this second case occurs.

Setup 10.8 gives a matching of H covering S. Remove all its edges incident with Z. At most δ edges are removed because they use distinct new vertices. Each removed edge has at most one endpoint in S, so at most δ required vertices become uncovered. The remaining edges form a matching of G. Therefore $u_S(G) \leq \delta$, proving the formula. □

Proposition 10.23 (Required odd components and prescribed exposed vertices)**.** *Let G be a finite simple graph and $S \subseteq V(G)$. Let $\delta = u_S(G)$. Choose X of maximum cardinality among the sets satisfying*

$$q_S(G - X) - |X| = \delta.$$

Such sets exist. Every odd component C of $G - X$ with $V(C) \subseteq S$ is factor-critical.

Prescribe an arbitrary vertex $z_C \in V(C)$ in every such required odd component. There is a matching W of cardinality $\nu(G)$ leaving exactly δ vertices of S uncovered and satisfying:

(i) Exactly $|X|$ required odd components have a W-edge to X; each has exactly one such edge, and these edges cover every vertex of X.

(ii) Every remaining required odd component has exactly one W-uncovered vertex, namely its prescribed z_C.

(iii) There are exactly δ remaining required odd components, and every other vertex of S is covered by W.

Proof. By Theorem 10.22,

$$\delta = \max_{Y \subseteq V(G)} \big(q_S(G-Y) - |Y|\big).$$

The family of attaining sets is finite and nonempty, so X exists.

Fix an odd component C of $G-X$ contained wholly in S. Apply Setup 10.9 to C, and choose an ordinary attaining set $Y \subseteq V(C)$ of maximum cardinality. Its identity is

$$o(C-Y) - |Y| = |V(C)| - 2\nu(C) \geq 1,$$

where the inequality follows from the odd order of C.

Deleting Y in addition to X leaves all other components of $G-X$ unchanged and replaces C by the components of $C-Y$. Every new component is still contained wholly in S, and no edges are introduced between components. Hence

$$q_S(G-(X\cup Y)) - |X\cup Y| = \delta - 1 + o(C-Y) - |Y| \geq \delta.$$

Theorem 10.22 forces equality. If $Y \neq \varnothing$, then $X \cup Y$ is a larger attaining set, contrary to the choice of X. Thus $Y = \varnothing$. The structural conclusion of Setup 10.9 says that every component of $C-Y$ is odd and factor-critical. Since C is connected and Y is empty, C is factor-critical.

By Setup 10.10, a matching minimizing uncovered required vertices can be augmented to maximum cardinality without losing coverage of any previously covered vertex. Let W_0 be such a maximum matching attaining δ. Put

$$p = q_S(G-X) = |X| + \delta,$$

and let m be the number of required odd components with a W_0-edge to X. Distinct reached components require distinct matching endpoints in X, so $m \leq |X|$.

A required odd component not reached by W_0 has no matching edge leaving it, because every edge leaving that component goes to X. Its internal matching leaves at least one required vertex uncovered. Consequently,

$$\delta \geq p - m \geq p - |X| = \delta.$$

Every inequality is equality. Thus $m = |X|$, every unreached required odd component has exactly one uncovered vertex, and no vertex of S outside those components is uncovered.

Each reached component has at least one W_0-edge to X. There are $|X|$ reached components and only $|X|$ matching endpoints available in X, so each has exactly one such edge, and these edges cover all of X. Every reached component is fully covered, with one vertex matched to X and all other vertices matched internally. Every unreached required component C contains exactly $(|V(C)|-1)/2$ internal matching edges.

For each unreached required odd component C, replace all its internal W_0-edges by a perfect matching of $C-z_C$. Factor-criticality gives this matching. The number of inserted edges is $(|V(C)|-1)/2$, equal to the number removed. No edge of W_0 joins C to its complement, so the replacement is disjoint from the unchanged matching edges. Replacements in different components are also disjoint.

The resulting matching W has cardinality $\nu(G)$, still leaves exactly δ required vertices uncovered, and preserves all cross edges to X. In each unreached required component its sole uncovered vertex is z_C. The number of these components is $p-m=\delta$, and every other vertex of S remains covered. □

Theorem 10.24 (Counting components forced by uncovered maximum-degree vertices)**.** *Let $D \in \mathbb{Z}_{\geq 3}$, and let H be a finite simple graph with $\Delta(H) \leq D$. Put*

$$S_D = \{x \in V(H) \mid d_H(x) = D\},$$

and assume that the minimum number δ of vertices of S_D left uncovered by a matching satisfies $\delta \geq 1$.

Let $S \subseteq V(H)$ satisfy

$$\nu(H) = |S| + \sum_B \left\lfloor \frac{|V(B)|}{2} \right\rfloor,$$

where B ranges over the connected components of $H - S$. Assume that every such component is odd and factor-critical. Put

$$w = \left\lfloor \frac{D+1}{2} \right\rfloor.$$

If D is odd, set

$$k = \left\lceil \frac{\delta}{D-1} \right\rceil.$$

If D is even, let $\kappa \in \mathbb{Z}_{\geq 0}$ count the whole connected components of H that are odd, D-regular, and factor-critical, and set

$$k = \max\left\{0, \left\lceil \frac{2(\delta-\kappa)}{D-2} \right\rceil \right\}.$$

Then

$$|S| \geq k, \qquad \nu(H) \geq k + w(\delta + k).$$

For every component C of $H - S$, put $h = \nu(H) - \nu(C)$. Then

$$h \geq k + w(\delta + k - 1).$$

If C has a vertex of ambient H-degree less than D, or if $|V(C)| \leq D$, then

$$h \geq k + w(\delta + k).$$

In particular, if D is odd, then

$$h \geq \frac{D+3}{2}$$

outside every component C of $H - S$.

No hypotheses concerning Ramsey arrowing, an edge budget, potential surplus, connectedness, or separation of exposed neighborhoods are imposed.

Proof. We describe maximum matchings relative to the specified set S, which need not be the separator in Proposition 10.23. An alternating exploration then identifies required components whose boundary degrees can be counted.

Step 1. Maximum matchings correspond to incidence matchings saturating S.

Each component B of $H - S$ is odd and factor-critical, so

$$\nu(B) = \frac{|V(B)| - 1}{2}.$$

The attaining equality becomes

$$\nu(H) = |S| + \sum_B \nu(B). \tag{10.3}$$

Any matching has at most $|S|$ edges incident with S, and at most $\nu(B)$ internal edges in each B. Equality in (10.3) forces all these bounds to be equalities for a maximum matching. Hence every vertex of S is covered, no matching edge has both endpoints in S, and every component B contains exactly $\nu(B)$ internal matching edges. Since $|V(B)| = 2\nu(B) + 1$, at most one matching edge joins B to S.

Consider the bipartite incidence graph whose classes are S and the components of $H - S$, with adjacency when an actual edge of H joins the vertex to the component. The crossing edges of a maximum matching give an incidence matching saturating S.

Conversely, every such incidence matching lifts to a maximum matching of H. Choose one actual crossing edge for each matched pair. In each matched component B, take a perfect matching after deleting its crossing endpoint. In each unmatched component, prescribe any exposed vertex and take a perfect matching after deleting it. Factor-criticality permits all these choices independently, and the total matching size is (10.3).

Call a component of $H-S$ required if every one of its vertices belongs to S_D. By Proposition 10.23, there is a maximum matching W attaining the minimum uncovered required count δ. In the incidence matching induced by W, an unmatched component that is not required can be made to expose a vertex of degree less than D, without changing any matching edge outside that component. Minimality of δ therefore implies that the degree-D vertices exposed by W occur precisely in its unmatched required components. There are exactly δ such components; denote their collection by $\mathcal{U}$.

Step 2. Alternating exploration reaches only required components.

Starting from $\mathcal{U}$, traverse every nonmatching incidence edge from a reached component to S, and traverse the matching edge from each reached vertex of S to its partner component. Let $\mathcal{Q}$ be the reached component class, and let $A \subseteq S$ be the reached vertex set.

Every component in $\mathcal{Q}$ is required. To prove this, take a simple alternating path from an unmatched component of $\mathcal{U}$ to a reached component B. The path has even length and can be toggled to leave B unmatched while matching its initial component. If B were not required, lift the new incidence matching while exposing a vertex of degree less than D in B, and retain the exposed choices in all other unmatched components. This would give a maximum matching with $\delta - 1$ uncovered vertices of S_D, a contradiction.

Every neighbor in S of a reached component belongs to A. Nonmatching incidences are traversed by definition, while the matching partner of a reached matched component has already been reached on the path to that component. Every vertex of A has its partner component in $\mathcal{Q}$. The only unmatched components in $\mathcal{Q}$ are the initial components $\mathcal{U}$, because every subsequently reached component is a matching partner.

Writing $a = |A|$, we obtain

$$|\mathcal{Q}| = a + \delta, \qquad \text{every edge leaving a component in } \mathcal{Q} \text{ ends in } A. \tag{10.4}$$

Step 3. Boundary parity forces many large components.

For $B \in \mathcal{Q}$, put $n_B = |V(B)|$, and let β_B count its edges to A. Every vertex of B has ambient degree D. Therefore

$$\beta_B = Dn_B - 2e(B), \qquad \beta_B \equiv D \pmod 2, \qquad \sum_{B \in \mathcal{Q}} \beta_B \le Da. \tag{10.5}$$

Here every n_B is odd, and the final inequality counts distinct edges incident with A.

If $n_B \le D$, simplicity gives

$$\beta_B \ge n_B(D - n_B + 1) \ge D, \tag{10.6}$$

because

$$n_B(D - n_B + 1) - D = (n_B - 1)(D - n_B) \ge 0.$$

Call B large if $n_B > D$, and let ℓ count the large members of $\mathcal{Q}$. Every large member has matching number at least w: for odd D, its odd order is at least $D+2$; for even D, its odd order is at least $D+1$.

Suppose D is odd. Every β_B is a positive odd integer. Equations (10.4) and (10.5) give

$$a + \delta \le Da,$$

so

$$a \ge \left\lceil \frac{\delta}{D-1} \right\rceil = k.$$

Using (10.6) on the $a + \delta - \ell$ small components and $\beta_B \ge 1$ on each large component gives

$$D(a + \delta - \ell) + \ell \le Da.$$

Thus

$$\ell \ge \left\lceil \frac{D\delta}{D-1} \right\rceil = \delta + k. \tag{10.7}$$

Suppose D is even, so $D \ge 4$. Let z count the members of $\mathcal{Q}$ with $\beta_B = 0$. Such a component has no edge to A, and (10.4) excludes every other outside edge. It is therefore a whole connected component of H, odd, D-regular, and factor-critical. Hence $z \le \kappa$. Every other β_B is a positive even integer and is at least two. Consequently,

$$2(a + \delta - z) \le Da,$$

and

$$(D-2)a \ge 2(\delta - z) \ge 2(\delta - \kappa). \tag{10.8}$$

Together with $a \ge 0$ and integrality, this gives $a \ge k$.

By (10.6), no small component has zero boundary. Using that inequality for the small components and the lower bound two for the large components with positive boundary gives

$$D(a + \delta - \ell) + 2(\ell - z) \le Da.$$

Thus

$$(D-2)\ell \ge D\delta - 2z \ge D\delta - 2\kappa. \tag{10.9}$$

Dropping all nonnegative large boundaries in (10.5) also gives $\ell \ge \delta$. Finally,

$$\left\lceil \frac{D\delta - 2\kappa}{D-2} \right\rceil = \delta + \left\lceil \frac{2(\delta - \kappa)}{D-2} \right\rceil.$$

Combining this identity with (10.9) and $\ell \ge \delta$ gives $\ell \ge \delta + k$.

Step 4. The large components give the matching lower bounds.

In both parity cases,

$$|S| \ge a \ge k, \qquad \ell \ge \delta + k.$$

The separator contribution and the matching contributions of the large components are separate terms in (10.3). Hence

$$\nu(H) \ge a + w\ell \ge k + w(\delta + k). \tag{10.10}$$

For a specified component C of $H - S$, at most one of the ℓ large members of $\mathcal{Q}$ is C. Therefore

$$\begin{aligned} h &= |S| + \sum_{B \ne C} \nu(B) \\ &\ge a + w(\ell - 1) \\ &\ge k + w(\delta + k - 1). \end{aligned}$$

The use of $\ell - 1$ is valid because $\ell \ge \delta + k \ge 1$.

If C contains a vertex of degree less than D, it is not required and hence is not in $\mathcal{Q}$. If $|V(C)| \le D$, it is not large. In either case, none of the ℓ large contributions is removed, so

$$h \ge a + w\ell \ge k + w(\delta + k).$$

For odd D, the assumptions $\delta \ge 1$ give $k \ge 1$ and $\delta + k - 1 \ge 1$. Since $w = (D+1)/2$,

$$h \ge 1 + \frac{D+1}{2} = \frac{D+3}{2}.$$

Only the stated graph and matching hypotheses were used. □

10.4. **Potential and covering bounds.** In this section, we relate incidence covers of maximum-degree vertices to Φ, and then derive structural consequences of equality in the covering and certificate bounds.

Proposition 10.25 (Fractional covering potential and equality)**.** *Let $D \in \mathbb{Z}_{\ge 2}$, and let H be a finite simple graph with $\Delta(H) \le D$. Put $S = S_D(H)$, and let w be the minimum total weight of a fractional matching cover of S. Then*

$$\Phi(H) \ge (D-1)w.$$

Let $p, r \in \mathbb{Z}_{\ge 1}$, assume

$$H \longrightarrow (pK_2, rK_{1,D}),$$

and put $k = p + r - 1$. Every edge set C covering S by incidence satisfies

$$\nu((V(H), C)) \ge k.$$

Consequently,

$$H \longrightarrow (kK_2, K_{1,D}), \qquad w \ge k.$$

If also $\Phi(H) = (D-1)k$, then $w = k$, and there is a k-edge matching N_0 covering S that is an optimal fractional cover. For any optimal vertex potential z from Setup 10.12, the following hold:

- *(i) Every $xy \in N_0$ satisfies $z_x + z_y = 1$.*
- *(ii) Every maximum matching W covers every vertex v with $z_v > 0$.*
- *(iii) Every maximum matching contains every edge xy with positive slack $1 - z_x - z_y$. In particular, the positive-slack edges form a matching.*
- *(iv) If a vertex v is uncovered by some maximum matching, then $z_v = 0$, and every neighbor of v has potential one.*

Proof. We first compare the total edge slack with the slack on a maximum matching. The Ramsey hypothesis then gives a lower bound for every incidence cover, which we apply to a half-integral optimum.

Step 1. The potential inequality holds without a Ramsey hypothesis.

Take an optimal potential z from Setup 10.12, and write

$$s_{xy} = 1 - z_x - z_y.$$

For any maximum matching W, disjointness of its endpoints gives

$$\begin{aligned} e(H) - Dw &\ge \sum_{xy \in W} s_{xy} \\ &= \nu(H) - \sum_{v \in V(W)} z_v \\ &\ge \nu(H) - w. \end{aligned}$$

Rearranging yields $\Phi(H) \ge (D-1)w$.

Step 2. Every incidence cover has matching number at least k.

Assume the Ramsey hypothesis, and let $C \subseteq E(H)$ cover S. Choose an inclusion-minimal subset $C_0 \subseteq C$ that still covers S. This exists because the edge set is finite.

For each $xy \in C_0$, at least one endpoint belongs to S and has degree one in $(V(H), C_0)$. Otherwise deleting xy would leave every required endpoint covered, contradicting minimality. Thus every edge of C_0 has an endpoint of degree one.

Every nontrivial component of C_0 is a star. A component containing a vertex v of degree at least two has all neighbors of v of degree one, since an edge cannot have two non-leaf endpoints. Those neighbors have no further edges, so they and v form the whole component. A nontrivial component with no vertex of degree at least two is a single edge.

Let $c \in \mathbb{Z}_{\geq 0}$ count these nontrivial components. Then

$$\nu(C_0) = c \leq \nu(C).$$

Suppose $\nu(C) < k$. Thus $c \leq p + r - 2$. Color $\min\{c, p-1\}$ star components of C_0 red, and color every other edge of H blue. The red graph is a disjoint union of at most $p-1$ stars, so its matching number is at most $p-1$.

The blue edges of C_0 form at most $r-1$ stars. If $c < p-1$, there are none. If $c \geq p-1$, their number is

$$c - (p-1) \leq r - 1.$$

A blue D-edge star would have a center of original degree exactly D, because $\Delta(H) \leq D$, and would use all original edges at that center. The center belongs to S, so an incident edge of C_0 exists and must be blue. Therefore r vertex-disjoint blue D-edge stars would give a matching of size r among the blue edges of C_0, which is impossible. This coloring avoids both original targets, a contradiction. Hence $\nu(C) \geq k$.

Now consider any coloring with no blue $K_{1,D}$. Every vertex of S has a red incident edge, since otherwise all its D incident edges would form a blue star. The red edges cover S, so their matching number is at least k. This proves

$$H \longrightarrow (kK_2, K_{1,D}).$$

Step 3. A half-integral optimum gives $w \geq k$.

By Setup 10.13, choose an optimal cover supported on weight-one matching edges and vertex-disjoint odd cycles of weight one-half. Its support covers S, since every required vertex has load one.

For an odd cycle

$$v_0 v_1 \cdots v_{2j} v_0, \qquad j \in \mathbb{Z}_{\geq 1},$$

replace the positive support by

$$v_0v_1, \quad v_1v_2, \quad v_3v_4, \quad v_5v_6, \quad \ldots, \quad v_{2j-1}v_{2j}.$$

This is a spanning edge cover of that cycle with one P_3 and $j-1$ single edges. Its matching number is j.

Perform this replacement on every odd cycle, and retain every weight-one matching edge. If $c_{\mathrm{odd}} \in \mathbb{Z}_{\geq 0}$ is the number of odd cycles, the resulting edge set covers S and has matching number

$$w - \frac{c_{\mathrm{odd}}}{2}.$$

The equality follows because the support components are mutually vertex-disjoint. Step 2 gives

$$w - \frac{c_{\mathrm{odd}}}{2} \geq k, \tag{10.11}$$

and therefore $w \geq k$.

Step 4. Equality forces an integral optimum and the stated rigidity.

Assume $\Phi(H) = (D-1)k$. Step 1 and $D-1>0$ give $w \le k$, so $w = k$. Equation (10.11) forces $c_{\mathrm{odd}} = 0$. The chosen optimum is therefore a matching N_0 with exactly k edges, covering S. It is still an optimal feasible fractional cover. By the positive-edge tightness statement in Setup 10.12,

$$z_x + z_y = 1 \qquad (xy \in N_0)$$

for any optimal potential z.

For every maximum matching W, the chain in Step 1 is also an equality, because

$$e(H) - Dw = \nu(H) - w.$$

Hence

$$\sum_{v \in V(W)} z_v = w, \qquad \sum_{e \in E(H)\setminus W} s_e = 0.$$

All potentials and slacks are nonnegative. The first equality says that every vertex of positive potential is covered by W. The second says that every positive-slack edge belongs to W. Since this holds for every maximum matching, the positive-slack edges form a matching.

Finally, let v be uncovered by a maximum matching W. Then $z_v = 0$. Every edge incident with v is absent from W, so it has zero slack. For every neighbor y,

$$1 - z_v - z_y = 0,$$

which gives $z_y = 1$. □

Proposition 10.26 (Well-coveredness and a star-forest degree count)**.** *Let $D \in \mathbb{Z}_{\ge 2}$, and let C be a finite simple graph with $\Delta(C) \le D$. Put $S_D = S_D(C)$. If*

$$\theta_D(C) = \alpha(C),$$

then C is well-covered and has a maximum independent set contained in S_D.

Assume next that C has odd order, is factor-critical, and is well-covered. Let I be any maximum independent set contained in S_D, put $p = |I|$, and assume $R = C - I$ is a star forest with isolated vertices permitted. Let $r \in \mathbb{Z}_{\ge 0}$ be the number of positive-edge star components. Choose a center in each, and let their edge sizes be $m_1, \dots, m_r \in \mathbb{Z}_{\ge 1}$. Let $z \in \mathbb{Z}_{\ge 0}$ count the isolated vertices of R, and put

$$t = \sum_{i=1}^{r} m_i, \qquad \beta = \sum_{v \in V(C)} (D - d_C(v)), \qquad \epsilon = \begin{cases} 1, & z > 0, \\ 0, & z = 0. \end{cases}$$

Then

$$1 \le m_i \le D-1, \qquad \beta \ge 0, \qquad \beta \equiv D \pmod 2,$$

and

$$D(p - z) = Dr + (D-2)t - \beta, \tag{10.12}$$

$$Dr - 2t \ge \beta, \tag{10.13}$$

$$Dr + (D-6)t + D\epsilon \le 3\beta. \tag{10.14}$$

If equality holds in (10.14), let B be the neighbors in I of the z isolated vertices of R. Then

$$|B| = z + \epsilon.$$

Every vertex of $I \setminus B$ meets exactly two positive star components, and within each its neighbors lie on only one of the center and leaf sides. Equivalently, under independent equiprobable choices of the center or all leaves in every positive star, together with all isolated vertices, each vertex of $I \setminus B$ has probability exactly one quarter of having no selected neighbor.

Proof. The covering threshold first forces all maximal independent sets to have the same size. For the second assertion, we combine an exact degree sum with a finite averaging argument.

Step 1. Equality of the covering threshold and independence number gives well-coveredness.

Let J be any inclusion-maximal independent set of C. The set of all edges incident with J covers S_D. A required vertex inside J has positive degree D; a required vertex outside J has a neighbor in J by maximality. In the selected edge graph, J is a vertex cover, so every matching has at most $|J|$ edges. Hence

$$\theta_D(C) \le |J| \le \alpha(C).$$

If $\theta_D(C) = \alpha(C)$, then $|J| = \alpha(C)$ for every maximal J, proving well-coveredness.

Choose an inclusion-maximal independent subset I of $C[S_D]$. Its incident edge set covers S_D: every unchosen vertex of S_D has an I-neighbor by maximality in the induced graph, and every chosen vertex has degree $D > 0$. This edge graph has matching number at most $|I|$. Therefore

$$\alpha(C) = \theta_D(C) \le |I| \le \alpha(C).$$

Thus I is maximum in C and contained in S_D. Finiteness gives all the choices, including when the graph is empty.

Step 2. The degrees give (10.12) and (10.13).

Assume the hypotheses of the second assertion. Since I is maximum, it is maximal, so every vertex of R has an I-neighbor. A positive star center of internal degree m_i therefore satisfies $m_i \le D - 1$.

Every vertex of I has degree D, and all its incident edges go to R. There are Dp edges between I and R. All deficit β occurs on R. Summing degrees there gives

$$D(z + t + r) - \beta = Dp + 2t,$$

which rearranges to (10.12).

All leaves of the chosen positive stars, together with the isolated vertices of R, form an independent set in C. Thus $z + t \le p$. Multiplying $p - z \ge t$ by D and using (10.12) gives (10.13).

The degree cap gives $\beta \ge 0$. Since C has odd order,

$$\beta = D|V(C)| - 2e(C) \equiv D \pmod 2.$$

Step 3. Factor-criticality controls the neighbors of the isolated vertices.

Let Z be the isolated vertices of R, and put $B = N_C(Z) \subseteq I$. If $z = 0$, then $B = \varnothing$ and $|B| = z + \epsilon = 0$.

If $z > 0$, maximality of I gives an I-neighbor for every vertex of Z, so $B \neq \varnothing$. Choose $w \in B$. Factor-criticality gives a perfect matching of $C - w$. Every vertex of Z remains and must be matched to a distinct vertex of $B \setminus \{w\}$, since it has no neighbor in R. Hence

$$|B| - 1 \ge z, \qquad |B| \ge z + 1 = z + \epsilon.$$

Step 4. Averaging maximal independent sets gives (10.14) and its equality conditions.

Independently in each positive star of R, choose its center with probability one-half or all its leaves with probability one-half. Include all vertices of Z, and call the resulting set A. It is a maximal independent set of R.

The set

$$A \cup \big(I \setminus N_C(A)\big)$$

is a maximal independent set of C. It is independent by construction. Every omitted vertex of R has a neighbor in A, and every omitted vertex of I has an A-neighbor by definition. Well-coveredness therefore gives

$$|I \setminus N_C(A)| = p - |A|.$$

Averaging over the finite 2^r choices yields

$$\mathbb{E}|I \setminus N_C(A)| = p - z - \frac{t + r}{2}. \tag{10.15}$$

Every vertex of B is always dominated by A. Let $v \in I \setminus B$. If v has both a center neighbor and a leaf neighbor in one star, then either choice in that star dominates v. Otherwise its neighbors in each star lie on only one of the two sides. Each side has at most $D-1$ vertices, but $d_C(v) = D$, and v has no isolated-vertex neighbor. Hence at least two positive star components meet its neighborhood. Avoiding the selected neighbors in each such component has probability one-half, independently. Therefore the probability that v is undominated is at most one quarter.

Summing these probabilities gives

$$\mathbb{E}|I \setminus N_C(A)| \le \frac{p - |B|}{4} \le \frac{p - z - \epsilon}{4}. \tag{10.16}$$

Comparison with (10.15) gives

$$3(p - z) + \epsilon \le 2(t + r).$$

Multiplying by D and substituting (10.12) yields

$$Dr + (D - 6)t + D\epsilon \le 3\beta.$$

If equality holds, the exact degree identity forces equality in

$$3(p - z) + \epsilon \le 2(t + r),$$

and hence in both inequalities in (10.16). The second forces $|B| = z+\epsilon$. The first forces every vertex of $I \setminus B$ to attain its individual probability bound one quarter. Such a vertex cannot meet both sides of any star. If it meets k positive components on single sides, its probability of being undominated is 2^{-k}. Thus $k = 2$, proving all equality conditions, including the vacuous case $I \setminus B = \varnothing$. □

Theorem 10.27 (An edge bound under zero potential slack)**.** *Let H be a finite simple graph, and write*

$$n = |V(H)|, \qquad \nu = \nu(H), \qquad \alpha = \alpha(H).$$

Let $D \in \mathbb{Z}_{\ge 3}$ satisfy $\Delta(H) \le D$. Use the generic slack and gap and the sharp slack of Setup 10.14. Assume either

$$\psi_{\mathrm{sharp}}(H) = 0,$$

or

$$\mathrm{ps}(H) = 0 \quad \text{and every induced subgraph of } H \text{ has generic gap at most } 1.$$

Then

$$e(H) \le (3 - D)n + (2D - 3)\nu + (D - 3)\alpha. \tag{10.17}$$

The same bound holds under generic $\mathrm{ps}(H) = 0$ without the hereditary hypothesis whenever $\Delta(H) \le 5$. In particular, every subcubic graph with $\mathrm{ps}(H) = 0$ satisfies

$$e(H) \le 3\nu(H),$$

with no bound on its matching number.

Assume $D \ge 4$ and equality in (10.17). Every process that repeatedly deletes an arbitrary maximum-degree vertex while the current maximum degree is at least 3 deletes an independent set T of exactly

$$t = 2\nu - n + \alpha$$

vertices, all of original and deletion-time degree D. Its terminal graph R has exactly

$$n - \nu - \alpha$$

triangle components and

$$2\alpha - n + \nu$$

isolated vertices, and $\alpha(R) = \alpha(H)$.

If also $t \le D - 2$, the original degree-D vertices do not contain an induced $2K_2$: no four of them have exactly two disjoint edges as their mutual edges.

Proof. Setup 10.15 permits repeated maximum-degree deletion, with each deletion lowering the matching number by one. We keep the edge cost of these deletions and calculate the remaining contribution component by component.

Let T be the deleted vertex set, put $t = |T|$, and let R be the terminal graph. Then

$$n(R) = n - t, \qquad \nu(R) = \nu - t, \qquad n(R) - \nu(R) = n - \nu.$$

Every deletion removes at most D edges, so

$$e(H) \le Dt + e(R) = Dn - \big(Dn(R) - e(R)\big). \tag{10.18}$$

Since R is induced, $\alpha(R) \le \alpha(H)$.

For a connected component C of R, define

$$F(C) = Dn(C) - e(C) - (2D-3)\big(n(C) - \nu(C)\big) + (D-3)\alpha(C).$$

Every component is a path or a cycle because $\Delta(R) \le 2$. Their matching and independence counts give the following values.

For a path on $2k$ vertices, $k \in \mathbb{Z}_{\ge 1}$,

$$e(C) = 2k - 1, \qquad \nu(C) = \alpha(C) = k,$$

and

$$F(C) = (D-2)k + 1 > 0.$$

For a path on $2k+1$ vertices, $k \in \mathbb{Z}_{\ge 0}$,

$$e(C) = 2k, \qquad \nu(C) = k, \qquad \alpha(C) = k + 1,$$

and

$$F(C) = (D-2)k.$$

Equality occurs exactly when $k = 0$, so the component is a singleton.

For an even cycle on $2k$ vertices, $k \in \mathbb{Z}_{\ge 2}$,

$$e(C) = 2k, \qquad \nu(C) = \alpha(C) = k,$$

and

$$F(C) = (D-2)k > 0.$$

For an odd cycle on $2k+1$ vertices, $k \in \mathbb{Z}_{\ge 1}$,

$$e(C) = 2k + 1, \qquad \nu(C) = \alpha(C) = k,$$

and

$$F(C) = (D-2)(k-1).$$

Equality occurs exactly when $k = 1$, so the component is a triangle.

Thus $F(C) \ge 0$, with equality only for triangles and singletons. Order, edge count, matching number, and independence number are additive over components. Summing gives

$$Dn(R) - e(R) \ge (2D-3)\big(n(R) - \nu(R)\big) - (D-3)\alpha(R).$$

Together with (10.18) and $D - 3 \ge 0$, this yields

$$\begin{aligned} e(H) &\le Dn - (2D-3)(n-\nu) + (D-3)\alpha(R) \\ &\le Dn - (2D-3)(n-\nu) + (D-3)\alpha(H) \\ &= (3-D)n + (2D-3)\nu + (D-3)\alpha. \end{aligned}$$

The unconditional generic conclusion for $\Delta(H) \le 5$ uses the corresponding clause of Setup 10.15. Taking $D = 3$ proves the subcubic assertion, including graphs of smaller actual maximum degree.

Assume now that $D \ge 4$ and equality holds. The deletion sequence was arbitrary, so all inequalities for that sequence are equalities. Equality in (10.18) makes every deleted degree exactly

D. Equality in the component sum makes every component of R a triangle or a singleton. Since $D-3>0$, equality in the independence comparison gives

$$\alpha(R)=\alpha(H).$$

No two vertices of T are adjacent in H. If an earlier deleted vertex were adjacent to a later one, the later vertex would have deletion-time degree at most its original degree minus one, and therefore at most $D-1$. This contradicts the degree-D equality. Thus T is independent, and every deleted vertex also had original degree D.

Let $q,b\in\mathbb{Z}_{\geq 0}$ count the triangles and singletons of R. The exact counts are

$$n=t+3q+b,\qquad \nu=t+q,\qquad \alpha=q+b.$$

Solving gives

$$t=2\nu-n+\alpha,\qquad q=n-\nu-\alpha,\qquad b=2\alpha-n+\nu.$$

This proves the equality structure for every permitted sequence.

Finally, assume $t\leq D-2$. Suppose that four original degree-D vertices v,w,x,y induce exactly the edges vw and xy. Delete v first; it is a maximum-degree vertex. Complete the permitted deletion process. Its deleted set has size t, is independent, and leaves maximum degree at most two.

The vertices x,y are nonadjacent to v. If x remained, it could lose at most $t-1$ incident edges in the remaining deletions. Its terminal degree would be at least

$$D-(t-1)\geq 3,$$

a contradiction. Hence x is deleted. The same degree count forces deletion of y. Since xy is an edge, this contradicts independence of the deleted set. □

10.5. **Regular factor-critical graphs of odd order.** The goal of this section is to use the independent-set certificate bound to control an attaining set after deleting a maximum independent set.

Proposition 10.28 (Independent-set deletion and the regular odd-order estimate)**.** *Use* ps, ν, *and* $\mathrm{gap}(Q)=\Sigma(Q)-e(Q)$ *from Setup 10.14. Let Q be a finite simple regular factor-critical graph of odd order $n\in\mathbb{Z}_{\geq 3}$ and positive degree. Let S be any maximum independent set of cardinality s, put $G=Q-S$, and set*

$$\eta=\nu(G)-\left(\frac{n+1}{2}-s\right).$$

Then $\eta\in\mathbb{Z}_{\geq 0}$,

$$\mathrm{ps}(G)+\eta\leq\mathrm{ps}(Q),$$

and every ordinary attaining set X of G of maximum cardinality satisfies

$$|X|\leq 1+2\eta.$$

In particular, $\mathrm{ps}(Q)=0$ *implies*

$$\mathrm{ps}(G)=0,\qquad \eta=0,\qquad |X|\leq 1.$$

Moreover, every finite simple 6-regular factor-critical graph Q on 17 vertices with $\alpha(Q)\leq 4$ satisfies

$$\mathrm{ps}(Q)\geq 1.$$

Proof. Factor-criticality gives $\nu(Q)=(n-1)/2$. By Setup 10.16,

$$\mathrm{gap}(Q)\leq c_{0,0}(Q)+1-s-\nu(G)-\mathrm{ps}(G).$$

Substituting (10.2) and rearranging yields

$$\begin{aligned}\mathrm{ps}(Q)&\geq s+\nu(G)+\mathrm{ps}(G)-\nu(Q)-1\\&=\eta+\mathrm{ps}(G).\end{aligned}\tag{10.19}$$

The uncovered vertices of a maximum matching of G form an independent set. Their number is $n - s - 2\nu(G)$, so

$$n - s - 2\nu(G) \leq \alpha(G) \leq s.$$

Since n is odd, rounding the resulting lower bound for the integer $\nu(G)$ gives

$$\nu(G) \geq \frac{n+1}{2} - s.$$

Thus $\eta \geq 0$.

Choose any ordinary attaining set X of G of maximum cardinality, and put $b = |X|$. By Setup 10.9, every component of $G - X$ is odd and factor-critical. If $p \in \mathbb{Z}_{\geq 0}$ is their number, the matching identity gives

$$p = b + |V(G)| - 2\nu(G) = b + s - 1 - 2\eta.$$

Choosing one vertex from each component gives an independent set of G, so $p \leq \alpha(G) \leq s$. Consequently,

$$b \leq 1 + 2\eta.$$

Nonnegativity of $\mathrm{ps}(G)$, together with (10.19), proves all conclusions when $\mathrm{ps}(Q) = 0$.

For the final assertion, suppose that Q is 6-regular and factor-critical on 17 vertices, with $\alpha(Q) \leq 4$ and $\mathrm{ps}(Q) = 0$. Put $s = \alpha(Q)$, and choose S, G, X as above. A greedy independent-set choice removes at most seven vertices at each step, namely the chosen vertex and its neighbors. After two choices, at most fourteen vertices have been removed, so a third choice is possible. Hence

$$3 \leq s \leq 4.$$

The first part gives

$$\eta = 0, \qquad \mathrm{ps}(G) = 0, \qquad b = |X| \leq 1.$$

Every vertex outside S has a neighbor in S, since S is maximum. Therefore $\Delta(G) \leq 5$. Each vertex loses at most s neighbors when S is deleted, so the minimum degree of G is at least $6 - s$. Also,

$$\alpha(G) \leq s, \qquad p = b + s - 1.$$

Case 1. Assume $s = 3$.

The graph G has fourteen vertices, minimum degree at least three, and maximum degree at most five.

If $b = 0$, then G has two odd components, each of order at least five. Their independence numbers sum to at most three, so at least one has independence number one and is complete. Its odd order and degree bounds force it to be K_5; the other component has nine vertices.

Every vertex of that K_5 has four neighbors in G and degree six in Q, so it has exactly two neighbors in S. This gives ten edges from the K_5 to S. Each of the other nine vertices has at least one neighbor in S. Thus at least nineteen edges meet S, whereas S is independent and Q is 6-regular, giving exactly $6|S| = 18$ such edges. This is a contradiction.

If $b = 1$, then $G - X$ has three odd components. Their minimum internal degree is at least two, so each has order at least three. Their independence numbers sum to at most three, forcing all three components to be complete. Their odd orders are at most five because they are subgraphs of a graph of maximum degree at most five. They total thirteen vertices, so their orders are $3, 5, 5$.

Let T be their vertex union, and put $U = S \cup X$. Then $|T| = 13$, $|U| = 4$, and

$$e(Q[T]) = 3 + 10 + 10 = 23.$$

The number of edges between T and U is

$$6 \cdot 13 - 2 \cdot 23 = 32.$$

The total degree at U is only $6 \cdot 4 = 24$, a contradiction.

Case 2. Assume $s = 4$.

The graph G has thirteen vertices, minimum degree at least two, and maximum degree at most five.

If $b = 0$, then G has three odd components, each of order at least three. Their orders are either $3, 3, 7$ or $3, 5, 5$.

In the $3, 3, 7$ case, the two components of order three are triangles, since their minimum degree is at least two. Each of their six vertices has four neighbors in S. They account for all $6|S| = 24$ edges incident with S. No edge from S to the component of order seven remains, contradicting maximality of S.

In the $3, 5, 5$ case, the triangle contributes twelve edges to S. Since $\alpha(G) \leq 4$ and there are three components, at least one component of order five has independence number one and is K_5. It contributes ten edges to S. Each of the remaining five vertices has an S-neighbor. Thus the number of edges incident with S is at least

$$12 + 10 + 5 = 27 > 24,$$

again a contradiction.

If $b = 1$, then $G - X$ has four odd components. Each has minimum internal degree at least one, and therefore odd order at least three. Their orders sum to twelve, so each has order three. They are factor-critical by Setup 10.9. A factor-critical graph on three vertices is a triangle, because deleting any vertex must leave an edge.

Thus $T = V(G) \setminus X$ induces four disjoint triangles. Every vertex of T has four neighbors outside T in Q, giving 48 edges from T to $U = S \cup X$. But U has five vertices and total degree 30, a contradiction.

All cases are excluded. Since $\mathrm{ps}(Q)$ is a nonnegative integer by Setup 10.14, it follows that $\mathrm{ps}(Q) \geq 1$. □

10.6. **A prescribed vertex in a large connected graph.** In this section, we construct an independent set through any prescribed vertex with a controlled neighborhood, and apply Setup 10.17.

Lemma 10.29 (A prescribed independent vertex and its neighborhood)**.** *Let $D \in \mathbb{Z}_{\geq 1}$ and $r \in \mathbb{Z}_{\geq 0}$. Let H be a connected finite simple graph with*

$$\Delta(H) \leq D, \qquad |V(H)| > rD + 1.$$

For every prescribed $x \in V(H)$, there is an independent set I of size $r + 1$ containing x such that

$$|N_H(I)| \leq d_H(x) + r(D - 1).$$

In particular, let J be a finite simple graph with at least one edge and no isolated vertices, and let $k \in \mathbb{Z}_{\geq 0}$. Assume

$$|V(H)| = |V(J)| + k + r$$

and that $H - X$ contains J for every k-element subset $X \subseteq V(H)$. Then every $x \in V(H)$ satisfies

$$d_H(x) \geq k + 1 - r(D - 1),$$

and consequently

$$k + 1 \leq (r + 1)D - r.$$

Proof. Fix x, and begin with $I_1 = \{x\}$. This independent set satisfies

$$|N_H(I_1)| = d_H(x).$$

We inductively construct independent sets I_j, for integers $1 \leq j \leq r + 1$, containing x, with $|I_j| = j$ and

$$|N_H(I_j)| \leq d_H(x) + (j - 1)(D - 1).$$

Assume $1 \le j \le r$ and I_j has been constructed. Independence makes I_j disjoint from its open neighborhood. Therefore

$$\begin{aligned} |N_H[I_j]| &\le j + d_H(x) + (j-1)(D-1) \\ &= d_H(x) + (j-1)D + 1 \\ &\le jD + 1 \\ &\le rD + 1 \\ &< |V(H)|. \end{aligned}$$

By connectedness, choose a path from a vertex of I_j to a vertex outside $N_H[I_j]$. Let y be the first vertex of the path outside $N_H[I_j]$, and let z precede it. Then $z \in N_H[I_j]$ and $zy \in E(H)$. The vertex z cannot belong to I_j, since that would put y in $N_H(I_j)$. Hence $z \in N_H(I_j)$.

The vertex y has no neighbor in I_j, so

$$I_{j+1} = I_j \cup \{y\}$$

is independent. Its new open neighborhood $N_H(y)$ shares z with $N_H(I_j)$. Since $d_H(y) \le D$,

$$\begin{aligned} |N_H(I_{j+1})| &\le |N_H(I_j)| + D - 1 \\ &\le d_H(x) + j(D-1). \end{aligned}$$

This completes the induction. When $r = 0$, the initial singleton already gives the first assertion.

Under the robustness hypotheses, Setup 10.17 says that every independent set of size $r+1$ has open neighborhood of size at least $k+1$. Apply this to the set just constructed:

$$k + 1 \le |N_H(I)| \le d_H(x) + r(D-1).$$

Rearranging gives

$$d_H(x) \ge k + 1 - r(D-1).$$

Since $d_H(x) \le D$,

$$k + 1 \le D + r(D-1) = (r+1)D - r.$$

These arguments use ordinary subgraph containment and count every vertex of the retained graph J. □

11. Closed lumps and LP-goodness at high degree

We study closed regular factor-critical graphs within the certificate framework established earlier in this series. Domination certificates and two level sums give LP-goodness in degree at least 30; covered deletions, uncovered barriers, and independence bounds describe the remaining obstructions. The final results restrict the quartic allocation profile and exclude degree 14 from a minimal induced failure of T_-.

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11.1. **Setup.** We assume, established earlier in this series, the certificate framework of Chapter 1, "Certificates, costs and the LP-goodness criterion," and the degree splitting and covering matching results of Chapter 3, "Degree splitting and covering matchings." The statements below record the forms of those results, and of the additional earlier results of Companion I, used in this chapter. Matching terminology follows [LP86].

[cite/blocker] The earlier results of Companion I have no supplied theorem numbers. Their provisional reference is [I, Theorem X]; no bibliographic entry for Companion I is invented. [math/blocker] [GAP: The complete predecessor statements and their locations in the earlier chapters were not supplied. The

forms used below are recorded explicitly as assumptions. Their hypotheses and their availability before this chapter require confirmation.]

Notation 11.1. All graphs are finite and simple. For a graph G, write

$$n(G) = |V(G)|, \qquad e(G) = |E(G)|,$$

and let $\nu(G)$, $\alpha(G)$, and $\tau(G)$ denote its matching number, independence number, and minimum vertex-cover size. The maximum degree of an edgeless graph is 0. Vertex deletion is denoted by $G - X$ for $X \subseteq V(G)$; deletion of an edge set $F \subseteq E(G)$ retains all vertices. The notation $\nu(F)$ or $\Delta(F)$ refers to the spanning graph with edge set F. For integers $a \le b$, put

$$[a, b]_{\mathbb{Z}} = \{a, a+1, \dots, b\}.$$

All capacities, matching budgets, levels, degrees, orders, and sequence indices are integers in the ranges specified. Probability parameters are real.

Definition 11.2 (Certificates and their costs)**.** Let $A, B, x, y \in \mathbb{Z}_{\ge 0}$. An $(A, B; x, y)$-certificate of G is an edge partition

$$E(G) = Q_R \sqcup Q_B \sqcup P_R \sqcup P_B$$

such that

$$\Delta(Q_R) \le A, \qquad \Delta(Q_B) \le B, \qquad \nu(P_R) \le x, \qquad \nu(P_B) \le y.$$

The cost $c_{A,B}(G)$ is the least integer $c \ge 0$ for which a certificate exists for every $x, y \in \mathbb{Z}_{\ge 0}$ with $x + y = c$. For $\theta \in \mathbb{Z}_{\ge 0}$, define

$$c^*_\theta(G) = \max_{\substack{A,B \in \mathbb{Z}_{\ge 0} \\ A+B=\theta}} c_{A,B}(G), \qquad \Sigma(G) = \sum_{\theta \ge 0} c^*_\theta(G).$$

In particular, $c^*_0 = c_{0,0}$. Define

$$\mathrm{ex}(G) = c_{0,0}(G) - \nu(G), \qquad \mathrm{ps}(G) = e(G) - \nu(G) - \sum_{\theta \ge 1} c^*_\theta(G),$$

and

$$\mathrm{gap}(G) = \Sigma(G) - e(G) = \mathrm{ex}(G) - \mathrm{ps}(G).$$

A graph is *LP-good* if $\Sigma(G) \le e(G)$, and *LP-bad* otherwise.

Setup 11.3 (Earlier certificate rules)**.** We assume the following results of Chapter 1, with provisional reference [Rec].

(i) The costs are nonnegative integers, $c^*_\theta(G)$ is nonincreasing in θ, and

$$c^*_\theta(G) \le c_{0,0}(G), \qquad \nu(G) \le c_{0,0}(G) \le \tau(G).$$

(ii) Let $\theta \in \mathbb{Z}_{\ge 1}$. If $\Delta(G) \le \theta$ and some matching covers every vertex of degree θ, then

$$c_{A,B}(G) = 0 \quad \text{for all } A, B \in \mathbb{Z}_{\ge 0} \text{ with } A + B = \theta.$$

Consequently $c^*_\theta(G) = 0$ for $\theta \ge \Delta(G) + 1$. All costs of an edgeless graph vanish.

(iii) For $v \in V(G)$ and $\theta \in \mathbb{Z}_{\ge 0}$,

$$c^*_\theta(G - v) \le c^*_\theta(G) \le c^*_\theta(G - v) + 1.$$

(iv) The slack satisfies $\mathrm{ps}(G) \ge 0$. Thus G is LP-good if and only if $\mathrm{ps}(G) \ge \mathrm{ex}(G)$.

Definition 11.4 (Split allocations and the top missing run)**.** A pair $(x, y) \in \mathbb{Z}^2_{\ge 0}$ is *split-achievable* in G if

$$E(G) = P_R \sqcup P_B, \qquad \nu(P_R) \le x, \qquad \nu(P_B) \le y.$$

For $c \in \mathbb{Z}_{\ge 0}$, put

$$Y_c(G) = \{x \in [0, c]_{\mathbb{Z}} \mid (x, c - x) \text{ is split-achievable in } G\}.$$

For $Y \subseteq [0,c]_{\mathbb{Z}}$, let $\mathrm{run}_c(Y)$ be the largest number of consecutive integers of $[0,c]_{\mathbb{Z}}$ absent from Y, with value 0 if there is no absent integer. If $c = c_{0,0}(G) \geq 1$, define

$$\mathrm{tr}(G) = \mathrm{run}_{c-1}\big(Y_{c-1}(G)\big).$$

Setup 11.5 (Earlier split bounds)**.** We assume the following results of Chapter 1, with provisional reference [Rec].

(i) If $c = c_{0,0}(G) \geq 1$, some pair of nonnegative integers with sum $c-1$ is not split-achievable.

(ii) For every $c \in \mathbb{Z}_{\geq 0}$,

$$c + \lceil c/2 \rceil + 1 \geq n(G) \quad \Longrightarrow \quad c_{0,0}(G) \leq c.$$

Every graph on at most 10 vertices has excess at most 1. In particular, $\mathrm{ex}(G) \geq 2$ implies $n(G) \geq 11$.

(iii) If the minimum degree of G is at least $c_{0,0}(G)$, then G is LP-good.

Definition 11.6 (Closed lumps and domination certificates)**.** Let $d \geq 2$ be even. A *closed d-lump* is a connected d-regular factor-critical graph H. Here factor-critical means that $H - v$ has a perfect matching for every $v \in V(H)$. Such a graph has

$$n(H) = 2k+1, \qquad \nu(H) = k, \qquad e(H) = \frac{dn(H)}{2}$$

for an integer $k \geq 0$.

Fix $v \in V(H)$, a perfect matching M_v of $H - v$, and $j \in [1, d-1]_{\mathbb{Z}}$. A set $Z \subseteq V(H)$ is (j, v, M_v)-*dominating* if $v \in Z$ and every vertex outside Z has at least j neighbours in Z in the spanning graph $H - M_v$. Let $\beta_j(H)$ be the minimum size of such a set over all choices of v, M_v, and Z. A closed quartic graph means a closed 4-lump.

Setup 11.7 (Earlier paired-level bounds)**.** We assume the following degree splitting results of Chapter 3, with provisional reference [Rec]. Let H be a closed d-lump of order $n = 2k+1$.

(i) If $a, b \in \mathbb{Z}_{\geq 1}$ and $a + b = d + 1$, then

$$c_a^*(H) + c_b^*(H) \leq n.$$

(ii) The perfect-matching form of the paired-level bound gives

$$c_a^*(H) + c_b^*(H) \leq n + 1$$

for $a, b \in \mathbb{Z}_{\geq 1}$ with $a + b = d$. Its hypotheses hold with degree bound $D = d$, since $H - v$ has a perfect matching.

Setup 11.8 (Earlier independent-vertex split test)**.** We assume the following specialization of the earlier split test, with provisional reference [Rec]. Let $d \geq 4$ be even, let H be a closed d-lump of order $2k+1$, and set

$$e_1 = \lceil \frac{(d-1)k}{3d+1} \rceil, \qquad x^* = \lfloor \frac{k + e_1}{2} \rfloor, \qquad t^* = x^* - 2e_1.$$

The parameters of the test are $D = d$, $N = k$, and $\epsilon = 1$. It gives

$$c_{0,0}(H) \leq k + e_1$$

if $t^* \leq 0$, or if $t^* \geq 1$, $dt^* \leq x^*$, and $\alpha(H) \geq t^*$.

Setup 11.9 (Earlier deletion certificate)**.** We assume the following result of Chapter 3, with provisional reference [Rec]. Let $\theta \in \mathbb{Z}_{\geq 1}$ and $U \subseteq V(H)$. If $H - U$ has maximum degree at most θ and a matching covering every vertex of degree θ, then

$$c_\theta^*(H) \leq |U|.$$

A graph satisfying the two stated conditions is called θ-good in the earlier development.

Definition 11.10 (Covered graphs and maximum-degree peelings)**.** A graph G is *covered* if it has an edge and some matching covers every vertex of degree $\Delta(G)$. A graph with an edge that is not covered is *uncovered*.

A maximum-degree peeling of a graph H with an edge is a sequence

$$H_1 = H, \qquad H_{i+1} = H_i - v_i \quad (1 \le i \le T),$$

where $T \in \mathbb{Z}_{\ge 1}$, each v_i has degree $\Delta(H_i) \ge 1$ in H_i, and H_{T+1} is edgeless. A deletion preserves matching number if $\nu(H_i - v_i) = \nu(H_i)$. A peeling is *economical* if every uncovered stage has a matching-preserving deletion.

Setup 11.11 (Earlier deletion and peeling identities)**.** We assume the following earlier results, with provisional reference [Rec].

(i) Let H have an edge and let v have degree $\Delta(H)$. Put

$$r = c_{0,0}(H) - c_{0,0}(H - v), \qquad h = \nu(H) - \nu(H - v),$$

and

$$g = \big|\{\theta \in [1, \Delta(H)]_{\mathbb{Z}} \mid c^*_\theta(H) = c^*_\theta(H - v)\}\big|.$$

Then $r, h \in \{0, 1\}$, and

$$\operatorname{ex}(H) - \operatorname{ex}(H - v) = r - h, \qquad \operatorname{gap}(H) - \operatorname{gap}(H - v) = r - g.$$

If H is covered, then $g \ge 1$, and

$$\operatorname{ps}(H) \ge \operatorname{ps}(H - v) + 1 - h.$$

(ii) Every graph with an edge has an economical peeling. For such a peeling, let R remain after k deletions, where $c_{0,0}(H) \le k \le T$. If z_{prefix} counts its covered matching-preserving deletions among the first k, then

$$\operatorname{ps}(H) \ge z_{\text{prefix}} + e(R) - \nu(R).$$

(iii) Every stage of a maximum-degree peeling of a noncomplete graph remains noncomplete. If a positive regular noncomplete graph of odd order loses one vertex and then a maximum-degree vertex of the residual graph, the resulting graph is nonregular.

(iv) Every nonregular factor-critical graph with an edge is covered.

Setup 11.12 (Earlier prescribed-vertex matching criterion)**.** We assume the following result of Chapter 3, with provisional reference [Rec]. For $S \subseteq V(G)$, let $u_S(G)$ be the minimum number of vertices of S left unmatched by a matching. For $X \subseteq V(G)$, let $q_S(G - X)$ count the odd components of $G - X$ whose vertex sets lie in S. Then

$$u_S(G) = \max_{X \subseteq V(G)} \big(q_S(G - X) - |X|\big).$$

Setup 11.13 (The ordinary matching deficiency formula)**.** We use the matching deficiency formula in [LP86, Section 3.1]:

$$n(G) - 2\nu(G) = \max_{X \subseteq V(G)} \big(o(G - X) - |X|\big),$$

where $o(J)$ denotes the number of odd components of J. In particular, if an even-order graph has no perfect matching, some X satisfies $o(G - X) > |X|$.

Setup 11.14 (Earlier component bounds)**.** We assume the following result of Chapter 1, with provisional reference [Rec]. If G has an edge and C ranges over its components with an edge, then

$$\operatorname{ex}(G) \le \max_C \operatorname{ex}(C), \qquad \operatorname{ps}(G) \ge \sum_C \operatorname{ps}(C).$$

Order, matching number, and independence number are additive over components.

Definition 11.15 (Hereditary gap bound)**.** A graph H has the *hereditary gap bound* if every induced subgraph of H, including H, has generic gap at most 1.

Setup 11.16 (Earlier inputs at excess two and slack zero)**.** We assume the following earlier results of Companion I, each with provisional reference [Rec].

(i) No graph of maximum degree at most 3, excess 2, and positive gap exists.

(ii) Let H be connected and uncovered, with the hereditary gap bound, $\operatorname{gap}(H) = 1$, and $\operatorname{ex}(H) = 2$. Put $D = \Delta(H) \geq 4$, let S be its degree-D set, and let $X \neq \varnothing$ be a maximum-cardinality maximizer in Setup 11.12. Assume that $H - X$ has an edge. Applied to all nonnegative capacity pairs at each total, the earlier barrier dichotomy gives $\operatorname{gap}(H) \leq 0$ when D is even. When D is odd, it gives a factor-critical component Z of $H - X$, wholly contained in S, with

$$e_H(V(Z), X) = 1, \qquad \operatorname{gap}(Z) = 1, \qquad \operatorname{ex}(Z) = \operatorname{ex}(H) = 2, \qquad \operatorname{tr}(Z) \geq \operatorname{tr}(H) + |X|.$$

(iii) If J has the hereditary gap bound, $\operatorname{ex}(J) = 1$, and $\operatorname{ps}(J) = 0$, then $\operatorname{tr}(J) \leq 2$.

(iv) If J has the hereditary gap bound and $\operatorname{ps}(J) = 0$, then

$$n(J) - \nu(J) \leq 2\alpha(J).$$

(v) A closed d-lump with even $d \geq 8$, excess 2, and gap at most 1 in every proper induced subgraph is LP-good.

(vi) For a closed 6-lump J with the hereditary gap bound, excess 2, gap 1, and slack 1, the earlier order formula gives

$$n(J) = 2\alpha(J) + 5, \qquad \nu(J) = \alpha(J) + 2.$$

Setup 11.17 (Colouring bound)**.** We use the colouring theorem of [Bro41]: a connected graph that is neither complete nor an odd cycle has chromatic number at most its maximum degree. In particular, a connected 4-regular graph other than K_5 has a proper four-colouring.

Setup 11.18 (Earlier quartic profile results)**.** We assume the following earlier classification and exchange results of Companion I, with provisional references [Rec]. Let H be a closed quartic graph with

$$\operatorname{ex}(H) = 3, \qquad \operatorname{gap}(H) \geq 1, \qquad \operatorname{tr}(H) \geq 2, \qquad k = \nu(H).$$

(i) The earlier profile classification gives exactly the alternatives called (D), (E), and (F). An earlier exclusion rules out (D).

(ii) In profile (E), the costs at levels $0, 1, 2, 3, 4$ are

$$(k+3, k+2, k, k-3, 1), \qquad 11 \leq k \leq 14.$$

The graph H is well-covered, with independence number $k-3$. For every maximal independent set S, the graph $G = H - S$ is subcubic and satisfies

$$\operatorname{ps}(G) = 0, \qquad c_3^*(G) = 0, \qquad \big(\nu(G), e(G), c_1^*(G), c_2^*(G)\big) = (6, 14, 5, 3).$$

The set of cubic vertices of G has independence number at most 2.

(iii) In profile (F), the costs are

$$(k+3, k+3, k, k-4, 1).$$

The graph H is well-covered, with $\alpha(H) = k - 4$, and $H - S'$ is subcubic for every maximal independent set S'.

(iv) Under (F), there is a maximal independent set S such that $G = H - S$ consists of two components Q_1, Q_2, each of type D_6 or diamond, two triangles T_1, T_2, one P_3, and isolated vertices. Each Q_i has designated cubic centres. Let B consist of the centre of the P_3 and the two outer vertices of each diamond. Every vertex in B has degree 2 in G, and $|B| \leq 5$.

For a vertex t in either triangle, put $P_t = N_H(t) \cap S$. These are two-element sets. They avoid the unique S-parents of all cubic centres of the Q_i. Within a triangle the pairs are equal

or disjoint. Pairs from different triangles are disjoint. Every distinct pair P_t is $N_H(x) \cap S$ for some $x \in B$.

(v) Under (F), let S' be any maximal independent set and let $\{u, v\}$ be any independent pair of cubic vertices of $H - S'$. In the earlier pair frame, put

$$R = H - (S' \cup \{u, v\}).$$

The vertex u is an endpoint of one of the designated W-edges, and $N_H(u) \cap V(R)$ contains the centre of a P_3 component of R.

[math/blocker] [GAP: The numerical definition of profile (D), the graph D_6, and the pair frame with its W-edges were not supplied. Their stated consequences are assumed here; their definitions must be supplied from the earlier development.]

Definition 11.19 (The property T_-)**.** For $D \in \mathbb{Z}_{\geq 0}$, put

$$h_-(D) = \max\{4, \lfloor D/2 \rfloor + 1\}.$$

A graph J satisfies T_- if

$$\operatorname{gap}(J) \geq 1 \quad \Longrightarrow \quad \operatorname{gap}(J) = 1 \ \text{ and } \ \operatorname{tr}(J) + \operatorname{ex}(J) \leq h_-(\Delta(J)).$$

A *minimal induced T_- failure* violates T_-, while every proper induced subgraph satisfies T_-.

Setup 11.20 (Earlier reductions for T_-)**.** We assume the following results of Companion I, with provisional references [Rec].

(i) Every graph violating T_- contains an induced minimal failure F that is a totally critical closed d-lump of even degree $4 \leq d \leq 24$, with

$$\operatorname{gap}(F) = 1, \qquad \operatorname{tr}(F) + \operatorname{ex}(F) \geq h_-(d) + 1.$$

Every minimal induced failure has this structure, and every proper induced subgraph of F satisfies T_-.

(ii) There is no minimal induced failure of degree 18. Closed lumps of degrees 20, 22, and 24 are LP-good. Consequently a minimal induced failure has even degree $4 \leq d \leq 16$.

(iii) There is no minimal induced failure of degree 16.

[math/blocker] [GAP: The definition of "totally critical" was not supplied. The term is retained with its earlier meaning and requires that definition before the reduction can be read as a complete statement.]

Setup 11.21 (Earlier degree-fourteen bounds)**.** We assume the following distinct earlier results of Companion I, each with provisional reference [Rec].

(i) Every 14-regular graph of odd order at most $2 \cdot 14 - 3 = 25$ is LP-good.

(ii) For a minimal induced T_- failure that is a closed 14-lump of order $2k + 1$, the nine earlier interval exclusions rule out, respectively,

$$\begin{aligned} 13 \leq k \leq 42, \qquad & 43 \leq k \leq 55, \qquad 56 \leq k \leq 62, \\ 63 \leq k \leq 76, \qquad & 77 \leq k \leq 88, \qquad 89 \leq k \leq 102, \\ 103 \leq k \leq 118, \qquad & 119 \leq k \leq 134, \qquad 135 \leq k \leq 147. \end{aligned}$$

(iii) The earlier tail bound gives $k \leq 147$ for such a failure.

Remark 11.22 (Automated assistance). The main result of this paper was obtained with the assistance of the Ansatz system, an automated proof-search and verification system. Because automated systems have limitations, it is possible that we have missed related references in the literature, and we welcome comments from experts. [ack/blocker] No verification statement was supplied; independent re-verification is not asserted.

11.2. **Domination certificates and level sums.** In this section, we prove the domination and level estimates that give LP-goodness for closed lumps of degree at least 30.

Proposition 11.23 (Domination certificates and the first level sum)**.** *Let $d \geq 2$ be even, and let H be a closed d-lump with $n = 2k+1$ vertices.*

(i) *For every integer $\theta \geq d+1$,*
$$c^*_\theta(H) = 0, \qquad c^*_d(H) \leq 1.$$

(ii) *For every $j \in [1, d-1]_{\mathbb{Z}}$,*
$$c^*_{d-j}(H) \leq \beta_j(H).$$

(iii) *For $p \in [0,1] \subseteq \mathbb{R}$, $j \in [1,d-1]_{\mathbb{Z}}$, and $m \in [0,d-1]_{\mathbb{Z}}$,*
$$\beta_j(H) \leq 1 + (n-1)\left[p + (1-p)\mathbb{P}\big(\mathrm{Bin}(m,p) \leq j-1\big)\right],$$
where $\mathrm{Bin}(m,p)$ has m independent trials of success probability p.

(iv) *For every $t \in [1, d/2-1]_{\mathbb{Z}}$,*
$$\mathrm{gap}(H) \leq (t+2)c_{0,0}(H) + 1 + \sum_{j=1}^{t} c^*_{d-j}(H) - (t+1)n. \tag{11.1}$$

Proof. Factor-criticality gives a perfect matching of $H - v$ for every vertex v. Thus $n - 1 = 2k$, $\nu(H) = k$, $\Delta(H) = d$, and $e(H) = dn/2$.

Step 1. We bound the top levels. Setup 11.3 gives $c^*_\theta(H) = 0$ for $\theta \geq d+1$. Fix v, and let M_v be a perfect matching of $H - v$. The graph $H - v$ has maximum degree at most d, and M_v covers all of its degree-d vertices. For every $A, B \in \mathbb{Z}_{\geq 0}$ with $A + B = d$, Setup 11.3 gives $c_{A,B}(H - v) = 0$. Hence
$$c^*_d(H) \leq c^*_d(H - v) + 1 = 1.$$

Step 2. A dominating set gives the asserted certificate. Let Z be (j, v, M_v)-dominating, and put $G = H - M_v$. Fix $A, B \in \mathbb{Z}_{\geq 0}$ with $A + B = d - j \geq 1$, and $x, y \in \mathbb{Z}_{\geq 0}$ with $x + y = |Z|$. Partition $Z = Z_R \sqcup Z_B$ with $|Z_R| = x$ and $|Z_B| = y$. Let P_R consist of all edges meeting Z_R. Let P_B consist of all edges meeting Z_B and avoiding Z_R. Every matching in P_R has at most $|Z_R|$ edges, and every matching in P_B has at most $|Z_B|$ edges. Therefore
$$\nu(P_R) \leq x, \qquad \nu(P_B) \leq y.$$

The remaining edges form $Q = H - Z$. For $u \notin Z$, the condition $v \in Z$ gives $u \neq v$, so
$$\deg_H(u) = d, \qquad \deg_G(u) = d - 1.$$
Since u has at least j neighbours in Z in G,
$$\deg_Q(u) = d - |N_H(u) \cap Z| \leq d - j.$$
If equality holds, then
$$|N_H(u) \cap Z| = j \leq |N_G(u) \cap Z| \leq |N_H(u) \cap Z|.$$
Thus $N_H(u) \cap Z = N_G(u) \cap Z$. The M_v-partner of u, which is an H-neighbour but not a G-neighbour, lies outside Z. Consequently $M_v \cap E(Q)$ covers every vertex of degree $d - j$ in Q.

Setup 11.3 now gives $c_{A,B}(Q) = 0$. Write
$$E(Q) = Q_R \sqcup Q_B, \qquad \Delta(Q_R) \leq A, \qquad \Delta(Q_B) \leq B.$$
Then
$$E(H) = Q_R \sqcup Q_B \sqcup P_R \sqcup P_B$$
is an $(A, B; x, y)$-certificate. This holds for every $x + y = |Z|$, so $c_{A,B}(H) \leq |Z|$. Taking the maximum over $A + B = d - j$, and then the minimum over Z, v, M_v, proves (II).

Step 3. Random selection bounds the size of a dominating set. Fix v, M_v, and $G = H - M_v$. Select every vertex independently with probability p to obtain Z_0. Let

$$B_0 = \{u \neq v \mid u \notin Z_0,\ |N_G(u) \cap Z_0| \leq j-1\}, \qquad Z = \{v\} \cup Z_0 \cup B_0.$$

Every vertex outside Z has at least j G-neighbours in $Z_0 \subseteq Z$. Thus Z is (j, v, M_v)-dominating.

For $u \neq v$, its $d-1$ neighbours in G are selected independently. The event $u \in Z_0$ is independent of these selections because G has no loops. Hence

$$\mathbb{P}(u \in Z) \leq p + (1-p)\mathbb{P}\big(\mathrm{Bin}(d-1,p) \leq j-1\big).$$

Couple $\mathrm{Bin}(m,p)$ with $\mathrm{Bin}(d-1,p)$ by using the first m of the $d-1$ indicators. The first sum is at most the second, so

$$\mathbb{P}\big(\mathrm{Bin}(d-1,p) \leq j-1\big) \leq \mathbb{P}\big(\mathrm{Bin}(m,p) \leq j-1\big).$$

Since $v \in Z$ always,

$$\mathbb{E}|Z| \leq 1 + (n-1)\left[p + (1-p)\mathbb{P}\big(\mathrm{Bin}(m,p) \leq j-1\big)\right].$$

Some outcome has size at most its expectation. This proves (III).

Step 4. We pair the levels with sum $d+1$. By (I),

$$\Sigma(H) = c_{0,0}(H) + \sum_{\theta=1}^{d} c^*_\theta(H).$$

Partition $1, \dots, d$ into the $d/2$ pairs

$$\{i, d+1-i\}, \qquad 1 \leq i \leq d/2.$$

The first pair costs at most $c_{0,0}(H)+1$. For $2 \leq i \leq t+1$, put $j = i-1$. Then

$$c^*_i(H) + c^*_{d-j}(H) \leq c_{0,0}(H) + c^*_{d-j}(H).$$

For $t+2 \leq i \leq d/2$, both levels are positive and sum to $d+1$. Setup 11.7 bounds each such pair by n. There are $d/2 - t - 1$ of these pairs. Therefore

$$\Sigma(H) \leq (t+2)c_{0,0}(H) + 1 + \sum_{j=1}^{t} c^*_{d-j}(H) + \left(\frac{d}{2} - t - 1\right) n.$$

Subtracting $e(H) = dn/2$ proves (11.1). □

Proposition 11.24 (Perfect-matching level sum and greedy domination)**.** *Let $d \geq 4$ be even, and let H be a closed d-lump of order $n = 2k+1$.*

(i) *For every $T \in [1, d/2-1]_{\mathbb{Z}}$,*

$$\mathrm{gap}(H) \leq (T+1)\,\mathrm{ex}(H) + \sum_{j=1}^{T} c^*_{d-j}(H) - Tk + \frac{d}{2} - 2T. \tag{11.2}$$

(ii) *Define*

$$E_{\mathrm{CL}}(k) = \min\{e \in \mathbb{Z}_{\geq 0} \mid e + \lceil (k+e)/2 \rceil \geq k\}, \qquad E_{\mathrm{iso}}(k,d) = \left\lceil \frac{(d-1)k}{3d+1} \right\rceil.$$

Then

$$\mathrm{ex}(H) \leq E_{\mathrm{CL}}(k), \qquad \mathrm{ex}(H) \leq E_{\mathrm{iso}}(k,d).$$

(iii) *Put $r_0 = n-1-d$, and, while $r_i > 0$, define*

$$r_{i+1} = \max\left\{0, r_i - \lceil \frac{dr_i}{n-1-i} \rceil\right\}.$$

Let s_1 be the least $i \in \mathbb{Z}_{\geq 0}$ with $r_i = 0$. Then

$$\beta_1(H) \leq 1 + s_1.$$

Put $\rho_0 = n$, and, while $\rho_i > 0$, define

$$\rho_{i+1} = \max\left\{0, \rho_i - \lceil \frac{(d+1)\rho_i}{n-i} \rceil\right\}.$$

Let σ_1 be the least $i \in \mathbb{Z}_{\geq 0}$ with $\rho_i = 0$. Then H has a dominating set with exactly σ_1 vertices.

(iv) *Let $j \in [1, d-1]_{\mathbb{Z}}$ be odd. If every vertex outside $Z \subseteq V(H)$ has at least j neighbours in Z, and*

$$n - |Z| \leq 2(d-j) + 4,$$

then

$$c^*_{d-j}(H) \leq |Z|.$$

In particular,

$$n - \sigma_1 \leq 2d + 2 \quad \Longrightarrow \quad c^*_{d-1}(H) \leq \sigma_1.$$

Proof. We have

$$\nu(H) = k, \qquad c_{0,0}(H) = k + \mathrm{ex}(H), \qquad e(H) = dk + d/2.$$

Step 1. We pair levels with sum d. Proposition 11.23(I) gives

$$\Sigma(H) = \sum_{\theta=0}^{d} c^*_\theta(H), \qquad c^*_d(H) \leq 1.$$

Fix $T \in [1, d/2-1]_{\mathbb{Z}}$. The levels are the disjoint union of

$$L = \{0, \dots, T\},$$
$$\{\theta, d-\theta\} \quad (T+1 \leq \theta \leq d/2-1),$$
$$\{d/2\}, \qquad U = \{d-T, \dots, d-1\}, \qquad \{d\}.$$

The paired second coordinates run from $d/2+1$ to $d-T-1$. Thus the underlying intervals

$$[0,T],\ [T+1, d/2-1],\ \{d/2\},\ [d/2+1, d-T-1],\ [d-T, d-1],\ \{d\}$$

are consecutive and disjoint; the two paired intervals are empty when $T = d/2-1$.

The levels in L contribute at most $(T+1)(k+\mathrm{ex}(H))$. We retain the levels in U as $\sum_{j=1}^{T} c^*_{d-j}(H)$. Each of the $d/2-1-T$ remaining pairs has positive levels with sum d. The perfect-matching bound in Setup 11.7, with $D = d$, gives cost at most $n+1 = 2k+2$. Applied with both levels equal to $d/2$, it also gives

$$2c^*_{d/2}(H) \leq 2k+2.$$

Consequently

$$\Sigma(H) \leq (T+1)(k+\mathrm{ex}(H)) + \sum_{j=1}^{T} c^*_{d-j}(H) + 1$$
$$+ \left(\frac{d}{2} - 1 - T\right)(2k+2) + (k+1).$$

Here

$$\left(\frac{d}{2} - 1 - T\right)(2k+2) = (d-2-2T)k + (d-2-2T).$$

After subtracting $dk + d/2$, the coefficient of k is

$$(T+1) + (d-2-2T) + 1 - d = -T,$$

and the constant is

$$1 + (d-2-2T) + 1 - d/2 = d/2 - 2T.$$

This proves (11.2).

Step 2. The earlier split tests give the two excess bounds. Setup 11.5 gives $c_{0,0}(H) \le c$ whenever

$$c + \lceil c/2 \rceil + 1 \ge 2k + 1.$$

For $c = k + e$, this condition is

$$e + \lceil (k+e)/2 \rceil \ge k.$$

Taking $e = E_{\mathrm{CL}}(k)$ proves the first bound.

For the second bound, set

$$e_1 = \lceil \frac{(d-1)k}{3d+1} \rceil, \qquad x^* = \lfloor \frac{k+e_1}{2} \rfloor, \qquad t^* = x^* - 2e_1.$$

These are the parameters of Setup 11.8, with $D = d$, $N = \lfloor n/2 \rfloor = k$, and $\epsilon = 1$. If $t^* \le 0$, that test gives $c_{0,0}(H) \le k + e_1$.

Assume $t^* \ge 1$. Since $(3d+1)e_1 \ge (d-1)k$,

$$(d-1)(k+e_1) \le (3d+1)e_1 + (d-1)e_1 = 4de_1.$$

Hence

$$(d-1)x^* \le \frac{(d-1)(k+e_1)}{2} \le 2de_1.$$

Equivalently,

$$dx^* - 2de_1 \le x^*, \qquad dt^* \le x^*.$$

Also,

$$e_1 < \frac{(d-1)k}{3d+1} + 1 < \frac{k}{3} + 1,$$

so

$$t^* \le \frac{x^*}{d} \le \frac{k+e_1}{2d} < \frac{4k+3}{6d}.$$

Repeatedly choose a remaining vertex of H and delete it with all its remaining neighbours. Each round removes at most $d+1$ vertices, and the chosen vertices are pairwise nonadjacent. Therefore

$$\alpha(H) \ge \frac{2k+1}{d+1}.$$

The comparison

$$\frac{2k+1}{d+1} \ge \frac{4k+3}{6d}$$

follows from

$$6d(2k+1) - (d+1)(4k+3) = 8dk + 3d - 4k - 3 \ge 0$$

for $d \ge 1$ and $k \ge 0$. Thus $\alpha(H) > t^* - 1$, and integrality gives $\alpha(H) \ge t^*$. All conditions of Setup 11.8 hold. It follows that

$$c_{0,0}(H) \le k + e_1, \qquad \mathrm{ex}(H) \le E_{\mathrm{iso}}(k,d).$$

Step 3. We justify the recurrences and the greedy bounds. For integers $N, c \ge 1$, define, on $r \in \mathbb{Z}_{\ge 0}$,

$$\phi(r) = \max\{0, r - \lceil cr/N \rceil\}.$$

This function is nondecreasing. If $c \ge N$, then $\lceil cr/N \rceil \ge r$, so ϕ is identically 0. If $c < N$, then

$$\lceil c(r+1)/N \rceil \le \lceil cr/N \rceil + \lceil c/N \rceil = \lceil cr/N \rceil + 1.$$

Therefore

$$(r+1) - \lceil c(r+1)/N \rceil \geq r - \lceil cr/N \rceil,$$

and taking the maximum with 0 preserves this inequality.

Since H is d-regular, $n \geq d+1$, so $r_0 = n-1-d \geq 0$. While $r_i > 0$, the recurrence decreases r_i by at least 1, and

$$r_i \leq r_0 - i < n - 1 - i.$$

Thus all denominators used are positive, and s_1 exists. For the second recurrence, while $\rho_i > 0$,

$$\rho_i \leq n - i, \qquad n - i \geq \rho_i \geq 1, \qquad \rho_{i+1} \leq \rho_i - 1.$$

Hence σ_1 exists and $\sigma_1 \leq n$.

Fix v and a perfect matching M_v of $H - v$. In $H' = H - M_v$,

$$\deg_{H'}(v) = d, \qquad \deg_{H'}(u) = d - 1 \quad (u \neq v).$$

Start with $Z_0 = \{v\}$. For a current set Z_i of size $i+1$, let U_i be the vertices outside Z_i with no H'-neighbour in Z_i. Then

$$|U_0| = n - 1 - d = r_0.$$

Suppose $U_i \neq \varnothing$ and $|U_i| \leq r_i$. For each $u \in U_i$, the closed H'-neighbourhood $N'[u]$ has size d and avoids Z_i. Counting pairs

$$(w, u), \qquad u \in U_i, \quad w \in N'[u],$$

gives $d|U_i|$ pairs whose first coordinate lies in the $n-1-i$ vertices outside Z_i. Some $w \notin Z_i$ therefore lies in at least

$$\lceil \frac{d|U_i|}{n-1-i} \rceil$$

of these closed neighbourhoods. Put $Z_{i+1} = Z_i \cup \{w\}$. Then

$$U_{i+1} = U_i \setminus N'[w],$$

and monotonicity of ϕ, with $c = d$ and $N = n - 1 - i$, gives

$$|U_{i+1}| \leq \phi(|U_i|) \leq \phi(r_i) = r_{i+1}.$$

Induction shows that the process stops at some $i \leq s_1$. Its final set contains v, has size at most $1 + s_1$, and is $(1, v, M_v)$-dominating. Hence $\beta_1(H) \leq 1 + s_1$.

For ordinary domination, start in H with $Z_0 = \varnothing$ and $U_0 = V(H)$. Each undominated u has a closed neighbourhood of size $d+1$, disjoint from Z_i. Pair counting over the $n - i$ vertices outside Z_i finds a vertex covering at least

$$\lceil \frac{(d+1)|U_i|}{n-i} \rceil$$

members of U_i. Monotonicity of ϕ, now with $c = d+1$ and $N = n - i$, gives $|U_i| \leq \rho_i$. The process stops with a dominating set of size at most σ_1. Adding vertices preserves domination, and $\sigma_1 \leq n$, so a dominating set of exactly σ_1 vertices exists.

Step 4. Dense residual graphs admit a matching covering their top degree. Put $G = H - Z$ and $\theta = d - j$. Since d is even and j is odd, θ is odd and $1 \leq \theta \leq d - 1$. Every vertex of G loses at least j neighbours, so $\Delta(G) \leq \theta$. Let S be the degree-θ set of G. If a matching covers S, Setup 11.9 gives $c^*_\theta(H) \leq |Z|$.

Suppose no matching covers S. If $\theta = 1$, the edges of G themselves form a matching covering all degree-1 vertices, a contradiction. Thus $\theta \geq 3$. Setup 11.12 supplies $X \subseteq V(G)$ and a family $\mathcal{K}$ of at least $|X| + 1$ odd components of $G - X$, each wholly contained in S. For $K \in \mathcal{K}$,

$$e_G(K, X) = \theta|V(K)| - 2e(G[V(K)]).$$

Both θ and $|V(K)|$ are odd, so $e_G(K, X)$ is odd and at least 1. In particular, $X \neq \varnothing$. Also,

$$e_G(K, X) \geq |V(K)|\big(\theta + 1 - |V(K)|\big).$$

Call K small if $|V(K)| \le \theta$, and large otherwise. For a small component,

$$|V(K)|\big(\theta + 1 - |V(K)|\big) - \theta = (|V(K)| - 1)(\theta - |V(K)|) \ge 0,$$

so its boundary has at least θ edges. A large component has odd order at least $\theta + 2$.

Let s and ℓ count the small and large components. Counting their edges at X gives

$$\theta s + \ell \le \theta |X|, \qquad s + \ell \ge |X| + 1.$$

Thus

$$\theta(|X| + 1 - \ell) + \ell \le \theta |X|, \qquad \theta \le (\theta - 1)\ell.$$

It follows that $\ell \ge 2$. Consequently

$$|V(G)| \ge |X| + 2(\theta + 2) \ge 2\theta + 5,$$

contrary to $|V(G)| = n - |Z| \le 2\theta + 4$. A matching covering S therefore exists, proving (IV).

Finally, take $j = 1$ and a dominating set Z with $|Z| = \sigma_1$, supplied by (III). The condition

$$n - \sigma_1 \le 2d + 2 = 2(d - 1) + 4$$

allows the preceding argument and gives $c^*_{d-1}(H) \le \sigma_1$. □

Remark 11.25 (Comparison of the two level sums). The low-level block in (11.2) contains one fewer copy of $c_{0,0}(H)$ than the block in (11.1). Rewriting both bounds in terms of k and $\mathrm{ex}(H)$ shows that their right sides differ by

$$-\mathrm{ex}(H) + \frac{d}{2} - T.$$

Thus the perfect-matching pairing trades one copy of the excess against $d/2 - T$ units. The supplied asymptotic comparison records the corresponding change

$$\iota < \frac{T - 2\sum q_j}{T + 2} \qquad \text{to} \qquad \iota < \frac{T - 2\sum q_j}{T + 1}.$$

[math/blocker] [GAP: The asymptotic parameter ι, the indexing of this sum of q_j, and the limiting regime were not defined. This comparison is retained as an unresolved explanatory statement and is not used in any proof.]

Theorem 11.26 (LP-goodness in degree at least thirty). *Let $d \ge 30$ be even, and let H be a closed d-lump of order $2k + 1$. Then H is LP-good. More precisely:*

(i) *if $d \ge k + \lceil k/3 \rceil$, the minimum-degree criterion gives LP-goodness;*
(ii) *otherwise $k \ge 23$, and*

$$\mathrm{gap}(H) < 4 - \frac{26k}{100} < 0.$$

Proof. Put $E = \lceil k/3 \rceil$ and $c = k + E$. Then

$$c + \lceil c/2 \rceil + 1 \ge \frac{3}{2}c + 1 \ge \frac{3}{2}\left(k + \frac{k}{3}\right) + 1 = 2k + 1.$$

Setup 11.5 gives $c_{0,0}(H) \le k + E$.

If $d \ge k + E$, then the minimum degree of H is at least $c_{0,0}(H)$. The minimum-degree criterion in Setup 11.5 proves LP-goodness.

Assume $d < k + E$. Since $d \ge 30$,

$$k + \lceil k/3 \rceil \ge 31.$$

Every integer $k \le 22$ satisfies

$$k + \lceil k/3 \rceil \le 22 + 8 = 30.$$

Hence $k \geq 23$. Also $4 \leq d/2 - 1$, so Proposition 11.23, with $t = 4$, gives

$$\operatorname{gap}(H) \leq 6c_{0,0}(H) + 1 + \sum_{j=1}^{4} \beta_j(H) - 5n.$$

Since $E \leq (k+2)/3$,

$$6c_{0,0}(H) \leq 8k + 4, \qquad 5n = 10k + 5.$$

Therefore

$$\operatorname{gap}(H) \leq \sum_{j=1}^{4} \beta_j(H) - 2k. \tag{11.3}$$

Apply Proposition 11.23(III) with $m = 29$, which is allowed because $29 \leq d-1$. For $j = 1, 2, 3, 4$, use respectively

$$p = \frac{1}{8}, \quad \frac{1}{6}, \quad \frac{1}{5}, \quad \frac{1}{4},$$

and put

$$q_j = p + (1-p)\mathbb{P}\big(\mathrm{Bin}(29, p) \leq j - 1\big).$$

Then $\beta_j(H) \leq 1 + 2kq_j$. Expanding the binomial tail gives

$$(1-p)\mathbb{P}\big(\mathrm{Bin}(29, p) \leq j-1\big) = \sum_{i=0}^{j-1} \binom{29}{i} p^i (1-p)^{30-i}.$$

Using

$$\binom{29}{1} = 29, \qquad \binom{29}{2} = 406, \qquad \binom{29}{3} = 3654,$$

we obtain

$$q_1 = \frac{1}{8} + \left(\frac{7}{8}\right)^{30},$$

$$\begin{aligned} q_2 &= \frac{1}{6} + \left(\frac{5}{6}\right)^{28}\left[\left(\frac{5}{6}\right)^2 + 29\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)\right] \\ &= \frac{1}{6} + \left(\frac{5}{6}\right)^{29}\left(\frac{5}{6} + \frac{29}{6}\right) = \frac{1}{6} + \frac{17}{3}\left(\frac{5}{6}\right)^{29}, \end{aligned}$$

$$\begin{aligned} q_3 &= \frac{1}{5} + \left(\frac{4}{5}\right)^{28}\left[\left(\frac{4}{5}\right)^2 + 29\left(\frac{1}{5}\right)\left(\frac{4}{5}\right) + 406\left(\frac{1}{5}\right)^2\right] \\ &= \frac{1}{5} + \frac{538}{25}\left(\frac{4}{5}\right)^{28}, \end{aligned}$$

and

$$\begin{aligned} q_4 = \frac{1}{4} + \left(\frac{3}{4}\right)^{27}\Bigg[&\left(\frac{3}{4}\right)^3 + 29\left(\frac{1}{4}\right)\left(\frac{3}{4}\right)^2 \\ &+406\left(\frac{1}{4}\right)^2\left(\frac{3}{4}\right) + 3654\left(\frac{1}{4}\right)^3\Bigg] \\ = \frac{1}{4} + \frac{5160}{64}\left(\frac{3}{4}\right)^{27} &= \frac{1}{4} + \frac{645}{8}\left(\frac{3}{4}\right)^{27}. \end{aligned}$$

The required rational estimates follow by integer cross-multiplication. First,

$$7^{30} = 22539340290692258087863249, \qquad 8^{30} = 1237940039285380274899124224,$$

and

$$7^{30} \cdot 10000 = 225393402906922580878632490000 < 226543027189224590306539732992 = 183 \cdot 8^{30}.$$

Thus $(7/8)^{30} < 183/10000$. Next,

$$5^{29} = 18626451492309570 3125, \qquad 6^{29} = 36845653286788892983296,$$

and

$$5^{29} \cdot 100000 = 1862645149230957031250000 0 < 1875443752297554652849766 4 = 509 \cdot 6^{29}.$$

Hence $(5/6)^{29} < 509/100000$. Also,

$$4^{28} = 72057594037927936, \qquad 5^{28} = 37252902984619140625,$$

and

$$4^{28} \cdot 100000 = 7205759403792793600000 < 7338821887969970703125 = 197 \cdot 5^{28}.$$

Thus $(4/5)^{28} < 197/100000$. Finally,

$$3^{27} = 7625597484987, \qquad 4^{27} = 18014398509481984,$$

and

$$3^{27} \cdot 500000 = 3812798742493500000 < 3945153273576554496 = 219 \cdot 4^{27}.$$

Therefore $(3/4)^{27} < 219/500000$.

Combining these estimates gives

$$\begin{aligned}\sum_{j=1}^{4} q_j <& \frac{1}{8} + \frac{183}{10000} + \frac{1}{6} + \frac{17}{3}\frac{509}{100000} \\ &+ \frac{1}{5} + \frac{538}{25}\frac{197}{100000} + \frac{1}{4} + \frac{645}{8}\frac{219}{500000} \\ =& \frac{1}{8} + \frac{183}{10000} + \frac{1}{6} + \frac{8653}{300000} + \frac{1}{5} + \frac{52993}{1250000} + \frac{1}{4} + \frac{28251}{800000} \\ =& \frac{17330363}{20000000} < \frac{87}{100}.\end{aligned}$$

Consequently

$$\sum_{j=1}^{4} \beta_j(H) \le 4 + 2k \sum_{j=1}^{4} q_j < 4 + \frac{174k}{100}.$$

Substitution in (11.3) yields

$$\operatorname{gap}(H) < 4 - \frac{26k}{100} \le 4 - \frac{598}{100} < 0.$$

This proves LP-goodness in both cases. □

11.3. **Covered maximum-degree deletions.** In this section, we prove that a positive gap survives covered maximum-degree deletions and reaches an uncovered induced graph with a controlled top missing run.

Theorem 11.27 (Covered deletions and the first uncovered graph)**.** *The following statements hold.*

(i) *For every graph G, every $c \in \mathbb{Z}_{\geq 1}$, and every $v \in V(G)$,*

$$Y_{c-1}(G-v) \cup \big(Y_{c-1}(G-v)+1\big) \subseteq Y_c(G),$$

and

$$Y_{c-1}(G) \cup \big(Y_{c-1}(G)+1\big) \subseteq Y_c(G).$$

If $c_{0,0}(G) \geq 1$, then $\mathrm{tr}(G) \geq 1$, and every missing integer at level $c_{0,0}(G)-1$ lies in

$$[\mathrm{ex}(G), \nu(G)-1]_{\mathbb{Z}}.$$

(ii) *Let H be covered, with $\mathrm{ex}(H) \geq 1$, and let v have degree $\Delta(H)$. Then*

$$c_{0,0}(H-v) \geq 1, \qquad \mathrm{gap}(H-v) \geq \mathrm{gap}(H).$$

If the gaps are equal, then

$$c_{0,0}(H-v) = c_{0,0}(H)-1, \qquad \mathrm{tr}(H-v) \geq \mathrm{tr}(H)+1, \qquad \mathrm{ex}(H-v) \geq \mathrm{ex}(H)-1.$$

(iii) *Let $\mathrm{gap}(B) \geq 1$, and fix any maximum-degree peeling*

$$B_1 = B, \qquad B_{i+1} = B_i - v_i \quad (1 \leq i \leq T).$$

Some B_i, with $i \leq T$, is not covered. Let i^* be the least such index and $B^* = B_{i^*}$. Then

$$\mathrm{gap}(B^*) \geq \mathrm{gap}(B),$$

and either

$$\mathrm{gap}(B^*) \geq \mathrm{gap}(B)+1$$

or

$$\mathrm{tr}(B^*) + \mathrm{ex}(B^*) \geq \mathrm{tr}(B) + \mathrm{ex}(B).$$

(iv) *Every connected graph C with $\mathrm{ex}(C) \geq 4$ and $\Sigma(C) > e(C)$ has, along any maximum-degree peeling, an induced subgraph*

$$B^* = C - \{v_1, \ldots, v_{i^*-1}\}$$

that is not covered and satisfies

$$\mathrm{gap}(B^*) \geq 1, \qquad [\mathrm{gap}(B^*) \geq 2 \quad \textit{or} \quad \mathrm{tr}(B^*) + \mathrm{ex}(B^*) \geq 5].$$

Consequently the assertion

$$(\mathrm{R4}^*): \quad \textit{every connected } C \textit{ with } \mathrm{ex}(C) \geq 4 \textit{ is LP-good}$$

holds if every graph G that is not covered and has $\mathrm{gap}(G) \geq 1$ satisfies

$$\mathrm{gap}(G) = 1, \qquad \mathrm{tr}(G) + \mathrm{ex}(G) \leq 4.$$

Proof. **Step 1.** Reinserted vertex edges extend split allocations in both directions. Let $x \in Y_{c-1}(G-v)$. There is a partition

$$E(G-v) = P_R \sqcup P_B, \qquad \nu(P_R) \leq x, \qquad \nu(P_B) \leq c-1-x.$$

Let S be the edges of G incident with v. A matching contains at most one member of S, so

$$\nu(P_B \cup S) \leq c-x, \qquad \nu(P_R \cup S) \leq x+1.$$

The partitions $(P_R, P_B \cup S)$ and $(P_R \cup S, P_B)$ show that $x, x+1 \in Y_c(G)$. Here $x+1 \leq c$. If $x \in Y_{c-1}(G)$, its existing partition satisfies the enlarged budgets

$$(x, c-x) \qquad \text{and} \qquad (x+1, c-1-x).$$

Thus $x, x+1 \in Y_c(G)$ in this case as well.

Put $c = c_{0,0}(G) \geq 1$. Setup 11.5 implies that $Y_{c-1}(G)$ misses an integer, so $\mathrm{tr}(G) \geq 1$. Write $\nu = \nu(G)$ and $\mathrm{ex} = \mathrm{ex}(G)$, so $c-1 = \nu + \mathrm{ex} - 1$. If $0 \leq x \leq \mathrm{ex}-1$, then $c-1-x \geq \nu$, and the partition with $P_R = \varnothing$, $P_B = E(G)$ realizes $(x, c-1-x)$. If $\nu \leq x \leq c-1$, the partition $P_R = E(G)$, $P_B = \varnothing$ realizes it. Every missing integer therefore lies in $[\mathrm{ex}, \nu-1]_{\mathbb{Z}}$.

Step 2. A covered deletion preserves the gap and enlarges the run when the gap is unchanged. Since H has an edge, $\nu(H) \geq 1$. Thus

$$c = c_{0,0}(H) = \nu(H) + \mathrm{ex}(H) \geq 2.$$

Let $R = \mathrm{tr}(H)$, and choose a missing run

$$x_0, x_0 + 1, \ldots, x_0 + R - 1$$

at level $c - 1$. Part (I) gives

$$x_0 \geq \mathrm{ex}(H) \geq 1, \qquad x_0 + R - 1 \leq \nu(H) - 1 \leq c - 2.$$

For each integer x in this run, neither x nor $x - 1$ belongs to $Y_{c-2}(H - v)$. Otherwise (I), applied with $c - 1$ in place of c, would put x in $Y_{c-1}(H)$. Consequently the $R + 1$ consecutive integers

$$x_0 - 1, x_0, \ldots, x_0 + R - 1$$

are absent from $Y_{c-2}(H - v)$, and all lie in $[0, c - 2]_{\mathbb{Z}}$.

Put

$$\mathrm{rise}_0 = c_{0,0}(H) - c_{0,0}(H - v), \qquad h = \nu(H) - \nu(H - v),$$

and let g count the equal thresholds in Setup 11.11. That setup gives

$$\mathrm{rise}_0 \in \{0, 1\}, \qquad \mathrm{ex}(H) - \mathrm{ex}(H - v) = \mathrm{rise}_0 - h,$$
$$\mathrm{gap}(H) - \mathrm{gap}(H - v) = \mathrm{rise}_0 - g, \qquad g \geq 1.$$

If $\mathrm{rise}_0 = 0$, then

$$c_{0,0}(H - v) = c \geq 2, \qquad \mathrm{gap}(H - v) = \mathrm{gap}(H) + g \geq \mathrm{gap}(H) + 1.$$

If $\mathrm{rise}_0 = 1$, then

$$c_{0,0}(H - v) = c - 1 \geq 1, \qquad \mathrm{gap}(H - v) = \mathrm{gap}(H) + g - 1 \geq \mathrm{gap}(H).$$

In this second case the top run of $H - v$ is measured at level $c - 2$, so

$$\mathrm{tr}(H - v) \geq R + 1.$$

Also,

$$\mathrm{ex}(H - v) = \mathrm{ex}(H) - 1 + h \geq \mathrm{ex}(H) - 1.$$

Equality of the gaps excludes the first case. This proves (II).

Step 3. The peeling must reach an uncovered graph. The final graph B_{T+1} is edgeless. Its two empty edge parts realize the split $(0, 0)$, and all its higher certificate costs vanish by Setup 11.3. Hence

$$\mathrm{gap}(B_{T+1}) = 0.$$

For every graph G,

$$\mathrm{gap}(G) = \mathrm{ex}(G) - \mathrm{ps}(G) \leq \mathrm{ex}(G).$$

Suppose $B_1, \ldots, B_k$ are covered, where $k \leq T$. We prove

$$\mathrm{gap}(B_i) \geq \mathrm{gap}(B) \qquad (1 \leq i \leq k + 1)$$

by induction. The assertion at $i = 1$ is equality. If it holds at $i \leq k$, then

$$\mathrm{ex}(B_i) \geq \mathrm{gap}(B_i) \geq \mathrm{gap}(B) \geq 1.$$

Part (II) applies to B_i and v_i, giving

$$\mathrm{gap}(B_{i+1}) \geq \mathrm{gap}(B_i) \geq \mathrm{gap}(B).$$

If all positive-degree stages were covered, this would give $\mathrm{gap}(B_{T+1}) \geq 1$, a contradiction. Thus $i^* \leq T$ exists, and the induction gives $\mathrm{gap}(B^*) \geq \mathrm{gap}(B)$.

Suppose $\mathrm{gap}(B^*) = \mathrm{gap}(B)$. The intermediate gaps are nondecreasing, so

$$\mathrm{gap}(B_{i+1}) = \mathrm{gap}(B_i) \qquad (i < i^*).$$

At each such step, part (II) gives

$$\operatorname{tr}(B_{i+1}) \geq \operatorname{tr}(B_i) + 1, \qquad \operatorname{ex}(B_{i+1}) \geq \operatorname{ex}(B_i) - 1.$$

The top runs are defined: the initial positive gap forces an edge and hence $c_{0,0}(B) \geq \nu(B) \geq 1$, and part (II) preserves positivity of $c_{0,0}$. Adding the two inequalities and iterating yields

$$\operatorname{tr}(B^*) + \operatorname{ex}(B^*) \geq \operatorname{tr}(B) + \operatorname{ex}(B).$$

If $i^* = 1$, this is equality without any iteration. If the final gap is not equal to the initial gap, integrality gives $\operatorname{gap}(B^*) \geq \operatorname{gap}(B) + 1$.

Step 4. We apply the peeling to the stated connected graph. A maximum-degree peeling exists by repeatedly deleting a maximum-degree vertex while an edge remains. Since $\Sigma(C) > e(C)$, integrality gives $\operatorname{gap}(C) \geq 1$. Part (III) supplies the induced, uncovered graph B^*, with

$$\operatorname{gap}(B^*) \geq \operatorname{gap}(C) \geq 1.$$

If its gap increases, then $\operatorname{gap}(B^*) \geq 2$. Otherwise

$$\operatorname{tr}(B^*) + \operatorname{ex}(B^*) \geq \operatorname{tr}(C) + \operatorname{ex}(C) \geq 1 + 4 = 5,$$

using (I).

Under the final hypothesis in (IV), this B^* would instead have gap 1 and $\operatorname{tr}(B^*) + \operatorname{ex}(B^*) \leq 4$. Both alternatives are contradicted. Therefore no connected graph with excess at least 4 has positive gap, proving (R4*). □

11.4. **Uncovered barrier accounting.** The goal of this section is to express the gap at an uncovered graph in terms of odd components exposed by a prescribed-vertex matching barrier.

Proposition 11.28 (Uncovered barrier accounting)**.** *Let G have maximum degree $\Delta \geq 1$, and let S be its degree-Δ set. Put*

$$\kappa = \min_M |S \setminus V(M)| = \max_{X \subseteq V(G)} \big(q_S(G - X) - |X|\big),$$

where M ranges over matchings. Let X attain the maximum. Let $\mathcal{K}$ be the odd components of $G - X$ whose vertex sets lie in S, so

$$|\mathcal{K}| = |X| + \kappa.$$

Let Y be the vertices outside X and outside every member of $\mathcal{K}$. For $K \in \mathcal{K}$, let $d(K)$ count the edges with exactly one endpoint in $V(K)$. Write $e(X)$ for the number of edges within X, and $e(X, Y)$ for the number between X and Y.

(i) *Every edge leaving $K \in \mathcal{K}$ ends in X, and*

$$d(K) \equiv \Delta \pmod 2.$$

If $d(K) \leq \Delta - 1$, then $|V(K)| \geq 3$.

(ii)
$$\sum_{K \in \mathcal{K}} d(K) = \sum_{x \in X} \deg_G(x) - 2e(X) - e(X, Y).$$

(iii) $c^*_\Delta(G) \leq \kappa$.

(iv)
$$\operatorname{gap}(G) \leq \operatorname{gap}(G - X) + \sum_{K \in \mathcal{K}} (\Delta - d(K)) - (\Delta - 1)\kappa - e(X) - e(X, Y). \tag{11.4}$$

(v) *If $G - X$ has an edge, let Z be a component with an edge and maximum excess. Then*

$$\begin{aligned} \operatorname{gap}(G) \leq\ & \operatorname{gap}(Z) - (\Delta - 1)\kappa \\ & + \sum_{\substack{K \in \mathcal{K} \\ K \neq Z}} \big(\Delta - d(K) - \operatorname{ps}(K)\big) + \mathbf{1}_{\{Z \in \mathcal{K}\}} \big(\Delta - d(Z)\big). \end{aligned} \tag{11.5}$$

The final term is 0 *when* $Z \notin \mathcal{K}$. *If* $G - X$ *is edgeless, then*

$$\operatorname{gap}(G) \le -(\Delta - 1)\kappa.$$

(vi) *If* X *has maximum cardinality among all maximizers, every member of* $\mathcal{K}$ *is factor-critical.*

Proof. **Step 1.** We count the boundary edges. A component K of $G - X$ has no edge to another component of $G - X$, so every edge leaving it ends in X. Since all its vertices have degree Δ in G,

$$\Delta|V(K)| = 2e(K) + d(K).$$

Its order is odd, and therefore $d(K) \equiv \Delta \pmod 2$. If K is a singleton, all Δ incident edges leave it. Thus $d(K) \le \Delta - 1$ forces odd order at least 3.

Summing degrees over X counts internal edges twice and other incident edges once:

$$\sum_{x \in X} \deg_G(x) = 2e(X) + e\left(X, \bigcup_{K \in \mathcal{K}} V(K)\right) + e(X, Y).$$

The middle term is $\sum_{K \in \mathcal{K}} d(K)$, proving (II).

Step 2. A matching with deficiency κ controls the top level. Choose a matching M leaving exactly $z_1, \dots, z_\kappa$ uncovered in S. For each i, choose an edge f_i incident with z_i, which exists because $\Delta \ge 1$. Put

$$F = \{f_1, \dots, f_\kappa\}.$$

Then $|F| \le \kappa$, and $F \cap M = \varnothing$. In $G' = G - F$, every remaining degree-Δ vertex lies in

$$S \setminus \{z_1, \dots, z_\kappa\}$$

and is covered by M.

For $A, B \in \mathbb{Z}_{\ge 0}$ with $A + B = \Delta$, Setup 11.3 gives

$$E(G') = Q_R \sqcup Q_B, \qquad \Delta(Q_R) \le A, \quad \Delta(Q_B) \le B.$$

For every $x, y \in \mathbb{Z}_{\ge 0}$ with $x + y = \kappa$, partition $F = F_R \sqcup F_B$ with $|F_R| \le x$, $|F_B| \le y$. Since matching number is at most edge count, these four parts form an $(A, B; x, y)$-certificate of G. Hence $c_{A,B}(G) \le \kappa$ for every such capacity pair, and $c^*_\Delta(G) \le \kappa$.

Step 3. We sum the costs of deleting X. By the definition of the gap,

$$\begin{aligned}\operatorname{gap}(G) - \operatorname{gap}(G - X) = &\sum_{\theta \ge 0} \big(c^*_\theta(G) - c^*_\theta(G - X)\big) \\ &- \big(e(G) - e(G - X)\big).\end{aligned}$$

Delete X one vertex at a time. Setup 11.3 gives a cost increase at most $|X|$ at each level $0 \le \theta \le \Delta - 1$. At level Δ, part (III) bounds the difference by κ. For $\theta > \Delta$, the cost of G is 0, so the difference is nonpositive. The number of deleted edges is

$$e(G) - e(G - X) = \sum_{x \in X} \deg_G(x) - e(X).$$

Consequently

$$\operatorname{gap}(G) - \operatorname{gap}(G - X) \le \Delta|X| + \kappa - \sum_{x \in X} \deg_G(x) + e(X).$$

Part (II) rewrites this as

$$\Delta|X| + \kappa - \sum_{K \in \mathcal{K}} d(K) - e(X) - e(X, Y).$$

Since $|X| = |\mathcal{K}| - \kappa$,

$$\Delta|X| + \kappa = \Delta|\mathcal{K}| - (\Delta - 1)\kappa.$$

This proves (11.4).

Step 4. We retain only the component with maximum excess. Assume $G - X$ has an edge. Setup 11.14 gives

$$\operatorname{ex}(G - X) \le \operatorname{ex}(Z), \qquad \operatorname{ps}(G - X) \ge \sum_{C} \operatorname{ps}(C),$$

where C ranges over its components with an edge. Hence

$$\operatorname{gap}(G - X) \le \operatorname{gap}(Z) - \sum_{C \ne Z} \operatorname{ps}(C).$$

An edgeless member K of $\mathcal{K}$ is a singleton, has $d(K) = \Delta$, and contributes 0 to $\sum_K (\Delta - d(K))$. Insert the preceding component bound into (11.4). Drop the nonpositive terms $-e(X) - e(X, Y)$, and drop $-\operatorname{ps}(C)$ for components outside $\mathcal{K}$. Pair each remaining $K \ne Z$ with its term $-\operatorname{ps}(K)$. If $Z \in \mathcal{K}$, its boundary contribution remains $\Delta - d(Z)$. This gives (11.5).

If $G - X$ is edgeless, its gap is 0 and every member of $\mathcal{K}$ is a singleton. Formula (11.4) then gives

$$\operatorname{gap}(G) \le -(\Delta - 1)\kappa.$$

Step 5. A largest maximizing barrier has factor-critical odd components. Choose X with maximum cardinality among the maximizers. Suppose a member $K \in \mathcal{K}$ is not factor-critical. Some $w \in V(K)$ has the property that $K - w$ has no perfect matching. The perfect matching criterion [LP86, Theorem 3.1.1], equivalently the consequence of Setup 11.13, supplies

$$Y' \subseteq V(K) \setminus \{w\}, \qquad o((K - w) - Y') > |Y'|.$$

The integer

$$|V(K)| - 1 - |Y'| - o((K - w) - Y')$$

is even. Therefore

$$o((K - w) - Y') \ge |Y'| + 2.$$

Put $X' = X \cup \{w\} \cup Y'$. The components of $G - X'$ are those of $G - X$ other than K, together with the components of $(K - w) - Y'$. All vertices of the latter lie in S. Thus

$$\begin{aligned} q_S(G - X') &\ge q_S(G - X) - 1 + |Y'| + 2 \\ &= |X| + \kappa + |Y'| + 1 = |X'| + \kappa. \end{aligned}$$

By maximality of the deficiency κ, X' is another maximizer. But $|X'| > |X|$, a contradiction. □

Remark 11.29 (The remaining boundary contributions). In (11.5), components other than Z can contribute positively only through

$$\Delta - d(K) - \operatorname{ps}(K) > 0.$$

The distinguished component has the separate term $\Delta - d(Z)$ when $Z \in \mathcal{K}$. Thus the boundary deficits of these odd components are the possible positive contributions left after the covered-deletion reduction of Theorem 11.27.

11.5. **Independence and order bounds.** In this section, we convert residual matching potential into slack and then use the barrier and deletion results to control graphs with slack at most one.

Proposition 11.30 (Independence bounds for residual potential)**.** *For every graph G, put $\Phi(G) = e(G) - \nu(G)$. Then*

$$\Phi(G) \ge 2n(G) - 4\alpha(G).$$

Let H have an edge, and fix a maximum-degree peeling that preserves matching number at every uncovered stage. Let T be its number of positive-degree deletions, let

$$k \in [c_{0,0}(H), T]_{\mathbb{Z}},$$

and let z_{prefix} count the covered matching-preserving deletions among its first k stages. Then

$$\operatorname{ps}(H) \ge z_{\text{prefix}} + \max\{0,\, 2(n(H) - k) - 4\alpha(H)\}.$$

In particular, every graph H satisfying

$$2(n(H) - c_{0,0}(H)) \geq \mathrm{ex}(H) + 4\alpha(H)$$

is LP-good.

Proof. Fix a maximum matching M of G, and write $n = n(G)$. From each edge of M, choose one endpoint with equal probabilities, independently between edges. Independently include each unmatched vertex with probability $1/2$. Let S be the chosen set. Every vertex is included with probability $1/2$, so

$$\mathbb{E}|S| = \frac{n}{2}.$$

No matching edge has both endpoints in S. The endpoints of a nonmatching edge lie in distinct choice blocks: in a simple graph, the edge joining the two vertices of one matching block is precisely the matching edge. Thus inclusion of the endpoints of a nonmatching edge is independent, with joint probability $1/4$. It follows that

$$\mathbb{E}e(G[S]) = \frac{e(G) - |M|}{4} = \frac{\Phi(G)}{4}.$$

For any outcome, select one endpoint of each edge of $G[S]$, and delete the union of the selected endpoints. This union has size at most $e(G[S])$ and meets every edge of $G[S]$. The remaining vertices are independent, so

$$\alpha(G) \geq |S| - e(G[S]).$$

Averaging over the finite probability space gives

$$\alpha(G) \geq \frac{n}{2} - \frac{\Phi(G)}{4}.$$

This is equivalent to the asserted potential bound. Also $\Phi(G) \geq 0$, since every matching is a subset of the edge set.

For the peeling of H, let R remain after k deletions. Then

$$n(R) = n(H) - k, \qquad \alpha(R) \leq \alpha(H).$$

Setup 11.11 gives

$$\mathrm{ps}(H) \geq z_{\mathrm{prefix}} + \Phi(R).$$

Apply the potential bound to R and use $\Phi(R) \geq 0$. This yields the second assertion.

Finally, assume H has an edge and satisfies the numerical condition. An economical peeling exists by Setup 11.11. The vertices deleted in a complete peeling form a vertex cover, because the remaining induced graph is edgeless. Therefore

$$T \geq \tau(H) \geq c_{0,0}(H),$$

using Setup 11.3. Apply the second assertion with $k = c_{0,0}(H)$, and discard $z_{\mathrm{prefix}} \geq 0$. The numerical hypothesis gives

$$\mathrm{ps}(H) \geq \mathrm{ex}(H),$$

so H is LP-good. If H is edgeless, the empty edge partition is a zero-budget certificate for every capacity pair. All its thresholds and its level sum vanish, so it is LP-good as well. □

Proposition 11.31 (Robust perfect matchings and covered stages)**.** *Let H be a noncomplete graph of order n. Let q be even, with $2 \leq q \leq n$, and assume every induced q-vertex subgraph has a perfect matching. Then every induced even-order subgraph of order at least q has a perfect matching, and every induced odd-order subgraph of order at least $q+1$ is factor-critical.*

In any maximum-degree peeling of H, let z count the covered stages whose chosen deletion preserves matching number. Then

$$z \geq \left\lfloor \frac{\lceil (n-q)/2 \rceil}{2} \right\rfloor.$$

Proof. We prove the even-order assertion by induction on the even integer $m \geq q$, simultaneously for all induced m-vertex subgraphs. The case $m = q$ is the hypothesis. Let $m \geq q+2$, and let $W \subseteq V(H)$ have size m. Any q-subset of W induces a graph with a perfect matching, and hence with an edge. Choose one such edge uv. The induction hypothesis gives a perfect matching of

$$H[W \setminus \{u, v\}],$$

whose order $m-2$ is even and at least q. Adding uv gives a perfect matching of $H[W]$.

If W has odd order at least $q+1$, deleting any $w \in W$ leaves even order at least q. The even-order assertion therefore makes $H[W]$ factor-critical.

Fix a maximum-degree peeling. Every induced graph of order at least q has an edge, because it contains a q-vertex induced graph with a perfect matching. The peeling therefore passes through every order down to q before terminating. It has exactly

$$K = \lceil (n-q)/2 \rceil$$

odd-order stages of order at least $q+1$. For odd n, their orders are $n, n-2, \ldots, q+1$. For even n, they are $n-1, n-3, \ldots, q+1$.

Each such stage Q, of odd order m, is factor-critical. Hence

$$\nu(Q) = \frac{m-1}{2}, \qquad \nu(Q-w) = \frac{m-1}{2} \quad \text{for every } w \in V(Q).$$

Its chosen deletion preserves matching number.

If Q is nonregular, choose w with $\deg_Q(w) < \Delta(Q)$. A perfect matching of $Q-w$ covers every vertex of Q except w, and hence every maximum-degree vertex. The stage is covered. If Q is regular, its degree is positive because $m \geq q+1$. No matching covers all of its odd number of vertices, so it is uncovered.

By Setup 11.11, every stage remains noncomplete. If one of the listed odd stages is regular and has a following listed odd stage, the next two deletions consist of one deletion from a positive regular noncomplete odd graph, followed by a maximum-degree deletion from its residual graph. The two-deletion result in that setup makes the next listed odd stage nonregular. Thus no consecutive positions in the list of K odd stages are both regular.

Pair positions $1, 2$, then $3, 4$, and continue. Each of the $\lfloor K/2 \rfloor$ pairs contains a nonregular stage. These stages are distinct, covered, and matching-preserving. Therefore

$$z \geq \lfloor K/2 \rfloor = \lfloor \frac{\lceil (n-q)/2 \rceil}{2} \rfloor.$$

□

Theorem 11.32 (Small slack under the hereditary gap bound)**.** *The following statements hold.*

(i) *Every connected uncovered graph H with the hereditary gap bound,*

$$\mathrm{gap}(H) \geq 1, \qquad \mathrm{ex}(H) = 2,$$

is a closed 4-lump or a closed 6-lump. Thus it is regular of degree 4 or 6, has odd order, and deleting any vertex leaves a perfect matching.

(ii) *Every graph H with the hereditary gap bound and $\mathrm{ps}(H) \leq 1$ satisfies*

$$n(H) - \nu(H) \leq 2\alpha(H) + \mathrm{ps}(H).$$

In particular, $\mathrm{ps}(H) = 1$ implies

$$n(H) - \nu(H) \leq 2\alpha(H) + 1.$$

Consequently every factor-critical graph Q with $\mathrm{ps}(Q) = 1$ and the hereditary gap bound satisfies

$$\nu(Q) \leq 2\alpha(Q).$$

No universal hereditary gap bound or universal assertion of T_- is made.

Proof. **Step 1.** We classify the connected uncovered graphs in (I). The hereditary gap bound gives $\operatorname{gap}(H) = 1$, so

$$\operatorname{ps}(H) = \operatorname{ex}(H) - \operatorname{gap}(H) = 1.$$

Setup 11.16(I) excludes maximum degree at most 3. Put $D = \Delta(H) \geq 4$, let S be the degree-D set, and let $\kappa \geq 1$ be its minimum unmatched count. By Proposition 11.28, choose a maximum-cardinality set X satisfying

$$q_S(H - X) - |X| = \kappa.$$

Suppose $X \neq \varnothing$. Every component of $H - X$ has gap at most 1. If $H - X$ is edgeless, Proposition 11.28(V) gives

$$\operatorname{gap}(H) \leq -(D-1)\kappa < 0,$$

a contradiction. Otherwise apply the earlier barrier dichotomy in Setup 11.16(II). It applies to all nonnegative capacity pairs at each total, whose generic profile c^*_θ is nonincreasing. For even D, it gives $\operatorname{gap}(H) \leq 0$. For odd D, it gives a factor-critical component Z of $H - X$, contained in S, with

$$e_H(V(Z), X) = 1, \qquad \operatorname{gap}(Z) = 1, \qquad \operatorname{ex}(Z) = 2,$$

and

$$\operatorname{tr}(Z) \geq \operatorname{tr}(H) + |X| \geq 2.$$

The last inequality uses Theorem 11.27(I).

This component Z is nonregular. Its excess 2 implies $|V(Z)| \geq 11$ by Setup 11.5. All its vertices have degree D in H, and exactly one edge leaves Z. Thus exactly one vertex has internal degree $D - 1$, while all the others have internal degree D. Since Z is nonregular and factor-critical, Setup 11.11 makes it covered.

Choose a maximum-degree vertex w of Z. Factor-criticality gives $\nu(Z - w) = \nu(Z)$. Theorem 11.27(II) gives

$$\operatorname{gap}(Z - w) \geq \operatorname{gap}(Z) = 1.$$

The hereditary gap bound makes the gap equal to 1. Therefore

$$c_{0,0}(Z - w) = c_{0,0}(Z) - 1, \qquad \operatorname{ex}(Z - w) = 1, \qquad \operatorname{tr}(Z - w) \geq \operatorname{tr}(Z) + 1 \geq 3.$$

Its slack is

$$\operatorname{ps}(Z - w) = \operatorname{ex}(Z - w) - \operatorname{gap}(Z - w) = 0.$$

It inherits the hereditary gap bound, so Setup 11.16(III) gives $\operatorname{tr}(Z - w) \leq 2$, a contradiction. Thus $X = \varnothing$.

Since H is connected and $q_S(H) = \kappa \geq 1$, the graph H has odd order and every vertex lies in S. Proposition 11.28(VI), using the maximum-cardinality choice of X, gives factor-criticality. Hence H is a closed D-lump. Its odd order and the degree-sum identity force D to be even.

If $D \geq 8$, Setup 11.16(V) makes H LP-good: its excess is 2, and every proper induced subgraph has gap at most 1. This contradicts $\operatorname{gap}(H) = 1$. The remaining degrees are 4 and 6, proving (I).

Step 2. A counterexample with slack one is connected. The case $\operatorname{ps}(H) = 0$ in (II) is Setup 11.16(IV). Suppose the assertion for $\operatorname{ps}(H) = 1$ fails, and choose a counterexample H of minimum order. For every graph J, put

$$\rho(J) = n(J) - \nu(J) - 2\alpha(J).$$

Then $\rho(H) \geq 2$. The graph H has an edge, since edgeless graphs have slack 0.

Apply Proposition 11.30 to an economical peeling at cutoff $c_{0,0}(H)$. Existence of this peeling and the fact that its length is at least $c_{0,0}(H)$ were justified in that proposition. Discarding the nonnegative prefix term gives

$$1 = \operatorname{ps}(H) \geq 2(n(H) - c_{0,0}(H)) - 4\alpha(H) = 2(\rho(H) - \operatorname{ex}(H)).$$

Both $\rho(H)$ and $\operatorname{ex}(H)$ are integers, so $\rho(H) \leq \operatorname{ex}(H)$. On the other hand,

$$\operatorname{ex}(H) = \operatorname{ps}(H) + \operatorname{gap}(H) \leq 2.$$

Hence

$$\rho(H) = \operatorname{ex}(H) = 2, \qquad \operatorname{gap}(H) = 1.$$

The quantity ρ is additive over components. Slack is nonnegative and superadditive by Setup 11.14. Since $\operatorname{ps}(H) = 1$, either every component has slack 0, or exactly one component has slack 1 and all others have slack 0. In the first case, Setup 11.16(IV) gives $\rho \le 0$ in every component, contrary to $\rho(H) = 2$. In the second case, all components other than the slack-one component have nonpositive ρ. The slack-one component therefore has $\rho \ge 2$. If H were disconnected, this component would be a smaller counterexample inheriting the hereditary gap bound. Thus H is connected.

Step 3. The counterexample cannot be covered. Suppose H is covered, and choose a maximum-degree vertex v. Put

$$t = \nu(H) - \nu(H - v) \in \{0, 1\}.$$

Setup 11.11 gives

$$1 = \operatorname{ps}(H) \ge \operatorname{ps}(H - v) + 1 - t.$$

If $t = 0$, then $\operatorname{ps}(H - v) = 0$. Since $\alpha(H - v) \le \alpha(H)$,

$$\begin{aligned}\rho(H - v) &= n(H) - 1 - \nu(H) - 2\alpha(H - v) \\ &\ge \rho(H) - 1 \ge 1.\end{aligned}$$

This contradicts the zero-slack bound in Setup 11.16.

If $t = 1$, then $\operatorname{ps}(H - v) \le 1$, and

$$\rho(H - v) \ge \rho(H) \ge 2.$$

Slack 0 again contradicts the zero-slack bound. Slack 1 gives a smaller counterexample. Both cases are impossible, so H is uncovered.

Step 4. Degrees four and six give contradictions. Part (I) makes H a closed 4-lump or a closed 6-lump.

In degree 4, the graph is not K_5, because Setup 11.5 bounds the excess of every graph on at most 10 vertices by 1. The colouring theorem [Bro41], in the form recorded in Setup 11.17, gives four independent colour classes. Thus

$$n(H) \le 4\alpha(H).$$

Factor-criticality gives $n(H) = 2\nu(H) + 1$. Since $\rho(H) \ge 2$,

$$\nu(H) + 1 - 2\alpha(H) \ge 2,$$

and therefore

$$n(H) \ge 4\alpha(H) + 3.$$

This is a contradiction.

In degree 6, Setup 11.16(VI) gives

$$n(H) = 2\alpha(H) + 5, \qquad \nu(H) = \alpha(H) + 2.$$

Hence

$$\rho(H) = 3 - \alpha(H).$$

The excess 2 gives $n(H) \ge 11$, so $\alpha(H) \ge 3$. Thus $\rho(H) \le 0$, contrary to $\rho(H) = 2$. This finishes the proof of (II).

Finally, if Q is factor-critical with slack 1 and the hereditary gap bound, then $n(Q) = 2\nu(Q) + 1$. Substitution in (II) gives

$$\nu(Q) + 1 \le 2\alpha(Q) + 1,$$

and hence $\nu(Q) \le 2\alpha(Q)$. □

11.6. **The remaining quartic profile.** In this section, we use the earlier quartic exchange structure to exclude profile (F), leaving the stated possibility (E).

Proposition 11.33 (Exclusion of the second quartic profile)**.** *Let H be a closed quartic graph with*

$$\mathrm{ex}(H) = 3, \qquad \mathrm{gap}(H) \geq 1, \qquad \mathrm{tr}(H) \geq 2, \qquad k = \nu(H).$$

(i) *The costs of H at levels $0, 1, 2, 3, 4$ are not*

$$(\mathrm{F}): \qquad (k+3, k+3, k, k-4, 1).$$

(ii) *Consequently its profile is*

$$(\mathrm{E}): \qquad (k+3, k+2, k, k-3, 1), \qquad 11 \leq k \leq 14.$$

It is well-covered with independence number $k-3$. For every maximal independent set S, the graph $G = H - S$ is subcubic and satisfies

$$\mathrm{ps}(G) = 0, \qquad c_3^*(G) = 0,$$

$$\big(\nu(G), e(G), c_1^*(G), c_2^*(G)\big) = (6, 14, 5, 3).$$

The cubic vertices of G have independence number at most 2.

No exclusion of profile (E) *is asserted.*

Proof. Assume H has profile (F), and put

$$a = k - 4 = \alpha(H).$$

Use Setup 11.18 to choose a maximal independent set S such that $G = H - S$ consists of two components Q_1, Q_2, each of type D_6 or diamond, two triangles T_1, T_2, one P_3, and isolated vertices. Let B and the parent pairs P_t be as in that setup. The pairs avoid all unique S-parents of cubic centres in Q_1, Q_2. Within each triangle they are equal or disjoint, and pairs from different triangles are disjoint. Every distinct pair is the S-neighbourhood of a vertex in B. Hence there are at most $|B| \leq 5$ distinct pairs.

Step 1. Every cubic vertex in an independent cubic pair has a cubic neighbour. Let S' be any maximal independent set of H, and let $\{u, v\}$ be any independent pair of cubic vertices of $H - S'$. Put

$$R = H - (S' \cup \{u, v\}).$$

The pair-frame result in Setup 11.18(V) makes u an endpoint of one of the designated W-edges and puts the centre z of a P_3 component of R in its R-neighbourhood. The vertex z has degree 2 in R and is adjacent to u in $H - S'$. Since $H - S'$ is subcubic, z has degree 3 there. Thus u has a cubic neighbour. The assertion holds for every cubic vertex that belongs to an independent cubic pair.

Step 2. Exactly two equal parent pairs in one triangle are impossible. Fix a triangle T with vertices t, b, c. Suppose, after relabelling, that

$$P_t = P_b = \{p, q\}, \qquad P_c \neq \{p, q\}.$$

The pair condition makes P_c disjoint from $\{p, q\}$. Choose $x \in B$ with $N_H(x) \cap S = \{p, q\}$. The set

$$S' = (S \setminus \{p, q\}) \cup \{t, x\}$$

is independent. Both t and x have no S-neighbour outside $\{p, q\}$, and they belong to different components of G. Its size is a, so it is a maximum independent set.

Consider $G' = H - S'$. The remaining neighbours of b are precisely c, p, q, so b is cubic. The vertex c has degree 1: its triangle neighbour t was selected, and its two S-neighbours remain selected because $P_c \cap \{p, q\} = \varnothing$. Only b remains adjacent to it in G'.

Each of p, q has degree 2 in G'. Each is adjacent in H to the three distinct vertices t, b, x, and degree 4 gives exactly one further neighbour y. Because S was independent, $y \in V(G)$. It is distinct

from t, b, x, and therefore was not selected in S'. After deleting t, x, the remaining neighbours are b, y. Hence none of the three G'-neighbours of b is cubic.

There is nevertheless an independent cubic partner for b. The vertex x is either the P_3-centre or an outer vertex of one of the Q_i. Choose a Q_i not containing x. Its cubic centres retain degree 3 in G': neither t nor x has a neighbour in that component, and the unique S-parents of its centres avoid $\{p, q\}$, so remain selected. Choose one such centre v. It is nonadjacent to b, whose G'-neighbours are c, p, q. Thus $\{b, v\}$ is an independent cubic pair. Step 1 gives a cubic neighbour of b, contradicting the degree counts.

Step 3. Three equal parent pairs in one triangle are impossible. Suppose all three pairs of T equal $\{p, q\}$. Again choose $x \in B$ with $N_H(x) \cap S = \{p, q\}$. Each of p, q is adjacent to all three vertices of T and to x. These four distinct vertices exhaust its neighbourhood in the quartic graph H.

Put

$$M = V(T) \cup \{p, q\}.$$

The induced graph $H[M]$ is K_5 with the edge pq removed, because S is independent. Its only edges to the complement are px and qx. The vertex x has degree 2 in G, as every vertex of B does. It has a G-neighbour $y \notin M$: its component is distinct from T, and its G-neighbours also lie outside S.

Let $R_0 = H - M$, and extend $\{y\}$ to a maximal independent set J of R_0. Since x is adjacent to y, we have $x \notin J$. The set $J \cup \{t\}$ is independent in H, because every H-neighbour of t lies in M. It is maximal: J dominates R_0, and t dominates M. Well-coveredness of H gives

$$|J| + 1 = a.$$

But $J \cup \{p, q\}$ is also independent. The vertices p, q are nonadjacent, and their only neighbour outside M is $x \notin J$. Its size is

$$|J| + 2 = a + 1,$$

contrary to $\alpha(H) = a$.

Step 4. Counting distinct pairs excludes (F). Within a triangle, every two parent pairs are equal or disjoint. Steps 2 and 3 exclude both possible patterns with a repetition. Thus each triangle has three distinct pairs. Pairs from different triangles are disjoint, so the six pairs on $T_1 \cup T_2$ are all distinct. This contradicts the earlier bound of at most $|B| \leq 5$ distinct pairs. Part (I) follows.

The earlier classification in Setup 11.18 gives exactly profiles (D), (E), and (F), with the stated uniform complement data. The earlier exclusion removes (D), and part (I) removes (F). Hence only (E) remains. Its bounds on k, well-coveredness, independence number, and complement properties are precisely the (E) conclusions recorded in that setup. □

11.7. **Closure of degree fourteen.** In this section, we combine the earlier degree-fourteen interval bounds and tail bound to restrict minimal induced T_- failures to degrees at most 12.

Theorem 11.34 (Degree fourteen is closed for T_-)**.** *The following statements hold.*

(i) *No minimal induced T_- failure has degree* 14.

(ii) *Every graph violating T_- contains an induced minimal failure F that is a totally critical closed d-lump of even degree*

$$d \in \{4, 6, 8, 10, 12\},$$

with

$$\mathrm{gap}(F) = 1, \qquad \mathrm{tr}(F) + \mathrm{ex}(F) \geq h_-(d) + 1,$$

and with every proper induced subgraph satisfying T_-.

The inherited term "totally critical" is subject to the definition blocker in Setup 11.20.

Proof. Let F be a minimal induced T_- failure of degree 14. Setup 11.20(I) makes F a totally critical closed 14-lump with $\operatorname{gap}(F) = 1$. Write

$$n(F) = 2k + 1, \qquad k \in \mathbb{Z}_{\geq 0}.$$

If $k \leq 12$, then

$$n(F) \leq 25 = 2 \cdot 14 - 3.$$

The small-order result in Setup 11.21(I) gives LP-goodness, contrary to $\operatorname{gap}(F) = 1$.

For $13 \leq k \leq 147$, the nine distinct earlier interval exclusions in Setup 11.21(II) apply in the following order:

range of k	earlier interval exclusion
$13 \leq k \leq 42$	first
$43 \leq k \leq 55$	second
$56 \leq k \leq 62$	third
$63 \leq k \leq 76$	fourth
$77 \leq k \leq 88$	fifth
$89 \leq k \leq 102$	sixth
$103 \leq k \leq 118$	seventh
$119 \leq k \leq 134$	eighth
$135 \leq k \leq 147$	ninth

Each excludes the assumed failure. If $k \geq 148$, the tail bound in Setup 11.21(III), which gives $k \leq 147$, is contradicted. These cases cover every integer $k \geq 0$, proving (I).

Let G violate T_-. By Setup 11.20(I), it contains an induced totally critical closed d-lump F, for an even integer $4 \leq d \leq 24$, such that

$$\operatorname{gap}(F) = 1, \qquad \operatorname{tr}(F) + \operatorname{ex}(F) \geq h_-(d) + 1,$$

and every proper induced subgraph satisfies T_-. The displayed inequality makes F itself violate T_-, so it is a minimal induced failure.

Setup 11.20(II) restricts its even degree to $4 \leq d \leq 16$. This uses the earlier exclusion of degree 18, together with LP-goodness of closed lumps of degrees 20, 22, and 24, which is incompatible with $\operatorname{gap}(F) = 1$. The earlier degree-16 exclusion removes $d = 16$. Part (I) removes $d = 14$. Therefore

$$d \in \{4, 6, 8, 10, 12\},$$

as required. □

12. The T_- criterion and induced-minimal failures

4.1. **The criterion and its minimal failures.** We use the certificate framework, degree splitting, and closed-lump results of the earlier development to study failures of the T_- criterion. The setup records the prerequisites, and the numbered theorems give the reductions and necessary conditions needed subsequently; the restricted top-run assertion remains a conditional input.

4.1.1. *Setup.* All graphs in this chapter are finite and simple. We assume, as established earlier in this series, the certificate framework of Chapter 1, the degree-splitting and matching results of Chapter 3, and the closed-lump, top-run, and covered-deletion results of Chapter 4. The more specialized earlier results are listed below and recalled in their required forms where they enter the proofs.

[author/blocker] No author, affiliation, or email was supplied. [addressed-partial: Existing authorship placeholders retained | deferred: Author, affiliation, and email require operator input.] [metadata/blocker] The subject classification and keywords require operator input. [addressed-partial: Existing classification and keyword placeholders retained | deferred: Their values require operator input.] [math/blocker] [GAP: Full statements and locations of the granular prerequisites were not supplied. The assumptions below record their applications in the supplied proofs. Their complete hypotheses and their availability in the

earlier chapters must be checked. In particular, the earlier definitions of the prescribed-set count q_W, the matching-aware quantity β_1, and economical peeling with a cutoff must be supplied.] [out-of-scope: thm-body-math; env=setup; label=setup:basic] [cite/blocker] [GAP: Companion I theorem numbers and its bibliographic entry were not supplied. References to its earlier results below require the corresponding [I, Theorem X] locations before insertion.] [deferred: to reference_verify] [instruction/conflict] The complete supplied proofs take precedence over the requested page target. The numerical specializations in Theorem 4.17 are retained because they are part of the supplied mathematical statements, despite the instruction to omit numerical examples. [acknowledged: The complete proof arguments and the numerical specializations are retained.] [prime/blocker] Compilation and reference resolution have not been tested in this text-only delivery. [acknowledged: Compilation and reference resolution remain pending at the external compile gate; no compile result is claimed.]

Notation 4.1. For a graph J, write $e(J) = |E(J)|$, $\nu(J)$ for its maximum matching size, $\alpha(J)$ for its independence number, and $\tau(J)$ for its minimum vertex-cover size. Its maximum and minimum degrees are $\Delta(J)$ and $\delta(J)$. We take $\Delta(\varnothing) = 0$. Vertex deletion removes all incident edges. A subgraph may be obtained by deleting vertices, edges, or both; it is proper if it differs from the original graph.

All degree capacities, matching budgets, thresholds, orders, cardinalities, and deletion counts are integers. Unless a stronger restriction is stated, they are nonnegative. In particular,

$$A, B, x, y, \theta, \ell \in \mathbb{Z}_{\geq 0}.$$

Intervals used as allocation sets are integer intervals. Disjoint unions of graphs are denoted by $\sqcup$.

Definition 4.2 (Certificates and allocation runs)**.** An $(A, B; x, y)$-certificate of J is an edge partition

$$E(J) = Q_R \sqcup Q_B \sqcup P_R \sqcup P_B$$

such that

$$\Delta(Q_R) \leq A, \qquad \Delta(Q_B) \leq B, \qquad \nu(P_R) \leq x, \qquad \nu(P_B) \leq y.$$

Here an edge set is viewed as a spanning subgraph when its degree or matching number is taken. The cost $c_{A,B}(J)$ is the least integer $m \geq 0$ for which every pair $x, y \geq 0$ with $x + y = m$ has such a certificate. Put

$$c^*_\theta(J) = \max_{A+B=\theta} c_{A,B}(J), \qquad c^*_0(J) = c_{0,0}(J),$$

and

$$\Sigma(J) = \sum_{\theta \geq 0} c^*_\theta(J), \qquad \operatorname{ex}(J) = c_{0,0}(J) - \nu(J),$$

$$\operatorname{ps}(J) = e(J) - \nu(J) - \sum_{\theta \geq 1} c^*_\theta(J), \qquad \operatorname{gap}(J) = \Sigma(J) - e(J) = \operatorname{ex}(J) - \operatorname{ps}(J).$$

A graph is *LP-good* if $\Sigma(J) \leq e(J)$.

For $\ell \geq 0$, define

$$\mathcal{Y}_\ell(J) = \left\{ x \in [0, \ell] \cap \mathbb{Z} : \begin{array}{ll} E(J) = P_R \sqcup P_B, & \nu(P_R) \leq x, \\ \nu(P_B) \leq \ell - x & \end{array} \right\}.$$

When $c_{0,0}(J) \geq 1$, the *top run* $\operatorname{tr}(J)$ is the longest run of consecutive integers missing from $\mathcal{Y}_{c_{0,0}(J)-1}(J)$. A set with no missing integer has missing-run length zero. Every occurrence of $\operatorname{tr}(J)$ below has $c_{0,0}(J) \geq 1$.

Definition 4.3 (The criterion and minimality)**.** For $D \in \mathbb{Z}_{\geq 0}$, put

$$h_-(D) = \max\left(4, \left\lfloor \frac{D}{2} \right\rfloor + 1\right).$$

A graph J satisfies T_- if

$$\operatorname{gap}(J) \geq 1 \quad \Longrightarrow \quad \operatorname{gap}(J) = 1 \quad \text{and} \quad \operatorname{tr}(J) + \operatorname{ex}(J) \leq h_-(\Delta(J)).$$

For an integer $B \geq 4$, the property T_B has the same implication with the constant bound B. We also use

$$T_0: \quad \operatorname{gap}(J) \geq 1 \Longrightarrow \left[\operatorname{gap}(J) = 1, \quad \operatorname{tr}(J) + \operatorname{ex}(J) \leq \max\left(4, \left\lceil \frac{\Delta(J)}{2} \right\rceil + 1\right)\right].$$

Thus the notation T_0 denotes this degree-dependent property, not the constant-bound property with $B = 0$. The property (S) is T_4.

An *induced-minimal T_- failure* violates T_-, while all its proper induced subgraphs satisfy T_-. A *subgraph-minimal T_- failure* violates T_-, while all its proper subgraphs satisfy T_-.

A graph has the *hereditary gap bound* if every induced subgraph, including itself, has gap at most one. The potential used below is

$$\Phi(J) = e(J) - \nu(J).$$

Definition 4.4 (Covered graphs and closed lumps)**.** A graph is *covered* if some matching covers every vertex of maximum degree. For an integer $D \geq 1$, it is *D-good* if its maximum degree is at most D and some matching covers every vertex of degree D.

A *closed d-lump* is a connected d-regular graph of odd order that is factor-critical: deleting any vertex leaves a perfect matching. A positive-gap graph is *totally critical* if deleting any maximum-degree vertex strictly lowers c^*_θ for every integer $1 \leq \theta \leq \Delta$. For a regular graph this condition concerns every vertex.

Setup 4.5 (Certificate and matching prerequisites)**.** We assume the following earlier results.

(i) Certificate costs are monotone under taking subgraphs. Reinserting one vertex increases each $c_{A,B}$, and each c^*_θ, by at most one. The profile c^*_θ is nonincreasing in θ. The costs vanish when $\theta > \Delta$, and also at $\theta = D$ on a D-good graph. Moreover,

$$c_{0,0}(J) \leq \tau(J), \qquad \operatorname{ps}(J) \geq 0,$$

and slack is superadditive over components. All costs and the gap vanish on an edgeless graph.

(ii) Let $D \geq 1$ and $\Delta(J) \leq D$. A matching covering every degree-D vertex yields a red-blue edge partition with degree capacities A, B for every $A + B = D$.

(iii) Positive gap implies positive excess, an edge, and $\operatorname{tr}(J) \geq 1$. If J is covered and w has maximum degree, then

$$\operatorname{gap}(J - w) \geq \operatorname{gap}(J).$$

If both gaps are positive and equal, then

$$c_{0,0}(J - w) = c_{0,0}(J) - 1, \qquad \operatorname{tr}(J - w) \geq \operatorname{tr}(J) + 1,$$

and $c_{0,0}(J - w) \geq 1$. For a covered maximum-degree deletion, with $h = \nu(J) - \nu(J - w)$,

$$\operatorname{ps}(J) \geq \operatorname{ps}(J - w) + 1 - h.$$

(iv) A graph with hereditary gap bound, zero slack, and maximum degree at least four is covered. For every zero-slack graph with hereditary gap bound, the allocation set $\mathcal{Y}_{\nu(J)}(J)$ has largest missing run at most two. The gluing and absorption rule allows a vertex-pool interval of length at least two to absorb such components at their matching budgets.

(v) For a maximum-cardinality set X attaining the ordinary matching-deficiency formula, all components Q of $J - X$ are odd and factor-critical, and

$$\nu(J) = |X| + \sum_Q \frac{|V(Q)| - 1}{2}.$$

If $r = |V(J)| - 2\nu(J)$, their number is $|X| + r$. We also assume the minimum-degree bound

$$\nu(J) \geq \min\left(\delta(J), \left\lfloor \frac{|V(J)|}{2} \right\rfloor\right)$$

and its earlier equality-case consequence used below: a graph of order $2r+2$, matching number r, and minimum degree at least r either has an independent set of size $r+2$, or belongs to the alternative that requires r even.

(vi) If $\Delta(J) \leq D$, there are at most $2(D+2)-1$ degree-D vertices, and no odd D-regular whole component occurs, then a matching covers all degree-D vertices. The excluded component cannot occur when D is odd.

(vii) For an uncovered graph, the prescribed-set matching criterion supplies a maximum-cardinality attaining set X with

$$q_W(J - X) - |X| = \kappa \geq 1,$$

where W is the maximum-degree vertex set and κ is its minimum unmatched count. The prescribed-set count vanishes on a connected even-order graph when $X = \varnothing$. We assume the exposure-transfer result in the precise form used in Theorem 4.16: if $X \neq \varnothing$, every residual component has gap at most one, and the generic nonincreasing profile is used, then even maximum degree forces $\mathrm{gap}(J) \leq 0$; odd maximum degree supplies a residual component Z with

$$\mathrm{gap}(Z) = 1, \quad \mathrm{ex}(Z) = \mathrm{ex}(J), \quad \mathrm{tr}(Z) \geq \mathrm{tr}(J) + |X|.$$

(viii) Let J be connected of order $2k+1$, with $\nu(J) = k$, $\Delta(J) \leq D$, $D \geq 3$, and $E = \mathrm{ex}(J) \geq 1$. Put

$$u = \max\left(0, \left\lfloor \frac{2E-3}{D-2} \right\rfloor\right).$$

Every missing top allocation lies in

$$[\,2E - 1 + u,\ k - E - u\,].$$

In particular,

$$\mathrm{tr}(J) \leq k - 3E + 2 - 2u.$$

More generally, for every integer B,

$$\mathrm{tr}(J) + E \geq B + 1 \quad \Longrightarrow \quad k \geq B + 2E - 1 + 2u.$$

[cite/blocker] The supplied provenance for the ordinary matching decomposition is [QLS+19b, Theorem 1.6]. Its bibliographic metadata and theorem attribution remain unverified. The proofs use the explicitly assumed earlier-series formulation in Setup 4.5. [deferred: to reference_verify]

Setup 4.6 (Closed-lump and residual prerequisites)**.** The following statements are also assumed from the earlier development.

(i) Complete regular graphs are LP-good under the earlier small-cost criterion. For a noncomplete closed d-lump and a maximal independent set S, write $a = |S|$ and $G = H - S$. Then

$$a \geq 2, \quad \Delta(G) \leq d - 1, \quad c_d^*(H) = 1,$$

and all higher costs vanish. If a component P of G is called big when $\lfloor |V(P)|/2 \rfloor \geq a + 2$, then

$$c_{0,0}(H) \leq a + \nu(G) + m, \qquad m = \max\big(\{0\} \cup \{\mathrm{ex}(P) : P \text{ big}\}\big).$$

(ii) For a totally critical closed d-lump, put $r_v = c_{0,0}(H) - c_{0,0}(H - v)$. The exact deletion ledger gives $r_v \in \{0, 1\}$, a drop of one at every positive threshold through d, and

$$\mathrm{gap}(H - v) = \mathrm{gap}(H) - r_v, \qquad \mathrm{ex}(H - v) = \mathrm{ex}(H) - r_v.$$

(iii) A closed 6- or 8-lump has slack at least one. At excess two and positive gap it has

$$\operatorname{gap}(H) = \operatorname{ps}(H) = 1, \qquad s = \alpha(H) \geq 3.$$

For every maximum independent set S, with $|V(H)| = 2k+1$ and $G = H - S$,

$$\operatorname{ps}(G) = 0, \qquad \nu(G) = k + 2 - s, \qquad c^*_{d-1}(H) = s + c^*_{d-1}(G).$$

Every graph of maximum degree at most five has gap at most one.

(iv) A zero-slack factor-critical graph of order at least three is one of C_3, C_5, C_7. In the zero-slack classification with matching deficiency equal to independence number, the graph is a disjoint union of K_1 and K_3. For the odd factor-critical residual components used below, the earlier small-independence estimate gives

$$\nu(Q) \leq 2\alpha(Q) + \operatorname{ps}(Q) - 1.$$

The strict nonregular factor-critical bound, in the degree-seven, slack-one case used below, is

$$e(Q) \leq 3\nu(Q) + 4\alpha(Q) + 6.$$

The other residual bounds used below are

$$\alpha(Q) \leq 3 \Longrightarrow c_{0,0}(Q) \leq \left\lceil \frac{2|V(Q)|}{3} \right\rceil - 1, \qquad \operatorname{ps}(Q) \geq 2\big(|V(Q)| - c_{0,0}(Q)\big) - 4\alpha(Q).$$

(v) For a closed 6-lump of excess two and positive gap,

$$|V(H)| = 2s + 5, \qquad \Delta(H - S) = 5,$$

and every maximum-independent-set complement has at most three degree-five vertices. Two maximum-degree deletions of degrees $5, 4$ leave either two triangles and $s-3$ singletons, or a graph with a degree-three vertex whose deletion leaves a triangle and $s-1$ singletons. The deleted side together with S has a three-edge core C_3, P_4, or $P_3 \sqcup K_2$. The triangle exchange used in Theorem 4.12 permits the two remaining triangle vertices to be the two maximum-degree deletions. The two-triangle terminal form contradicts $c^*_5(H) = s$.

(vi) For the hypothetical closed 6-lump in Theorem 4.14, the earlier rigidity results give well-coveredness,

$$|V(H)| = 2s + 9, \quad s \geq 5, \quad \nu(H - S) = 7, \quad \operatorname{ps}(H - S) = 0$$

for every maximum independent set S. Every maximum-degree deletion process has list $(5, 5, 4, 4, 3)$, preserves zero slack, lowers matching number by one at each deletion, and ends in two triangles and isolates. The central partition and the universal private-edge statement recalled in that proof are assumed as part of the same earlier rigidity result.

(vii) For a closed 8-lump of gap one and excess three, with all proper induced subgraphs satisfying T_-, write $|V(H)| = 2k+1$ and $a = \alpha(H)$. The earlier order bounds give

$$k \geq 8, \qquad k \leq 2a + 2.$$

For a maximum-independent-set complement, the barrier-rank bounds used below are

$$\eta = 0: \quad 2a + 9b \leq 9 \quad \text{and some residual component has order at least } 2a + 2b + 5,$$

$$\eta = 1: \quad 2a + 9b \leq 23 \quad \text{and some residual component has matching number at least } a + b + 2.$$

If every such complement has zero slack and matching number $k + 3 - a$, the earlier zero-slack density bound gives $\nu(H - S) \leq 7$.

(viii) The degree-seven domination certificates for a closed 8-lump of order $2k+1$ give

$$c_7^*(H) \le 1 + 2k\big(p + (1-p)^8\big) \quad \text{at } p = \tfrac{1}{4}, \qquad c_7^*(H) \le \beta_1(H).$$

The ordinary domination recurrence, its dense-certificate consequence when the undeleted order is at most 18, and the matching-aware recurrence

$$r_0 = 2k - 8, \qquad r_{i+1} = \max\left(0, r_i - \left\lceil \frac{8r_i}{2k-i} \right\rceil\right)$$

are assumed in the forms evaluated in Theorem 4.13.

Setup 4.7 (Earlier reductions and independence estimates)**.** We assume the following remaining prerequisites.

(i) Every T_- violation contains an induced-minimal violation that is a totally critical closed d-lump of even degree $4 \le d \le 24$, with gap in $\{1, 2\}$. For its gap-two case, the independent-set ledger and numerical bounds are precisely those recalled in (4.19)–(4.21). A hereditary T_- graph of excess three has top run at most five.

(ii) Universal T_- is equivalent, by the earlier reduction, to universal T_0. It implies LP-goodness whenever

$$\operatorname{ex}(C) \ge \max\left(4, \left\lceil \frac{\Delta(C)}{2} \right\rceil + 1\right).$$

We assume the earlier size-Ramsey implication in the following form: if every proper-subgraph-minimal failure of the restricted criterion at degrees $4, 6, 8, 10, 12$ is excluded, then the full size-Ramsey star-forest formula follows. We also assume the hereditary constant-bound consequence: a graph all of whose induced subgraphs satisfy T_- has excess at most three whenever its gap is positive.

(iii) If $E = \operatorname{ex}(H) \ge 4$, $4\alpha(H) \le 3E + 4$, and H is not LP-good, the earlier exceptional-case reduction forces

$$|V(H)| = 23, \qquad E = 4, \qquad \alpha(H) = 4.$$

The exceptional structural result gives connectedness, 8-regularity,

$$e(H) = 92, \quad c_{0,0}(H) = 15, \quad \nu(H) = 11, \quad \operatorname{ps}(H) = 3,$$

and an induced cutoff-15 residual $4K_2$ for every economical maximum-degree peeling. Such a peeling exists. Every induced 16-vertex subgraph has a perfect matching; the earlier matching-extension criterion with order 23 and deletion target 16 then gives factor-criticality.

(iv) An uncovered graph of order 19, maximum degree 7, and independence number at most four, in the exceptional reduction just described, has

$$e(G) = 7 \cdot 9 + \frac{h}{2}, \qquad h \in \{2, 4, 6\}.$$

For the resulting graph A of order 15, independence number at most four, maximum degree at most six, and 32 edges, the terminal bound is

$$\nu(A) + \operatorname{ps}(A) \ge 8.$$

The independent-set surplus inequalities used with this reduction are

$$\nu(H) + \operatorname{ps}(H) \ge |I| - 1 + \nu(H - I) + \operatorname{ps}(H - I)$$

for the regular factor-critical parent, and

$$\nu(G) + \operatorname{ps}(G) \ge |S| + \nu(G - S) + \operatorname{ps}(G - S)$$

for a covered graph G and an independent set S of maximum-degree vertices.

(v) For a closed d-lump the top cost satisfies $c_d^*(H) \le 1$.

4.1.2. *The deletion potential.* We first relate failure of degree splitting to a maximum-degree deletion.

Theorem 4.8 (Potential drop under splitting failure)**.** *For every finite simple graph H,*

$$\Phi(H) = e(H) - \nu(H) \geq 0, \qquad \Phi(H - v) \leq \Phi(H) \quad (v \in V(H)).$$

Let $A, B \in \mathbb{Z}_{\geq 0}$. If H has no red-blue edge coloring with red maximum degree at most A and blue maximum degree at most B, then some maximum-degree vertex v satisfies

$$\Phi(H) - \Phi(H - v) \geq A + B.$$

More precisely, if $\Delta(H) \geq A + B + 1$, every maximum-degree vertex has this property. Otherwise, $\Delta(H) = A + B \geq 1$, and one can choose a maximum-degree vertex v uncovered by a maximum matching, so that

$$\nu(H - v) = \nu(H), \qquad \Phi(H) - \Phi(H - v) = A + B.$$

Proof. A matching is an edge subset, so $\nu(H) \leq e(H)$. Every matching of $H - v$ is a matching of H. Deleting the possible edge at v from a maximum matching of H leaves at least $\nu(H) - 1$ edges. Hence

$$\nu(H) - \nu(H - v) \in \{0, 1\}.$$

For an isolated vertex this difference is zero. Since vertex deletion removes exactly $d_H(v)$ edges,

$$\Phi(H) - \Phi(H - v) = d_H(v) - \big(\nu(H) - \nu(H - v)\big). \tag{4.1}$$

Both terms vanish when $d_H(v) = 0$. Otherwise the matching-number difference is at most $1 \leq d_H(v)$. This proves monotonicity.

Put $D = A + B$. If $\Delta(H) \geq D + 1$, then every maximum-degree vertex satisfies, by (4.1),

$$\Phi(H) - \Phi(H - v) \geq \Delta(H) - 1 \geq D.$$

It remains to consider $\Delta(H) \leq D$. If $D = 0$, then H is edgeless and its empty coloring satisfies both capacities. Thus $D \geq 1$.

By Setup 4.5(ii), a matching covering every degree-D vertex would give the required coloring. No such matching exists. The degree-D set is nonempty, since otherwise the empty matching would cover it. Therefore $\Delta(H) = D$.

Choose a maximum matching M. It leaves some degree-D vertex v uncovered. Every edge of M remains in $H - v$, so

$$\nu(H - v) \geq |M| = \nu(H).$$

The reverse inequality follows from subgraph containment. Formula (4.1) now gives the exact drop D. □

4.1.3. *Perfect matchings and exact independent-set profiles.* The next two results turn covered deletions and long top runs into exact certificate identities.

Theorem 4.9 (Perfect-matching descendants and full criticality)**.** *Fix an integer $B \geq 4$.*

(I) Let J have a perfect matching and $\operatorname{gap}(J) \geq 1$. Assume every proper induced subgraph of J satisfies T_B. Then

$$\operatorname{gap}(J) = 1, \qquad \operatorname{tr}(J) + \operatorname{ex}(J) \leq B - 1, \qquad \operatorname{ex}(J) \leq B - 2.$$

(II) Let H be a totally critical closed d-lump, where $d \geq 2$ is even. Assume $\operatorname{gap}(H) \geq 1$, and every proper induced subgraph satisfies T_B. Then $\operatorname{gap}(H) \leq 2$. If $\operatorname{gap}(H) = 2$, then $\operatorname{ex}(H) \leq B-1$. If $\operatorname{gap}(H) = 1$ and $\operatorname{ex}(H) \geq B - 1$, then, for every vertex v and every integer $0 \leq \theta \leq d$,

$$c^*_\theta(H - v) = c^*_\theta(H) - 1, \qquad \operatorname{ex}(H - v) = \operatorname{ex}(H) - 1, \qquad \operatorname{gap}(H - v) = 0.$$

In particular, if $\operatorname{ex}(H) \geq B$, all these criticality conclusions hold and $\operatorname{gap}(H) = 1$.

Proof. For (I), the identity gap $=$ ex $-$ ps, with ps ≥ 0, gives $\mathrm{ex}(J) \geq 1$. The graph has an edge because an edgeless graph has gap zero. Let w have maximum degree. A perfect matching covers every maximum-degree vertex, so Setup 4.5(iii) gives

$$\mathrm{gap}(J-w) \geq \mathrm{gap}(J) \geq 1.$$

The property T_B on $J-w$ makes its gap one. Thus

$$\mathrm{gap}(J-w) = \mathrm{gap}(J) = 1,$$

and the equality clause gives

$$c_{0,0}(J-w) = c_{0,0}(J) - 1, \qquad \mathrm{tr}(J-w) \geq \mathrm{tr}(J) + 1.$$

Write $|V(J)| = 2q$. The perfect matching has q edges. Removing its edge at w leaves a matching of size $q-1$, and the order $2q-1$ bounds every matching of $J-w$ by $q-1$. Hence

$$\nu(J-w) = \nu(J) - 1, \qquad \mathrm{ex}(J-w) = \mathrm{ex}(J).$$

Applying T_B to $J-w$ yields

$$\mathrm{tr}(J) + 1 + \mathrm{ex}(J) \leq \mathrm{tr}(J-w) + \mathrm{ex}(J-w) \leq B.$$

Since $\mathrm{tr}(J) \geq 1$, the excess is at most $B-2$. All runs here are defined: positive gap gives $c_{0,0}(J) \geq 1$, and the covered-deletion statement gives $c_{0,0}(J-w) \geq 1$.

For (II), fix v and put

$$r_v = c_{0,0}(H) - c_{0,0}(H-v) \in \{0,1\}.$$

Total criticality and the one-vertex cost bound give a drop of one at every threshold $1, \ldots, d$. Write $|V(H)| = 2k+1$. The perfect matching of $H-v$ has size k, so $\nu(H-v) = \nu(H) = k$. There are no non-rising positive thresholds and no matching-number loss. The exact ledger in Setup 4.6(ii) gives

$$\mathrm{gap}(H-v) = \mathrm{gap}(H) - r_v, \qquad \mathrm{ex}(H-v) = \mathrm{ex}(H) - r_v. \tag{4.2}$$

The property T_B implies $\mathrm{gap}(H-v) \leq 1$, whether its gap is positive or not. Since $r_v \leq 1$, we obtain $\mathrm{gap}(H) \leq 2$.

If $\mathrm{gap}(H) = 2$, then (4.2) forces $r_v = 1$ and $\mathrm{gap}(H-v) = 1$ for every v. The graph $H-v$ has a perfect matching, and all its proper induced subgraphs satisfy T_B. Part (I) gives

$$\mathrm{ex}(H-v) \leq B-2.$$

Therefore $\mathrm{ex}(H) = \mathrm{ex}(H-v) + 1 \leq B-1$.

Suppose $\mathrm{gap}(H) = 1$ and $\mathrm{ex}(H) \geq B-1$. If $r_v = 0$ for some vertex, then (4.2) gives

$$\mathrm{gap}(H-v) = 1, \qquad \mathrm{ex}(H-v) = \mathrm{ex}(H) \geq B-1,$$

contrary to part (I). Thus every $r_v = 1$. Together with the positive-threshold drops, this proves the full profile decrease, and (4.2) gives the excess decrease and zero child gap.

Finally, $\mathrm{ex}(H) \geq B$ excludes gap two by the bound already proved. Positive gap is therefore one, and the full-criticality conclusion applies. □

Theorem 4.10 (Independent-set profiles at a long top run)**.** *Let H be a connected d-regular factor-critical graph of odd order, where $d \geq 4$ is even. Assume $\mathrm{gap}(H) = 1$, and every proper induced subgraph satisfies T_-. Write $|V(H)| = 2k+1$ and $E = \mathrm{ex}(H)$. For any maximal independent set S, put*

$$a = |S|, \qquad G = H - S, \qquad \eta = \nu(G) - (k+1-a), \qquad \epsilon = \mathrm{ps}(H) - \mathrm{ps}(G) - \eta.$$

(I) We have

$$a \geq 2, \quad \Delta(G) \leq d-1, \quad 0 \leq \eta \leq \left\lfloor \frac{a-1}{2} \right\rfloor, \quad \epsilon \in \{0,1\}.$$

If $\epsilon = 1$, then

$$\mathrm{tr}(H) + E \leq h_-(d-1) - \left\lfloor \frac{a}{2} \right\rfloor.$$

(II) If $\epsilon = 0$, then

$$\mathrm{ps}(H) = \eta + \mathrm{ps}(G), \qquad c^*_\theta(H) = a + c^*_\theta(G) \quad (1 \le \theta \le d-1), \qquad c^*_d(H) = 1,$$
$$c_{0,0}(H) = a + \nu(G) + \mathrm{ps}(G), \qquad \mathrm{gap}(G) \in \{0,1\}, \qquad c_{0,0}(H) = a + c_{0,0}(G) - \mathrm{gap}(G).$$

(III) If $\mathrm{tr}(H) + E \ge h_-(d-1)$, the conclusions in (II) hold for every maximal independent set S. More generally, they hold for any such S satisfying

$$\mathrm{tr}(H) + E > h_-(d-1) - \left\lfloor \frac{|S|}{2} \right\rfloor.$$

No minimal-failure assumption or upper degree bound is imposed.

Proof. The graph H is not complete. For a complete d-regular graph, $c_{0,0}(H) \le \tau(H) = d$, and the earlier small-cost criterion in Setup 4.6(i) gives LP-goodness, contrary to $\mathrm{gap}(H) = 1$. Thus the independent-set statements in that setup apply to every maximal S:

$$a \ge 2, \qquad \Delta(G) \le d-1, \qquad c^*_d(H) = 1,$$

all higher costs vanish, and

$$c_{0,0}(H) \le a + \nu(G) + m, \tag{4.3}$$

where m is the maximum of zero and the excesses of the big components P of G, with $\lfloor |V(P)|/2 \rfloor \ge a+2$.

Factor-criticality gives $\nu(H) = k$. Choose $v \in S$ and a perfect matching of $H - v$. Each of the $a-1$ vertices of $S \setminus \{v\}$ is matched outside S. Removing these $a-1$ distinct edges leaves a matching of G of size $k+1-a$. Hence $\eta \ge 0$. The order bound gives

$$\nu(G) \le \left\lfloor \frac{2k+1-a}{2} \right\rfloor, \qquad \eta \le \left\lfloor \frac{a-1}{2} \right\rfloor.$$

For $1 \le \theta \le d-1$, put $r_\theta = c^*_\theta(H) - c^*_\theta(G)$. Monotonicity and reinsertion give $0 \le r_\theta \le a$. Since S is independent and H is d-regular, $e(H) - e(G) = da$. All costs of G at thresholds at least d vanish. Subtracting the slack definitions, using $c^*_d(H) = 1$, gives

$$\epsilon = \mathrm{ps}(H) - \mathrm{ps}(G) - \eta = \sum_{\theta=1}^{d-1} (a - r_\theta) \ge 0. \tag{4.4}$$

The identity $\mathrm{gap}(H) = 1$ gives $\mathrm{ps}(H) = E - 1$. From (4.3),

$$E \le 1 + \eta + m, \qquad \epsilon \le m - \mathrm{ps}(G).$$

If $m = 0$, nonnegative slack gives $\epsilon = 0$. If $m \ge 1$, choose a maximizing big component P^*. It is proper induced in H, so $\mathrm{gap}(P^*) \le 1$, and $\mathrm{ps}(P^*) \ge m-1$. Slack superadditivity gives $\mathrm{ps}(G) \ge m-1$. Consequently $\epsilon \le 1$, proving $\epsilon \in \{0,1\}$.

Suppose $\epsilon = 1$. Equality holds throughout these bounds:

$$\mathrm{ps}(G) = m-1, \quad E = 1+\eta+m, \quad \mathrm{ps}(P^*) = m-1, \quad \mathrm{gap}(P^*) = 1,$$

and every other component Q of G has $\mathrm{ps}(Q) = 0$. Also $c_{0,0}(H) = a + \nu(G) + m$. Each induced subgraph of such a Q is proper induced in H, and hence has gap at most one. By Setup 4.5(iv), the largest missing run of $\mathcal{Y}_{\nu(Q)}(Q)$ is at most two.

The vertex-pool interval $[0,a]$ has length at least two, so the gluing and absorption rule absorbs all components other than P^* at their matching budgets. Put

$$L = a + \sum_{Q \ne P^*} \nu(Q).$$

It follows that

$$[0,L] + \mathcal{Y}_{c_{0,0}(P^*)-1}(P^*) \subseteq \mathcal{Y}_{c_{0,0}(H)-1}(H), \tag{4.5}$$

because

$$L + c_{0,0}(P^*) - 1 = a + \nu(G) + m - 1 = c_{0,0}(H) - 1.$$

The allocation set for P^* contains both endpoints, since $c_{0,0}(P^*) - 1 = \nu(P^*) + m - 1 \geq \nu(P^*)$.

Let $Y \subseteq [0, \ell] \cap \mathbb{Z}$ contain both endpoints, and let its largest missing run be r. Writing its successive elements as $0 = y_0 < \cdots < y_b = \ell$, the sumset $[0, L] + Y$ is the union of $[y_i, y_i + L]$. Its consecutive missing gaps have lengths

$$\max(0, y_{i+1} - y_i - L - 1).$$

Thus its largest missing run is $\max(0, r - L)$. The same formula gives zero when $\ell = 0$.

Apply this to (4.5). Positive gap gives $\mathrm{tr}(H) \geq 1$, so

$$\mathrm{tr}(H) + L \leq \mathrm{tr}(P^*).$$

Since P^* satisfies T_-,

$$\mathrm{tr}(P^*) + m \leq h_-(\Delta(P^*)) \leq h_-(d-1).$$

Therefore

$$\begin{aligned} \mathrm{tr}(H) + E &\leq \mathrm{tr}(P^*) - L + 1 + \eta + m \\ &\leq h_-(d-1) - L + 1 + \eta \\ &\leq h_-(d-1) - a + 1 + \left\lfloor \frac{a-1}{2} \right\rfloor \\ &= h_-(d-1) - \left\lfloor \frac{a}{2} \right\rfloor. \end{aligned}$$

This proves (I).

If $\epsilon = 0$, every summand in (4.4) is zero. Hence every positive-threshold equality in (II) holds. The slack equality is the definition of ϵ. Since $\mathrm{ps}(H) = E - 1$,

$$E = 1 + \eta + \mathrm{ps}(G),$$

and consequently

$$c_{0,0}(H) = k + E = a + \nu(G) + \mathrm{ps}(G).$$

Reinsertion gives $c_{0,0}(H) \leq a + c_{0,0}(G)$, whence $\mathrm{ps}(G) \leq \mathrm{ex}(G)$. Thus $\mathrm{gap}(G) \geq 0$. The property T_- on the proper induced graph G gives $\mathrm{gap}(G) \leq 1$. Rearranging now gives $c_{0,0}(H) = a + c_{0,0}(G) - \mathrm{gap}(G)$. The top cost was obtained at the start of the proof.

Finally, $a \geq 2$ implies $\lfloor a/2 \rfloor \geq 1$. Either hypothesis in (III) excludes the $\epsilon = 1$ alternative of (I). Part (II) applies. □

4.1.4. *Zero slack and excess two in degree eight.* We use matching deficiency to bound a zero-slack deletion process and then construct a cheaper degree-seven certificate.

Theorem 4.11 (Zero-slack peeling and the degree-eight excess-two case)**.** *(I) Let G have the hereditary gap bound and $\mathrm{ps}(G) = 0$. Put*

$$a = \alpha(G), \qquad r = |V(G)| - 2\nu(G), \qquad D = \Delta(G).$$

Successively delete an arbitrary maximum-degree vertex whenever the current maximum degree is at least four, and stop when it is at most three. The process stops after $t \leq a - r$ deletions. If T is the deleted set and $B = G - T$, then

$$\mathrm{ps}(B) = 0, \qquad \nu(B) = \nu(G) - t, \qquad \Delta(B) \leq 3.$$

The slack and matching equalities hold at every intermediate step. If $D \geq 4$, then

$$2e(G) \leq 7\nu(G) + (2D - 7)t.$$

Every subcubic graph Q satisfies $2e(Q) \leq 7\nu(Q)$.

(II) Let $d \in \{6, 8\}$, and let H be a closed d-lump with $\mathrm{ex}(H) = 2$, $\mathrm{gap}(H) \geq 1$, and gap at most one on every proper induced subgraph. For every maximum independent set S, the graph $G = H - S$ satisfies

$$\nu(G) \leq 5,$$

and the process in (I) has $t \leq 3$.

(III) Every closed 8-lump H with $\mathrm{ex}(H) \leq 2$ is LP-good if every proper induced subgraph has gap at most one. In particular, an induced-minimal failure of either (S) or T_0 cannot be a closed 8-lump of excess at most two.

Proof. We first prove the subcubic bound. Choose a maximum-cardinality attaining set Y as in Setup 4.5(v), and put $b = |Y|$. For each component C of $Q - Y$, write $\mu_C = (|V(C)| - 1)/2$. Then

$$\nu(Q) = b + \sum_C \mu_C.$$

A singleton has $e(C) = \mu_C = 0$. A component of order three has $\mu_C = 1$ and at most three edges, so $e(C) \leq 7\mu_C/2$. If $\mu_C \geq 2$, then

$$e(C) \leq \left\lfloor \frac{3(2\mu_C + 1)}{2} \right\rfloor = 3\mu_C + 1 \leq \frac{7}{2}\mu_C.$$

The number of edges incident with Y is at most $3b$; edges inside Y are counted twice in the degree sum at Y, which preserves this upper bound. Hence

$$e(Q) \leq 3b + \sum_C e(C) \leq 3b + \frac{7}{2}\sum_C \mu_C \leq \frac{7}{2}\nu(Q).$$

Now consider a stage J of the process in (I), with $\mathrm{ps}(J) = 0$ and $\Delta(J) \geq 4$. The hereditary gap bound passes to J, so J is covered by Setup 4.5(iv). Delete any maximum-degree vertex v, and put $h = \nu(J) - \nu(J - v) \in \{0, 1\}$. The covered-deletion inequality gives

$$0 = \mathrm{ps}(J) \geq \mathrm{ps}(J - v) + 1 - h.$$

Nonnegative slack forces $h = 1$ and $\mathrm{ps}(J - v) = 0$. Induction proves the matching and slack equalities at every stage.

After j deletions, the matching deficiency is

$$(|V(G)| - j) - 2(\nu(G) - j) = r + j.$$

Unmatched vertices of a maximum matching are independent: an edge between two of them could be added to the matching. Thus $r + j$ is at most the independence number of the current graph, which is at most a. Therefore $j \leq a - r$. The process deletes a vertex at each step. It cannot still have maximum degree at least four after $a - r$ deletions, since the next step would contradict this bound. It stops with $t \leq a - r$ and $\Delta(B) \leq 3$. If the empty graph is reached, it also satisfies the asserted equalities.

Every deleted degree is at most D. Consequently

$$e(G) - e(B) \leq Dt.$$

The subcubic bound gives

$$2e(G) \leq 2Dt + 7\nu(B) = 7\nu(G) + (2D - 7)t.$$

This proves (I).

For (II), fix any maximum independent set S, and put $s = |S|$, $k = \nu(H) = (|V(H)| - 1)/2$, and $G = H - S$. Setup 4.6(iii) gives

$$s \geq 3, \qquad \mathrm{ps}(G) = 0, \qquad \nu(G) = k + 2 - s, \qquad r = |V(G)| - 2\nu(G) = s - 3.$$

Every induced subgraph of G has gap at most one, and $\alpha(G) \le s$. Part (I) gives

$$t \le \alpha(G) - r \le 3, \qquad \mathrm{ps}(B) = 0, \qquad \nu(B) = \nu(G) - t, \qquad \Delta(B) \le 3.$$

Since S is maximal, every vertex outside S has an S-neighbor. Thus $\Delta(G) \le d-1$. Counting edges at S gives

$$e(G) = e(H) - ds = \frac{d(2k+1-2s)}{2} = d\nu(G) - \frac{3d}{2}. \tag{4.6}$$

The edge estimate in (I), with $d-1$ bounding every deleted degree, gives, also for $t = 0$,

$$2e(G) \le 7\nu(G) + (2d-9)t.$$

If $d = 6$, then (4.6) and $t \le 3$ give

$$12\nu(G) - 18 \le 7\nu(G) + 9, \qquad 5\nu(G) \le 27.$$

If $d = 8$, they give

$$16\nu(G) - 24 \le 7\nu(G) + 21, \qquad 9\nu(G) \le 45.$$

Integrality implies $\nu(G) \le 5$ in both cases.

For (III), excess at most one gives LP-goodness because a closed 8-lump has $\mathrm{ps}(H) \ge 1$, so $\mathrm{gap}(H) = \mathrm{ex}(H) - \mathrm{ps}(H) \le 0$. Suppose instead that $\mathrm{ex}(H) = 2$ and $\mathrm{gap}(H) \ge 1$. Retain G, s, T, t, B from (II). We have

$$s \ge 3, \quad 0 \le t \le 3, \quad \Delta(B) \le 3, \quad \mathrm{ps}(B) = 0, \quad \nu(B) = \nu(G) - t \le 5 - t,$$

and

$$|V(B)| - 2\nu(B) = s - 3 + t, \qquad \alpha(B) \le s. \tag{4.7}$$

The excess-two identity gives

$$c_7^*(H) = s + c_7^*(G). \tag{4.8}$$

If $\Delta(G) \le 6$, then $c_7^*(G) = 0$. If $\Delta(G) = 7$, its hereditary gap bound and zero slack make it covered, so the same vanishing criterion applies. Thus $c_7^*(H) = s$.

We construct a seven-good remainder by deleting $s-1$ vertices. For $g \in G$, let $N_S(g)$ denote its neighbors in the original set S. We seek $g \in B$ with $|N_S(g)| \ge t+2$, with the additional internal-degree properties specified below. The graph B is nonempty: every vertex of the nonempty set S has eight distinct neighbors in G, so $|V(G)| \ge 8 > t$.

If $t = 0$ or 1, choose any $g \in B$. It has at most three neighbors in B and at most t in T. Hence

$$|N_S(g)| \ge 8 - (3+t) = 5 - t \ge t + 2.$$

Suppose $t = 2$. We show that B has a vertex of degree at most two. Otherwise B is 3-regular. Put

$$r_B = |V(B)| - 2\nu(B) = s - 1 \ge 2, \qquad a_B = \alpha(B) \le s = r_B + 1.$$

Choose a maximum-cardinality attaining set Y in B, and write $b = |Y|$. The number of odd factor-critical components is $p = b + r_B$. Choosing one vertex from each component gives $p \le a_B$, so $b \le 1$.

If $b = 0$, the components are odd-order whole components of a 3-regular graph. Their degree sums would be odd, contradicting the degree-sum identity. Thus $b = 1$ and $p = r_B + 1$. Since $p \le a_B \le r_B + 1$, the sum of component independence numbers is at most p, so each component has independence number one and is a clique. Each vertex outside Y has at most one neighbor in Y and total degree three. The component cliques therefore have order between three and four. Their odd order is three. Every vertex in every triangle must be adjacent to the unique vertex of Y, giving that vertex at least $3p \ge 9$ neighbors. This contradicts maximum degree three. Choose g of degree at most two in B. Then

$$d_G(g) \le 2 + t = 4, \qquad |N_S(g)| \ge 4 = t + 2.$$

Suppose $t = 3$. Now (4.7) gives $r_B = s$. Since $r_B \le \alpha(B) \le s$, the independence number is also s. The zero-slack classification in Setup 4.6(iv) makes B a disjoint union of K_1 and K_3, with exactly s components. Also $\nu(B) \le 2$. If every component were a triangle, then $\nu(B) = s \ge 3$. Thus B has a singleton g. All its G-neighbors lie in T, so

$$|N_S(g)| \ge 5 = t + 2.$$

In every case choose $R \subseteq N_S(g)$ with $|R| = t + 2$, and set

$$U = (S \setminus R) \cup T \cup \{g\}, \qquad J = H - U.$$

The three deleted parts are disjoint, and

$$|U| = s - (t + 2) + t + 1 = s - 1.$$

The retained vertices are $R \cup (V(B) \setminus \{g\})$. Every vertex of R loses its neighbor g, so has degree at most seven. Every vertex outside R has at most $t + 2$ neighbors in R. For $t \le 2$, its internal degree is at most three, giving total degree at most $t + 5 \le 7$. For $t = 3$, the components of B are triangles and singletons, so its internal degree is at most two and its total degree is at most $2 + 5 = 7$. Therefore $\Delta(J) \le 7$.

We bound the number of degree-seven vertices. For $t \le 1$, vertices outside R have degree at most six, leaving at most $|R| \le 3$ degree-seven vertices. For $t = 2$, a degree-seven vertex outside R has degree three in B. Since $\nu(B) \le 3$, the subcubic bound gives

$$e(B) \le \left\lfloor \frac{21}{2} \right\rfloor = 10.$$

The number of degree-three vertices is at most $\lfloor 2e(B)/3 \rfloor \le 6$. Including the four vertices of R gives at most ten. For $t = 3$, only triangle vertices outside R can have degree seven. There are $3\nu(B) \le 6$ of them, so the total is at most $6 + 5 = 11$.

By Setup 4.5(vi), a graph of maximum degree at most seven with at most $2(7 + 2) - 1 = 17$ degree-seven vertices is seven-good. The odd-regular-component exclusion is automatic. Our graph has at most eleven such vertices. Thus $c_7^*(J) = 0$, and reinsertion gives

$$c_7^*(H) \le |U| + c_7^*(J) = s - 1,$$

contrary to (4.8).

An induced-minimal failure of either (S) or T_0 has gap at most one on every proper induced subgraph. The LP-goodness assertion therefore excludes a closed 8-lump of excess at most two from either class. □

4.1.5. *Excess two in degree six.* We next use the universal maximum-independent-set structure to bound the remaining degree-six case.

Theorem 4.12 (Closed degree-six lumps of excess two)**.** *Let H be a closed 6-lump with $\mathrm{ex}(H) = 2$ and $\mathrm{gap}(H) \ge 1$. Then*

$$\mathrm{gap}(H) = \mathrm{ps}(H) = 1, \qquad s = \alpha(H) \in \{3, 4\},$$
$$|V(H)| = 2s + 5 \in \{11, 13\}, \qquad \mathrm{tr}(H) \le s - 2 \le 2.$$

Consequently every closed 6-lump of excess at most two satisfies (S). No minimality or total-criticality hypothesis is required. No LP-goodness assertion is made for the possible orders 11, 13 at excess two.

Proof. The excess-two prerequisites give

$$\mathrm{gap}(H) = \mathrm{ps}(H) = 1, \qquad s = \alpha(H) \ge 3.$$

For every maximum independent set S, with $G = H - S$, they give

$$\mathrm{ps}(G) = 0, \qquad c_5^*(H) = s + c_5^*(G), \qquad |V(H)| = 2s + 5, \qquad \Delta(G) = 5.$$

Every induced subgraph of G has maximum degree at most five, and hence gap at most one. The zero-slack coverage rule therefore gives $c_5^*(G) = 0$. Thus

$$c_5^*(H) = s. \tag{4.9}$$

We prove $s \le 4$ by constructing, if $s \ge 5$, a set Z of size $s-1$ such that $H - Z$ is five-good. The vanishing and reinsertion rules would then give $c_5^*(H) \le s-1$, contradicting (4.9).

Fix a maximum independent set S. The earlier degree-six structure in Setup 4.6(v) supplies two maximum-degree deletions T, of degrees $5, 4$, from $H - S$. The graph on $S \cup T$, of order $s+2$, has a three-edge core C_3, P_4, or $P_3 \sqcup K_2$. The remaining graph B has either two triangles and $s-3$ singletons, or a degree-three vertex y_0 such that $B - y_0$ is a triangle and $s-1$ singletons. The two-triangle alternative contradicts (4.9) by the earlier exclusion in that setup.

In the second alternative, let S' consist of all singletons of $B - y_0$ and one triangle vertex. It is independent of size s, hence maximum. The two remaining triangle vertices are adjacent and each has exactly one neighbor in S'. Both have degree five in $H - S'$. The triangle-exchange statement permits these two vertices to be the maximum-degree deletion set. For this choice, the side formed by S' and the deletions is exactly $V(B - y_0)$. Thus

$$H[U] = C_3 \sqcup (s-1)K_1. \tag{4.10}$$

The complementary side again has one of the two terminal forms. The two-triangle form is again excluded.

Write C for the triangle in (4.10), A for its $s-1$ singletons, and y for the degree-three separator on the other side. We obtain

$$U = C \sqcup A, \qquad V = \{y\} \sqcup D \sqcup I,$$

where D is a triangle, $|I| = s-1$, and $H[V]-y$ consists exactly of that triangle and the independent set I. The degree of y inside $H[V]$ is three. For a vertex v, write $N_A(v)$ and $N_C(v)$ for its neighbors in A and C. Put $r = |N_H(y) \cap D|$. Exactly $3-r$ vertices of I are adjacent to y; call them leaves. The other $s+r-4$ vertices are nonleaves. A leaf has five neighbors in U, and a nonleaf has six. Every I-vertex therefore has at least two A-neighbors. Every nonleaf has at least three, and a nonleaf with exactly three A-neighbors is adjacent to every vertex of C.

We use the following deletion construction in each case. Choose $z \in D \cup I$ and put $P = \{y, z\}$. There are two allowed choices:

(a) Choose $R \subseteq N_A(y) \cup N_A(z)$ of size two, require $N_C(y) \cup N_C(z) = C$, and put $Z = (A \setminus R) \cup P$.
(b) Choose $R \subseteq N_A(y) \cup N_A(z)$ of size three, choose $c \in C$, and put $Z = (A \setminus R) \cup P \cup \{c\}$.

Both choices have $|Z| = s-1$. Require additionally that either $z \in D$, or every vertex of D has a neighbor in $A \setminus R$.

We verify the construction immediately. Every retained A-vertex belongs to R and has a neighbor in P. In (a), every retained C-vertex is dominated by P. In (b), it is dominated by c. If $z \in D$, the other two vertices of D lose their neighbor z. Otherwise all three have a neighbor in $A \setminus R$. Every leaf loses y. A retained nonleaf has at least three A-neighbors. In (a), these cannot all lie in the two-element set R. In (b), if they all lie in R, the nonleaf has exactly three A-neighbors and sees every vertex of C, including c. Otherwise it loses an $A \setminus R$ neighbor. Hence every retained vertex loses a neighbor and $J = H - Z$ has maximum degree at most five.

Exactly five U-vertices remain: three in C and two in A in (a), or two in C and three in A in (b). On the V-side, deleting y, z leaves an edge and singletons, or a triangle and singletons. Its internal maximum degree is at most two. Each degree-five vertex on that side has at least three neighbors among the five retained U-vertices. Their total degree in J is at most 25, so there are at most $\lfloor 25/3 \rfloor = 8$ degree-five vertices on the V-side. There are at most five on the U-side. Thus J has at most thirteen degree-five vertices. Setup 4.5(vi), with $D = 5$, supplies a matching covering them,

since $13 = 2(5+2)-1$ and an odd 5-regular component is impossible. Every allowed construction contradicts (4.9).

Assume $s \geq 5$, put $a = |N_A(y)|$, and let $Y = N_A(y)$. The vertex y has three neighbors on the U-side, so

$$|N_C(y)| = 3 - a, \qquad 0 \leq a \leq 3.$$

First try $z = d$ for each $d \in D$. The additional domination condition on D is automatic. A construction is available if

$$|Y \cup N_A(d)| \geq 3, \quad \text{or} \quad \big[\,|Y \cup N_A(d)| \geq 2 \text{ and } N_C(y) \cup N_C(d) = C\,\big]. \tag{4.11}$$

Assume (4.11) fails for every $d \in D$. A vertex d has four U-neighbors if it is not adjacent to y, and three otherwise. We also use the earlier universal bound: the complement of every maximum independent set has at most three degree-five vertices.

Case $a = 3$**.** The first condition in (4.11) holds for every d, a contradiction.

Case $a = 1$**.** Write $Y = \{b\}$, and let c^* be the vertex of C omitted by $N_C(y)$. We show every $d \in D$ is adjacent to b. Suppose d is not. If d is not adjacent to y, it has four U-neighbors and at least one A-neighbor. Two A-neighbors give a union with $\{b\}$ of size at least three. Exactly one A-neighbor forces adjacency to all of C, and gives the second condition in (4.11). Both are excluded.

Now let d be adjacent to y but not b. It has three U-neighbors. At least two A-neighbors again give the first condition. If it has one A-neighbor, then it has two C-neighbors. Unless they are exactly $N_C(y)$, their union with $N_C(y)$ is C, giving the second condition. In the remaining case, $A \cup \{c^*\}$ is a maximum independent set of size s. The two other C-vertices, y, and d each have exactly one neighbor in it: respectively c^*, c^*, b, and the unique A-neighbor of d. Its complement therefore has four degree-five vertices, a contradiction. If d has no A-neighbor, it sees all of C, and the same independent set gives it the unique neighbor c^*. The same four-vertex contradiction applies.

Thus b is adjacent to y and all three vertices of D. Its degree six leaves at most two neighbors in I. Since $|I| = s - 1 \geq 4$, choose $w \in I$ not adjacent to b. If w has at least three A-neighbors, choose three as R and use construction (b). If it has exactly two, it is a leaf with five U-neighbors and sees all of C; choose those two as R and use (a). In both choices $b \notin R$, so every vertex of D loses its neighbor $b \in A \setminus R$. Both constructions give a contradiction.

Case $a = 2$**.** Write $Y = \{b_1, b_2\}$ and $N_C(y) = \{c_0\}$. Failure of (4.11) gives

$$\varnothing \neq N_A(d) \subseteq Y \qquad (d \in D). \tag{4.12}$$

An A-neighbor outside Y would make the union have size at least three. An empty A-neighborhood is impossible for a vertex not adjacent to y, since it has four U-neighbors. For a vertex adjacent to y, an empty A-neighborhood gives all of C and the second condition in (4.11).

If d is not adjacent to y, its four U-neighbors force either one A-neighbor and all of C, which is excluded, or two A-neighbors and two C-neighbors. Thus $N_A(d) = Y$. Its two C-neighbors include c_0, since otherwise their union with c_0 would be C.

Seek a nonleaf $w \in I$ satisfying

$$|N_A(w) \setminus Y| \geq 3, \tag{4.13}$$

or

$$|N_A(w) \setminus Y| \geq 2, \qquad \{c_0\} \cup N_C(w) = C. \tag{4.14}$$

In (4.13), choose three neighbors outside Y as R and use (b). In (4.14), choose two and use (a). Both avoid Y, so (4.12) ensures that each D-vertex loses an $A \setminus R$ neighbor.

Suppose neither choice exists. Every nonleaf sees c_0. To verify this, put $q = |N_A(w) \setminus Y|$. The vertex w has six U-neighbors. If $q \leq 1$, then $|N_A(w)| \leq 3$, forcing exactly three A-neighbors and all three C-neighbors. If $q = 2$, then $|N_A(w)| \leq 4$. At most three A-neighbors would force all of C

and satisfy (4.14). Thus it has four A-neighbors and two C-neighbors. Failure of (4.14) forces c_0 into that pair. The case $q \geq 3$ is (4.13).

There are $s+r-4$ nonleaves. The vertex c_0 is also adjacent to y and the $3-r$ vertices of D not adjacent to y. These are

$$1 + (3-r) + (s+r-4) = s$$

distinct V-neighbors. But c_0 already has two neighbors in its triangle and total degree six, leaving exactly four V-neighbors. This contradicts $s \geq 5$.

Case $a = 0$. The vertex y sees all of C. Failure of (4.11) gives $|N_A(d)| \leq 1$ for every $d \in D$. Suppose d is adjacent to y. Its three U-neighbors are either all of C, or one A-vertex and two C-vertices. In the first case choose any $c \in C$; in the second choose the omitted C-vertex c. The maximum independent set $A \cup \{c\}$ has a complement in which the two unchosen C-vertices and y have degree five. The vertex d also has degree five, since its unique neighbor in the independent set is c in the first case and its A-neighbor in the second. This is a contradiction.

Hence $r = 0$. Each $d \in D$ has four U-neighbors, exactly one in A, and all three in C. Each C-vertex has its four cross-neighbors exactly at y and the three vertices of D. There are no edges from C to I. Every $w \in I$ therefore has at least five A-neighbors. Let L be the union of the three singleton A-neighbor sets of the vertices of D, so $|L| \leq 3$. Choose $w \in I$ and two vertices $R \subseteq N_A(w) \setminus L$. Use construction (a). The vertex y dominates C, and each D-vertex loses its neighbor in $L \subseteq A \setminus R$. This is the final contradiction.

All four cases have been excluded under $s \geq 5$. Thus $s \in \{3, 4\}$, and $|V(H)| = 2s + 5$. Put $k = \nu(H) = s + 2$. The top-window result with $D = 6, E = 2$ has

$$u = \max\left(0, \left\lfloor \frac{2E-3}{D-2} \right\rfloor\right) = 0.$$

It gives

$$\mathrm{tr}(H) \leq k - 3E + 2 = s - 2 \leq 2.$$

Together with gap one, this proves $\mathrm{tr}(H) + \mathrm{ex}(H) \leq 4$.

For excess at most one, a closed 6-lump has $\mathrm{ps}(H) \geq 1$, so its gap is nonpositive. The excess-two case therefore proves (S) for all excesses at most two. □

4.1.6. *Excess three in degree eight.* The next proof combines residual matching bounds with the earlier domination certificates.

Theorem 4.13 (Closed degree-eight lumps of excess three)**.** *Let H be a closed 8-lump with*

$$\mathrm{gap}(H) = 1, \qquad \mathrm{ex}(H) = 3,$$

and assume every proper induced subgraph satisfies T_-. Then

$$|V(H)| = 17, \qquad \alpha(H) \in \{3, 4\}, \qquad \mathrm{tr}(H) = 1.$$

No existence assertion for such a graph, or universal T_- assertion, is made.

Proof. Write $|V(H)| = 2k + 1$ and $a = \alpha(H)$. Then

$$\nu(H) = k, \qquad c_{0,0}(H) = k + 3, \qquad \mathrm{ps}(H) = 2.$$

The earlier order bounds give $k \geq 8$ and $k \leq 2a + 2$. Suppose $k \geq 9$, so $a \geq 4$.

Fix any maximum independent set S, and put

$$G = H - S, \qquad \eta = \nu(G) - (k + 1 - a).$$

Since $\mathrm{tr}(H) \geq 1$, we have $\mathrm{tr}(H) + 3 \geq 4 = h_-(7)$. Theorem 4.10 gives

$$\mathrm{ps}(G) = 2 - \eta, \qquad \eta \geq 0, \qquad c^*_\theta(H) = a + c^*_\theta(G) \quad (1 \leq \theta \leq 7).$$

Thus $\eta \in \{0, 1, 2\}$. Maximality and regularity give

$$\Delta(G) \leq 7, \quad \delta(G) \geq 8 - a, \quad |V(G)| = 2k + 1 - a, \quad e(G) = 8(k - a) + 4.$$

Every induced subgraph of G has gap at most one.

Choose a maximum-cardinality ordinary attaining set X in G, and put $b = |X|$. Its residual components are odd and factor-critical. Their number is $a + b - 1 - 2\eta$, and

$$\nu(G) = b + \sum_{Q} \nu(Q).$$

Their independence numbers sum to at most a. Each component therefore has independence number at most $2\eta + 2 - b$.

If $\eta = 0$, the earlier barrier-rank result gives

$$2a + 9b \le 9$$

and a component of order at least $2a + 2b + 5$. Thus $b = 0, a = 4$, and $k \in \{9, 10\}$. There are three whole components of G, each of minimum degree at least four and odd order at least five. Since $|V(G)| = 2k-3 \le 17$, each has order at most seven. This contradicts the required component order at least thirteen.

Suppose $\eta = 1$. Then $\mathrm{ps}(G) = 1$ and $\nu(G) = k + 2 - a$. The barrier-rank result gives

$$2a + 9b \le 23$$

and a residual component Q with $\nu(Q) \ge a + b + 2$. Hence $b \in \{0, 1\}$, and $b = 1$ implies $a \le 7$.

First take $b = 0$. There are $a - 3$ whole odd factor-critical components, each of independence number at most four. Slack superadditivity bounds each component slack by one. The small-independence estimate gives

$$\nu(R) \le 2\alpha(R) + \mathrm{ps}(R) - 1 \le 8.$$

For $a \ge 7$, this contradicts the required rank $a + 2 \ge 9$.

If $a = 4$, then $k = 9$ or 10, and G is factor-critical. It has order 15 or 17, matching number $k - 2$, and 44 or 52 edges. It is nonregular: 88 is not divisible by 15, and 104 is not divisible by 17. The strict nonregular factor-critical bound gives

$$e(G) \le 3(k - 2) + 4\alpha(G) + 6 \le 3k + 16.$$

This is 43 or 46, below 44 or 52.

If $a = 5$, there are two whole factor-critical components, each of minimum degree at least three. Neither can have slack zero, since the nontrivial zero-slack factor-critical graphs are C_3, C_5, C_7. Their slacks sum to at least two, contrary to $\mathrm{ps}(G) = 1$.

If $a = 6$, the required component Q has rank at least eight. The bound

$$\nu(Q) \le 2\alpha(Q) + \mathrm{ps}(Q) - 1, \qquad \alpha(Q) \le 4, \quad \mathrm{ps}(Q) \le 1$$

forces

$$\nu(Q) = 8, \qquad \alpha(Q) = 4, \qquad \mathrm{ps}(Q) = 1.$$

The other two components have slack zero and independence numbers summing to at most two. Both have independence number one. Their minimum degree is at least two, so the zero-slack classification makes both triangles. Thus $\nu(G) = 8 + 1 + 1 = 10$, whence $k = 14$, and Q has seventeen vertices. Each of the six triangle vertices has two G-neighbors and is adjacent to all six vertices of S. They require 36 edges incident with S. Maximality of S requires at least one further S-edge from each of the seventeen vertices of Q. This requires at least 53 edges at S, contrary to $8|S| = 48$. These cases exhaust $b = 0$.

Now take $b = 1$, with $X = \{x\}$. There are $a - 2$ odd factor-critical components of $G - x$, each of independence number at most three. Some Q has order at least $2a + 7$. Every residual component has minimum degree at least $7-a$. For $a = 4$, Q has at least fifteen vertices and the other component at least five, whereas

$$|V(G - x)| = 2k - a \le 3a + 4 = 16.$$

For $a = 5$, the required orders are at least seventeen and three for each of the other two components, exceeding the total bound nineteen. For $a = 6$, they are at least nineteen and three for each of the other three components, exceeding the total bound twenty-two.

For $a = 7$, Q has at least twenty-one vertices and there are four other components. The total order is at least twenty-five and at most $2k - a \le 25$. Thus $k = 16$, and $G - x$ consists of Q on twenty-one vertices and four singletons. The earlier certificate and slack bounds give

$$c_{0,0}(Q) \le \left\lceil \frac{42}{3} \right\rceil - 1 = 13,$$

$$\mathrm{ps}(Q) \ge 2(21 - c_{0,0}(Q)) - 4\alpha(Q) \ge 4.$$

Hence $\mathrm{ps}(G - x) \ge 4$.

Each singleton has all eight H-neighbors in $S \cup \{x\}$, and is therefore adjacent to x. Put $h = d_G(x) \ge 4$. The matching decomposition gives $\nu(G) - \nu(G - x) = 1$. All costs above seven vanish, and each of the seven positive costs can drop by at most one. The slack definition yields

$$1 = \mathrm{ps}(G) \ge \mathrm{ps}(G - x) + h - 1 - 7 \ge h - 4,$$

so $h \le 5$. If $h = 4$, the only neighbors of x are the four singletons, and Q is a whole component of G. Then $\mathrm{ps}(G) \ge \mathrm{ps}(Q) \ge 4$, a contradiction. Thus $h = 5$, and x has exactly one neighbor $w \in Q$. A perfect matching of $Q - w$, together with xw, covers $Q \cup \{x\}$. Only the four degree-one vertices remain unmatched. This matching covers every degree-seven vertex, so $c_7^*(G) = 0$. Monotonicity gives $c_7^*(G - x) = 0$. Only six positive costs can now drop, and

$$\mathrm{ps}(G) \ge \mathrm{ps}(G - x) + 5 - 1 - 6 \ge 2,$$

contrary to $\mathrm{ps}(G) = 1$. This excludes $\eta = 1$.

We have proved $\eta = 2$ and $\mathrm{ps}(G) = 0$ for every maximum independent set S. Thus $\nu(G) = k + 3 - a$. If $\Delta(G) = 7$, zero slack and the hereditary gap bound make G covered. If $\Delta(G) < 7$, the degree vanishing rule applies directly. In either case $c_7^*(G) = 0$, and exact additivity gives

$$c_7^*(H) = a.$$

The earlier zero-slack density bound with its parameters $p = 2, d = 8$ gives $\nu(G) \le 7$. Therefore $a \ge k - 4$.

The first-moment certificate with parameters $j = 1, m = 7, p = 1/4$ gives

$$c_7^*(H) \le 1 + 2kq, \qquad q = \frac{1}{4} + \left(\frac{3}{4}\right)^8 = \frac{22945}{65536} < \frac{6}{17},$$

where the strict comparison is $390065 < 393216$. If $k \ge 17$, then

$$a = c_7^*(H) < 1 + \frac{12k}{17} \le k - 4,$$

a contradiction. Thus $9 \le k \le 16$.

For $k = 9, 10$, the ordinary domination recurrence gives respectively

$$19, 10, 5, 2, 0, \qquad 21, 12, 6, 3, 1, 0.$$

The resulting dominating-set sizes are at most four and five. The dense certificate applies because $19 - 4 \le 18$ and $21 - 5 \le 18$. Thus $c_7^*(H) \le 4$ or 5, below $a \ge k - 4$.

For $11 \le k \le 16$, the matching-aware recurrence in Setup 4.6(viii) gives the full lists

k	(r_i)	β_1 bound
11	$14, 8, 4, 2, 1, 0$	6
12	$16, 10, 6, 3, 1, 0$	6
13	$18, 12, 8, 5, 3, 1, 0$	7
14	$20, 14, 9, 6, 4, 2, 1, 0$	8
15	$22, 16, 11, 7, 4, 2, 1, 0$	8
16	$24, 18, 13, 9, 6, 4, 2, 1, 0$	9

Each bound is less than $k-4$. Since $c_7^*(H) \le \beta_1(H)$, every case is a contradiction. All $k \ge 9$ are excluded.

Therefore $k = 8$, so $|V(H)| = 17$. The inequality $k \le 2a+2$ gives $a \ge 3$. At order seventeen the ordinary recurrence gives

$$17, 8, 3, 1, 0,$$

and hence $c_7^*(H) \le 4$. Exact profile additivity gives $c_7^*(H) \ge a$, so $a \le 4$. Finally the top-window bound with $E = 3, D = 8$ gives

$$\mathrm{tr}(H) \le k - 7 = 1.$$

Positive gap gives $\mathrm{tr}(H) \ge 1$, completing the proof. □

4.1.7. *Excess three in degree six.* We use the exact second-threshold cost to exclude the remaining long-run configuration.

Theorem 4.14 (Exclusion at excess three in degree six)**.** *There is no closed* 6*-lump* H *such that*

$$\mathrm{gap}(H) = 1, \qquad \mathrm{ex}(H) = 3, \qquad \mathrm{tr}(H) \ge 2,$$

and every proper induced subgraph satisfies T_-*. Equivalently, every closed* 6*-lump of gap one and excess three, with all proper induced subgraphs satisfying* T_-*, has*

$$\mathrm{tr}(H) \le 1, \qquad \mathrm{tr}(H) + \mathrm{ex}(H) \le 4.$$

Proof. Suppose such H exists, and put $s = \alpha(H)$. The earlier rigidity results in Setup 4.6(vi) give well-coveredness,

$$|V(H)| = 2s + 9, \qquad s \ge 5,$$

and residual matching number seven for every maximum independent set. Here well-coveredness means that every maximal independent set is maximum. Every maximum-degree process on the complement has list $(5, 5, 4, 4, 3)$.

The central partition from the same earlier result is

$$V(H) = A \sqcup B \sqcup \{w\}, \qquad H[A] \cong H[B] \cong 2C_3 \sqcup (s-2)K_1.$$

The root w is adjacent to exactly three isolates on each side and to no triangle vertex. For every maximum independent set I, the private vertices

$$P_I = \{v \notin I : |N_H(v) \cap I| = 1\}$$

consist of exactly two disjoint edges. The endpoints of each edge have the same unique I-neighbor, called their parent, and the two parents are distinct. Thus every private vertex has an adjacent private companion with the same parent, and two private vertices with the same parent are adjacent.

We first establish

$$c_2^*(H) = s + 7. \tag{4.15}$$

For a maximum independent set I, put $G = H - I$. Its slack is zero. The five deletions preserve zero slack and lower matching number by one at each step. The terminal graph is two triangles and isolates, with six edges and matching number two. Deleting one vertex of each triangle leaves a matching. Its first and second costs are zero, so the terminal first and second costs are at most

two. All higher costs vanish. Zero slack forces the sum of positive costs to be $6-2=4$, so both the first and second costs equal two.

At a deletion of maximum degree $d \geq 3$, zero slack and matching loss one give total positive-cost drop $d-1$. The $d-1$ thresholds below d each drop by at most one. For $d \geq 4$, the zero-slack coverage rule in Setup 4.5(iv) supplies a matching covering every degree-d vertex. At the last deletion, $d=3$. Immediately before it, the graph has nine edges: the deletion removes three edges and leaves two triangles and isolates, with six edges in total. Its degree sum is eighteen, so it has at most six degree-three vertices. Let v be the vertex deleted at this stage. Every vertex other than v and its three neighbors keeps its degree when v is deleted. The remaining graph consists of two triangles and isolates, so every such vertex has degree at most two before deletion. Consequently, only v and its three neighbors can have degree three, giving at most four such vertices. Since $4 \leq 6 \leq 2(3+2)-1=9$, the small-required-set criterion in Setup 4.5(vi) applies with $D=3$; an odd 3-regular whole component is impossible by the degree-sum identity. It supplies a matching covering every degree-three vertex. Thus the pre-deletion graph is d-good in every case, so the cost at d, and all higher costs, vanish by Setup 4.5(i). They also vanish after deletion by monotonicity. Every threshold below d must therefore drop by exactly one. In particular, each of the five deletions lowers the second cost by one, giving $c_2^*(G)=7$. Since

$$\operatorname{tr}(H)+\operatorname{ex}(H) \geq 5 > \max\left(4, \left\lfloor \frac{5}{2} \right\rfloor + 1\right),$$

Theorem 4.10 gives exact additivity at threshold two. This proves (4.15).

Let X, Y be the isolate sets of A, B, and put

$$K = N_H(w) \cap X, \qquad L = N_H(w) \cap Y, \qquad Y_0 = Y \setminus L.$$

Let T_1, T_2 be the two triangles of B. For $p \in T_1, q \in T_2$, define

$$J(p,q) = Y_0 \cup \{w, p, q\}.$$

This set is independent and maximal in $R = H[B \cup \{w\}]$: w dominates L, and p, q dominate the two triangles. Its size is $(s-5)+3 = s-2$. Extending it to a maximal independent set of H, well-coveredness requires exactly two independent vertices of A. Call an A-vertex available for (p,q) if it has no neighbor in $J(p,q)$.

No A-triangle vertex u is available. Suppose one is. It has internal A-degree two, no w-neighbor, and four neighbors in R, all in

$$L \cup (T_1 \setminus \{p\}) \cup (T_2 \setminus \{q\}).$$

Each triangle contributes at most two neighbors.

If u meets both triangles, delete w, one u-neighbor in T_1, and one u-neighbor in T_2 from $R+u$. The triangles become edges and the former claw leaves become isolated. The vertex u has two remaining neighbors, with at most one in each remaining triangle edge. All nonisolated components are paths: its neighbors cannot be the two ends of one edge, so no cycle is formed.

If u meets only T_1, its four neighbors comprise either two vertices of L and two of T_1, or three vertices of L and one of T_1. It cannot meet neither triangle because $|L|=3$. Delete a neighbor $\ell \in L$, a neighbor $t \in T_1$, and any vertex of T_2. The old claw becomes a three-vertex path with endpoints $L \setminus \{\ell\}$, and the triangles become edges. In the first case, u joins an endpoint of that path to an endpoint of the T_1-edge, giving a path. In the second, its remaining neighbors are the two endpoints of the shortened claw, giving a four-cycle. Thus every component is a path, an even cycle, or an isolate.

Such a graph has a matching covering every degree-two vertex: match even-order paths and even cycles perfectly, and leave an endpoint unmatched in each odd-order path. Its second cost is zero. Restoring the three deleted vertices gives

$$c_2^*(R+u) \leq 3.$$

But $|A| = s + 4$, so deleting $A \setminus \{u\}$ removes $s + 3$ vertices. Reinsertion and (4.15) give

$$s + 7 = c_2^*(H) \le s + 3 + c_2^*(R + u),$$

forcing $c_2^*(R + u) \ge 4$. This contradiction excludes every available triangle vertex.

Each $J(p, q)$ must therefore extend by two A-isolates. An available isolate x has all six neighbors in the seven-element set

$$L \cup (T_1 \setminus \{p\}) \cup (T_2 \setminus \{q\}).$$

Its neighborhood has one of two forms:

(i) all three vertices of L, two vertices in one triangle, and one in the other;
(ii) two vertices of L and two vertices in each triangle.

The first form is impossible. After interchanging the triangle names if necessary, write

$$N_H(x) = L \cup (T_1 \setminus \{p\}) \cup \{q_0\}, \qquad q_0 \in T_2.$$

Extend $J(p, q_0)$ to a maximum independent set I. The vertex x is unselected and has the unique I-neighbor q_0. It has no edge to any A-extender because it is an A-isolate. Every neighbor of x in L has the selected neighbor w, and every neighbor in $T_1 \setminus \{p\}$ has the selected neighbor p. These parents differ from q_0. The remaining neighbor q_0 is selected. Thus x has no private companion with parent q_0, a contradiction.

Fix (p, q). Its two available A-isolates x, x' therefore have the second form, with the same triangle-neighbor sets $T_1 \setminus \{p\}$ and $T_2 \setminus \{q\}$. Choose $p' \in T_1 \setminus \{p\}$, and extend $J(p', q)$ to a maximum independent set I'. Both x, x' are unselected and have the unique selected neighbor p': they avoid w, Y_0, q and have no A-neighbors. They are nonadjacent private vertices with the same parent, contrary to the private-edge structure.

Thus $J(p, q)$ cannot have its required extension. The hypothetical H does not exist. Since $\mathrm{tr}(H)$ is an integer, exclusion of $\mathrm{tr}(H) \ge 2$ gives the equivalent bound $\mathrm{tr}(H) \le 1$. □

4.1.8. *An independence criterion for LP-goodness.* The next theorem removes the exceptional small-independence case using two successive independent-set deletions.

Theorem 4.15 (LP-goodness from independence and excess)**.** *Let H be a finite simple graph with*

$$E = \mathrm{ex}(H) \ge 4, \qquad 4\alpha(H) \le 3E + 4.$$

Then H is LP-good:

$$\Sigma(H) \le e(H).$$

In particular, every graph with $\mathrm{ex}(H) \ge 4$ and $\alpha(H) \le 4$ is LP-good, with no restriction on order, degree, or connectedness.

Proof. Suppose H satisfies the hypotheses and is not LP-good. The earlier exceptional reduction in Setup 4.7(iii) forces

$$|V(H)| = 23, \qquad E = 4, \qquad \alpha(H) = 4.$$

Its structural refinement makes H connected and 8-regular, with

$$e(H) = 92, \quad c_{0,0}(H) = 15, \quad \nu(H) = 11, \quad \mathrm{ps}(H) = 3.$$

Every economical maximum-degree peeling has an induced cutoff-15 residual $R = 4K_2$, and such a peeling exists. Every induced 16-vertex subgraph has a perfect matching. The earlier matching-extension criterion with $n = 23, q = 16$ makes H factor-critical. It is noncomplete because $\alpha(H) = 4$.

Choose one endpoint of each edge of R to form I, and let S consist of the opposite endpoints. Both are independent four-sets, hence maximum independent sets of H. Put

$$G = H - I, \qquad A = H - (I \cup S).$$

Every vertex outside I has an I-neighbor, since otherwise it could be added to I. Each vertex of S has exactly one I-neighbor because R is induced $4K_2$. Thus G has nineteen vertices and maximum degree exactly seven. Since I is independent,

$$e(G) = 92 - 4 \cdot 8 = 60, \qquad \alpha(G) \le 4.$$

If G were uncovered, Setup 4.7(iv) would give

$$e(G) = 7 \cdot 9 + \frac{h}{2}, \qquad h \in \{2, 4, 6\},$$

so $e(G) \in \{64, 65, 66\}$, contrary to $e(G) = 60$. Therefore G is covered.

Every vertex of the independent set S has degree seven in G. Hence

$$e(A) = 60 - 4 \cdot 7 = 32.$$

The graph A has fifteen vertices and independence number at most four. Each of its vertices has a neighbor in each of the disjoint maximum independent sets I, S. It loses at least two of its original eight neighbors, so $\Delta(A) \le 6$. The earlier terminal estimate gives

$$\nu(A) + \mathrm{ps}(A) \ge 8. \tag{4.16}$$

Apply the regular factor-critical independent-set inequality to H, I. Using the identity gap $= c_{0,0} - \nu - \mathrm{ps}$, its stated surplus form is

$$\nu(H) + \mathrm{ps}(H) \ge |I| - 1 + \nu(G) + \mathrm{ps}(G) = 3 + \nu(G) + \mathrm{ps}(G). \tag{4.17}$$

Apply the covered independent-set inequality to G, S. It gives

$$\nu(G) + \mathrm{ps}(G) \ge |S| + \nu(A) + \mathrm{ps}(A) \ge 4 + 8 = 12. \tag{4.18}$$

Combining (4.17) and (4.18),

$$\nu(H) + \mathrm{ps}(H) \ge 15,$$

contrary to $11 + 3 = 14$. The exceptional graph cannot exist.

Finally, if $E \ge 4$ and $\alpha(H) \le 4$, then

$$4\alpha(H) \le 16 \le 3E + 4.$$

The first assertion applies. □

4.1.9. *Eliminating gap two from minimal failures.* We now sharpen the earlier minimal-failure reduction to gap one.

Theorem 4.16 (The restricted criterion). *(I) Every induced-minimal T_- failure supplied by the earlier reduction in Setup 4.7(i) has gap one. Its gap-two alternative is impossible.*

(II) Every finite simple graph violating T_- contains an induced totally critical closed d-lump F, for an even integer $4 \le d \le 24$, such that

$$\mathrm{gap}(F) = 1, \qquad \mathrm{tr}(F) + \mathrm{ex}(F) \ge h_-(d) + 1,$$

and every proper induced subgraph of F satisfies T_-.

(III) Universal T_- is equivalent to the following restricted assertion: every totally critical closed d-lump F, with even $4 \le d \le 24$ and $\mathrm{gap}(F) = 1$, satisfies

$$\mathrm{tr}(F) + \mathrm{ex}(F) \le \max(4, d/2 + 1).$$

Under this restricted assertion, every graph C with

$$\mathrm{ex}(C) \ge \max\left(4, \left\lceil \frac{\Delta(C)}{2} \right\rceil + 1\right)$$

is LP-good. The size-Ramsey conclusion is conditional: if every proper-subgraph-minimal failure of the restricted criterion at degrees $4, 6, 8, 10, 12$ is excluded, then the full size-Ramsey

star-forest formula follows. The restricted top-run assertion and these minimal-failure exclusions are not proved here.

Proof. Let F be an induced-minimal failure supplied by the earlier reduction. Its gap belongs to $\{1,2\}$. Suppose $\operatorname{gap}(F)=2$. For any maximum independent set S, write

$$|V(F)|=2k+1,\quad E=\operatorname{ex}(F),\quad a=|S|,\quad G=F-S,\quad \eta=\nu(G)-(k+1-a).$$

The earlier gap-two independent-set ledger gives a component P^* and an integer m such that

$$a\in\{5,6,7\},\qquad d\in\{20,22,24\},\qquad m\ge 4,\qquad a+m+1\le d/2,$$

$$E=1+m+\eta,\qquad 0\le\eta\le\left\lfloor\frac{a-1}{2}\right\rfloor, \tag{4.19}$$

$$\nu(G)=k+1-a+\eta,\qquad |V(G)|=2k+1-a,\qquad \delta(G)\ge d-a>0.$$

It also gives $\operatorname{gap}(P^*)=1,\operatorname{ex}(P^*)=m$, and, with $L=a+\sum_{Q\neq P^*}\nu(Q)$,

$$L+1\le\operatorname{tr}(P^*),\qquad \operatorname{tr}(P^*)+m\le d/2. \tag{4.20}$$

The numerical bounds from the earlier reduction are

$$E\le\left\lfloor\frac{4a-5}{3}\right\rfloor,\qquad \max(3E-1,d+1-E)\le k\le\left\lfloor\frac{3E+4a-4}{2}\right\rfloor. \tag{4.21}$$

If $a=5$, then (4.19) and (4.21) force $E=5,m=4,\eta=0$. The upper bound gives

$$k\le\left\lfloor\frac{15+20-4}{2}\right\rfloor=15,$$

whereas the lower bound gives $k\ge d-4\ge 16$.

Suppose $a=6$. The bounds $m\ge 4$ and $a+m+1\le d/2$ give $d\ge 22$. Also $5\le E\le 6$. If $E=5$, then

$$k\le\left\lfloor\frac{15+24-4}{2}\right\rfloor=17,\qquad k\ge d-4\ge 18.$$

Thus $E=6$. Now $k\le 19$ and $k\ge\max(17,d-5)$. For $d=22$, this gives $17\le k\le 19$. For $d=24$, it gives $k=19$. In both cases,

$$|V(G)|=2k-5,\qquad \delta(G)\ge d-6\ge 16,\qquad \left\lfloor\frac{|V(G)|}{2}\right\rfloor=k-3\le 16.$$

The minimum-degree matching bound gives $\nu(G)=k-3$. Hence

$$\eta=(k-3)-(k+1-6)=2,\qquad m=E-1-\eta=3,$$

contrary to $m\ge 4$.

It remains to consider $a=7$. Then $a+m+1\le d/2\le 12$ and $m\ge 4$ force $d=24,m=4$. Thus $E=5+\eta$, with $5\le E\le 7$. If $E=5$, then

$$k\le\left\lfloor\frac{15+28-4}{2}\right\rfloor=19,\qquad k\ge 25-5=20.$$

If $E=6$, then $\eta=1$ and $19\le k\le 21$. We would have

$$\nu(G)=k-5,\quad |V(G)|=2k-6,\quad \delta(G)\ge 17.$$

The matching bound gives $\nu(G)\ge\min(17,k-3)$. For $k=19,20,21$, these lower bounds are $16,17,17$, each greater than $k-5=14,15,16$.

Thus $E=7,\eta=2$, and $20\le k\le 22$. Now $\nu(G)=k-4$. If $k=20$, then G has thirty-four vertices and minimum degree at least seventeen, so it has a perfect matching of size seventeen, contrary to $\nu(G)=16$. If $k=21$, then G has thirty-six vertices, matching number seventeen, and minimum degree at least seventeen. Apply the equality-case matching result with $r=17,N=36=2r+2$.

Its first alternative gives an independent set of size $N - r = 19$, contrary to $\alpha(G) \le 7$. Its second requires r even, contrary to $r = 17$.

The only remaining numerical possibility is

$$a = 7, \quad d = 24, \quad E = 7, \quad k = 22, \quad m = 4, \quad \eta = 2, \quad |V(G)| = 38, \quad \nu(G) = 18. \tag{4.22}$$

By (4.20),

$$L \ge 7, \qquad L + 1 \le \mathrm{tr}(P^*) \le 12 - 4 = 8.$$

Thus $L = 7, \mathrm{tr}(P^*) = 8$, and $\sum_{Q \ne P^*} \nu(Q) = 0$. Since $\delta(G) \ge 17$, every component has an edge and positive matching number. There is no component other than P^*. Consequently

$$G = P^* \text{ is connected}, \quad \mathrm{gap}(G) = 1, \quad \mathrm{ex}(G) = 4, \quad \mathrm{tr}(G) = 8, \quad 17 \le D := \Delta(G) \le 23. \tag{4.23}$$

The upper degree bound follows from maximality of S, since every G-vertex loses an S-neighbor from its original degree twenty-four. Every induced subgraph of G satisfies T_-.

Suppose G is covered. Delete a maximum-degree vertex v. The covered-deletion rule gives $\mathrm{gap}(G - v) \ge 1$. The property T_- makes this gap one, and the equality clause gives

$$c_{0,0}(G - v) = c_{0,0}(G) - 1, \qquad \mathrm{tr}(G - v) \ge 9.$$

Put $t = \nu(G) - \nu(G - v) \in \{0, 1\}$. Then $\mathrm{ex}(G - v) = 3 + t$. If $t = 1$, the run-plus-excess is at least thirteen, while

$$h_-(\Delta(G - v)) \le h_-(23) = 12.$$

If $t = 0$, the excess is three. The earlier hereditary T_- excess-three bound gives $\mathrm{tr}(G - v) \le 5$, contrary to $\mathrm{tr}(G - v) \ge 9$. Thus G is uncovered.

Let W be the degree-D vertex set, let $\kappa \ge 1$ be its minimum unmatched count, and choose a maximum-cardinality attaining set X with

$$q_W(G - X) - |X| = \kappa.$$

If $X \ne \varnothing$, every component of $G - X$ has gap at most one. The generic profile is nonincreasing, so the exposure-transfer result in Setup 4.5(vii) applies. For even D, it gives $\mathrm{gap}(G) \le 0$, contrary to (4.23). For odd D, it gives a component Z with

$$\mathrm{gap}(Z) = 1, \quad \mathrm{ex}(Z) = 4, \quad \mathrm{tr}(Z) \ge 8 + |X| \ge 9.$$

But Z satisfies T_-, and

$$\mathrm{tr}(Z) + \mathrm{ex}(Z) \le h_-(\Delta(Z)) \le h_-(23) = 12,$$

contrary to the lower bound thirteen. Therefore $X = \varnothing$. The connected graph G has even order thirty-eight, so $q_W(G) = 0$. This contradicts $q_W(G) = \kappa \ge 1$.

All cases in (4.19) are excluded. This proves (I). For (II), apply the earlier induced-minimal reduction to any violation, and then apply (I). With gap one, failure is exactly

$$\mathrm{tr}(F) + \mathrm{ex}(F) > h_-(d),$$

which is the stated integer lower bound. The remaining structural properties come from that reduction.

For (III), universal T_- implies the restriction because $h_-(d) = \max(4, d/2 + 1)$ for even d. Conversely, if the restriction holds and some graph violates T_-, part (II) supplies a totally critical closed lump in the specified degree range that violates the restriction. Thus T_- holds universally.

Setup 4.7(ii) then gives universal T_0 and the stated LP-goodness threshold. For the size-Ramsey conclusion, assume that every proper-subgraph-minimal failure of the restricted criterion at degrees $4, 6, 8, 10, 12$ is excluded. The earlier conditional implication assumed in Setup 4.7(ii) then gives the full size-Ramsey star-forest formula. These exclusions are not established here. □

[math/blocker] [GAP: The full size-Ramsey star-forest formula, its parameter domains, and the precise earlier implication were not included in the supplied statements. The final clause of Theorem 4.16 records that conditional implication without supplying an invented formula.] [addressed-partial: The supplied conditional exclusion hypothesis at degrees $4, 6, 8, 10, 12$ is now explicit in the prerequisite, theorem, and proof | deferred: The formula and its parameter domains still require a supplied statement.]

4.1.10. *The excess window and the low-degree restrictions.* The top allocation window now gives an order restriction, and the preceding small-excess results give exact necessary conditions in degrees six and eight.

Theorem 4.17 (Excess window for minimal failures)**.** *Let $d \geq 3$ be an integer, and let F be a connected graph of odd order $2k+1$ with*

$$\nu(F) = k, \qquad \Delta(F) \leq d, \qquad E = \mathrm{ex}(F) \geq 1, \qquad \mathrm{tr}(F) + E \geq h_-(d) + 1.$$

Put

$$u = \max\left(0, \left\lfloor \frac{2E-3}{d-2} \right\rfloor\right).$$

Then

$$k \geq h_-(d) + 2E - 1 + 2u,$$

or equivalently,

$$E \leq \left\lfloor \frac{k - h_-(d) + 1 - 2u}{2} \right\rfloor \leq \left\lfloor \frac{k - h_-(d) + 1}{2} \right\rfloor.$$

This applies to every induced-minimal failure in Theorem 4.16(II), with d its regular degree. For even $d \geq 6$, it reads

$$E \leq \left\lfloor \frac{k - d/2 - 2u}{2} \right\rfloor.$$

At $d = 18$, a minimal failure with $k = 23$ has $E \leq 7$, and one with $k = 26$ has $E \leq 8$.

Proof. The top-window result in Setup 4.5(viii), applied with degree bound $D = d$, puts every missing top split in

$$[\,2E - 1 + u,\ k - E - u\,].$$

Its general integer-bound conclusion says that

$$\mathrm{tr}(F) + E \geq B + 1 \quad \Longrightarrow \quad k \geq B + 2E - 1 + 2u.$$

Take $B = h_-(d)$. Rearranging gives

$$2E \leq k - h_-(d) + 1 - 2u.$$

Since E is an integer and $u \geq 0$, this is equivalent to the first floor bound and implies the second.

For a failure in Theorem 4.16(II), closed-lump structure gives connectedness, odd order $2k+1$, matching number k, and maximum degree d. Also

$$1 = \mathrm{gap}(F) = \mathrm{ex}(F) - \mathrm{ps}(F) \leq \mathrm{ex}(F),$$

so $E \geq 1$. The required run-plus-excess lower bound is part of that theorem. Thus all hypotheses hold.

For even $d \geq 6$,

$$h_-(d) = d/2 + 1, \qquad k - h_-(d) + 1 - 2u = k - d/2 - 2u.$$

At $d = 18$, $h_-(18) = 10$. For $k = 23$,

$$2E \leq 23 - 9 - 2u \leq 14,$$

giving $E \leq 7$. For $k = 26$,

$$2E \leq 26 - 9 - 2u \leq 17,$$

giving $E \leq 8$. □

Theorem 4.18 (Induced-minimal restrictions in degrees six and eight)**.** *Let H be an induced-minimal T_- failure with maximum degree $d \in \{6, 8\}$. Then H is a totally critical closed d-lump with* $\operatorname{gap}(H) = 1$. *Write*

$$|V(H)| = 2k + 1, \qquad E = \operatorname{ex}(H).$$

Then

$$E \geq 4, \qquad \alpha(H) \geq \left\lceil \frac{3E + 5}{4} \right\rceil \geq 5.$$

For every vertex v and every integer $0 \leq \theta \leq d$,

$$c^*_\theta(H - v) = c^*_\theta(H) - 1, \qquad \operatorname{ex}(H - v) = E - 1, \qquad \operatorname{gap}(H - v) = 0.$$

With

$$u = \max\left(0, \left\lfloor \frac{2E - 3}{d - 2} \right\rfloor\right),$$

we have

$$k \geq h_-(d) + 2E - 1 + 2u.$$

For every maximal independent set S, put

$$a = |S|, \qquad G = H - S, \qquad \eta = \nu(G) - (k + 1 - a).$$

Then

$$a \geq 2, \qquad \Delta(G) \leq d - 1, \qquad 0 \leq \eta \leq \min\left(E - 1, \left\lfloor \frac{a - 1}{2} \right\rfloor\right),$$

$$\operatorname{ps}(G) = E - 1 - \eta, \qquad c^*_\theta(H) = a + c^*_\theta(G) \quad (1 \leq \theta \leq d - 1), \qquad c^*_d(H) = 1,$$

$$c_{0,0}(H) = a + \nu(G) + \operatorname{ps}(G), \qquad \operatorname{gap}(G) \in \{0, 1\}.$$

These are necessary conditions only. No full degree-six or degree-eight closure is asserted.

Proof. Theorem 4.16(II) supplies an induced totally critical closed even-degree lump $F \subseteq H$ that violates T_- and has gap one. Induced minimality forces $F = H$. Thus H is a closed d-lump of gap one and

$$\operatorname{tr}(H) + E \geq h_-(d) + 1.$$

Factor-criticality gives $\nu(H) = k$.

We exclude $E \leq 3$. For $d = 6$, Theorem 4.12 gives (S) at excess at most two, with bound $4 = h_-(6)$. If $E = 3$, failure requires $\operatorname{tr}(H) \geq 2$, which Theorem 4.14 excludes. For $d = 8$, every proper induced subgraph has gap at most one. Theorem 4.11(III) gives LP-goodness at excess at most two. If $E = 3$, Theorem 4.13 gives $\operatorname{tr}(H) = 1$, so

$$\operatorname{tr}(H) + E = 4 \leq h_-(8) = 5,$$

again excluding failure. Hence $E \geq 4$.

The graph H is not LP-good. The contrapositive of Theorem 4.15 gives $4\alpha(H) > 3E + 4$. Integrality implies

$$\alpha(H) \geq \left\lceil \frac{3E + 5}{4} \right\rceil \geq 5.$$

Put $B = h_-(d)$, equal to four or five. For every proper induced $J \subset H$,

$$h_-(\Delta(J)) \leq h_-(d) = B.$$

Thus its T_- property implies T_B. The graph H is totally critical, has gap one, and $E \geq 4 \geq B - 1$. Theorem 4.9(II) gives all threshold drops through d, the excess decrease, and zero child gap.

Theorem 4.17 applies because H is connected of order $2k+1$, has matching number k, maximum degree d, positive excess, and the required run-plus-excess lower bound. It gives the stated lower bound on k.

Finally fix any maximal independent set S. Theorem 4.10(I) gives

$$a \ge 2, \qquad \Delta(G) \le d-1, \qquad 0 \le \eta \le \left\lfloor \frac{a-1}{2} \right\rfloor.$$

Since

$$\operatorname{tr}(H) + E \ge h_-(d) + 1 > h_-(d-1),$$

part (III) of that theorem gives all exact-profile conclusions in its part (II). In particular,

$$\operatorname{ps}(H) = \eta + \operatorname{ps}(G).$$

The identity $\operatorname{gap}(H) = E - \operatorname{ps}(H) = 1$ gives $\operatorname{ps}(H) = E - 1$. Therefore

$$\operatorname{ps}(G) = E - 1 - \eta.$$

Its nonnegativity gives $\eta \le E - 1$. All assertions hold for every maximal independent set S. □

4.1.11. *Subgraph minimality and critical edges.* We finally strengthen the minimality choice to include edge deletions and identify the resulting certificate drops.

Theorem 4.19 (Proper subgraphs and critical edges)**.** *(I) If some finite simple graph violates T_-, there is a subgraph-minimal T_- failure F. It can be chosen by minimizing $e(F)$, and then $|V(F)|$. Every such F is a totally critical closed d-lump of odd order $2k+1$, for an even integer $4 \le d \le 24$, with*

$$\operatorname{gap}(F) = 1, \qquad \operatorname{tr}(F) + \operatorname{ex}(F) \ge h_-(d) + 1.$$

(II) Every proper subgraph H of such an F with $\operatorname{ex}(H) \ge 4$ is LP-good.

(III) Assume additionally $\operatorname{ex}(F) \ge 4$, and let $e \in E(F)$. For every integer $\theta \ge 0$, put

$$r_\theta(e) = c^*_\theta(F) - c^*_\theta(F-e), \qquad c^*_0 = c_{0,0}.$$

Then

$$r_\theta(e) \in \{0,1\}, \qquad r_\theta(e) = 0 \quad (\theta \ge d+1), \qquad r_d(e) = 1,$$
$$\nu(F-e) = k, \qquad \operatorname{ex}(F-e) = \operatorname{ex}(F) - r_0(e).$$

Moreover,

$$R(e) := \Sigma(F) - \Sigma(F-e) = \sum_{\theta=0}^{d} r_\theta(e), \qquad \operatorname{gap}(F-e) = 2 - R(e),$$

and $R(e) \ge 2$. Thus $F-e$ is LP-good, and for some integer $0 \le \theta \le d-1$,

$$c^*_\theta(F-e) = c^*_\theta(F) - 1.$$

Proof. For (I), choose a violating graph F with the fewest edges and, subject to that, the fewest vertices. If $H \subset F$ is proper, either $e(H) < e(F)$, or the edge numbers agree and H differs only by missing vertices. In the latter case $|V(H)| < |V(F)|$. The choice of F makes every such H satisfy T_-. In particular, every proper induced subgraph satisfies T_-.

Theorem 4.16(II) supplies an induced $F_0 \subseteq F$ that is a totally critical closed d-lump, with even $4 \le d \le 24$,

$$\operatorname{gap}(F_0) = 1, \qquad \operatorname{tr}(F_0) + \operatorname{ex}(F_0) \ge h_-(d) + 1.$$

It violates T_-, so it cannot be a proper induced subgraph of F. Thus $F_0 = F$, proving the structure. The same argument applies to every subgraph-minimal failure.

For (II), every induced subgraph of a proper subgraph $H \subset F$, including H, is itself a proper subgraph of F. All satisfy T_-. The hereditary constant-bound consequence in Setup 4.7(ii) says that positive gap would force $\operatorname{ex}(H) \le 3$. Thus $\operatorname{ex}(H) \ge 4$ implies $\operatorname{gap}(H) \le 0$, which is LP-goodness.

For (III), fix $e = uv$, write $r_\theta = r_\theta(e)$, and put $n = 2k+1$ and $E = \operatorname{ex}(F) \ge 4$. We prove the profile and gap identities in six steps.

Step 1. Every cost drops by zero or one. Monotonicity gives

$$c_{A,B}(F-e) \le c_{A,B}(F), \qquad c^*_\theta(F-e) \le c^*_\theta(F).$$

Conversely, fix $A, B \ge 0$, put $c = c_{A,B}(F-e)$, and let $x, y \ge 0$ with $x+y = c+1$. If $x \ge 1$, there is an $(A, B; x-1, y)$-certificate

$$E(F-e) = Q_R \sqcup Q_B \sqcup P_R \sqcup P_B.$$

Place e in P_R. Removing e from any matching of $P_R \cup \{e\}$ gives a matching of P_R, so

$$\nu(P_R \cup \{e\}) \le \nu(P_R) + 1 \le x.$$

This is an $(A, B; x, y)$-certificate of F. If $x = 0$, then $y = c+1 \ge 1$; place e in P_B of an $(A, B; 0, y-1)$-certificate. Thus every allocation of sum $c+1$ is certified, and

$$c_{A,B}(F) \le c_{A,B}(F-e) + 1.$$

Taking maxima over $A + B = \theta$ gives $r_\theta \in \{0, 1\}$, including $\theta = 0$.

Step 2. Every threshold above d vanishes. Since both maximum degrees are at most d,

$$c^*_\theta(F) = c^*_\theta(F-e) = 0 \qquad (\theta \ge d+1).$$

Hence $r_\theta = 0$ there.

Step 3. The top threshold drops by one. The earlier closed-lump bound gives $c^*_d(F) \le 1$. If $c_{1,d-1}(F) = 0$, the zero-budget certificate has $\nu(P_R) = \nu(P_B) = 0$, so both matching-pool edge sets are empty. Thus

$$E(F) = Q_R \sqcup Q_B, \qquad \Delta(Q_R) \le 1, \qquad \Delta(Q_B) \le d-1.$$

Every vertex has total degree d, forcing exactly one Q_R-edge at each vertex. This would be a perfect matching of the odd-order graph F. Therefore $c_{1,d-1}(F) \ge 1$ and $c^*_d(F) = 1$.

The degree-d vertices of $F - e$ are exactly those other than u, v. Factor-criticality supplies a perfect matching M' of $F - u$. It avoids e and covers every vertex other than u, hence every degree-d vertex of $F - e$. The degree-splitting vanishing rule gives

$$c_{A,B}(F-e) = 0 \quad (A+B=d), \qquad c^*_d(F-e) = 0.$$

Thus $r_d = 1$.

Step 4. The matching number is unchanged. The matching M' has k edges and lies in $F - e$. The odd order bounds its matching number by k. Hence

$$\nu(F-e) = k, \qquad \mathrm{ex}(F-e) = c_{0,0}(F-e) - k = E - r_0.$$

Step 5. The total cost drop determines the child gap. Steps 1 and 2 give

$$\Sigma(F) - \Sigma(F-e) = \sum_{\theta=0}^{d} r_\theta = R(e).$$

Since $e(F-e) = e(F) - 1$ and $\mathrm{gap}(F) = 1$,

$$\mathrm{gap}(F-e) = \Sigma(F) - R(e) - e(F) + 1 = 2 - R(e).$$

Step 6. There is a second drop below the top. Step 3 gives $R(e) \ge 1$. Suppose $R(e) = 1$. Then $r_d = 1$ is the only nonzero term, so $r_0 = 0$. By Step 4,

$$\mathrm{ex}(F-e) = E \ge 4.$$

Part (II) makes the proper subgraph $F-e$ LP-good. But Step 5 gives $\mathrm{gap}(F-e) = 1$, a contradiction. Thus $R(e) \ge 2$, and $\mathrm{gap}(F-e) \le 0$. Finally,

$$R(e) = 1 + \sum_{\theta=0}^{d-1} r_\theta \ge 2$$

forces $r_\theta = 1$ for some $0 \le \theta \le d-1$. □

5. Technical proofs for tails, pools, and tight hosts

In this section, we give the supplied supporting arguments in dependency order. References to pending setups indicate the exact premises whose technical proofs remain to be inserted.

5.1. **Exact pool residuals.**

Proof of Theorem 4.2. We first prove commutation, then establish the exchange argument for an arbitrary core, and finally apply it separately to the two colors.

Step 1. The deletion operators commute on every positive integer multiset.

Fix integers $b \le d$. If no entry is at most b, then D_b does nothing. Any entry deleted by D_d exceeds b, so D_b still does nothing after that deletion.

Otherwise, let x be a largest entry at most b, and let y be a largest entry at most d. If $y > x$, the choice of x implies $y > b$. Deleting one copy of x leaves the maximum y in the d-range unchanged. Deleting y leaves the maximum x in the b-range unchanged. The two orders therefore delete the same values.

If $y = x$, there is no entry in $(b, d]$. Both operators remove a largest entry from the same set of entries at most b. Their two orders remove the same two largest such entries, or all of them if fewer than two exist. Thus

$$D_b D_d = D_d D_b.$$

Every permutation of finitely many capacities is a composition of adjacent swaps. Hence the final multiset is independent of their order. Equal-sized entries are indistinguishable at the multiset level, which also proves independence of choices between equal components.

Step 2. Every feasible pool assignment can be changed into the greedy assignment without making its core requirement harder.

Temporarily label the components of F, including components of equal sizes. A positive-edge target component embedded in $P(c)$ lies in one pool star. Every positive-edge subgraph of a host star contains its designated center, including a unit edge. Two vertex-disjoint target components therefore cannot share one pool star.

A copy of F in $J \sqcup P(c)$ consequently specifies an assignment of some labeled target components to distinct capacity positions that fit their sizes. It also specifies actual disjoint copies in J of the remaining components.

Process the capacity positions in the order $1, \dots, N$, keeping all previous assignments fixed. At position k, let g be a largest remaining component whose size fits c_k. If none exists, an assignment agreeing at earlier positions cannot assign a remaining component to position k, so leave that position unused.

Assume that g exists. If position k already contains g, leave the assignment unchanged. Otherwise, let h be its currently assigned component, if any. The size of h is at most the size of g.

If g is assigned to a later position l, move g to position k. When h exists, move h to position l. This is feasible because

$$|E(h)| \le |E(g)| \le c_l.$$

When h does not exist, leave position l unused. The core requirements are unchanged.

If g is assigned to the core, move it to position k. When h exists, realize h in the old core location of g by retaining the required number of spokes. When h does not exist, discard that core component. Every other core component retains its copy. Thus vertex disjointness is preserved, and earlier pool positions remain fixed.

Induction over the positions gives the greedy assignment together with an actual core copy of all remaining components. Its remaining multiset is exactly $\mathrm{res}_c(F)$.

Starting this argument with a feasible complementary forest $F - I$ itself as the core shows that $\mathrm{res}_c(F)$ embeds in every feasible complement by trimming and discarding components. The greedy assignment is feasible, since each selected component fits its assigned capacity and distinct selections use distinct positions. Hence $\mathrm{res}_c(F)$ is itself a feasible complement.

Step 3. A core copy of the greedy residual extends to a full copy.

Assume that J contains $\mathrm{res}_c(F)$. Assign all other labeled target components by the greedy rule. In every used pool star retain exactly the number of spokes required by its assigned component. Different assigned components use different disjoint pool stars, and all pool stars are disjoint from J. Their union with the given core copy is a copy of the entire F.

This proves both directions in part (ii). If F is empty, every deletion is vacuous and both sides contain the empty forest. If $N = 0$ or all capacities are zero, then $\mathrm{res}_c(F) = F$, and the same conclusion holds.

Step 4. The residual has the stated greatest-common property.

Assume that F is nonempty and $P(c)$ avoids F. No feasible assignment covers every target component, so $\mathrm{res}_c(F)$ is nonempty. By part (ii), it embeds in every feasible complement and is itself a feasible complement.

Any star forest embedded in every feasible complement therefore embeds in $\mathrm{res}_c(F)$. Conversely, $\mathrm{res}_c(F)$ embeds in all those complements. This is exactly the greatest-common characterization of $\mathrm{rem}(F, P(c))$, and it identifies the two forests up to isomorphism. The greedy construction gives attainment as an actual complement. The core equivalence was already proved in Steps 2 and 3.

Step 5. The one-color equivalence gives the prescribed-color arrowing equivalence.

Fix a capacity vector r in part (iv). In any coloring of Q with these red capacities, its red graph is $P(r)$ together with isolated vertices. Its blue graph is $P(d - r)$ together with isolated vertices. Isolated vertices do not affect containment of positive-edge star forests.

Apply part (ii) separately to the two color graphs of any coloring of H. The full graph contains a red F_R exactly when the red core contains $F_R(r)$. It contains a blue F_B exactly when the blue core contains $F_B(r)$. Thus a core coloring avoids both residuals if and only if any extension with pool capacities r avoids both original targets.

If G arrows the original pair, every r with two nonempty residuals forces H to arrow that actual residual pair. Conversely, assume that H arrows every pair with two nonempty residuals. Read the vector r from an arbitrary coloring of G. If either residual is empty, part (ii), applied with an empty core, shows that the pool already contains the corresponding full target. Otherwise, residual arrowing in H gives a full target by the colorwise equivalence.

If no vector has two nonempty residuals, the same reasoning with an empty core shows that Q arrows the original pair. If a vector does have two nonempty residuals and admits an avoiding core coloring, choose any r_i red spokes in pool star i. The resulting full coloring avoids both targets. Every unit component remains an actual assigned or residual component throughout. □

5.2. **Tail persistence.**

Proof of Lemma 4.3. The specified edge categories partition all edges incident with X. Thus the extension is defined and retains every color on $G - X$.

Every red edge incident with B has its other endpoint in A, since all edges within B and from B to $G - X$ are blue. Consider a connected red subgraph containing an edge and meeting X. If it meets A, the required assertion holds. Otherwise, a vertex of B in that nontrivial connected subgraph is incident with a red edge of the subgraph and therefore has a neighbor in A. Hence the subgraph contains a vertex of A.

Every blue edge incident with A has its other endpoint in B, since all edges within A and from A to $G - X$ are red. The same connectedness argument proves that every nontrivial connected

blue subgraph meeting X contains a vertex of B. Neither argument restricts the colors of the edges between A and B.

Suppose that the extension contains a red F_1. Its components are vertex-disjoint. Each component meeting X contains a distinct vertex of A, so at most α components meet X. Let $r \le \alpha$ be their number, and list the indices of the untouched components as

$$i_1 < \cdots < i_{s-r}.$$

At most r indices have been omitted. Hence

$$i_k \le k + r \le k + \alpha \qquad (1 \le k \le s - \alpha).$$

Monotonicity of the original list gives

$$n_{i_k} \ge n_{\alpha+k}.$$

Keep the first $s - \alpha$ untouched components and trim the kth to exactly $n_{\alpha+k}$ edges. This gives a forbidden red F_1^α in $G - X$.

Suppose instead that the extension contains a blue F_2. At most β components meet X. If their number is r_B and the untouched indices are

$$j_1 < \cdots < j_{t-r_B},$$

then

$$j_k \le k + r_B \le k + \beta \qquad (1 \le k \le t - \beta).$$

Therefore

$$m_{j_k} \ge m_{\beta+k}.$$

Trimming the first $t - \beta$ untouched stars gives a forbidden blue F_2^β in the core.

Both full targets are absent. An avoiding coloring of the suffix pair would consequently contradict arrowing of the full pair, which proves tail persistence. No argument assumed that a deleted vertex was a star center. Positivity ensures that every target component used in the connectedness argument contains an edge. □

5.3. **The potential bound.**

Proof of Theorem 4.4. Put

$$\Phi(J) = e(J) - \nu(J).$$

We first describe the vertex-deletion property used by the argument. For every vertex v,

$$\nu(J) - \nu(J - v) \in \{0, 1\},$$

because restricting a maximum matching loses at most its edge incident with v. Hence

$$\Phi(J) - \Phi(J - v) = d_J(v) - \big(\nu(J) - \nu(J - v)\big). \tag{5.1}$$

Also $\Phi(J) \ge 0$, since every matching uses edges of J.

Let $A, B \in \mathbb{Z}_{\ge 0}$, and assume that J has no red-blue degree coloring with capacities A, B. Set $D = A + B$. Subject to the pending split in Setup 5.1, there is a maximum-degree vertex v with

$$\Phi(J) - \Phi(J - v) \ge D. \tag{5.2}$$

To see the mechanism, an edgeless graph admits the required coloring, so J has an edge. If $\Delta(J) \ge D + 1$, any maximum-degree vertex works by (5.1). If $\Delta(J) < D$, the empty matching covers every degree-D vertex, since none exists, and Setup 5.1 would give the forbidden coloring. The remaining case is $\Delta(J) = D$. Fix a maximum matching. If it covers every degree-D vertex, the same setup gives the forbidden coloring. Otherwise, choose a degree-D vertex missed by that matching. The matching survives its deletion, so the matching number is preserved and (5.1) gives (5.2).

Each $\overline{\ell}_k$ is well-defined. Choose γ with $s_\gamma + t_\gamma - 1 = N$. The valid sums $i + j$ for this pair run through all integers from 2 to $N + 1$, because the two index sets are nonempty consecutive integer intervals. Thus every finite maximum defining $\overline{\ell}_k$ has a candidate and is attained.

We recursively delete N distinct vertices. Set

$$X_0 = \varnothing, \qquad H_0 = G.$$

At stage q, where $0 \le q < N$, let X_q be the set of previously deleted vertices and let $H_q = G - X_q$. Choose γ and valid indices i, j satisfying

$$i + j = q + 2, \qquad n_i^\gamma + m_j^\gamma - 1 = \overline{\ell}_{q+2}.$$

The cuts $i-1$ and $j-1$ are legal and total q. Lemma 4.3, applied to the original pair and the entire set X_q, gives

$$H_q \to \left(\bigsqcup_{a=i}^{s_\gamma} K_{1,n_a^\gamma}, \bigsqcup_{b=j}^{t_\gamma} K_{1,m_b^\gamma} \right).$$

For $q = 0$, this is the original arrowing assumption.

Both tails are nonempty. Put

$$A = n_i^\gamma - 1, \qquad B = m_j^\gamma - 1.$$

A red-blue coloring of H_q with red maximum degree at most A and blue maximum degree at most B would contain neither leading star. It would therefore avoid both entire tails, contradicting their arrowing. By (5.2), a maximum-degree vertex v_q satisfies

$$\Phi(H_q) - \Phi(H_q - v_q) \ge A + B = \overline{\ell}_{q+2} - 1.$$

The impossibility of the degree coloring also ensures that H_q has an edge. Set

$$X_{q+1} = X_q \cup \{v_q\}, \qquad H_{q+1} = H_q - v_q.$$

The selected vertex belongs to H_q, so the deleted vertices are distinct. At each stage, the argument uses tail persistence for the full deletion set X_q. It does not require one fixed pair of tails to work at every stage.

Summing the N inequalities gives

$$\Phi(G) - \Phi(H_N) \ge \sum_{k=2}^{N+1} (\overline{\ell}_k - 1).$$

Since $\Phi(H_N) \ge 0$,

$$e(G) - \nu(G) \ge \sum_{k=2}^{N+1} (\overline{\ell}_k - 1).$$

For a single pair, $N = s + t - 1 = M$, and the sum on the right is $L - M$. Rearranging yields

$$e(G) \ge L + \nu(G) - M.$$

If $\nu(G) \ge M$, then $e(G) \ge L$. If $e(G) < L$, integrality gives

$$1 \le L - e(G) \le M - \nu(G),$$

so $\nu(G) \le M - 1 = s + t - 2$. Only positivity and ordering were used; no restriction on repetitions, gaps, parities, or units enters the argument. □

5.4. **A fixed matching along a tight chain.**

Proof of Theorem 4.5. For an arbitrary vertex set X of size $q \le M$, write

$$H_X = G - X, \qquad Q_X = \{xy \in Q : x, y \notin X\}, \qquad J_X = (G - Q) - X.$$

We require the all-vertex residual inequality

$$\Phi(H_X) \ge P_q. \tag{5.3}$$

For $q = M$, this is $\Phi(H_X) \ge 0$. For $q < M$, Lemma 4.3 makes H_X arrow every member of $\mathcal{A}_q$. The envelope of their local diagonal a is ℓ_{q+a}. Indeed, a tail diagonal corresponds to original indices with $i + j = q + a$. Conversely, for such indices choose

$$\alpha = \max(0, q - (j - 1)), \qquad \beta = q - \alpha.$$

Then $\alpha \le i - 1$, $\beta \le j - 1$, and the cuts are legal. The original entry therefore occurs on local diagonal a of a member of $\mathcal{A}_q$. Applying Theorem 4.4 to this family proves (5.3).

The edges of H_X are the disjoint union of Q_X and $E(J_X)$. Since Q_X is a matching,

$$\begin{aligned} e(J_X) &= e(H_X) - |Q_X| \\ &= \Phi(H_X) + \nu(H_X) - |Q_X| \\ &\ge P_q. \end{aligned}$$

Both differences

$$\Phi(H_X) - P_q, \qquad \nu(H_X) - |Q_X|$$

are nonnegative. Consequently,

$$e(J_X) = P_q \implies \Phi(H_X) = P_q, \qquad |Q_X| = \nu(H_X). \tag{5.4}$$

At $X = \varnothing$, maximality of the size of Q gives

$$e(J_\varnothing) = e(G) - \nu(G) = L - M = P_0.$$

Assume inductively that $|X| = q < M$ and $e(J_X) = P_q$. By (5.4), Q_X is maximum in H_X and $\Phi(H_X) = P_q$.

For every $v \in V(H_X)$, apply the general inequality to $X \cup \{v\}$. It gives

$$e(J_X) - d_{J_X}(v) \ge P_{q+1}.$$

Thus

$$d_{J_X}(v) \le D := P_q - P_{q+1} = \ell_{q+2} - 1.$$

Choose valid indices i, j attaining ℓ_{q+2} with $i + j = q + 2$. By Lemma 4.3, H_X arrows the nonempty actual suffix pair starting at i, j. An edgeless graph avoids both positive-edge targets, so H_X has an edge and a nonempty vertex set.

If $D = 0$, the degree upper bound gives $\Delta(J_X) = 0 = D$. Assume that $D \ge 1$, and suppose for a contradiction that $\Delta(J_X) < D$. Every degree in H_X exceeds the corresponding degree in J_X by zero or one, according to whether Q_X covers that vertex. Hence

$$\Delta(H_X) \le D,$$

and every degree-D vertex of H_X is covered by Q_X.

Apply the pending implication in Setup 5.1 with

$$A = n_i - 1, \qquad B = m_j - 1, \qquad A + B = D.$$

The resulting coloring has no red leading star and no blue leading star of the arrowed suffixes. Thus it avoids both suffix targets, a contradiction. Therefore

$$\Delta(J_X) = D.$$

Any maximum-degree vertex v of J_X has degree D, and deleting it gives

$$e(J_{X\cup\{v\}}) = P_q - D = P_{q+1}.$$

The implication (5.4) simultaneously shows that the new residual potential is P_{q+1} and that the surviving restriction of Q is maximum. This proves the induction through $q = M$ for every tie choice. The positive-edge suffix argument establishes the required vertex existence whenever $q < M$.

The matching Q_{q+1} is obtained from Q_q by deleting its unique edge incident with v_q if that vertex is covered, and by deleting no edge otherwise. Since both restrictions are maximum,

$$\nu(H_q) - \nu(H_{q+1}) = \epsilon_q.$$

The edge partition of H_q also gives

$$d_{H_q}(v_q) = D + \epsilon_q.$$

Every vertex of H_q has degree at most $D+1$. If some maximum-degree vertex of J_q is covered by Q_q, that vertex has degree $D+1$ in H_q, which is maximum. If none is covered, a degree-$D+1$ vertex of H_q would be covered and have degree D in J_q, a contradiction. In that case $\Delta(H_q) \le D$, and a selected degree-D vertex of J_q attains the maximum in H_q. This proves the tie-preference assertion while keeping the original matching fixed.

At the final stage,

$$e(J_M) = P_M = 0.$$

Thus H_M has exactly the edges of Q_M, with every other vertex isolated. Telescoping gives

$$\nu(G) = |Q| = |Q_M| + \sum_{q=0}^{M-1} \epsilon_q.$$

Subtracting this equality from M proves

$$M - \nu(G) = \big|\{q : \epsilon_q = 0\}\big| - |Q_M|.$$

This is an accounting identity. It imposes no sign on its right-hand side. □

5.5. **Degree-two residuals.**

Proof of Theorem 4.7. By Lemma 4.3, H arrows every nonempty actual suffix pair with legal cuts a, b satisfying $a + b = q$. Put

$$a_0 = \max(0, q - t + 1), \qquad b_0 = \max(0, q - s + 1).$$

These are the minimum feasible values of a and b, respectively. The condition $b \le t - 1$ gives $a \ge q - t + 1$. Conversely, $a = a_0$ permits $b = q - a_0$ with $0 \le b \le t - 1$. Interchanging the roles of the lists proves the assertion for b_0.

Since $q \le s + t - 2$,

$$a_0 \le s - 1, \qquad b_0 \le t - 1, \qquad a_0 + b_0 \le q.$$

Consider the feasible pair with $a = a_0$. In the all-red coloring of H, no nonempty blue target occurs. Hence H contains the corresponding red target. Its largest star has size n_{a_0+1}, so $\Delta(H) \le 2$ gives

$$n_{a_0+1} \le 2.$$

All later entries are positive and no larger. Thus

$$n_{a_0+1}, \dots, n_s \in \{1, 2\}.$$

The feasible pair with $b = b_0$ and the all-blue coloring gives

$$m_{b_0+1}, \dots, m_t \in \{1, 2\}.$$

Put

$$S = s - a_0, \qquad T = t - b_0, \qquad q' = q - a_0 - b_0.$$

Then $S, T \geq 1$ and $0 \leq q' \leq S + T - 2$. Every feasible original cut has $a \geq a_0$ and $b \geq b_0$. Writing

$$a' = a - a_0, \qquad b' = b - b_0$$

gives

$$a' + b' = q', \qquad 0 \leq a' < S, \qquad 0 \leq b' < T.$$

Conversely, every such pair gives a feasible original cut. The tail family arrowed by H is therefore precisely the natural q'-deletion family of these two binary suffix lists.

The envelope-attainment input in Setup 4.6 gives one member whose profile attains the coordinatewise envelope of the family. That envelope is

$$(\ell_{q+2}, \ldots, \ell_{M+1}).$$

For a direct verification, a local diagonal a of an original tail corresponds to full indices i, j with $i + j = q + a$. Conversely, for every such pair choose

$$a_{\mathrm{cut}} = \max(0, q - (j-1)), \qquad b_{\mathrm{cut}} = q - a_{\mathrm{cut}}.$$

These satisfy

$$a_{\mathrm{cut}} \leq i - 1, \qquad b_{\mathrm{cut}} \leq j - 1,$$

and are feasible cuts totaling q. Thus the original entry occurs on local diagonal a of that tail. The envelope is exactly ℓ_{q+a}.

The selected member consists of nonempty binary forests and is arrowed by H. The binary lower-bound input in Setup 4.6 gives

$$e(H) \geq \sum_{k=q+2}^{M+1} \ell_k.$$

Adding the assumed payment for the deleted edges gives $e(G) \geq L$. No tail is exhausted because $q < M$. At $q = M$, the corresponding residual inequality would only be $e(H) \geq 0$.

For the final assertion, consider a zero-surplus maximum-degree chain and put

$$\epsilon_i = \nu(H_i) - \nu(H_{i+1}) \in \{0, 1\}.$$

The potential identity (5.1) gives

$$d_{H_i}(v_i) = \ell_{i+2} - 1 + \epsilon_i.$$

The terminal potential is zero, so its edge count equals its matching number. Telescoping the matching losses gives

$$M - \nu(G) = \big|\{i : \epsilon_i = 0\}\big| - \nu(H_M).$$

Since $M - \nu(G) > 0$, at least one ϵ_i is zero.

Let q be the first such index. For every $i < q$, $\epsilon_i = 1$, and the deleted degree is exactly ℓ_{i+2}. The edge sets removed by successive vertex deletions are disjoint, so

$$e(G) - e(H_q) = \sum_{k=2}^{q+1} \ell_k.$$

If $\Delta(H_q) \leq 2$, the residual inequality just proved would give $e(G) \geq L$. But

$$e(G) - \nu(G) = L - M, \qquad \nu(G) < M$$

give $e(G) < L$. Therefore $\Delta(H_q) \geq 3$. At this step $\epsilon_q = 0$, so

$$\Delta(H_q) = \ell_{q+2} - 1 \geq 3,$$

as required. □

5.6. **Canonical prefix pools and exact normalization.**

Proof of Theorem 4.8. We first prove the constructive pool equivalence. We then describe the pullback across an actual vertex set and finally apply the tight matching chain.

Step 1. An avoiding pool coloring selects a nonempty actual suffix pair avoided by the unchanged core coloring.

Fix an avoiding coloring of $H \sqcup T_q$. Scan the q pool stars in their displayed order. Before star h, where $1 \le h \le q$, assume that α red heads and β blue heads have been assigned to earlier pool stars, with

$$\alpha + \beta = h - 1.$$

The assigned components use spokes of their chosen color and lie in distinct pool stars. Neither original target has been exhausted, because its assigned components would then form a full monochromatic copy in the pool.

Thus the current largest remaining sizes $n_{\alpha+1}$ and $m_{\beta+1}$ exist. Their index sum is $h+1$, and

$$\ell_{h+1} \ge n_{\alpha+1} + m_{\beta+1} - 1.$$

If star h had fewer than $n_{\alpha+1}$ red spokes and fewer than $m_{\beta+1}$ blue spokes, its capacity would be at most $n_{\alpha+1} + m_{\beta+1} - 2$, a contradiction.

Choose a color with enough spokes and assign its current largest remaining component to this star. Increase that color's prefix count by one. Avoidance again prevents target exhaustion.

After all q stars have been scanned,

$$\alpha + \beta = q, \qquad \alpha < s, \qquad \beta < t.$$

If the red core contained the suffix starting at $\alpha + 1$, it would combine with the assigned red pool components to form the full red target. The required disjointness holds because different pool stars are disjoint and the entire pool is disjoint from the core. Thus that red suffix is absent. The corresponding blue suffix is absent for the same disjoint-union reason. The unchanged core coloring avoids the resulting member of $\mathcal{A}_q$.

Step 2. An avoiding coloring of any actual suffix pair extends using monochromatic pool stars.

Assume that the core coloring avoids the pair with legal cuts α, β totaling q. Choose any sequence of q color labels with α red labels and β blue labels. Make the successive pool stars entirely the designated colors.

At a step with preceding counts α_0, β_0, the index-sum calculation gives

$$\ell_{\alpha_0+\beta_0+2} \ge n_{\alpha_0+1} + m_{\beta_0+1} - 1.$$

The capacity is therefore at least the current largest remaining size in the designated color. Neither target is exhausted, since the final counts are strictly below the respective component counts.

Under the exact residual operation of Theorem 4.2, this monochromatic star deletes exactly the largest remaining component in its designated color and none in the other color. After all stars have been processed, the residual pair is precisely the chosen actual suffix pair. The full-copy equivalence proves avoidance of both full original targets.

Applying this construction to the labels selected in Step 1 makes every pool star monochromatic while retaining all core colors. The two implications prove the arrowing equivalence. If $q = 0$, the pool is empty and the sole suffix pair is the original pair.

Step 3. The pool replacement has an explicit avoiding-color pullback to the original graph.

Let $X \subseteq V(G)$ have size $q < M$. If G arrows the original pair, Lemma 4.3 gives arrowing of every member of $\mathcal{A}_q$ by $G - X$. The pool equivalence proves that $G^\star_X$ arrows the original pair.

Start instead with an avoiding coloring of $G^\star_X$. Step 1 retains its core colors and gives legal cuts α, β for which these core colors avoid the actual suffixes. Partition

$$X = X_R \sqcup X_B, \qquad |X_R| = \alpha, \qquad |X_B| = \beta.$$

Keep every core color. Color every original edge having an endpoint in X_R red. Color every remaining original edge having an endpoint in X_B blue. These prescriptions partition all edges outside the core, with red priority on cross edges.

Suppose that a full red target occurs. At most α of its vertex-disjoint components meet X_R. Discard those components. Every remaining positive red component avoids X_B, because every edge from X_B to a vertex outside X_R is blue. Hence all remaining components lie in the core.

Deleting at most α components from an ordered star forest leaves a forest containing the suffix after its α largest components. To verify this, order the surviving original indices; the jth is at most $\alpha + j$, so its size is at least the corresponding suffix size. Trimming surplus spokes gives the forbidden red core suffix.

No blue component can meet X_R, since all edges incident with X_R are red. Discarding the at most β blue components meeting X_B therefore leaves a core forest containing the forbidden blue suffix. Both alleged full copies are impossible. The pullback includes unit stars.

Step 4. The replacement has the stated exact edge and matching counts.

The pool T_q is the union of q positive-edge stars. Therefore

$$e(T_q) = \sum_{j=2}^{q+1} \ell_j, \qquad \nu(T_q) = q.$$

Edge count and matching number are additive over disjoint components. It follows that

$$e(G_X^\star) = e(G) - c_X + \sum_{j=2}^{q+1} \ell_j, \qquad \nu(G_X^\star) = \nu(G) - h_X + q.$$

Step 5. A tight host admits successive full-degree deletions preserving the prescribed matching.

Fix the maximum matching Q in the final assertion. At a stage h of the chain permitted by Theorem 4.5, write

$$H_h = G - X_h, \qquad Q_h = Q|_{H_h}, \qquad J_h = H_h - Q_h.$$

That theorem gives

$$Q_h \text{ maximum in } H_h, \qquad e(J_h) = P_h, \qquad \Delta(J_h) = \ell_{h+2} - 1 \quad (h < M).$$

A covered vertex has its H_h-degree one larger than its J_h-degree, and an uncovered vertex has the same degree in both graphs. Hence every degree in H_h is at most ℓ_{h+2}.

If a vertex has degree exactly ℓ_{h+2} in H_h, it must be covered by Q_h, since an uncovered vertex of that degree would exceed $\Delta(J_h)$. Its degree in J_h is $\ell_{h+2} - 1$, so it is an allowed greedy choice. Delete any such vertex whenever one exists. Each such step removes exactly ℓ_{h+2} residual edges, loses one matching edge, and keeps the restriction of Q maximum.

The process stops at an index $q < \nu(G)$. Otherwise, after $\nu(G)$ steps the residual matching number would be zero, so the residual graph would be edgeless. But $\nu(G) < M$, and Lemma 4.3 makes that residual arrow a nonempty actual suffix pair, which an edgeless graph cannot do.

At the stopping index q, no vertex of H_q has degree ℓ_{q+2}. Thus

$$\Delta(H_q) \le \ell_{q+2} - 1.$$

Since J_q is a subgraph of H_q and $\Delta(J_q) = \ell_{q+2} - 1$, equality holds. Summing the full-degree edge losses and the matching losses gives

$$e(H_q) = e(G) - \sum_{j=2}^{q+1} \ell_j, \qquad \nu(H_q) = \nu(G) - q.$$

The surviving restriction of Q is maximum.

The potential equality and $\nu(G) < M$ imply $e(G) < L$. If $\Delta(H_q) \le 2$, Theorem 4.7, applied to the original deletion set X_q, would pay the full residual sum. Adding the already paid prefix would give $e(G) \ge L$, a contradiction. Therefore

$$\Delta(H_q) = \ell_{q+2} - 1 \ge 3.$$

In Step 4, substitute

$$c_X = \sum_{j=2}^{q+1} \ell_j, \qquad h_X = q.$$

The replacement has the same edge count and matching number as G, and therefore the same potential. Its arrowing and pullback were proved for every X. At each intermediate stage, the replacement consists of the surviving core and the already extracted stars. These replacements realize the successive whole-star detachments and correspond exactly to the full actual-tail families. No additional matching-rank conclusion is used. □

5.7. **Proper subgraphs at degree four.**

Proof of Theorem 6.2. Fix a proper subgraph J of Q. Its maximum degree is at most four. By the pending sharp-profile input in Setup 6.1, its sharp profile agrees with its generic profile: the restricted sharp thresholds begin at six, and all costs above $\Delta(J)$ vanish by Setup 3.4. The sharp-profile gap bound gives

$$\operatorname{gap}(J) \le 1.$$

If the gap is positive, integrality gives $\operatorname{gap}(J) = 1$. Also

$$\operatorname{gap}(J) = \operatorname{ex}(J) - \operatorname{ps}(J), \qquad \operatorname{ps}(J) \ge 0,$$

so $\operatorname{ex}(J) \ge 1$. Thus $c_{0,0}(J) \ge 1$ and $\operatorname{tr}(J)$ is defined.

Suppose that

$$\operatorname{tr}(J) + \operatorname{ex}(J) \ge 4.$$

Setup 6.1 gives a nonempty induced subgraph F of J which is a closed 4-lump. The threshold excludes its C_5 and C_7 alternatives.

The graph F is a 4-regular subgraph of Q, whether or not J is induced in Q. For every $u \in V(F)$, its four incident F-edges are Q-edges, and $d_Q(u) = 4$. These are therefore all the Q-edges incident with u. No edge of Q joins $V(F)$ to its complement.

Since Q is connected and F is nonempty, $V(F) = V(Q)$. The incident-edge equality then also gives $E(F) = E(Q)$. The containments

$$F \subseteq J \subseteq Q$$

force $J = Q$, contrary to properness. Hence

$$\operatorname{tr}(J) + \operatorname{ex}(J) < 4,$$

and integrality gives the bound three.

Assume that a proper J has $\operatorname{ex}(J) \ge 3$ and is not LP-good. Its integer gap is at least one, so the bound just proved applies. Setup 3.4 gives $\operatorname{tr}(J) \ge 1$, since positive excess implies $c_{0,0}(J) \ge 1$. Consequently,

$$\operatorname{tr}(J) + \operatorname{ex}(J) \ge 4,$$

a contradiction. This proves LP-goodness at excess at least three.

For a proper J of nonpositive gap, T_- is vacuous. For positive gap, the proved bound is

$$\operatorname{gap}(J) = 1, \qquad \operatorname{tr}(J) + \operatorname{ex}(J) \le 3 < h_-(\Delta(J)).$$

Thus every proper subgraph satisfies T_-.

Finally, a closed 4-lump Q with gap one and $\operatorname{tr}(Q) + \operatorname{ex}(Q) \ge 5$ violates T_-, since $h_-(4) = 4$. Every proper subgraph satisfies the property by the preceding argument. □

6. Assembly of the five-degree reduction

The goal of this section is to assemble the chapter inputs into the universal criterion and the actual-host obstruction theorems. The pending premises remain those explicitly identified in Sections 3–7.

6.1. **The exact gap-one graph input.**

Proof of Theorem 7.6. Setup 7.1 gives the equivalence of the two universal assertions with exactly the displayed floor and ceiling thresholds. It also gives the inclusive LP-goodness consequence. We use that equivalence only for universal assertions.

Assume that every finite simple graph satisfies T_-. Let F satisfy the hypotheses in B_5. Its gap is one, so

$$\mathrm{tr}(F) + \mathrm{ex}(F) \le h_-(\Delta(F)).$$

Since F is d-regular and d is even,

$$h_-(\Delta(F)) = h_-(d) = \max(4, d/2 + 1).$$

Thus B_5 holds.

Conversely, assume B_5, and suppose that the universal T_- assertion fails. Let J violate T_-. By Setup 7.2, J contains an induced-minimal failure F which is a totally critical closed d-lump with

$$d \in \{4, 6, 8, 10, 12\}, \qquad \mathrm{gap}(F) = 1, \qquad \mathrm{tr}(F) + \mathrm{ex}(F) \ge h_-(d) + 1.$$

The degree d is even. Hence

$$\mathrm{tr}(F) + \mathrm{ex}(F) \ge \max(4, d/2 + 1) + 1,$$

contradicting B_5. This proves all three equivalences.

Substitution of $d = 4, 6, 8, 10, 12$ into $\max(4, d/2+1)$ gives $4, 4, 5, 6, 7$. The inclusive LP-goodness conclusion follows from Setup 7.1. No five-degree inequality or emptiness assertion has been assumed outside the indicated conditional implications. □

6.2. **Induced obstructions in unrestricted hosts.**

Theorem 6.1 (Unrestricted induced five-degree obstruction)**.** *Use Setup 2.2, the generic universal-allocation certificate quantities, and Definitions 3.2 and 3.3. Fix an integer $0 \le q < M$, and put*

$$K = M - q, \qquad L_q = \sum_{k=q+2}^{s+t} \ell_k.$$

The family $\mathcal{A}_q$ contains every ordered nonempty actual suffix pair after legal leading cuts totaling q.

(i) *If a finite simple graph H arrows every entire prescribed-color member of $\mathcal{A}_q$ and satisfies $e(H) < L_q$, then H contains an induced subgraph F which is a totally critical closed d-lump with*

$$d \in \{4, 6, 8, 10, 12\}, \qquad d \le \Delta(H), \qquad \mathrm{gap}(F) = 1,$$
$$\mathrm{tr}(F) + \mathrm{ex}(F) \ge \max(4, d/2 + 1) + 1.$$

Every proper induced subgraph of F satisfies T_-.

The witness may be chosen inside an actual connected component C of H satisfying

$$\mathrm{ex}(C) \ge K - \nu(H) \ge 4, \qquad \mathrm{gap}(C) \ge 1.$$

No maximum-degree bound, exact boundary equality, potential bound, row-surplus condition, head restriction, or core-class restriction on H is required. Neither failure of T_- by H itself nor equality $d = \Delta(H)$ is asserted.

(ii) *Assume that every induced totally critical closed d-lump F of H with*

$$d \in \{4, 6, 8, 10, 12\}, \qquad \operatorname{gap}(F) = 1$$

satisfies

$$\operatorname{tr}(F) + \operatorname{ex}(F) \le \max(4, d/2 + 1).$$

If $e(H) < L_q$, then some actual member of $\mathcal{A}_q$ has a coloring of H avoiding both of its entire prescribed targets.

Every such coloring extends over the unchanged canonical q-star pool. It also extends across every specified actual outer q-vertex set with residual H, preserving every H-edge color and avoiding both unchanged full originals. Unit components and multiplicities are retained.

This is an obstruction theorem and a conditional avoidance consequence. It does not establish the universal five-degree bounds or the unrestricted size Ramsey formula.

Proof. Assume the hypotheses in part (i). Apply Setup 7.4 to the literal graph H, the unchanged full positive lists, the legal total q, and the complete actual family $\mathcal{A}_q$. Its assumptions are exactly the strict edge inequality and arrowing of every entire family member. It gives an actual connected component C with

$$\operatorname{ex}(C) \ge K - \nu(H) \ge 4, \qquad \operatorname{gap}(C) \ge 1,$$

and an induced subgraph J of C violating T_-.

Setup 7.2, applied to J, gives an induced-minimal failure F which is a totally critical closed d-lump with

$$d \in \{4, 6, 8, 10, 12\}, \qquad \operatorname{gap}(F) = 1, \qquad \operatorname{tr}(F) + \operatorname{ex}(F) \ge h_-(d) + 1,$$

and whose every proper induced subgraph satisfies T_-. Since d is even,

$$h_-(d) = \max(4, d/2 + 1).$$

A connected component is induced in its host. Since J is induced in C and F is induced in J, the graph F is induced in H and lies inside the stated component C. Every vertex of F has degree d in F and at least that degree in H. Hence $d \le \Delta(H)$. This gives no equality of degrees and does not require H itself to violate T_-.

For part (ii), assume the local bound and $e(H) < L_q$. If H arrowed every entire member of $\mathcal{A}_q$, part (i) would give an induced totally critical closed lump satisfying all the hypotheses of the local bound but violating its conclusion. Therefore some actual member is not arrowed. By prescribed-color arrowing, there is an edge coloring avoiding both entire targets of that member.

Let α, β be its legal leading cuts. Theorem 4.8 extends any such coloring over the canonical pool of capacities $\ell_2, \dots, \ell_{q+1}$ while retaining all core colors. For a specified graph G and outer set X with $G - X = H$ and $|X| = q$, partition X into parts of sizes α, β and apply Lemma 4.3. That lemma permits arbitrary cross-edge colors and treats copies meeting X at leaves. When $q = 0$, both extensions are identities. No component of either full target is discarded, and the actual suffix family is unchanged. □

6.3. **Contained proper-subgraph-minimal obstructions.** We now prove Theorem A.

Proof of Theorem A. Assume the below-budget common-host hypotheses in part (i). Theorem 6.1 gives an actual connected component C with

$$\operatorname{ex}(C) \ge K - \nu(H) \ge 4, \qquad \operatorname{gap}(C) \ge 1,$$

and an induced subgraph J_0 of C violating T_-.

The finite graph J_0 has finitely many subgraphs. Among its subgraphs violating T_-, choose F with least edge count and, subject to that, least vertex count. If J is a proper subgraph of F, it has either fewer edges or the same edges and fewer vertices. It is also a subgraph of J_0. The minimizing choice therefore implies that every proper subgraph J satisfies T_-.

The every-minimal-failure assertion in Setup 7.3 gives that F is a totally critical closed even d-lump with

$$\operatorname{gap}(F) = 1, \qquad \operatorname{tr}(F) + \operatorname{ex}(F) \geq h_-(d) + 1.$$

It also gives LP-goodness of every proper subgraph with excess at least four.

Apply Setup 7.2 to this same F. It gives an induced T_--failure F_0 whose degree belongs to $\{4, 6, 8, 10, 12\}$. The graph F_0 cannot be proper, because every proper subgraph of F satisfies T_-. Thus $F_0 = F$, and the degree of F belongs to the asserted five-element set.

Both vertex and edge containment in J_0 and C are retained by the minimization. Hence F is an actual subgraph of H. Its d edges incident with any vertex are also edges of H, so $d \leq \Delta(H)$. Edge deletion during minimization may destroy inducedness. The excess bound remains a bound on C and is not transferred to F.

If $\operatorname{ex}(F) \geq 4$, apply part (iii) of Setup 7.3 to the same graph F. It gives all the displayed row drops, matching identities, excess identities, and gap identities for every edge. In particular, $R(a) \geq 2$ while the top row contributes one, so some row below d drops. This argument does not select one common row for different edges.

If $d = 4$, Theorem 6.2 applies to the connected 4-regular graph F. It concerns every proper subgraph, including spanning edge deletions. It gives the additional bound $\operatorname{tr} + \operatorname{ex} \leq 3$ under positive gap and LP-goodness at excess at least three. This proves part (i).

For part (ii), assume the local subgraph bound and $e(H) < L_q$. If H arrowed every entire member of $\mathcal{A}_q$, part (i) would give a contained graph F with exactly the degree, closed-lump, total-criticality, gap, and proper-subgraph properties quantified by that bound, but with

$$\operatorname{tr}(F) + \operatorname{ex}(F) > h_-(d).$$

This is a contradiction. Hence some legal actual member of $\mathcal{A}_q$ admits a coloring avoiding both complete prescribed targets.

Theorem 4.8 extends every such coloring over the unchanged canonical pool. The chosen member's legal cuts partition any specified actual outer q-vertex set, so Lemma 4.3 extends across that set. It preserves every residual color and excludes full copies meeting the outer set through leaves. At $q = 0$, both extensions are identities. All units and repetitions remain in their original targets. □

6.4. **Vertex and incident-edge deletion identities.**

Theorem 6.2 (Deletion rigidity of every minimal failure)**.** *Use the generic certificate quantities and T_- from Section 3.*

(i) *Let F be any finite simple graph violating T_- such that every proper subgraph satisfies T_-. Then F is a totally critical closed d-lump with*

$$d \in \{4, 6, 8, 10, 12\}, \qquad \operatorname{gap}(F) = 1, \qquad \operatorname{tr}(F) + \operatorname{ex}(F) \geq h_-(d) + 1.$$

Write

$$|V(F)| = 2k + 1, \qquad E = \operatorname{ex}(F).$$

Then

$$E \geq 4 \quad (d = 4), \qquad E \geq 5 \quad (d = 6, 8), \qquad E \geq 6 \quad (d = 10, 12),$$

and $E \neq 7$ if $d = 12$.

Every proper subgraph J with $\operatorname{ex}(J) \geq 4$ is LP-good. If $d = 4$, every proper subgraph with $\operatorname{ex}(J) \geq 3$ is LP-good.

(ii) *For every $v \in V(F)$ and every integer $0 \leq \theta \leq d$,*

$$c^*_\theta(F - v) = c^*_\theta(F) - 1.$$

Moreover,

$$\nu(F - v) = k, \qquad \operatorname{ex}(F - v) = E - 1, \qquad \operatorname{gap}(F - v) = 0.$$

(iii) *For every $v \in V(F)$ and every nonempty subset S of the edges incident with v, let $F - S$ retain all vertices and delete exactly S. Put*

$$r_\theta(S) = c^*_\theta(F) - c^*_\theta(F - S).$$

For every integer $\theta \geq 0$,

$$r_\theta(S) \in \{0, 1\}, \qquad r_\theta(S) = 0 \quad (\theta \geq d + 1), \qquad r_d(S) = 1.$$

Also,

$$\nu(F - S) = k, \qquad \mathrm{ex}(F - S) = E - r_0(S),$$

and $F - S$ is LP-good.

Writing

$$R(S) = \sum_{\theta=0}^{d} r_\theta(S),$$

the exact gap identity gives

$$\mathrm{gap}(F - S) = 1 + |S| - R(S) \leq 0.$$

Consequently,

$$R(S) \geq |S| + 1, \qquad \sum_{\theta=0}^{d-1} r_\theta(S) \geq |S|.$$

Thus every nonempty incident-edge batch strictly lowers at least as many distinct rows below the top as its number of edges. The rows may depend on the entire batch.

(iv) *Use Setup 2.2, and fix an integer $0 \leq q < M$. If a finite simple graph H arrows every entire prescribed-color member of $\mathcal{A}_q$ and satisfies $e(H) < L_q$, then some actual connected component C of H contains a subgraph F satisfying all of parts* (i)–(iii)*, and*

$$\mathrm{ex}(C) \geq K - \nu(H) \geq 4, \qquad \mathrm{gap}(C) \geq 1.$$

In particular, $d \leq \Delta(H)$.

The contained F need not be induced. No inequality $\mathrm{ex}(F) \geq K - \nu(H)$ is asserted. If H contains no such subgraph, then $e(H) < L_q$ guarantees an avoiding coloring of both entire targets of some actual member of $\mathcal{A}_q$. Every such coloring extends over the unchanged canonical q-star pool and across any specified actual outer q-vertex set, preserving every residual edge color and avoiding both unchanged full originals.

The remaining five-degree bounds and the unrestricted Ramsey formula are not asserted.

Proof. We prove the structural and excess restrictions first, then use them to make the deletion estimates exact.

Step 1. Every proper-subgraph-minimal failure has the stated degree and excess restrictions.

Apply Setup 7.3 to F. Its assertion applies to every graph with the displayed minimality, not merely to a globally least-edge failure. Thus F is a totally critical closed even d-lump with gap one and

$$\mathrm{tr}(F) + \mathrm{ex}(F) \geq h_-(d) + 1.$$

Setup 7.2 gives an induced failure F_0 in F with degree in $\{4, 6, 8, 10, 12\}$. If F_0 were proper, it would satisfy T_- by the hypothesis on F, a contradiction. Hence $F_0 = F$, and $d \in \{4, 6, 8, 10, 12\}$. Every proper induced subgraph of F satisfies T_-, so F is also induced-minimal. Its excess E is a nonnegative integer by Setup 3.4.

If $d = 4$, the possibility $E \leq 2$ is excluded by the first input in Setup 6.3, since that input gives the T_- bound at threshold four. If $E = 3$, the same setup gives $\mathrm{tr}(F) \leq 1$, so

$$\mathrm{tr}(F) + E \leq 4,$$

again contradicting failure. Therefore $E \geq 4$.

For $d = 6$, the induced-minimal degree-six exclusion gives $E \geq 5$. For $d = 8$, the degree-eight exclusion gives $E \geq 5$. For $d = 10$ or 12, the corresponding exclusion gives $E \geq 6$. The degree-twelve excess-seven exclusion applies because every proper induced subgraph satisfies T_-. It gives $E \neq 7$ at degree twelve.

The excess-four LP-goodness statement for proper subgraphs is part (ii) of Setup 7.3. At degree four, Theorem 6.2 gives the stronger threshold three.

Step 2. Every vertex deletion lowers each row through d by exactly one.

Fix $v \in V(F)$. Factor-criticality gives a perfect matching of $F - v$ with k edges. Odd order bounds every matching of F by k, so

$$\nu(F - v) = k = \nu(F).$$

For every positive row $1 \leq \theta \leq d$, total criticality gives a strict drop. The one-vertex bound and subgraph monotonicity in Setup 3.4 place the integer drop in $\{0, 1\}$. Therefore each positive-row drop is one.

The same bounds put

$$r_0 := c_{0,0}(F) - c_{0,0}(F - v) \in \{0, 1\}.$$

All profiles vanish above row d. Deleting v removes exactly d edges, and the positive-row drops sum to d. Hence

$$\operatorname{gap}(F - v) = \operatorname{gap}(F) - r_0 = 1 - r_0, \qquad \operatorname{ex}(F - v) = E - r_0.$$

If $r_0 = 0$, the proper subgraph $F - v$ has excess $E \geq 4$. Step 1 makes it LP-good, contrary to its gap one. Thus $r_0 = 1$. Every row $0, \ldots, d$ drops by one, and

$$\operatorname{ex}(F - v) = E - 1, \qquad \operatorname{gap}(F - v) = 0.$$

Step 3. Every incident-edge batch has the stated matching number and row quota.

Fix v and a nonempty incident-edge set S. The subgraph containments

$$F - v \subseteq F - S \subseteq F$$

and Step 2 give

$$c^*_\theta(F) - 1 \leq c^*_\theta(F - S) \leq c^*_\theta(F) \qquad (0 \leq \theta \leq d).$$

Therefore every such drop is zero or one. Both profiles vanish for $\theta \geq d + 1$.

A perfect matching of $F - v$ uses no edge incident with v, so it survives in $F - S$. Matching monotonicity and the odd order give

$$\nu(F - S) = k.$$

It follows that

$$\operatorname{ex}(F - S) = E - r_0(S).$$

For the top drop, take that perfect matching of $F - v$. It covers every vertex except v. In $F - S$, the degree of v is at most $d - 1$, because S is nonempty. Thus the matching covers every remaining degree-d vertex. Setup 3.4 gives

$$c^*_d(F - S) = 0.$$

Applied to $F - v$ and its perfect matching, it also gives $c^*_d(F - v) = 0$. Step 2 therefore gives $c^*_d(F) = 1$, and consequently $r_d(S) = 1$.

If $d \neq 4$, Step 1 gives $E \geq 5$, so

$$\operatorname{ex}(F - S) \geq E - 1 \geq 4.$$

Since $F - S$ is proper, it is LP-good by Step 1. If $d = 4$, the same estimate gives $\operatorname{ex}(F - S) \geq 3$, and the stronger quartic conclusion applies. Thus $F - S$ is LP-good in every case.

Its edge count is $e(F) - |S|$, and its profile sum is $\Sigma(F) - R(S)$. Therefore

$$\begin{aligned}\operatorname{gap}(F - S) &= \Sigma(F) - R(S) - e(F) + |S| \\ &= 1 + |S| - R(S).\end{aligned}$$

Nonpositive gap gives $R(S) \geq |S| + 1$. Subtracting $r_d(S) = 1$ yields

$$\sum_{\theta=0}^{d-1} r_\theta(S) \geq |S|.$$

Every individual drop is zero or one, so the sum counts at least $|S|$ distinct lower rows. It does not assign fixed rows to individual edges.

Step 4. The same conclusions hold for a contained witness in every below-budget actual-family host.

Apply Theorem A(i) to the unchanged original lists, the complete actual family, and H. It supplies an actual connected component C with the stated excess and gap bounds, and a contained subgraph F violating T_- whose every proper subgraph satisfies T_-.

Steps 1–3 apply to that graph without any additional premise $\operatorname{ex}(F) \geq 4$. Degree monotonicity gives $d \leq \Delta(H)$. Theorem A does not assert inducedness or transfer the component excess bound to F, and neither conclusion is needed.

If no such subgraph exists, all-family arrowing with $e(H) < L_q$ is impossible. Some entire actual member therefore has an avoiding coloring. Theorem 4.8 gives the unchanged canonical-pool extension. Lemma 4.3, using that member's exact legal cuts, gives the extension across any specified actual outer set. Both preserve all residual colors and include copies meeting the outer set through leaves. At $q = 0$, both are identities. No external theorem is used in these assembly steps. □

6.5. **The vertex-set row ledger.**

Theorem 6.3 (A conditional ledger for vertex-set deletion)**.** *Use the generic certificate quantities and T_- from Section 3. Let F be any finite simple graph violating T_- such that every proper subgraph satisfies T_-. Put*

$$d = \Delta(F), \qquad k = \nu(F), \qquad E = \operatorname{ex}(F).$$

By Theorem 6.2, F is a connected d-regular factor-critical graph of order $2k + 1$, where

$$d \in \{4, 6, 8, 10, 12\}, \qquad \operatorname{gap}(F) = 1.$$

Put

$$p_d = \begin{cases} 3, & d = 4, \\ 4, & d \neq 4. \end{cases}$$

Let $\varnothing \neq X \subseteq V(F)$, and put

$$j = |X|, \qquad J = F - X, \qquad b = j + \nu(J) - k.$$

Assume the explicit matching-gain inequality

$$b \leq E - p_d.$$

Then $b \geq 1$, the graph J is LP-good, and the following exact ledger holds.

For every integer $0 \leq \theta < d$, put

$$r_\theta = c^*_\theta(F) - c^*_\theta(J), \qquad t_\theta = j - r_\theta.$$

Then

$$r_\theta \in \{1, \ldots, j\}, \qquad t_\theta \in \mathbb{Z}_{\geq 0}.$$

The top costs satisfy

$$c^*_d(F) = 1, \qquad c^*_d(J) = 0,$$

and all costs above d vanish. Moreover,

$$\operatorname{gap}(J) = \sum_{\theta=0}^{d-1} t_\theta - e(F[X]) \le 0,$$

$$\operatorname{ex}(J) = E - b + t_0 \ge p_d.$$

In particular,

$$\sum_{\theta=0}^{d-1} t_\theta \le e(F[X]).$$

If X is independent, every lower row $0 \le \theta < d$ drops exactly j, and

$$\operatorname{gap}(J) = 0, \qquad \operatorname{ex}(J) = E - b.$$

The matching-gain inequality is a hypothesis. No unrestricted assertion for all vertex sets or existence of such a graph F is made.

Proof. Theorem 6.2(i),(ii) gives the structure of F, exact drops of one in every row $0, \dots, d$ under each single vertex deletion, and LP-goodness of each proper subgraph whose excess is at least p_d.

Fix $v \in X$. Factor-criticality supplies a perfect matching N of $F - v$ with k edges. Restrict N to $J = F - X$, deleting its edges incident with $X \setminus \{v\}$. At most $j - 1$ matching edges are removed. Hence

$$\nu(J) \ge k - j + 1,$$

and

$$b = j + \nu(J) - k \ge 1.$$

Successively deleting the j vertices and applying the one-vertex bound gives

$$c^*_\theta(F) \le c^*_\theta(J) + j \qquad (\theta \ge 0).$$

For $0 \le \theta \le d$, subgraph monotonicity and $J \subseteq F - v$ give

$$c^*_\theta(J) \le c^*_\theta(F - v) = c^*_\theta(F) - 1.$$

Thus

$$1 \le c^*_\theta(F) - c^*_\theta(J) \le j \qquad (0 \le \theta \le d).$$

Every displayed t_θ is therefore a nonnegative integer.

The restricted matching also covers every vertex of degree d in J. Indeed, N covers every vertex of F other than v. A remaining vertex w uncovered by its restriction had its matching partner in $X \setminus \{v\}$. It was adjacent to that deleted partner in F, so its degree in J is at most $d - 1$.

The covering-matching vanishing property in Setup 3.4, at total capacity d, gives

$$c^*_d(J) = 0.$$

The perfect matching of $F - v$ also gives $c^*_d(F - v) = 0$. Its known single-vertex drop yields

$$c^*_d(F) = 1.$$

All costs above d vanish by the same setup.

At row zero,

$$c_{0,0}(J) \ge c_{0,0}(F) - j.$$

Using $\nu(J) = k - j + b$, we obtain

$$\begin{aligned} \operatorname{ex}(J) &\ge c_{0,0}(F) - j - (k - j + b) \\ &= E - b \\ &\ge p_d. \end{aligned}$$

The set X is nonempty, so J is proper. Its excess licenses the proper-subgraph LP-goodness conclusion of Theorem 6.2: threshold three at degree four, and threshold four at the other degrees. Thus $\operatorname{gap}(J) \le 0$.

Since F is d-regular, the number of edges removed by deleting X is

$$dj - e(F[X]).$$

The degree sum over X counts each internal edge twice. Subtracting one copy of the internal edge count therefore counts each removed edge exactly once.

The sum of the lower-row drops is

$$\sum_{\theta=0}^{d-1}(j - t_\theta) = dj - \sum_{\theta=0}^{d-1} t_\theta.$$

The top drop is one and all higher drops vanish. Hence

$$\Sigma(F) - \Sigma(J) = dj + 1 - \sum_{\theta=0}^{d-1} t_\theta.$$

Since $\operatorname{gap}(F) = 1$,

$$\begin{aligned}\operatorname{gap}(J) &= 1 - \left(dj + 1 - \sum_{\theta=0}^{d-1} t_\theta\right) + (dj - e(F[X])) \\ &= \sum_{\theta=0}^{d-1} t_\theta - e(F[X]).\end{aligned}$$

Its nonpositivity proves the defect-sum inequality.

At row zero, $r_0 = j - t_0$, so

$$\begin{aligned}\operatorname{ex}(J) &= c_{0,0}(F) - j + t_0 - (k - j + b) \\ &= E - b + t_0.\end{aligned}$$

This is the exact excess identity.

If X is independent, $e(F[X]) = 0$. The sum of the nonnegative integers t_θ is then at most zero, so every t_θ vanishes. All lower rows drop by j, the gap is zero, and the excess is $E - b$. The proof retains the displayed matching-gain hypothesis. □

6.6. **The universal criterion on the restricted class.** We now prove Theorem B.

Proof of Theorem B. Theorem 7.6 gives the equivalence of assertions (i), (ii), and (iii), together with the conditional inclusive LP-goodness conclusion. We prove their equivalence with (iv) and (v).

Assume (i). Every $F \in \mathcal{L}_{\mathrm{rig}}$ is a finite simple graph with gap one. Therefore

$$\operatorname{tr}(F) + \operatorname{ex}(F) \le h_-(\Delta(F)) = h_-(d),$$

which proves (v).

To prove (iv), suppose that some $F \in \mathcal{L}_{\mathrm{rig}}$ exists. Under (i), every induced subgraph of F, including F itself, satisfies T_-. The hereditary input in Setup 7.5 gives

$$\operatorname{ex}(F) \le 3,$$

because $\operatorname{gap}(F) = 1$. But the degree-specific floors in Definition 7.7 give $\operatorname{ex}(F) \ge 4$. This contradiction proves $\mathcal{L}_{\mathrm{rig}} = \varnothing$.

Conversely, suppose that (i) fails. Choose a finite simple graph violating T_-. Among its subgraphs violating T_-, minimize edge count and then vertex count. This finite choice gives a graph F violating T_- whose every proper subgraph satisfies T_-. The choice uses no assumption about the original Ramsey problem.

Theorem 6.2(i)–(iii) gives precisely:

$$d \in \{4, 6, 8, 10, 12\}, \qquad \operatorname{gap}(F) = 1,$$

closed regularity and factor-criticality, every stated excess floor, the degree-twelve exclusion of excess seven, and the vertex and incident-edge conditions (a) and (b) in Definition 7.7. It also gives

$$\operatorname{tr}(F) + \operatorname{ex}(F) \geq h_{-}(d) + 1.$$

For every nonempty X whose matching gain satisfies the hypothesis in condition (c), apply Theorem 6.3 to the same graph F and set X. Proper-subgraph minimality and the explicit gain bound are exactly its assumptions. It gives top cost zero, the lower-row bounds, and the defect-sum inequality. Thus condition (c) holds for every qualifying set.

Consequently $F \in \mathcal{L}_{\mathrm{rig}}$, and it violates the bound in (v). Failure of (i) therefore contradicts either (iv) or (v). This proves equivalence of (i), (iv), and (v). Together with Theorem 7.6, it proves all five equivalences and the conditional LP-goodness conclusion. Since the allowed degrees are even, their thresholds are $4, 4, 5, 6, 7$.

For the local statement, assume that H satisfies the strict edge bound and arrows every entire actual member of $\mathcal{A}_q$. Theorem 6.2(iv) gives a contained subgraph F violating T_{-} whose every proper subgraph satisfies T_{-}, with all the properties in parts (i)–(iii) of that theorem.

This graph satisfies every requirement for $\mathcal{L}_{\mathrm{rig}}$ except condition (c) before that condition is checked. Applying Theorem 6.3 to every qualifying X verifies (c). Hence the actual contained graph belongs to $\mathcal{L}_{\mathrm{rig}}$. Subgraph degree monotonicity gives

$$\Delta(F) \leq \Delta(H).$$

The argument does not require inducedness.

The chosen F violates the run-plus-excess bound. Therefore, if every contained member of $\mathcal{L}_{\mathrm{rig}}$ obeyed that bound, the existence of the chosen F would be a contradiction. Absence of all such members is a sufficient special case.

Negating arrowing of every member gives a coloring avoiding both entire prescribed targets of some actual member of $\mathcal{A}_q$. Theorem 4.8 extends any such coloring over the unchanged canonical pool. Lemma 4.3 extends it across any specified actual outer q-vertex set using the selected member's legal cut allocation. All residual colors are retained, and full copies meeting that set through leaves as well as centers are excluded. When $q = 0$, both extensions are identities.

No universal condition or remaining degree bound has been assumed except within its stated conditional implication. □

7. Concluding remarks

Theorem A supplies an actual contained obstruction with minimality under both vertex and edge deletion. Theorem 6.2 adds exact vertex drops and a lower-row quota for each incident-edge batch. Theorem 6.3 gives the corresponding constraint for a vertex set when its matching gain leaves enough excess. These are the deletion statements available for use in the second companion.

Theorem B shows that restricting the five-degree graph input to $\mathcal{L}_{\mathrm{rig}}$ loses no universal failure. It also retains an actual-host implication: a below-budget graph arrowing the complete actual suffix family contains a member of that class. The universal bounds and the emptiness of $\mathcal{L}_{\mathrm{rig}}$ remain additional conclusions to be established.

Theorem 4.8 identifies canonical pool replacement with the full family of actual suffixes. Lemma 4.3 transfers every resulting avoiding coloring back across an actual outer vertex set. These transfers preserve unit components, multiplicities, and every residual edge color. They are the interfaces through which the intended main paper can use obstruction exclusion to address (1.1).

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